The Assets menu, screen by screen

Every screen behind the **Assets** rail item in SMARTBillHub — what exists, whether a person can get to it, what runs behind each field, and what the product itself says it cannot do. Read from the code at one commit, driven in a browser, and measured — the spacing anchors, the field rhythm, every KPI and every description, read off the running app.

READ AT	COMMIT	READ · MEASURED	ADDRESSES IN SCOPE	SCREENS	WIREFRAMES DRAWN	FIELDS TRACED
2.234.17	**79c659d7**	**09-10 · 09-11**	**7**	**35**	**56**	**137**

NAMING NO DOOR	JOURNEYS WALKED	JOURNEY ISSUES	RULE VIOLATIONS	STATED ABSENCES	DEAD CONTROLS	SPECIFICATIONS
0 of 35	**6 of 6**	**9**	**12**	**49**	**3**	**24**

WITHDRAWN, NOT REUSED	RULES STATED HERE	OUT OF SCOPE	REMEDY DRAWINGS	THINGS A REMEDY DROPS
4	**16**	**35**	**6**	**2**

DESIGN SYSTEM FROM

`styles/tokens.css`, `components/AppShell.css`, `PageHeader.css`, `KPIStrip.css`, `DataTable.css`, `FormModal.css`, `StateFrame.css` — plus `getComputedStyle` on the running app for every spacing anchor

IS THIS THE PRODUCT NOW?

Could not check. `origin/HEAD` does not resolve in this checkout; local HEAD is the tip of `claude/fervent-carson-0uksv0`. Whether that branch is what a customer runs is not determinable from here.

DESCRIPTION — GREY RAIL

A statement about **what the code does today**, including an honest statement that something is missing. Every section of this document runs down a grey rail, and everything on that rail is a description. **It commits nobody to anything.**

COMMITMENT — ACCENT CARD

A **change specification**: a named piece of work, what it waits on, the screens it changes, and what a person would be able to do and see once it is done. A card **interrupts** the rail rather than sitting on it — that is the whole visual argument.

If a thing is not in a commitment card, nobody has undertaken to build it.

Read at one moment · re-verify before acting

Every as-built statement here was true at **2.234.17**, commit **79c659d7**, read on **2026-09-10** and re-measured on **2026-09-11**. Line numbers move; paths and values do not, so each claim is cited by `file:line` and the value is quoted beside it.

Whether the product has moved on: could not check. There is no resolvable default branch in this checkout and no deployment to compare against.

This document has been revised forty-eight times, and every revision is on the record. Section 13 carries one row per instruction, in the order they arrived, including **the five findings an earlier pass made and withdrew** and the three figures it invented — struck and corrected rather than deleted, so the same claim is not made again by somebody reading the same source files.

What goes stale here, and what does not. This warning covers **descriptions** — every sentence about what the code does today, every field table, every route, every measured figure, and every drawing of a screen as it is. **It does not cover rulings.** A decision recorded in this document stays decided until another decision supersedes it, and a question stays open until somebody answers it — **not until the code changes**. Appendix D is therefore not subject to this band at all, and neither is any `BR-` rule quoted here: those are the repository's law, and a stale description of a screen does not weaken one.

Whether the product has moved on: checked again on 2026-09-11, and the answer is still that it cannot be determined from here. `origin/HEAD` does not resolve — *"fatal: ref refs/remotes/origin/HEAD is not a symbolic ref"* — and this checkout knows of exactly **two** remote branches, `claude/fervent-carson-0uksv0` (which local HEAD is the tip of, at `79c659d7`) and `claude/confirm-understanding-3ktufc`. `package.json` still reads **2.234.17**. **Neither of those is a default branch**, so nothing here says which of them a customer runs, and no claim in this document has been quietly refreshed against either.

Scope, stated once: the registry draws **seven** tabs and this document covers **six**. The Imports tab and the five-stage import runner behind it left this document by instruction on 2026-09-11. That is scope, not a defect, and nothing here describes them.

How much of this was seen, how much was measured, and how much was read

The app was built, seeded and driven in a real browser. PostgreSQL provisioned with the repository's own script, 278 migrations applied, `seed_demo` produced the tenant **brazosbend** with **126 assets**, the SPA built with `vite build` and served through the product's own same-origin join. Every screen marked *walked* was opened at 1440×1000 and clicked.

And a third pass measured rather than clicked: `getComputedStyle` over the ten R11 spacing anchors, and the DOM order and pixel gaps of **eighteen form fields** across four screens for R12. Those runs produced three rows in section 06 and two in section 08.

Everything in sections 02–05, 07, 08 and 10 is a **read** of the source at that commit. Section 06 is the only section whose findings could not have been produced without running the product, and it says for every row what was done and what happened.

Description
A statement about what the code does today, including an honest statement that something is missing. It commits nobody to anything.

Commitment
A change specification: a named piece of work, what it waits on, the screens it changes, and what a person could do and see once it is done.

If a thing is not in a commitment card, nobody has undertaken to build it. Four state chips, and only four: EXISTS BEHIND | NEW READ | NEW BUILD | BLOCKED. A screen no specification covers carries NO COMMITMENT — that absence is the finding, not an oversight. **The IDs here are a new series, `AST-B-##`, opened for this document.** The repository's live series are `US-AST-###`, `BR-AST-###`, `OQ-###` and `DEF-###`; borrowing one would make this analysis look like scheduled work, which it is not.

01

The standards every screen here is drawn to

Nothing in this list is new design philosophy. Each item is either a rule this repository has already ratified, a failure it has already paid for, or a constraint of the drawing itself. Where a rule is cited, its authority is named — because a violation is only a violation against a stated authority.

A · The rule set, and its precedence

B14 wins	`04-docs/02-decisions/14-ui-conventions.md` — **38 binding UI rules**. This is the authority. Where it diverges from anything below, B14 wins and the divergence is stamped in B14 itself.
r10 is the source	`05-prototype/r10-2026-08-16/UI-RULES.md` — twenty rules in five groups. B14's *source*, not its authority.
The build file	The repository's own React/Tailwind build skill. Not a third rule set: it is how the two above are obeyed in this tree, and it is what turns a violation from arguable into nameable.
Green proves nothing	Most of B14 is **RATIFIED and UNENFORCED** — no gate in this repository reads rendered output. Section 09 carries an *Enforced?* column for exactly this reason.

B · How the drawings are made

Desktop only	One drawing per screen at a fixed **1440** viewport, scaled to fit its box as a unit and never reflowed. No phone screens, ever. The customer portal is mobile-first in the product and nothing here changes that — there is no portal screen in the Assets menu at all.

customer portal is mobile-first in the product and nothing here changes that — there is no portal screen in the Assets menu at all.

Side rail, never top nav	Every signed-in drawing shows the left rail. It carries **thirteen** items, read from `SHELL_NAV` and confirmed by counting them in the running app.
The product's own literals	Accent **#ec3013** · ground **#f3f2f2** · surface **#eae9e9** · ink **#201e1d** · **Archivo** at 400/600/800 · table body **13.5px**, headers **12px** uppercase · rail **240px**, topbar **56px** · space scale 4/8/12/16/20/24/32 · **radius 0 at every size**. Pinned as literals, not as `var()`, so a picture of a light product does not restyle itself in a dark viewer.
Real copy	Every label, button and message in a drawing is the string the product actually renders, taken from the English catalogue. A reader has to be able to tell whether the wording is right, and nobody can do that from a grey rectangle.
As-built, not as-should-be	Where a screen breaks a rule, the drawing shows the broken control. A tidied as-built wireframe would delete every finding in section 09 from the picture.

C · Product rules these screens are judged against

Tenant scoping	No tenant table is touched without `app.tenant_id`; RLS `FORCE` is on. Verified incidentally in this pass: querying `assets` without the session GUC returns **0 rows** on a database holding 126.
Money	Stored to 4 decimal places, percentages to 6, both **displayed at 2**. `CHAR(3)` currency, never a float. Every amount on these screens goes through the shared formatter.
Consumption	Always with an explicit unit; never assumed. Unit of measure is a required field on a measuring asset for this reason.
An address has one home	An address is stored on `premises` and nowhere else; every other entity derives it by walking. **Checked field by field in this pass: `assets` declares no address column.** `location_note` is a note; `assigned_address` is derived through the metering seam.
No hard caps	`NFR-5` — no per-tenant cap, no account maximum, no unpaginated list. `NFR-3` — universal server pagination. The in-scope customer is up to **40,000** accounts, and the flexibility is the deliverable, not the tidiness. Section 09 has one row on this.
Financial immutability	No `UPDATE` or `DELETE` on issued invoices, invoice lines, payments or journal entries. Nothing in the Assets menu writes any of them — which is why an edit control here is a feature and not a finding, unlike on a payment.

02

The list — one page, six tabs

Everything the Assets rail item opens onto. One address, `/:tenant/assets`, carries six tabs and two dialogs. Three of the tabs are the asset list partitioned by type (BR-AST-004); three are their own reads. The tab is URL state, so every one is linkable — but only the bare address is in the sidebar, which is what section 09 turns out to be about.

> **What changed in this section, and why**
>
> **The Imports tab and the import runner are out of scope for this document** by instruction (2026-09-11) and every screen, finding and specification that belonged to them has been removed rather than left dangling. The registry draws **six** tabs here and the product draws seven; that gap is the document's scope, not a defect, and section 13 lists exactly what went.
>
> **Screen 05 gains three sub-screens.** The Lifecycle tab stacks three tables on one page today with no toolbar over any of them; 05a, 05b and 05c draw it as three tabs, each with the product's own table toolkit. Those three are a **new build**, drawn to the rules rather than to the code, and screen 05 keeps the as-built so the carry-over can reconcile.
>
> **Screen 18 is a full page, not a dialog**, by the same instruction — and that is a change of kind, so it is drawn to the rules and carries its own carry-over table.

---

01 Meters

EXISTS BEHIND · NEW READ · COMMITMENT 01 · COMMITMENT 15 · COMMITMENT 20

`/:tenant/assets?tab=meters` · also `/:tenant/assets` (the bare address)

> **Seen in the browser: the KPI strip has two states, and a real tenant only ever meets the wrong one**
>
> There *is* a served figure behind every card. Asking for the whole result set on one page — `?tab=meters&per=200` — the strip reads **Meters in service 103 · 7 masters · 9 subs · 1% AMI**, **Needs repair 0**, **In inventory 6**, **Past certification age 0**. Both states are drawn above, in that order.
>
> **On first load it renders four em dashes**, because `kpiRowsComplete` (`assetsShared.tsx:252-264`) will only let a figure render when the page holds the *whole* result set, and the page size is 50 against 109 meters. **Filtering does not help either** — measured: filtering to the six uninstalled meters still shows four dashes, because the same rule refuses any state with a filter applied. So a tenant meets the numbers only by changing the page size by hand. Specification `AST-B-01`.

> **The export is broken on the deployed stack, and it is one missing line of `docker-compose.yml`**
>
> Reported from `smartbillhub.bynry.com` on 2026-09-11 (Ottumwa Water Works, v2.234.17): the export accepts, the notice says *“Your export is ready.*

Download the CSV", and the browser saves **`file.json`** and says **"File wasn't available on site."**

The filename is the diagnosis. The link is an `<a download>` pointing at `/api/exports/{job_id}/file` — a URL whose last segment is `file`, with no extension, because the real name arrives in `Content-Disposition`. When the response is a JSON error instead of a CSV, Chrome names the download from the URL plus the MIME type and gets `file.json`. So the request **reached Django and Django refused it**: the SPA proxy never falls back to `index.html` for `/api/*`, which rules out an HTML body.

Why Django refuses it. `export_file` answers one plain 404 for five different noes, and the one that fires here is the last: `artifact_within_dir()` ends with `candidate.is_file()`, and the file is not on that container.

```
POST /exports          → backend container → job accepted, 202
celery worker          → runs the body   → writes /app/var/artifacts/ottumwa_….csv  ← HERE
GET  /exports/{id}/file → backend container → looks in /app/var/artifacts  ← EMPTY
                                            → is_file() False → 404 JSON → file.json
```

The two halves are each written down and nobody added them up. `06-backend/.env.example:173` sets `JOBS_EAGER=false`, so the job body runs on the `celery` container and never on the web one. And `docker-compose.yml` mounts **no volume on any service** — there is no top-level `volumes:` key at all — so `/app/var/artifacts` is private container disk in six separate containers. The compose file even reasons about the risk and reaches the wrong conclusion: *"they survive a restart, and they do not survive `docker compose down` or a rebuild … today's only job kind is a CSV export fetched within seconds, so that is a real but small consequence."* **That sentence assumes the writer and the reader are the same container.** With `JOBS_EAGER=false` they never are, so it is not small and it is not occasional — **every export fails, every time**.

⚠ **And the runbook's mitigation is gated on a trigger that will not fire.** §10.6 is open and correct about the mechanism — *"an export written on replica A 404s when the download lands on replica B"* — but it makes the shared volume *"Required, not optional, the moment web `desiredCount > 1`."* **The true condition is the moment `JOBS_EAGER` is false**, which production sets by default, so the fix stays parked at one replica while the feature is broken.

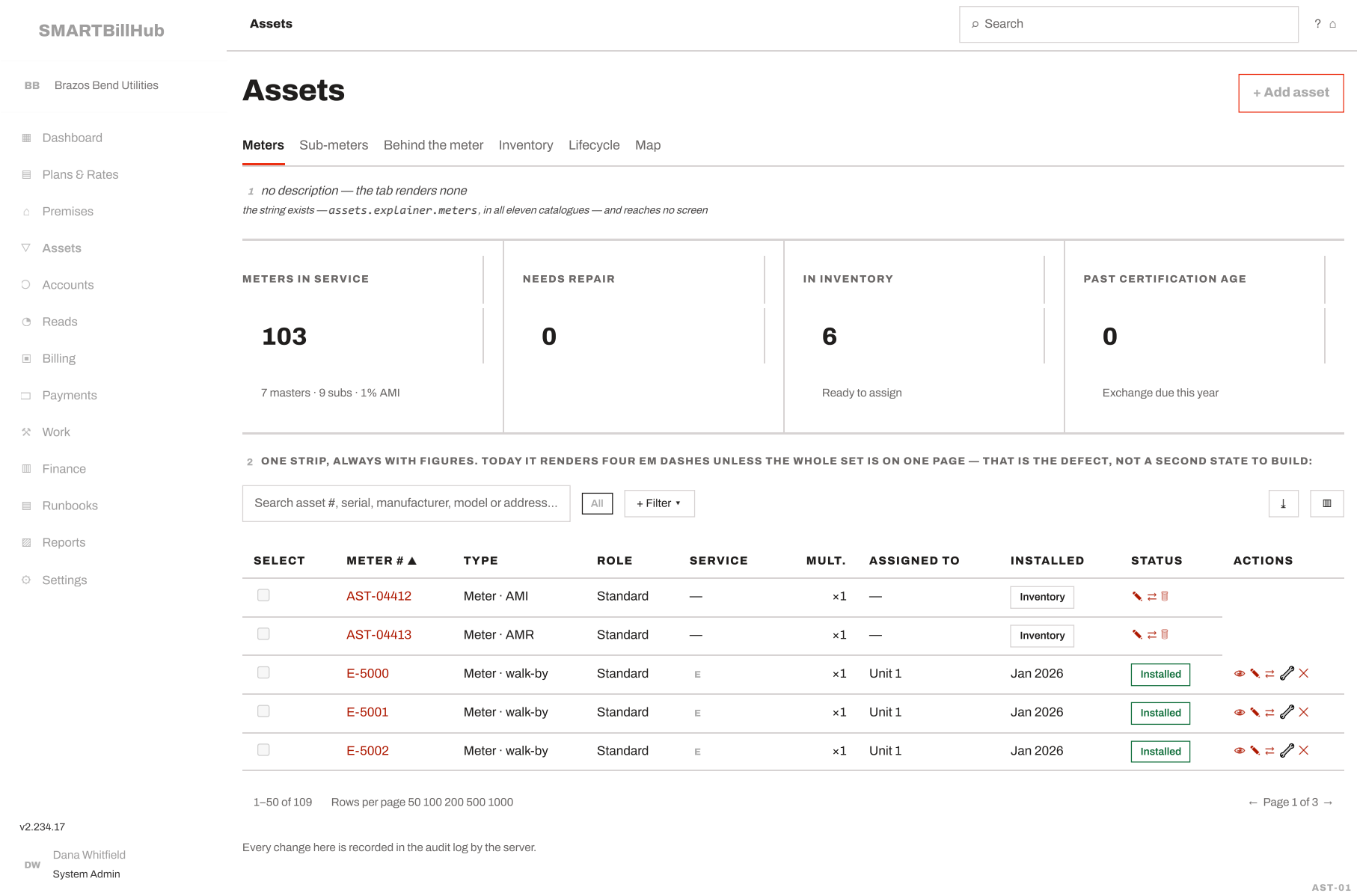
Two more downloads break the same way and neither has been reported yet: the bill **print set** (`celery-billing` writes a ZIP through the same `finish_artifact` contract) and the **audit-log export**. All three register through `register_export_kind` and all three read from the same directory. Specification `AST-B-20`.

And this tab has no description at all

The catalogue holds one — `assets.explainer.meters`, *"Meters and behind-the-meter devices · assign to units · track lifecycle"* — and the screen never renders it: `AssetRegistryPage.tsx:801` passes an explainer only when the tab is `sub_meters`. Behind the meter is the same. That was a ruling — the KPI grid replaced the prose on those two tabs — taken when the grid showed numbers; the grid now shows dashes, so those two tabs carry neither. **Two dead strings, and two screens with nothing under the title.** Specification `AST-B-15`.

I have a meter number on a work order and I need to know where that meter is and whether it is working.

FIGURE 1 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	A clerk or admin holding a meter number, a serial off the side of a device, or an address.
How you get here	**Assets** in the left rail (`AppShell.tsx:211`, rendered at `:752`), the global search's *view all* for the assets entity

	(`searchDestinations.ts:49`), and ← `Assets` from every screen below it. Audited, all read from source.
Primary action	**+ Add asset** — the tab's own label, so the Sub-meters tab reads *+ Add sub-meter*. It navigates to the four-step wizard rather than opening a dialog.
Tabs	Six drawn here, all URL state: Meters · Sub-meters · Behind the meter · Inventory · Lifecycle · Map. The three non-list tabs spend no registry read at all.
Table controls	**Search** — Server. `q=` reaches `GET /assets` and is applied as a queryset `.filter()` over six legs — `asset_no · serial_no · manufacturer · model · parent_asset__asset_no` and a sub-select of assets on units whose address matches (`selectors.py:474-492`). **Walked:** typing `E-500` put `q=E-500` in the URL and issued `/assets?q=E-500&sort=asset_no&dir=asc&page=1&per=50&f.tab=meters`, 10 rows back. **Filter** — Server, and all seven declared filters are read by the handler: `status · type · meter_type · service · premises_id · assigned · test_due`. Four of them — `assigned`, `test_due`, `premises_id`, `service` — are applied in **Python** after every matching row is materialised, each with a per-row seam call (`selectors.py:496-512`). **Columns** — Client. Per-tab column sets with a show/hide, reorder and Reset chooser; the choice is **persisted server-side** through `GET|PUT /user-preferences/{key}` (`DataTable.tsx:252-285`). **Sort** — Server: `.order_by()` in SQL with a stable secondary on `id`. **Walked:** three clicks on Meter # gave `dir=desc` → `dir=asc` → `dir=desc`; it never clears. **Pagination** — Enveloped on the server (50/100/200/500/1000, default 50) — but `asset_list` returns `list(qs.order_by(...))`, a materialised Python list, so **no LIMIT or OFFSET reaches SQL** (`selectors.py:495`).
Write surfaces	**Detail** — the asset # links to `/assets/{id}`. **Add** — the header primary. **Edit** — the pencil, now a full page (screen 18). **Update** — `PATCH /assets/{id}`. **Delete** — the trash, only on a never-referenced inventory row.
Walked	Opened at 1440 on the seeded Brazos Bend tenant. 50 rows of 109, zero failed requests, and **all four KPI cards rendered an em dash** — and, re-measured on 2026-09-11 with `per=200`, real figures. Row-action cells drew 5 controls on installed rows and 3 on inventory rows, with no overflow menu on any.
Spacing	Measured 2026-09-11: screen body **24/16/32**, tabs margin 16 gap 16, KPI cell 16×20 gap 6, `th/td` 8×12, search box 6×10 min-height 36. Every one is on the R11 scale. The rail item's **9px 12px** and the status pill's **2px 9px** are not — see section 09.
Cognitive load	It takes *which of my 126 devices is this* and answers it from any of five things a person might be holding — our number, the number stamped on it, its maker, its model, or the address it sits at.
States	**Loading** — skeleton rows. **Empty** — *"Add one at a time, or bring a whole batch in from a CSV"*. **Failed** — the reason plus Retry. **Read-only role** — every write control visible and disabled, titled *"Your role can view this but not change it."*
Words	**Meter #**, never *ID*. **Retire** and **Delete** are two glyphs in one cell on purpose: a retired asset still exists and is still reportable.
Components	`PageHeader · KPIStrip · DataTable · FilterBar · RowActions · StatusPill · GuardedButton`

Components	`PageHeader` · `KPIStrip` · `DataTable` · `FilterBar` · `RowActions` · `StatusPill` · `GuardedButton`.
Rules	BR-AST-004 · 005 · 006 · 013 · 026 · 036 · 041 · 044 · BR-SHL-010/011/025/044 · NFR-3 · B14 rules 2, 4, 12, 13, 14, 15, 16, 17, 19, 21.
Audience	Staff.
Health	Worst field is DERIVED. 11 SERVED · 2 DERIVED · 0 NOT SERVED.

Data flow

```
assets.asset_no ──────→ selectors.asset_dict          (apps/assets)
                    └→ GET /assets?f.tab=meters       api.py:306
                       └→ getAssets()  →  DataTable   field ②   WHOLE

premises.address_line1 ─→ assets/seams.py:96 unit_ref()["premise_address"]
                          └→ the "Assigned to" cell    ⇄ assets → metering   WHOLE

(rows on this page) ──────→ metersKpis(rows, complete)   assetsShared.tsx:278
                          └→ the four KPI cards          DERIVED — no endpoint
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Select	a client-side checkbox driving bulk assign	—	N/A
②	Meter #	`assets.asset_no`	assets	SERVED
③	Type	`assets.type.name` folded with `meter_type.name`	assets	SERVED
④	Role	`row.sub_count` / `parent_asset_no`, served	assets	SERVED
⑤	Service	`service_point.service`	assets → metering ⇄	SERVED
⑥	Mult.	`assets.multiplier`, trailing zeros trimmed	assets	SERVED
⑦	Assigned to	`premises.address_line1` via the unit seam	assets → metering ⇄	SERVED

⑧	Installed	`assets.install_date`	assets	SERVED
⑨	Status	`asset_lifecycle_stages.name`	assets	SERVED
⑩	Actions	`row.actions` — the **server's** set for this row's state	assets	SERVED
⑪	Meters in service	**103** — counted in the browser over the rows on this page, so a dash unless the page holds all of them	—	DERIVED
⑫	Past certification age	**0** — same client-side count, same dash rule	—	DERIVED
⑬	1–50 of 109	`count` from the DRF envelope	assets	SERVED
⑭	Description	*nothing renders* — `assets.explainer.meters` exists in all eleven catalogues and reaches no screen	—	DEAD

AST-B-20

Pressing export gives a person the CSV

STATUS Unverified

WAITS ON Shared storage for `JOB_ARTIFACTS_DIR`, mounted at the same path on **EVERY** web and worker task. On compose that is a named volume on `backend`, `celery` and `celery-billing`; on ECS it is the EFS mount §2.5 already specifies. No application code changes.

SCREENS 01, 02, 03, 04

WHAT DONE LOOKS LIKE ON SCREEN On Ottumwa Water Works, pressing the export control on Meters and following *Download the CSV* saves a file named `ottumwa-water-works_assets_…_current-state_….csv` that opens in a spreadsheet — not `file.json`.

ALSO FIXED BY THE SAME MOUNT The bill print set and the audit-log export, which write to the same directory through the same contract and fail the same way.

WORTH DOING BESIDE IT The notice promises a download the anchor cannot deliver. `exportOutcome` already has a *succeeded but no file* branch; nothing checks whether the file is still *there*, so a browser-level failure is the only thing the user sees. A product sentence beats `file.json`.

AST-B-01

A clerk opening Assets sees how many meters are in service

STATUS Unverified

WAITS ON An aggregate read. Nothing blocks it: four `COUNT`s on the predicate `asset_list` already builds.

SCREENS 01, 02, 03, 04

WHAT DONE LOOKS LIKE ON SCREEN On Brazos Bend the Meters tab reads **METERS IN SERVICE 103 · 7 MASTERS · 9 SUBS · 1% AMI**, **NEEDS REPAIR 0**, **IN INVENTORY 6 · READY TO ASSIGN**, **PAST CERTIFICATION AGE 0** — on first load, on page 2, and with a filter applied.

---

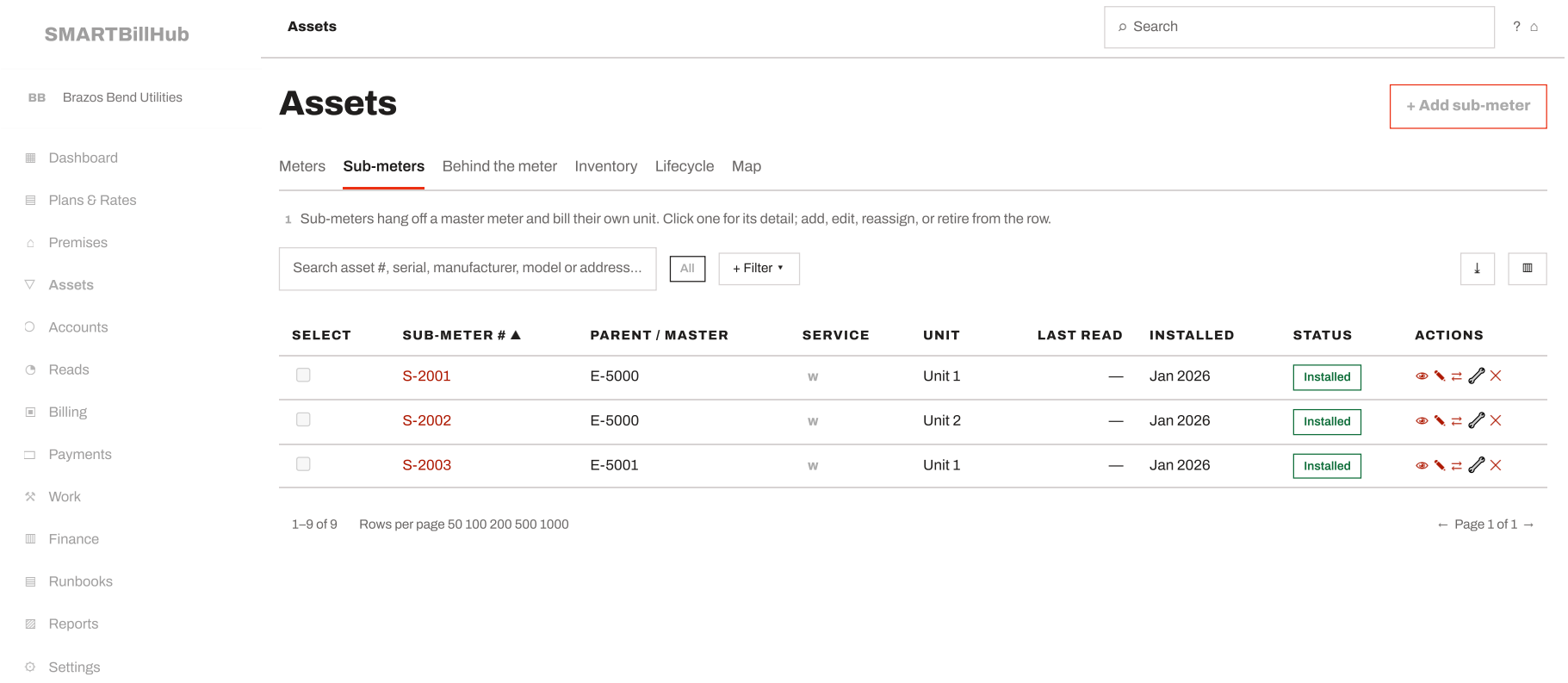
02 Sub-meters

EXISTS BEHIND · COMMITMENT 01 · COMMITMENT 15 · COMMITMENT 20

`/:tenant/assets?tab=sub_meters`

This tenant is billed for part of what the master meter reads — which sub-meter is theirs, and what did it last read?

FIGURE 2 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



DW Dana Whitfield
System Admin

The screen

For	A clerk reconciling a master's total against the sub-meters allocated under it.
How you get here	The tab strip on screen 01, and a direct `?tab=sub_meters` link.
Primary action	**+ Add sub-meter** → `/assets/new?class=sub_meters`.
Tabs	The same six. This is the tab, not a screen of its own.
Table controls	Identical mechanism to screen 01 — same endpoint, same `f.tab` filter, same unbounded query underneath. The column set differs: **Parent / master**, **Unit** and a right-aligned **Last read** replace Role, Mult. and Assigned to.
Write surfaces	Detail · Add · Edit (screen 18) · Update · Delete — all as screen 01.
Walked	Opened, 3 rows, zero failed requests. **Last read** rendered – on all three: the batch fill found no committed reads on the seeded tenant.
Spacing	As screen 01 — the same shared table and header.
Cognitive load	It keeps the master–sub relationship on the row, so nobody opens two screens to see which meter a sub hangs off.
States	As screen 01. The empty state reads *"No sub-meters yet"* with the tab's own Add.
Words	**Parent / master**, both words, because the schema calls it a parent and a clerk calls it the master.
Components	As screen 01.
Rules	BR-AST-003 (a sub-meter requires an ACTIVE master) · BR-AST-004 · DEF-592.
Audience	Staff.
Health	Worst is SERVED — the KPI strip is absent on this tab by ruling, so screen 01's dashes do not apply.

Data flow

```
assets.parent_asset_id ─→ asset_dict["parent_asset_no"]     (apps/assets)
                          └→ the Parent / master cell           field ③  WHOLE
meter_reads (partitioned) → asset_page_rows_batch_fill  selectors.py:384
```

meter_reads (partitioned) → asset_page_rows batch fill selectors.py:384
                          ↳ the Last read cell   ⇄ assets → metering WHOLE (empty here)

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Select	client-side checkbox	—	N/A
②	Sub-meter #	`assets.asset_no`	assets	SERVED
③	Parent / master	`assets.parent_asset.asset_no`	assets	SERVED
④	Service	`service_point.service`	assets → metering ⇄	SERVED
⑤	Unit	`units.label` via the seam	assets → metering ⇄	SERVED
⑥	Last read	`meter_reads`, batch-filled per page	assets → metering ⇄	SERVED
⑦	Installed	`assets.install_date`	assets	SERVED
⑧	Status	`asset_lifecycle_stages.name`	assets	SERVED

03 Behind the meter

EXISTS BEHIND · NEW READ · COMMITMENT 01 · COMMITMENT 15 · COMMITMENT 20

`/:tenant/assets?tab=behind_the_meter`

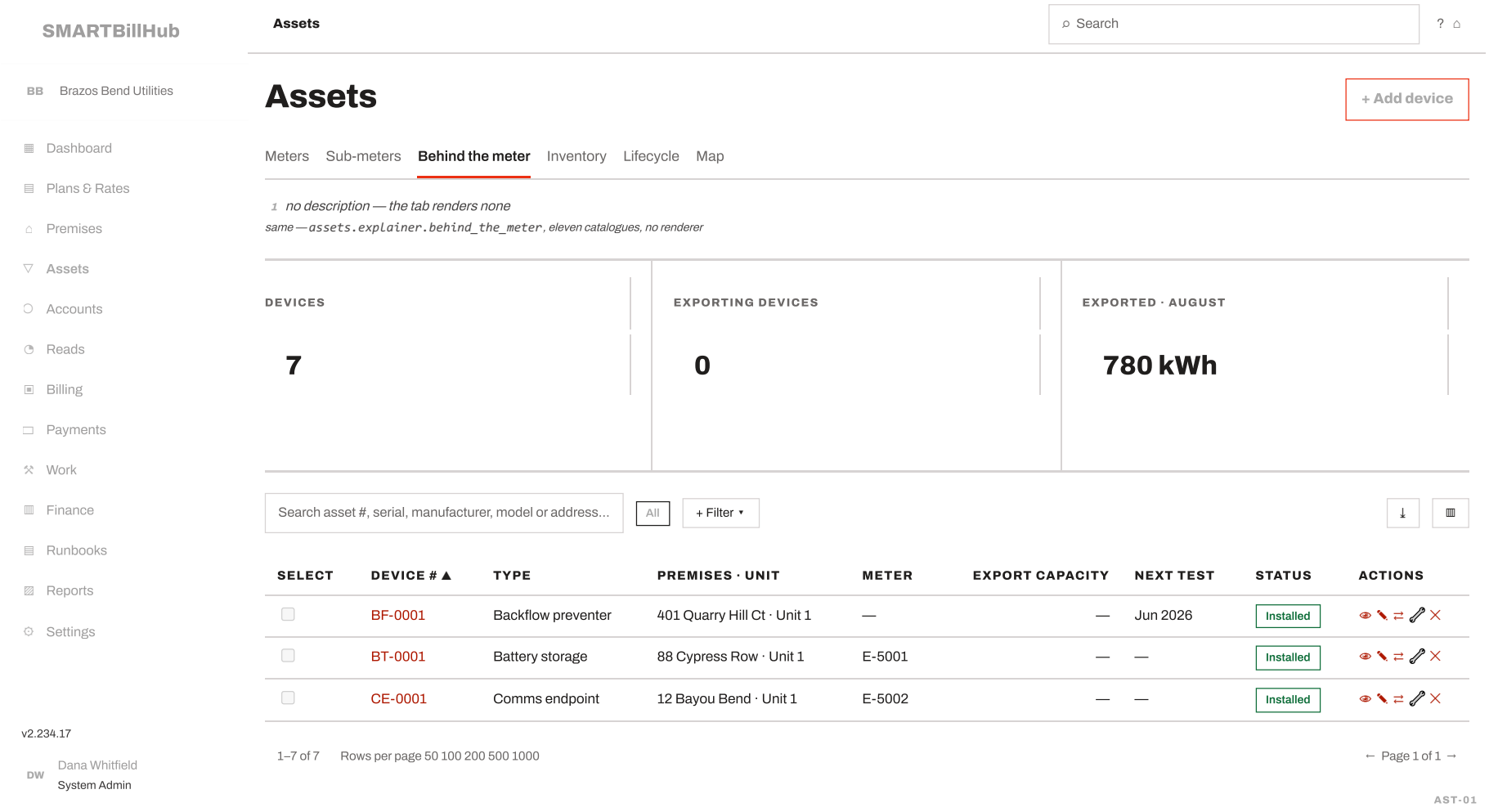
The file's own notes say this card is always a dash. The code six hundred lines above says otherwise.

`AssetRegistryPage.tsx` ends with an `exceptions:` footer, and one of its entries reads verbatim: *"E3 the BTM 'Exported · month' KPI VALUE is always the stated dash — no export quantity rides the list payload; the card is drawn and labeled so the absence is stated, not implied"*. That was true when it was written. **It is not true now** — line 1000 of the same file renders `exportRollup.totals`, and the comment beside it says *"D4 (DEF closed): the served rollup renders quantity WITH its unit"*. The absence was closed and the footer that recorded it was never amended.

This is the sentence that produced the worst error in version one of this document. The card was described as permanently empty on the authority of a comment, and the running app shows **780 kWh**. Hard-won law 1 — *a comment claiming a guarantee is not the guarantee* — in its mirror form: a comment claiming an *absence* that the code no longer has. Nothing in this repository checks a footer against the file it sits in, so it goes stale silently and reads as authority.

A customer has put solar or a battery on their roof — what is it, what can it push back, and when was it last tested?

FIGURE 3 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	A clerk handling net metering, backflow devices and anything else on the customer's side of the meter.
How you get here	The tab strip.
Primary action	**+ Add device**.
Tabs	The same six.

Table controls	As screen 01. Columns: Device # · Type · Premises · unit · Meter · Export capacity · Next test · Warranty (hidden by default) · Status.
Write surfaces	As screen 01.
Walked	Opened, and re-measured on 2026-09-11 with the whole set on one page: **7 devices · 0 exporting · Exported · August 780 kWh**. ⚠ **The third card is a real served figure** — the first version of this document said it was always a dash, which was true of the default page and not of the card. Corrected.
Spacing	As screen 01.
Cognitive load	It puts the linked meter on the device's own row, so *which meter measures this inverter's export* is answered without opening anything.
States	As screen 01.
Words	**Export capacity** is a capacity in kW and never money. BR-AST-025 keeps this tab free of tariffs and credited dollars, by ruling — a decision argued on the page rather than an omission.
Components	As screen 01.
Rules	BR-AST-025 · BR-AST-044 · the D2 ruling of 2026-08-22.
Audience	Staff.
Health	Worst is DERIVED for the two counted cards; **Exported** is the one served rollup and carries a real quantity with its unit.

Data flow

```
assets.export_capacity_kw ─→ the Export capacity cell     field ⑥  WHOLE
metering export channels ──→ seams.export_rollup_last_month()  ⇄ assets → metering
                           └↳ the "Exported · <month>" card    api.py:127-134  WHOLE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
②	Device #	`assets.asset_no`	assets	SERVED
③	Asset type	`asset_types.name` — four groups on this tenant	assets	SERVED
④	Premises · unit	`premises.address_line1` + `units.label`	assets → metering ⇄	SERVED

⑤	Meter	the linked meter's `asset_no`	assets	SERVED
⑥	Export capacity	`assets.export_capacity_kw`	assets	SERVED
⑦	Next test	derived from `last_test_date` + `test_interval_months`	assets	SERVED
⑧	Devices / Exporting devices	*counted in the browser over this page*	—	DERIVED
⑨	Exported · August	`export_rollup.totals` — **780 kWh**, one entry per unit, never summed	assets → metering ⇄	SERVED

04 Inventory

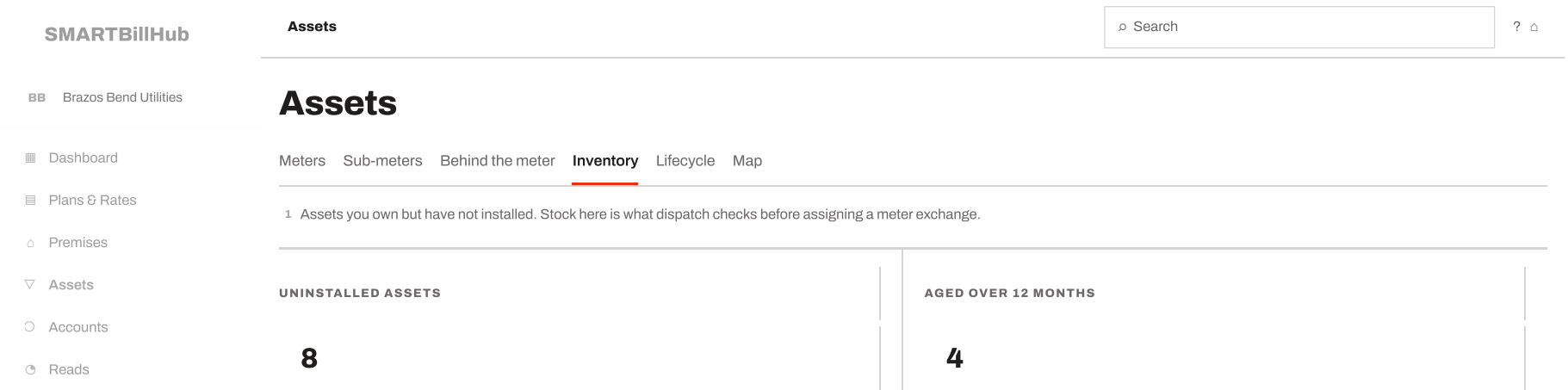
EXISTS BEHIND · COMMITMENT 01 · COMMITMENT 15 · COMMITMENT 20

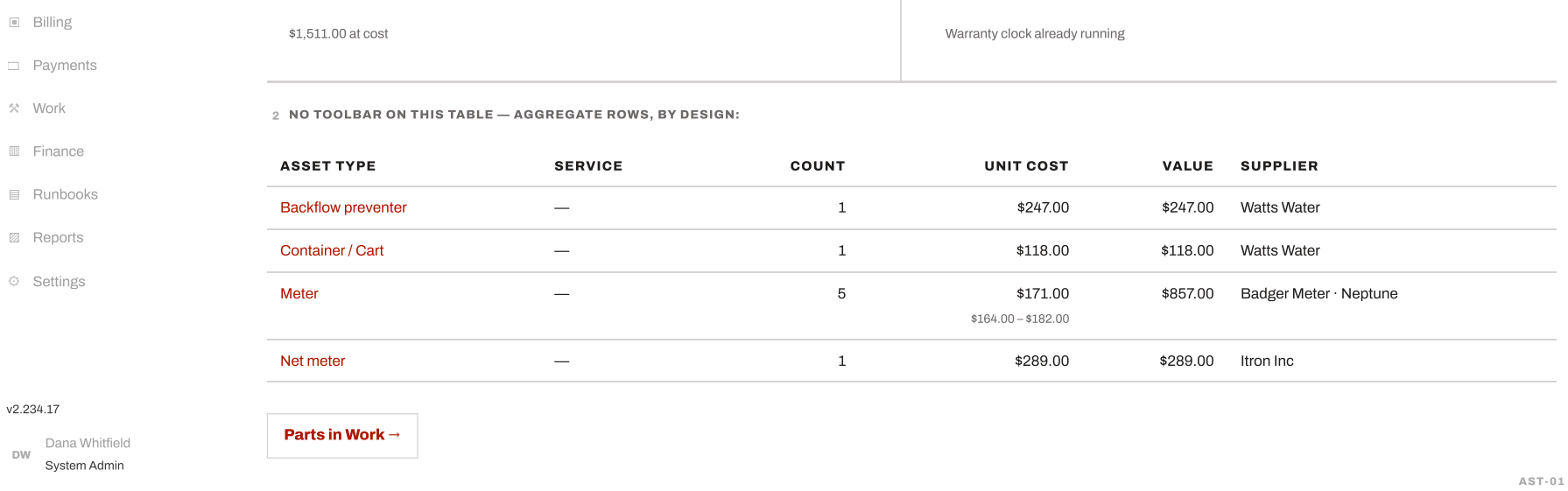
`/:tenant/assets?tab=inventory`

> **What is deliberately not here, and who ruled it**
>
> r10 draws a Yard / On-vans split, a *Reorder at* column with three statuses, a **Raise reorder** button and a stock-count import. The PO cut all four on 2026-08-22: *"Launch with counts & costs only."* The tests assert each absence by name. Out of scope, ruled — not a gap.
>
> The **Parts in Work** → button was disabled for two waves with the two words *"Work R5"* after the screen it points at had already shipped. It is live now, and the file says so about itself — the worked example of hard-won law 2's mirror clause in this module.

How many spare meters do we have on the shelf, and what are they worth?

FIGURE 4 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	Whoever has to answer *can we connect this house next week* before ordering anything.
How you get here	The tab strip only.
Primary action	None — deliberately. These rows are aggregates, not creatable things.
Tabs	The same six.
Table controls	**None, by design.** This is the detail-panel table, not the `DataTable` — four aggregate lines have nothing to search, sort or page. The drill is the row's own affordance: clicking a type opens screen 01 preset to that type's uninstalled rows.
Write surfaces	**Detail** — the type name drills to the filtered registry. **Add · Edit · Update · Delete** — **NOTHING**, and correctly: there is no such thing as editing a stock line.
Walked	Opened, and **re-measured on 2026-09-11**: **8 uninstalled assets · $1,511.00 at cost · 4 aged over 12 months**, over four type rows. Real numbers, because this tab has a served aggregate, `GET /assets/stock`. ⚠ **The first version of this document printed 3 and $546.00 here and called them measured. They were not — they were written from a screenshot that was never opened.** Corrected, and recorded in section 13.
Spacing	KPI cell 16×20 gap 6, on the scale. The table is the panel table at 8×12.
Cognitive load	Two numbers and a four-row table replace *go and count the shelf*. The unit-cost cell carries a range under the average where a group spans prices — **$171.00** over **$164.00 – $182.00** — which is the kind of detail a drawing invents and the running app actually has.
States	**Empty** — *"Nothing in stock — every asset is installed or retired."*

States	**Empty** — *"Nothing in stock — every asset is installed or retired."*
Words	**Uninstalled**, not *available*: the behaviour class is what the server filters on.
Components	`KPIStrip` · the detail-panel table · `MoneyCell` · `ServiceTags`.
Rules	§10-D5 (*"Launch with counts & costs only"*) · BR-AST-031 · hard rule 2.
Audience	Staff.
Health	SERVED throughout — the only registry tab where that is true.

Data flow

```
assets (status class = available) ─→ selectors.stock_view   selectors.py:1796
                                   └→ GET /assets/stock       api.py:312
                                      └→ kpis + per-type rows  WHOLE
                                         ⚠ materialises every available asset (:1812)
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Uninstalled assets	`stock_view.kpis.uninstalled` — **8**	assets	SERVED
②	$1,511.00 at cost	`kpis.value_at_cost`, 4 dp stored, 2 shown	assets	SERVED
③	Aged over 12 months	`kpis.aged_over_12_months` — **4**, detailed *"Warranty clock already running"*	assets	SERVED
④	Asset type	`asset_types.name` — four groups on this tenant	assets	SERVED
⑤	Service	`service_point.service`, honest-empty for never-assigned stock	assets → metering ⇄	SERVED
⑥	Count / Unit cost / Value	`stock_view` per-type rows; the unit cost shows a **range** beneath the average where the group spans prices	assets	SERVED
⑦	Parts in Work →	a navigation to `/work/parts`	assets → work ⇄	SERVED

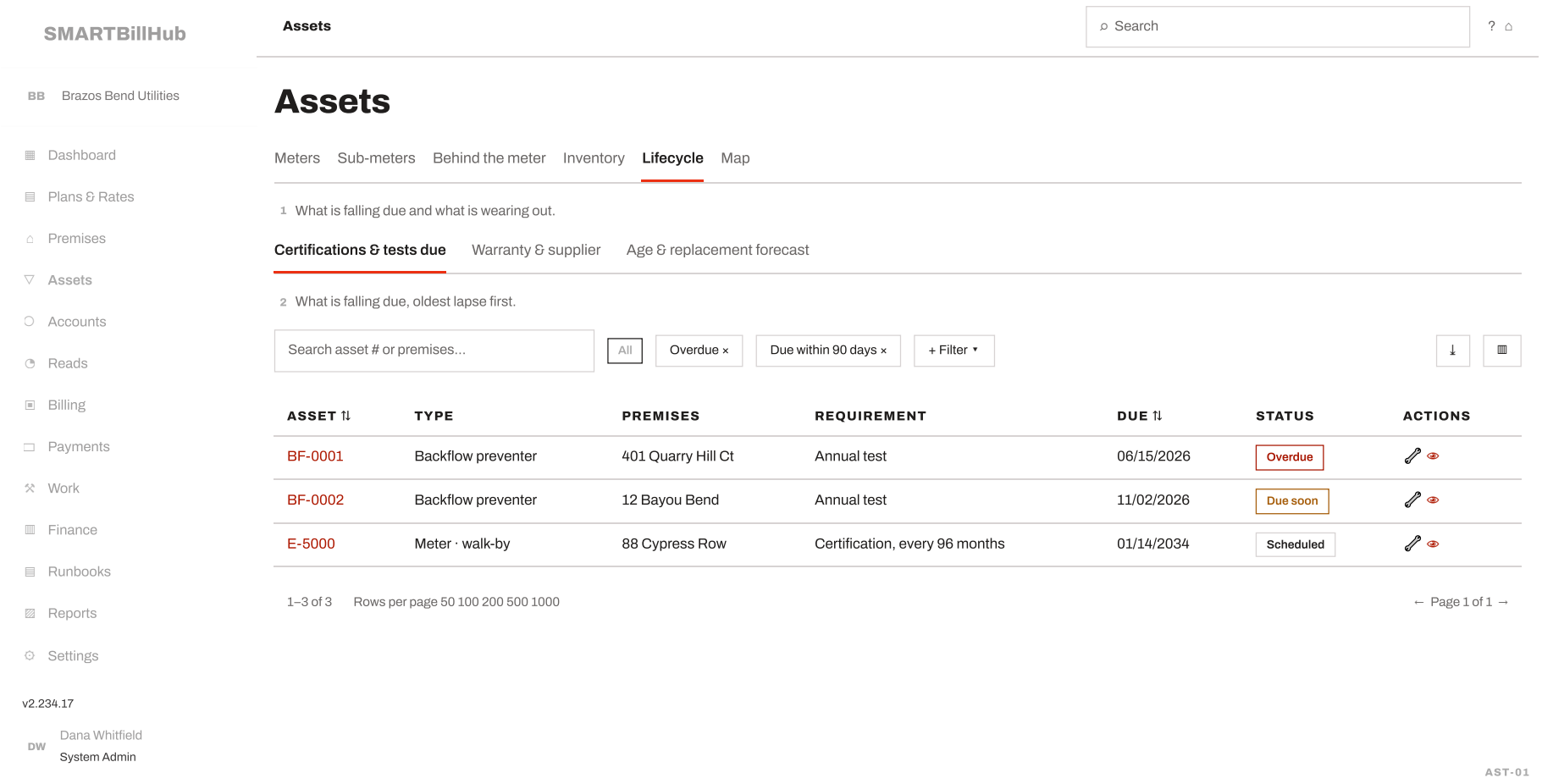
05a Lifecycle · Certifications & tests due

NEW BUILD | COMMITMENT 11 | COMMITMENT 15

`/:tenant/assets?tab=lifecycle&section=certifications`

Which tests are overdue, and can I work through them in the order they fell due?

FIGURE 5 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	Whoever is responsible for a lapsed backflow test when the regulator asks.
How you get here	The Lifecycle tab's own sub-tab strip. It is the default section, so `?tab=lifecycle` lands here.
Primary action	None — this is a worklist. The row's 🔧 is the action.
Tabs	Three: Certifications & tests due · Warranty & supplier · Age & replacement forecast, in the order the as-built stacks them. Same words,

	same sequence.
Table controls	**Search** — asset # or premises. **Filter** — *Overdue*, *Due within 90 days*, and the status vocabulary the server already computes. **Columns** — the shared chooser. **Sort** — Asset and Due, with **Due ascending as the default**, because the oldest lapse is the one that matters. **Pagination** — the standard footer.
Write surfaces	**Detail** — 👁 to the asset. **Add · Edit · Update · Delete** — **NOTHING**; a certification date is a property of the asset and is edited on screen 18. **Raise work order** — the wrench, unchanged.
Walked	**Not walked — this screen does not exist.** It is drawn to the rules, from the payload the as-built already receives. The three rows shown are illustrations consistent with the seeded tenant's one real overdue row.
Spacing	Toolbar 16 above / 12 below, table 8×12, pager 12 — the same values every other list in the product uses, which is the point of using the shared component.
Cognitive load	It turns three stacked tables into one question at a time, and lets the answer be sorted by the thing that decides urgency.
States	**Empty** — the as-built's own sentence survives: *"No certifications or tests are on the calendar."* **Loading** — skeleton rows at table height. **Failed** — the reason plus Retry.
Words	**Requirement**, not *rule* or *cadence* — it is derived from the type's interval and reads as a sentence a person would say. Unchanged from the as-built.
Components	`PageHeader` with a second-level tab strip · `DataTable` · `FilterBar` · `RowActions` · `StatusPill`.
Rules	B14 rules 12, 13, 14, 15, 16, 17, 19 · BR-SHL-011 (the sub-tab is URL state) · BR-AST-007/027.
Audience	Staff.
Health	SERVED — every column already arrives in the existing payload. Nothing here needs a new read.

Data flow

```
GET /assets/lifecycle —→ the SAME single call the as-built already makes
   └→ certifications[]  →  DataTable rows            WHOLE
      ⚠ search, sort, filter and paging are CLIENT-SIDE over this payload unless
        and until the endpoint grows params — which is honest at ten rows and is
        the same NFR-3 question the registry has. Stated rather than assumed.
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Asset	`assets.asset_no`	assets	SERVED
②	Type	`asset_types.name`	assets	SERVED
③	Premises	`premises.address_line1` by walking	assets → metering ⇄	SERVED
④	Requirement	derived from the type's cadence	assets	SERVED
⑤	Due	`certification_due` or `last_test_date + interval`	assets	SERVED
⑥	Status	computed from `days_until`	assets	SERVED
⑦	🔧 Raise work order	`/assets/{id}/work-orders/new`	assets → work ⇄	SERVED
⑧	👁 View	`/assets/{id}`	assets	SERVED

1 **Sub-tab strip** · the same three section titles, now navigable · `?tab=lifecycle&section=`, URL state like every other tab · **B14 rule 12** · **NEW**

2 **Search** · posts q= against the same `GET /assets/lifecycle` payload · **B14 rule 19** (“Search {entities}…” per module) · **NEW**

3 **Filter chips** · removable, `All` is the default · **B14 rule 12** · **NEW**

4 ▥ **Columns** · show / hide / reorder / Reset, persisted through `PUT /user-preferences` like every other list · **B14 rule 14** · **NEW**

5 **Sortable headers** · ⇅, cycling asc → desc → clear · **B14 rule 13** · **NEW**

6 **Row actions as icons** · the wrench that already exists, plus 👁 to the asset · max two visible · **B14 rule 15** · **CHANGED**

7 **Pagination footer** · range, rows-per-page, first / prev / next / last · **B14 rule 16** · **NEW**

Carry-over

As-built element	Disposition	Note
Section title *Certifications & tests due*	RELABELLED	Becomes a tab label. Same words, same order, same position in the sequence — a person who knew where it was still knows.
Asset · Type · Premises · Requirement · Due · Status	UNCHANGED	Six columns, same order.
🔧 wrench	UNCHANGED	Same glyph, same destination.
Empty-state sentence	UNCHANGE	—

	D	
Search · filter chips · ▥ Columns	NEW	Answers the B14 12/14 finding.
Sortable ⇅ on Asset and Due	NEW	Answers B14 13; Due ascending is the new default.
👁 View	NEW	Two row actions, which is the rule's cap.
Pagination footer	NEW	Answers B14 16.

```
As-built elements    9
  unchanged          7
  moved              0
  relabelled         1   ← section heading becomes a tab label
  removed            0
Remedy elements     17   = 9 − 0 removed + 8 new    ✓ reconciles
```

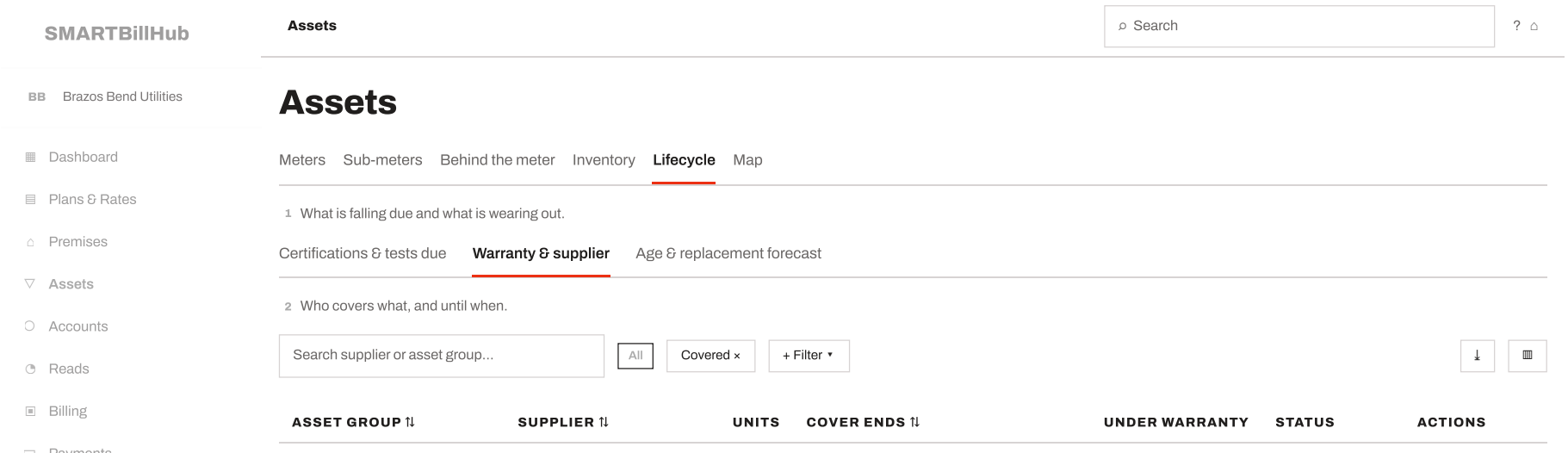
05b Lifecycle · Warranty & supplier

NEW BUILD · COMMITMENT 11 · COMMITMENT 15

`/:tenant/assets?tab=lifecycle&section=warranty`

Before I pay for this repair — is it still under warranty, and who do I ring?

FIGURE 6 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





AST-01

The screen

For	Whoever is about to authorise a repair that the supplier might owe them.
How you get here	The Lifecycle tab's sub-tab strip.
Primary action	None — a reference list.
Tabs	The same three, in the same order.
Table controls	**Search** — supplier or asset group, which is how a person arrives at this table (they have a supplier's name on an invoice). **Filter** — *Covered* / *Expired* / *No cover recorded*, the three states the status column already computes. **Columns** — the shared chooser. **Sort** — Asset group, Supplier and Cover ends, the last being the one that answers *what falls out of warranty this year*. **Pagination** — the standard footer over nine rows today.
Write surfaces	**Detail** — 👁 to the group's assets, filtered. **Add · Edit · Update · Delete** — **NOTHING**; a warranty end date is a property of the asset and is edited on screen 18.
Walked	**Not walked — this screen does not exist.** The five rows drawn are the real nine, abbreviated; the suppliers, counts and cover dates are the seeded values read from the running app.
Spacing	As 05a — the shared toolbar and table, on the scale.
Cognitive load	It answers a money question with a name and a date, which is all the person on the phone needs.
States	**Empty** — the as-built's sentence survives: *"No supplier is recorded on any asset yet."*
Words	**Cover ends**, not *expiry* — the product already says it this way and a warranty ending is not the same event as a certification expiring, which is the next tab along.
Components	As 05a.

Rules	B14 rules 12, 13, 14, 16, 17, 19 · hard rule 2 where a replacement cost is shown.
Audience	Staff.
Health	SERVED — every column is in the existing payload.

Data flow

```
GET /assets/lifecycle —→ warranty[]  →  DataTable rows        WHOLE
   grouped server-side by (asset type, supplier); assets.warranty_end decides
   the Status column. No new read.
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Asset group	`asset_types.name`	assets	SERVED
②	Supplier	`assets.supplier` — free text, no supplier list exists	assets	SERVED
③	Units	count of assets in the group	assets	SERVED
④	Cover ends	`assets.warranty_end`	assets	SERVED
⑤	Under warranty	count where `warranty_end` is in the future	assets	SERVED
⑥	Status	Covered / Expired / —	assets	SERVED
⑦	👁 View	the registry, filtered to that group	assets	SERVED

1 **Sub-tab strip** · the same three section titles, now navigable · `?tab=lifecycle&section=`, URL state like every other tab · **B14 rule 12** · **NEW**

2 **Search** · posts q= against the same `GET /assets/lifecycle` payload · **B14 rule 19** (“Search {entities}…” per module) · **NEW**

3 **Filter chips** · removable, `All` is the default · **B14 rule 12** · **NEW**

4 ▥ **Columns** · show / hide / reorder / Reset, persisted through `PUT /user-preferences` like every other list · **B14 rule 14** · **NEW**

5 **Sortable headers** · ⇅, cycling asc → desc → clear · **B14 rule 13** · **NEW**

6 **Row actions as icons** · the wrench that already exists, plus 👁 to the asset · max two visible · **B14 rule 15** · **CHANGED**

7 **Pagination footer** · range, rows-per-page, first / prev / next / last · **B14 rule 16** · **NEW**

Carry-over

As-built element	Disposition	Note
Section title *Warranty & supplier*	RELABELLED	Becomes a tab label; same words, same place in the sequence.
Asset group · Supplier · Units · Cover ends · Under warranty · Status	UNCHANGED	Six columns, same order, same alignment.
Empty-state sentence	UNCHANGED	—
Search · filter chips · ▥ Columns	NEW	—
Sortable ⇅ on three columns	NEW	Cover ends is the one that answers the year-ahead question.
👁 View	NEW	This table had no row action at all.
Pagination footer	NEW	—

```
As-built elements    8
  unchanged          7
  moved              0
  relabelled         1
  removed            0
Remedy elements      15   = 8 − 0 removed + 7 new    ✓ reconciles
```

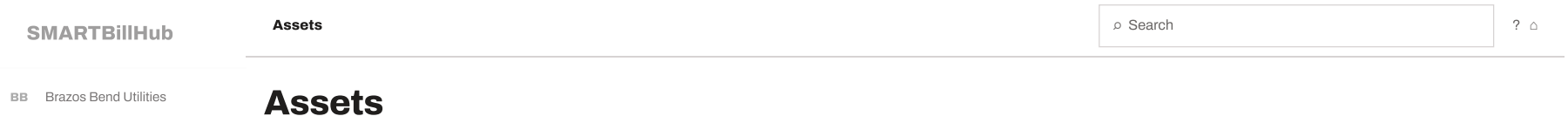
05c Lifecycle · Age & replacement forecast

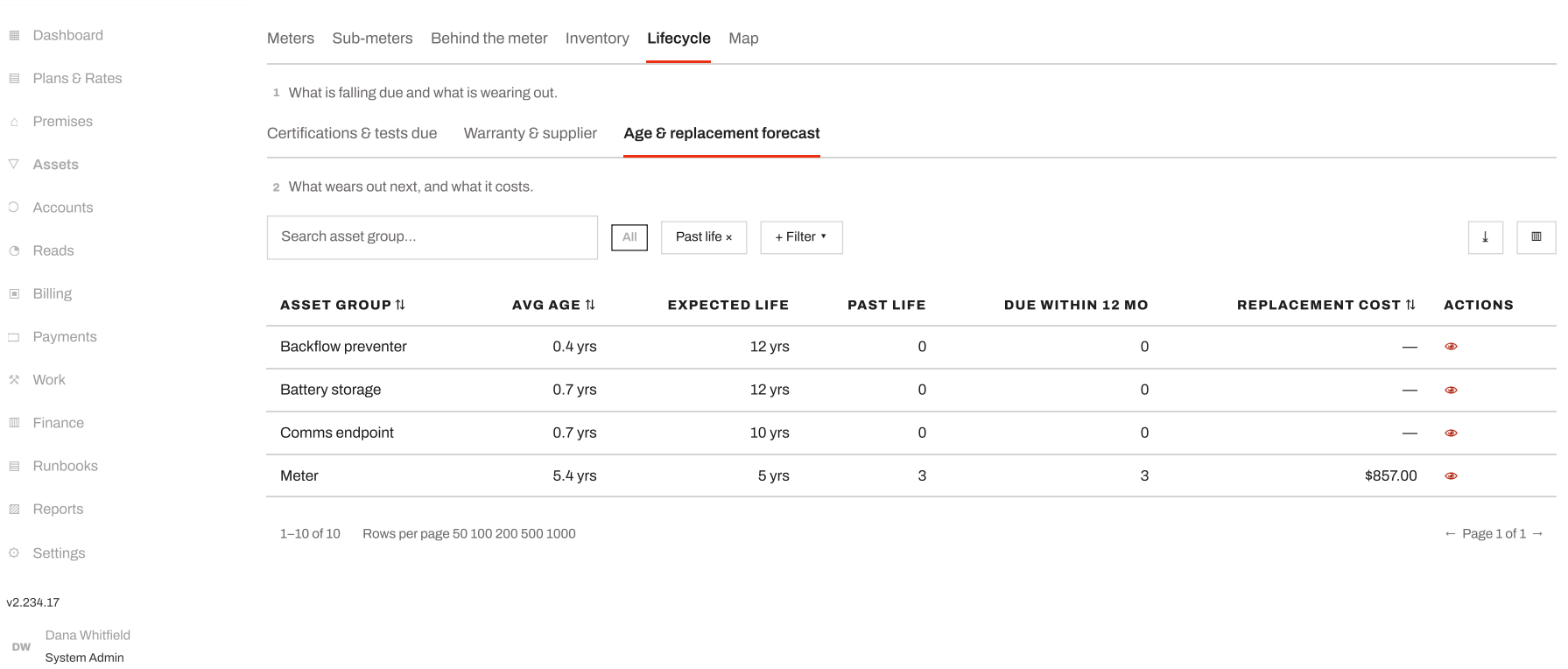
NEW BUILD · COMMITMENT 11 · COMMITMENT 15

`/:tenant/assets?tab=lifecycle&section=forecast`

What am I going to have to buy next year, and what will it cost?

FIGURE 7 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	Whoever writes the capital line in next year's budget.
How you get here	The Lifecycle tab's sub-tab strip.
Primary action	None — a planning table.
Tabs	The same three, in the same order.
Table controls	**Search** — asset group. **Filter** — *Past life* and *Due within 12 months*, the two columns a budget is built from. **Columns** — the shared chooser. **Sort** — Asset group, Avg age and, most usefully, **Replacement cost descending**. **Pagination** — the standard footer over ten rows.
Write surfaces	**Detail** — 👁 to the group's assets. Everything else **NOTHING**.
Walked	**Not walked — this screen does not exist.** Four of the ten real rows are drawn; the ages and lives are seeded values, the $546.00 is the real stock figure from screen 04.
Spacing	As 05a. With the Asset value panel gone the table takes the full width, and the toolbar, table and pager sit on the same 16 / 12 / 8×12 / 12 rhythm as every other list in this section.

Cognitive load	It is the only screen in the product that answers a budget question, and sorting by replacement cost is what makes it usable as one.
States	**Empty** — no asset has an expected life recorded, which is a real state on a young tenant.
Words	**Past life** — assets already older than their expected life. Blunt, and the right word: they are still in service and should not be.
Components	As 05a, plus the **Asset value** panel carried over whole.
Rules	B14 rules 12, 13, 14, 16, 19 · hard rule 2 (money 2 dp on display) · §10-D3 (no GL account anywhere) · BR-AST-047.
Audience	Staff.
Health	SERVED.

Data flow

```
GET /assets/lifecycle —→ forecast[] + asset_value          WHOLE
   straight-line over assets.purchase_cost / purchase_date / expected_life_years
   ⚠ reaches NO journal line, by ruling — the panel says so on its own face
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Asset group	`asset_types.name`	assets	SERVED
②	Avg age	mean `age_months` for the group	assets	SERVED
③	Expected life	`assets.expected_life_years`	assets	SERVED
④	Past life	count already beyond expected life	assets	SERVED
⑤	Due within 12 mo	count reaching expected life inside a year	assets	SERVED
⑥	Replacement cost	summed `purchase_cost`, 4 dp stored / 2 shown	assets	SERVED
⑦	~~Buy value / Present value~~	~~server straight-line across the tenant~~ — the panel is **REMOVED** by instruction; appendix K says what it costs	assets	REMOVED

⑧	👁 View	the registry, filtered to that group	assets	SERVED

1. **Sub-tab strip** · the same three section titles, now navigable · `?tab=lifecycle&section=`, URL state like every other tab · **B14 rule 12** · **NEW**
2. **Search** · posts `q=` against the same `GET /assets/lifecycle` payload · **B14 rule 19** (“Search {entities}…” per module) · **NEW**
3. **Filter chips** · removable, `All` is the default · **B14 rule 12** · **NEW**
4. 𝍐 **Columns** · show / hide / reorder / Reset, persisted through `PUT /user-preferences` like every other list · **B14 rule 14** · **NEW**
5. **Sortable headers** · ⇅, cycling asc → desc → clear · **B14 rule 13** · **NEW**
6. **Row actions as icons** · the wrench that already exists, plus 👁 to the asset · max two visible · **B14 rule 15** · **CHANGED**
7. **Pagination footer** · range, rows-per-page, first / prev / next / last · **B14 rule 16** · **NEW**

Carry-over

As-built element	Disposition	Note
Section title *Age & replacement forecast*	RELABELLED	Becomes a tab label; same words, same place.
Asset group · Avg age · Expected life · Past life · Due within 12 mo · Replacement cost	UNCHANGED	Six columns, same order, same right alignment.
Asset value panel — Buy value $3,381.00, Present value $2,608.02, and its *“Nothing here posts anywhere”* line	**REMOVED**	⚠ **Instruction, 2026-09-11:** *“remove this - Asset value”*, quoting the panel in full. It is the only field-bearing removal in this document. What it costs: the two figures are computed nowhere else in the product, so a person who used them loses them entirely — appendix K says so rather than letting it pass as a tidy-up.
Search · filter chips · 𝍐 Columns	NEW	—
Sortable ⇅	NEW	Replacement cost descending is what makes this a budget tool.
👁 View	NEW	—
Pagination footer	NEW	—

```
As-built elements   10
  unchanged          9
  moved              0
```

```
  relabelled          1
  removed             1   ← the Asset value panel, on a quoted instruction
Remedy elements      16   = 10 − 1 removed + 7 new   ✓ reconciles
```

AST-B-11

A clerk can sort and search what is falling due, instead of reading three stacked tables

STATUS Unverified

WAITS ON Nothing on the server — all three tables already arrive in one `GET /assets/lifecycle`. It is the shared `DataTable` in place of the panel table, plus a second-level tab strip carried in the URL.

SCREENS 05a, 05b, 05c

WHAT DONE LOOKS LIKE ON SCREEN Opening **LIFECYCLE** on Brazos Bend lands on **CERTIFICATIONS & TESTS DUE** with **BF-0001** at the top under a **DUE** ⇅ header sorted ascending, a search box reading *Search asset # or premises…*, an **OVERDUE** × chip, and a footer reading **1–3 OF 3**. The other two sections are tabs beside it, and nothing that was on the old page is missing from the three.

---

06 Map

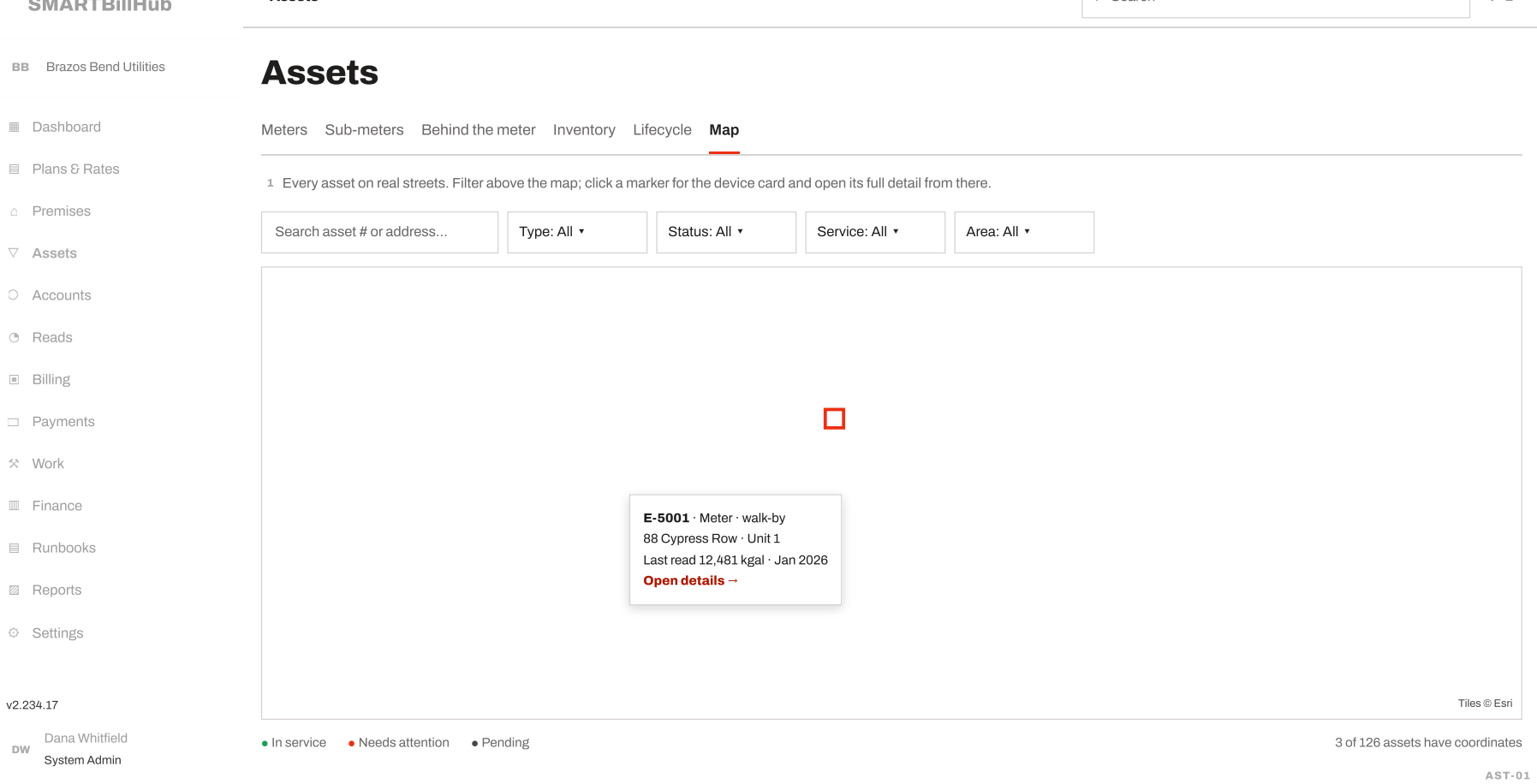
EXISTS BEHIND | COMMITMENT 06 | COMMITMENT 15

`/:tenant/assets?tab=map`

> **A finding this document made and then withdrew — and the withdrawal is the point**
>
> From the source alone these four dropdowns look like a B14 rule-8 violation: raw `<select>` elements whose class does not set `appearance`. **In the browser they compute to `appearance: none`** and draw the product's own chevron, because an ancestor carries `.view` and `FormModal.css` declares `.view select`. Measured, not assumed.
>
> So it is not a violation — it is **conforming by inheritance from a class this component never asked for**. Rename that wrapper and four dropdowns silently revert to the browser's control with every check green. A fragility, recorded as `AST-B-06`.

Where actually is it? I have a crew in the truck and an address that does not match the note.

FIGURE 8 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	Anyone sending someone to a device rather than reading about it.
How you get here	The tab strip only.
Primary action	None. The popup's **Open details** → is the one navigation.
Tabs	The registry's six. Inside: a filter bar of a search box and four dropdowns.
Table controls	No table. The four dropdowns filter **in the browser** over the whole `GET /assets/geo` payload — a single-shot answer with no page and no bound.
Write surfaces	**Detail** — the popup's link to screen 08. Everything else **NOTHING**.
Walked	Opened. Leaflet mounted, filter bar and legend rendered, and the four dropdowns computed `appearance: none` — the product's own chevron, not the browser's. See the note below, a correction to this document's own earlier reading.

Spacing	The filter bar is hand-built rather than the shared toolbar: `gap 8` and controls at 6×10 min-height 36. On the scale, and not the same bar every other list uses.
Cognitive load	It turns *which of these is the one with the broken lid* into a thing you point at.
States	**Empty** — *"No mapped assets yet · Premises gain coordinates from the geocoding run; until then the map has no markers to draw."* **No tile host** — *"Background map is switched off."* and the markers still draw.
Words	Three marker states and no fourth: in service · needs attention · pending.
Components	Leaflet (dynamic import) · a hand-built filter bar · the legend.
Rules	§10-D6 (the Esri tile egress is the ACCEPTED third-party choice) · BR-AST-026 · BR-AST-031.
Audience	Staff.
Health	SERVED, with one caveat on the seam — the tile host is the tenant's own.

Data flow

```
premises.lat / premises.lng ─→ geo_view                      api.py:216
   (or units.lat/lng where the  └→ GET /assets/geo
    unit carries its own —          └→ filterGeoAssets() in the browser
    BR-MET-1162)                       └→ Leaflet markers          WHOLE
integrations.page_policy.tile_host ─→ the basemap URL  ⇄ assets → integrations  WHOLE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Search	filters the loaded payload in the browser	—	DERIVED
②	Type / Status / Service / Area	four selects, filtering client-side	—	DERIVED
③	Marker position	`premises.lat/lng`, or the unit's own	assets → metering ⇄	SERVED
④	Popup: asset # · type	`assets.asset_no`, `asset_types.name`	assets	SERVED
⑤	Popup: last read	`meter_reads`	assets → metering ⇄	SERVED

⑥	Basemap tiles	`tile_host` from the page policy	assets → integrations ⇄	SERVED
⑦	“3 of 126 assets have coordinates”	counted in the browser	—	DERIVED

AST-B-06

The Map tab's dropdowns look like the product because they say so, not because of where they sit

STATUS Unverified

WAITS ON Nothing. Use the shared `SelectField`, or add `appearance-none` and the chevron to the local class.

SCREENS 06

WHAT DONE LOOKS LIKE ON SCREEN Rendering the Map tab's filter bar outside any `.view` wrapper still shows four dropdowns with the product's small square chevron.

---

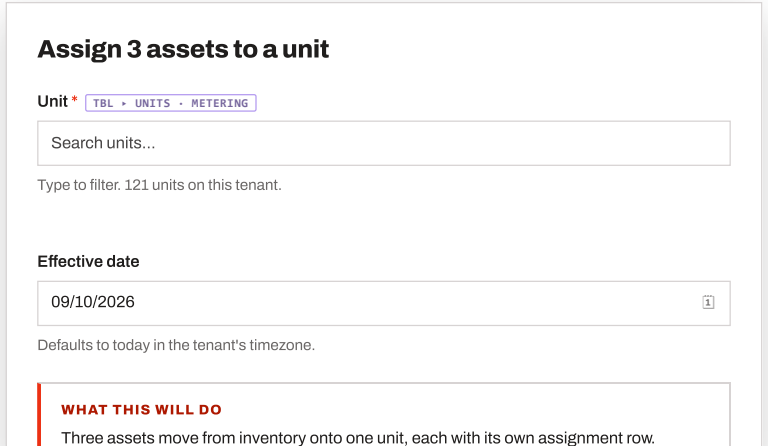
07 Assign 3 assets to a unit

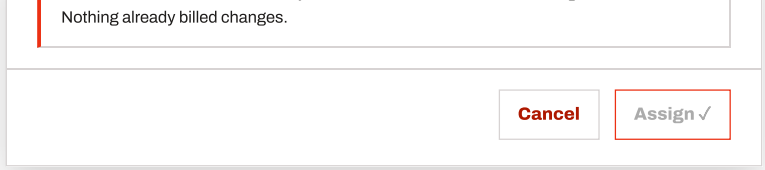
EXISTS BEHIND · NO COMMITMENT

`no address of its own — a dialog on /:tenant/assets`

A crew just fitted three meters at the same address. Do I have to do them one at a time?

FIGURE 9 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	A clerk closing out a multi-meter install.
How you get here	Tick rows in the registry; a selection bar appears above the table with **Assign to a unit** and **Clear**.
Primary action	**Assign ✓** → `POST /assets/bulk-assign`, returning `{assigned, failed}`.
Tabs	None.
Table controls	N/A.
Write surfaces	The bulk write. Inventory → unit **only**.
Walked	Not opened — the selection bar needs row ticks.
Spacing	R12 inside the dialog: the unit picker's helper sits under its input, and the two fields are 20px apart.
Cognitive load	One dialog instead of three trips through a full-page flow. **This one stays a dialog** — it is two fields and a consequence, and it acts on a selection that only exists on the screen behind it.
States	**Partial** — a unit→unit row in the batch is refused individually and reported, and the rest still land.
Words	The bar says the limit out loud: *"Bulk assignment moves inventory to units only. A unit-to-unit move goes one at a time, or by file import."*
Components	`FormModal` · `SearchSelectField` · `DatePickerField` · `Note`.
Rules	BR-AST-036/038 · B14 rule 9 (a 121-option list is a searchable picker) · B14 rule 11.
Audience	Staff, System Admin or Billing Clerk.
Health	SERVED.

Data flow

```
POST /assets/bulk-assign ─→ services.bulk_assign        services.py:1867
   ├→ inventory → unit ONLY; a unit→unit row is refused individually
   └→ {assigned, failed}                                    WHOLE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Unit	options from `GET /units?per=500`, filtered in the browser	assets → metering ⇄	SERVED, capped at 500
②	Effective date	the assignment boundary; defaults to tenant today	assets	SERVED

03

One asset — read it, and change it

One template for every kind of asset, and every write that acts on **one** of them. The tab set varies by kind and the structure never does — **six tabs** on a meter or sub-meter, six on an exporting inverter where Export replaces Reads, and **five** on a plain device, which stores no reading. Then the four steps that put one in, then the two that change and remove it. Bulk assignment is not here: it starts by ticking rows in the list, so it stays with the list.

08 **Overview** EXISTS BEHIND NEW BUILD COMMITMENT 12 COMMITMENT 16 COMMITMENT 22

`/:tenant/assets/{id}`

The whole redesign is a rearrangement of a payload that is already complete

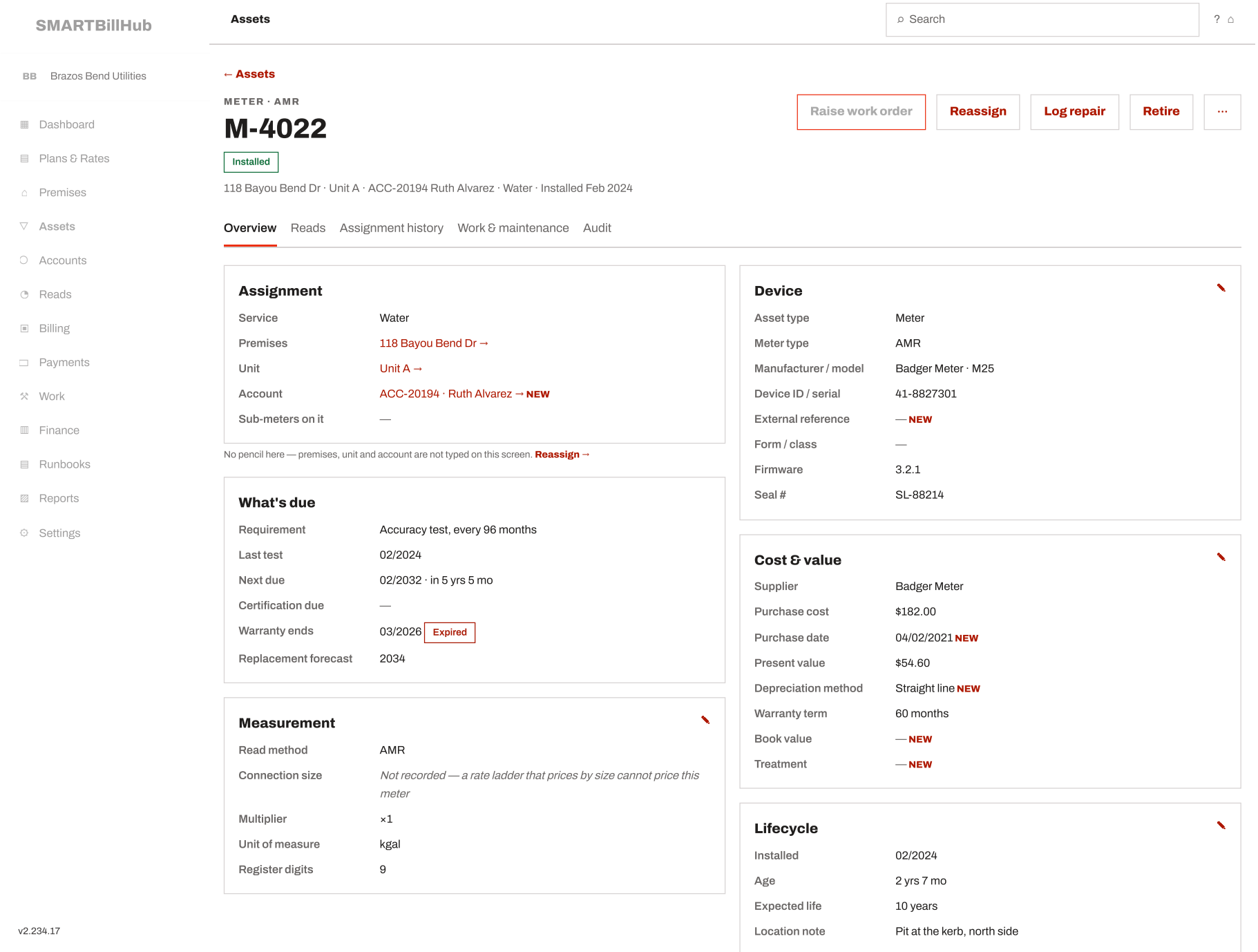
Thirty-eight facts arrive at this page today and thirty-one are drawn. The redesign draws all thirty-eight, adds the account link and the master link from ids already on the payload, and builds the attention strip from five fields the page already receives — `test_due`, `next_test`, `certification_due`, `warranty_covered` and `status`.

No new endpoint, no new column, no new query. The one thing that is not free is the absence sentences: each is a catalogue key in eleven languages, and B14 rule 35 says a catalogue is written in its own language rather than translated from English. That is a translation cost and it is why `AST-`

`R-03` says *the handful that matter* rather than *every dash*.

What is this thing, where is it, what is it attached to, and is anything wrong with it?

FIGURE 10 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



DW Dana Whitfield
System Admin

AST-02

1 **The header, five slots** · kicker without the number · the number once, in the title · the status once, in the pill · the place line leading with the **address** and carrying the **account**, which is served today and shown nowhere · and the multiplier **gone** from it, because a ×1 is configuration rather than identity.

2 **Six panels, in the order they are read** · Assignment · What's due · Measurement · Device · Cost & value · Lifecycle. The first two answer the questions that bring somebody here; the rest is reference. **All 38 facts carry over** and the carry-over below reconciles every one.

3 **Seven rows that were already on the wire** — account, external reference, purchase date, depreciation method, allocation rule, loss tolerance, book value and its treatment. Marked **NEW** on the drawing.

4 **A pencil per panel**, not one Edit for twenty-three fields. *What's due* has none: nothing on it is typed, every row is derived.

The header, in every state it has

Eight worked examples, header only. The last two are the ones that matter most: a strip in the quieter register when there is nothing to *do* about the fact, and — the whole point — **no strip at all** when nothing is due.

AN INSTALLED METER

← Assets

METER · AMR

M-4022

Raise work order | Reassign | Log repair | Retire | ···

Installed

118 Bayou Bend Dr · Unit A · ACC-20194 Ruth Alvarez · Water · Installed Feb 2024

A METER IN INVENTORY — NO CHAIN TO SHOW, AND IT SAYS SO RATHER THAN DRAWING FOUR DASHES

← Assets

METER · AMI

AST-04412

Assign | Edit | Delete

Inventory

In inventory · unassigned · never installed

A SUB-METER — ITS MASTER IS THE FACT THAT MATTERS, AND IT IS A LINK

← Assets

SUB-METER · WALK-BY

S-2001

Raise work order | Reassign | Log repair | Retire | ···

Installed

Sub of E-5000 · 118 Bayou Bend Dr · Unit 1 · ACC-20194 Ruth Alvarez · Water

A BACKFLOW PREVENTER WITH A LAPSED TEST — THIS IS WHAT THE ATTENTION STRIP IS FOR

← Assets

BACKFLOW PREVENTER

BF-0001

Raise work order | Reassign | Log repair | Retire | ···

Installed

⚠ Annual test overdue 11 days · due 06/15/2026

Raise work order →

401 Quarry Hill Ct · Unit 1 · ACC-20188 Cole Whitfield · Water

A MASTER METER — THE SUB COUNT IS THE FACT A CLERK NEEDS

← Assets

METER · MASTER

M-MASTER-01

Raise work order | Reassign | Log repair | Retire | ···

Installed

Apt 101 · 3 sub-meters on it · ACC-20201 Bayou Bend HOA · Water · Installed Jan 2026

A WARRANTY THAT HAS RUN OUT — THE SAME STRIP, THE QUIETER REGISTER, AND NO ACTION TO OFFER

← Assets

METER · WALK-BY

W-5002

Raise work order | Reassign | Log repair | Retire | ···

Installed

Warranty ended 03/2026

118 Bayou Bend Dr · Unit 1 · ACC-20194 Ruth Alvarez · Water · Installed Jan 2026

A RETIRED METER — THE PILL CARRIES IT AND THE ACTIONS COLLAPSE TO NONE

← Assets

METER · WALK-BY

W-5004

···

Retired

Retired 08/14/2026 · Failed accuracy test · Replaced by W-5021

Last at 118 Bayou Bend Dr · Unit 1

NOTHING DUE — THE STRIP IS ABSENT, WHICH IS WHAT MAKES IT WORTH READING WHEN IT APPEARS

← Assets

METER · AMI

W-5001

Raise work order | Reassign | Log repair | Retire | ···

Installed

118 Bayou Bend Dr · Unit 1 · ACC-20194 Ruth Alvarez · Water · Installed Jan 2026

What changes when the kind changes

Figure 11 · The structure never moves. The panel SET does, and a panel a kind has no data for is **absent, not empty** — AST-B-02, and the same

Figure 11 The structure never moves. The panel SET does, and a panel a kind has no data for is **absent, not empty** — `AST-R-02`, and the same reasoning the product already applies to the Export column on the Reads table.

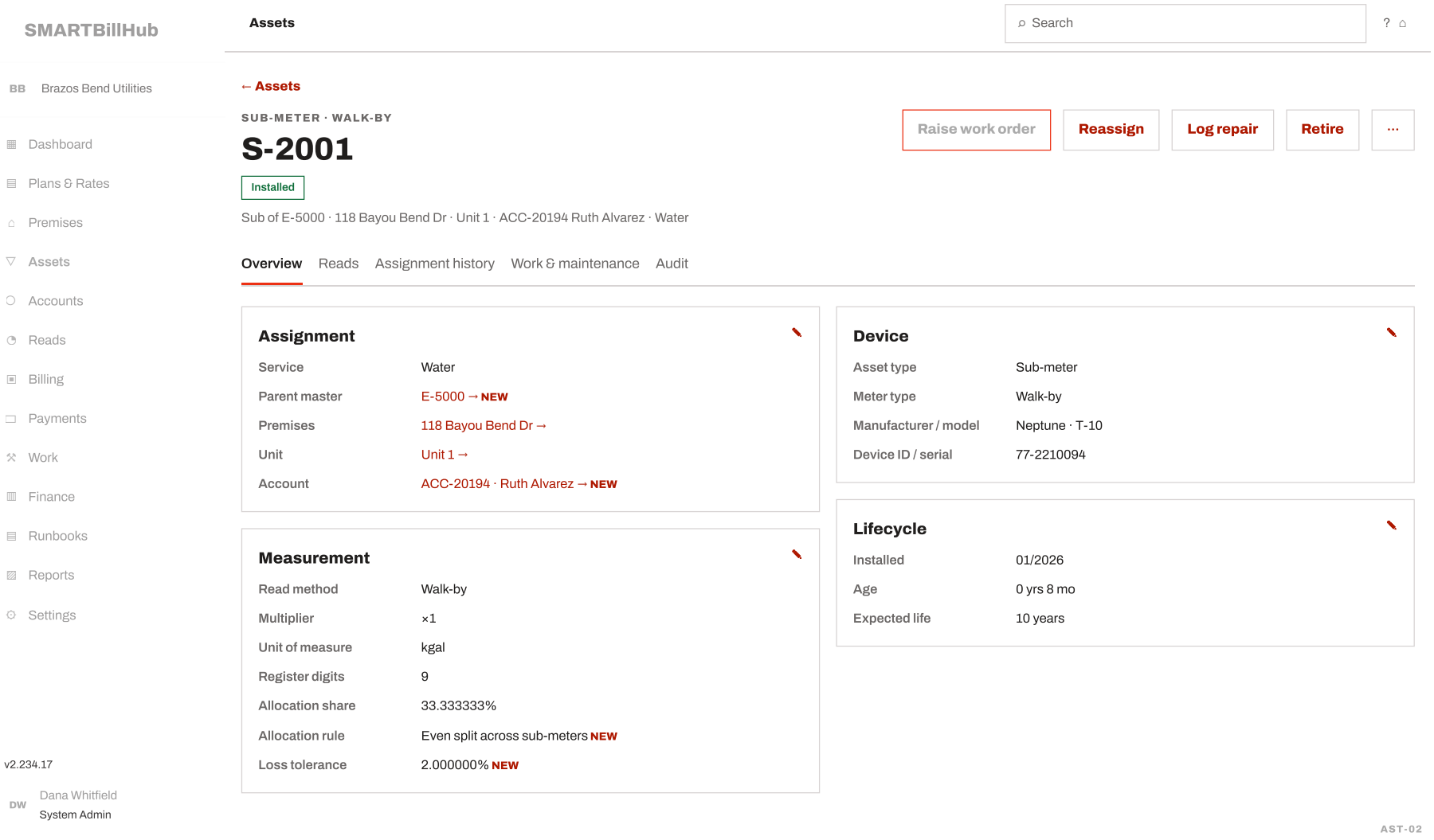
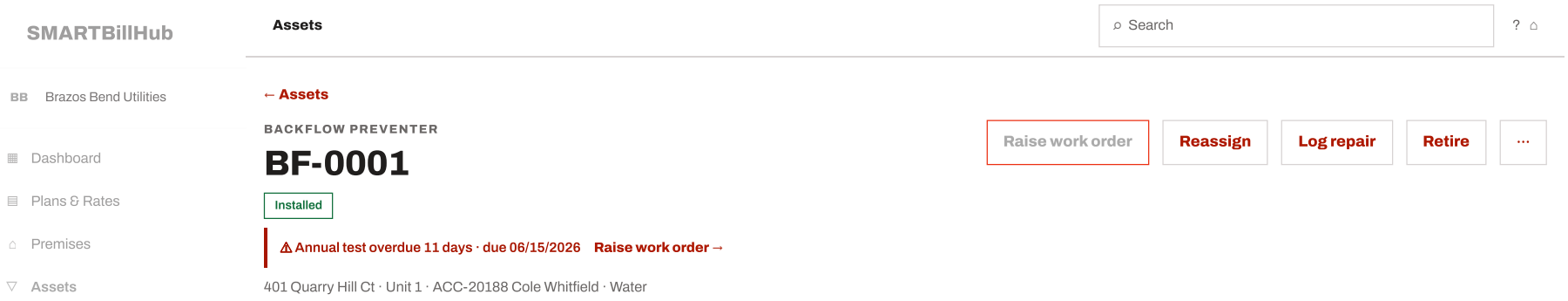
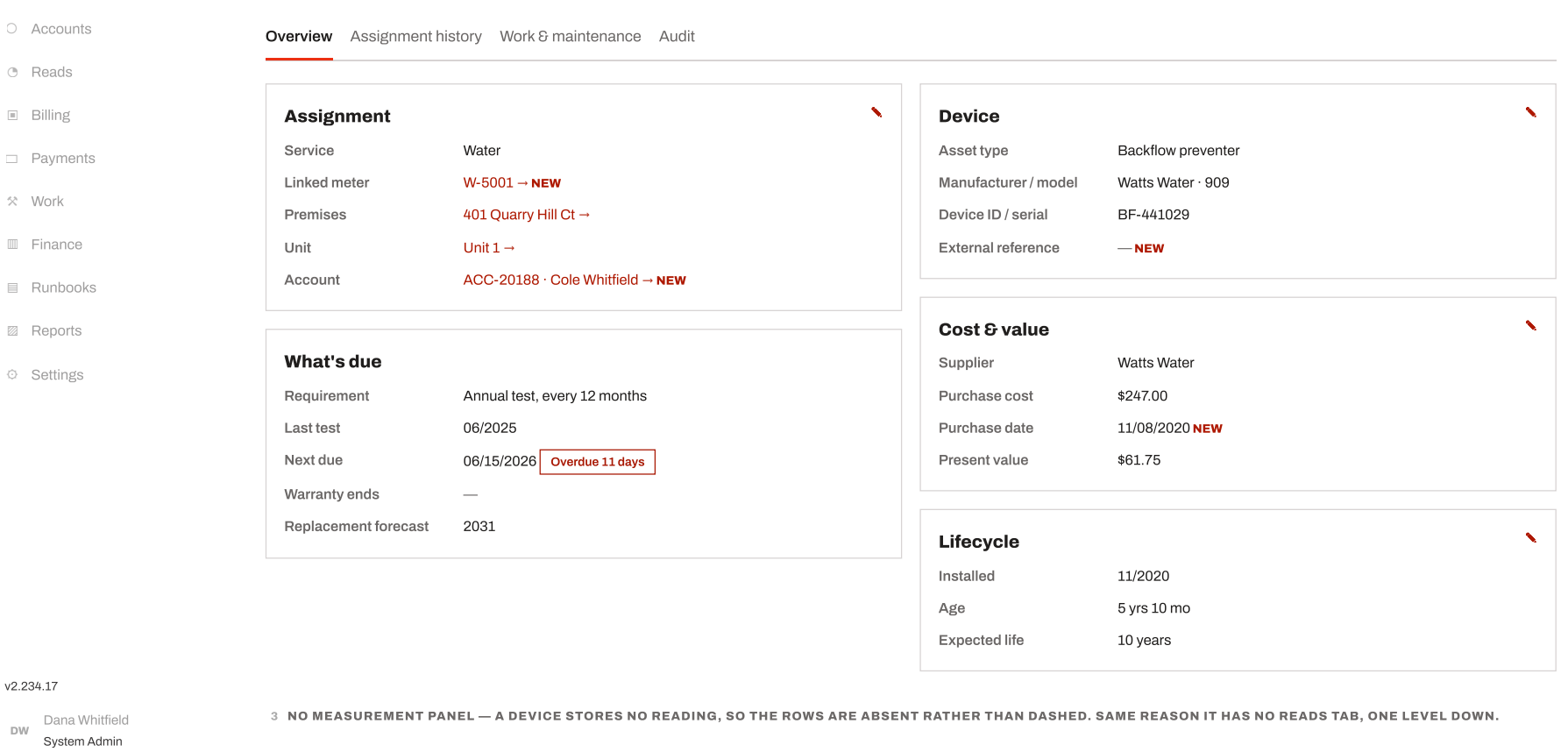


FIGURE 12 · 5 A SUB-METER: ASSIGNMENT NAMES ITS MASTER AND LINKS IT, MEASUREMENT GAINS ALLOCATION SHARE, RULE AND TOLERANCE — THE THREE ROWS NO OTHER KIND HAS.





Carry-over — all 38 facts, and where each one went

As-built element	Disposition	Note
Specifications, 23 rows	SPLIT	Into five of the six panels. It was one list mixing identity, classification, metering configuration, physical build, install facts, location and end-of-life — nothing in it was grouped by the question it answers.
Asset # (row 1)	MOVED	To the header title, where it already was. It was printed twice, adjacent.
Status (row 21)	MOVED	To the header pill, where it already was. The pill carries the complete value, so a labelled row adds nothing.
Parent (rows 10 and 27)	MERGED	It rendered in **two** panels. One row now, in Assignment, and a **link** — `parent_asset_id` is served and the cell was plain text.
Last test date (rows 18 and 39)	MERGED	The second duplication. One row, in What's due.
Warranty & supplier, 5 rows	SPLIT	The warranty *term* is a purchase fact and goes to Cost & value; the *date* it ends is a thing that expires and goes to What's due.

Lifecycle, 4 rows	SPLIT	Age and expected life stay; certification due and replacement forecast are things that fall due.
Test & certification, 3 rows	MOVED	Into What's due, whole.
Account, premises, unit	UNCHANGED · now linked	Premises and unit were already links. **The account was not**, so the chain ran Premises → Unit → dead end, and the dead end was the money.
Every label	UNCHANGED	Not one row is renamed. Only the **panel** names change, and each is now at most two words.
Anything at all	**REMOVED — none**	All 38 distinct facts carry over. The only two things that stop appearing twice are Parent and Last test date, and each still appears once.

```
Facts on the as-built      40 rendered rows − 2 duplicates = 38 distinct
  carried over             38
  removed                   0
  added                     7   (+ export capacity, on an inverter)
Panels                      5 → 6
Rows per panel, worst      23 → 9                                ✓ reconciles
```

The screen

For	Everyone. It is the screen every other asset screen returns to.
How you get here	The asset number in any registry tab, the map popup, a premises or a unit, and the address bar.
Primary action	**Raise work order** — which `BR-WRK-056` names and the as-built does not build. Then Reassign, Log repair, Retire, and Edit and Delete under the overflow.
Tabs	**Five** on a meter or sub-meter · five on an exporting inverter, Export replacing Reads · **four** on a plain device. The tab SET varies and the structure never does.
Table controls	None — this screen holds no table.
Write surfaces	**Edit** — a pencil per panel, opening the form at that group rather than one control over twenty-three fields. **Delete** — under the overflow, and refused with its reason on an asset with history.
Walked	The **as-built** was opened and every label read from the source. **The drawings above are illustrations** (B14 rule 37): the shape, the panel each fact lands in, the rows each kind does and does not have, and every label are read from the code — **the figures are not measured**, and this tenant's seeded assets carry almost none of them.

Spacing	Two columns at 16, panels at the standard 20, the attention strip 8×12 on a 3px rule.
Cognitive load	It answers *whose meter is this* and *does anything need doing* before anything else, and it stops asking a reader to find a 23-row list's structure for themselves.
States	**Assigned · in inventory · retired**, each with its own place line and its own action set. **Attention** — present only when something is due, expired or broken; **absent** otherwise, which is what makes it worth reading.
Words	**Assignment**, **What's due**, **Measurement**, **Device**, **Cost & value**, **Lifecycle** — six labels, none longer than two words, each naming the question rather than the table the rows came from.
Components	`PageHeader` + a new attention strip · `StatusPill` · `DateCell` · `MoneyCell` · the fact list · `GuardedButton`.
Rules	`AST-R-01–13`, below · `BR-AST-026` (the kicker names the kind) · `BR-AST-039` (the live billing chain) · `BR-AST-044` · `BR-WRK-056` · B14 rules 18, 31, 32.
Audience	Staff.
Health	SERVED throughout. **Every row on every drawing above is already on the wire** — the seven marked NEW included. This screen costs nothing on the server.

Data flow

```
GET /assets/{id} ──→ asset_dict()        38 flat keys        SERVED
                  ├→ specs{}             11 keys             SERVED
                  ├→ classifications{}   3                   SERVED
                  ├→ billing_chain{}     premises·unit·account  ⇄ metering, customers
                  ├→ valuation{}         supplier·cost·date·value  SERVED
                  ├→ lifecycle{}         age·life·forecast   DERIVED, server-side
                  └↳ sub_assets[]        ⇄ assets            SERVED
the attention strip ← test_due · next_test · certification_due
                      · warranty_covered · status           ALL SERVED, RENDERED NOWHERE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	The header place line	`assigned_address` + `unit_label` + `billing_chain.account`	assets → metering, customers ⇄	SERVED

②	The attention strip	`test_due` · `next_test` · `certification_due` · `warranty_covered`	assets	SERVED
③	Account	`billing_chain.account` — served, and **plain text** today	assets → customers ⇄	SERVED
④	Parent master	`parent_asset_no` + `parent_asset_id` — the id is served and the cell is plain text	assets	SERVED
⑤	External reference	`assets.external_ref` — served on every row, rendered nowhere in the module	assets	SERVED
⑥	Purchase date	`assets.purchase_date` — age and present value are computed from it and both are shown; the date is not	assets	SERVED
⑦	Depreciation method	`assets.depreciation_method` — present value is computed from it and the method is never named	assets	SERVED
⑧	Allocation rule · loss tolerance	`specs.allocation_rule` · `specs.loss_tolerance_pct`	assets	SERVED
⑨	Book value · treatment	`book_value_amount` · `book_value_treatment` — written at retire, never readable again	assets	SERVED
⑩	Export capacity	`assets.export_capacity_kw` — a registry column, absent from the device's own page	assets	SERVED

AST-B-22

Somebody opening an asset sees what it is, where it is and whether anything is wrong, without reading a twenty-three row list

STATUS Unverified

WAITS ON NOTHING. Every field is on the wire, both link targets are on the payload, and the attention strip is five served fields and a condition.

SCREENS 08

WHAT DONE LOOKS LIKE ON SCREEN Opening `BF-0001` shows *"⚠ Annual test overdue 11 days · due 06/15/2026"* under the title with a **RAISE WORK ORDER** control beside it, and **NO MEASUREMENT PANEL AT ALL** — a backflow preventer stores no reading. Opening a healthy meter shows **NO STRIP**.

The rules this screen is built to

Fifteen rules for the detail view, minted here because no rule set in this repository covers a screen whose shape changes with the kind of thing it is showing. **They are PROPOSED, not ratified** — B14 is the authority and none of these has been put to the PO yet. The last column is the point: **almost all of them are checkable by a machine**, and none of them by a gate. Nothing in this repository reads rendered output — but `vitest` does, so these belong as component tests beside `AssetDetail.test.tsx`, not as a merge gate.

Rule	What it says	How it is checked
AST-R-01	**One template, every kind.** The detail view is one component. The **tab set** and the **panel set** vary; the structure, the order and the panel names never do. This restates the rule the code already states in its own words — *"ONE template, EVERY kind — the tab SET is what varies, the structure never does"* — and extends it from tabs to panels.	Human review. A snapshot of panel order per kind makes drift visible.
AST-R-02	**A panel or row a kind cannot have is ABSENT, not empty.** Never draw a row of dashes for a fact the kind has no source for. The precedent is already in the product, on the Reads table: *"a column of dashes on a meter that cannot export would read as measured nothing, which nobody did."*	**Machine.** Render a device and assert the Measurement panel and its labels are absent from the DOM.
AST-R-03	**Absence by KIND is silent; absence by DATA is stated.** The two are different facts and the screen must not render them the same way. *This kind has no register* → the row is not drawn. *This meter has no connection size recorded* → the row IS drawn, with a dash, and where the absence has a consequence the consequence is named. The product already does the second half: *"Not recorded — a rate ladder that prices by size cannot price this meter."*	**Machine.** Two fixtures per contested row: the wrong kind (absent) and the right kind with a null (present, dashed).
AST-R-04	**Measurement belongs to measuring kinds only.** Multiplier, unit of measure, register digits and read method render on a meter and a sub-meter and nowhere else — the same reasoning that gives a plain device no Reads tab, applied one level down.	**Machine.** Per-kind render test.
AST-R-05	**Allocation belongs to sub-meters only.** Allocation share, allocation rule and loss tolerance render on a sub-meter and on nothing else. A null share means *metered, not allocated* and is served as null — **never as a zero**, which would read as *allocated nothing*.	**Machine.** Per-kind render test, plus one asserting a null share does not render `0%`.
AST-R-06	**Export capacity belongs to an exporting device only**, and it is a capacity in **kW**, never money. `BR-AST-025` keeps tariffs and credited dollars off this whole kind.	**Machine.** Render test, plus the currency-symbol assertion `BR-AST-025` already earns.
AST-R-07	**Every kind names what it hangs off, and names it as a link.** A sub-meter names its master; a behind-the-meter device names its linked meter; a master names the sub-meters on it. Today the master is **plain text** while `parent_asset_id` is served — a dead end with the id sitting right beside it.	**Machine.** Assert an anchor, not a string, for each of the three.

AST-R-08	**The tab set and the panel set are DERIVED, never written per screen.** One function decides each from the asset, so a new kind changes that function and nothing else. `assetDetailTabsFor()` already does this for tabs; panels need its twin.	**Machine.** A grep test: no tab id or panel id literal outside the two deciding functions.
AST-R-09	**A fact is said once.** The status is the pill; the number is the title; every other fact lives in exactly one panel. Where the header *abbreviates* a chain — *118 Bayou Bend Dr · Unit A · ACC-20194* — the panel carries it in full: that is a summary, not a repeat. The test for the difference: **does the header carry the complete value?** A pill does; an address line does not. **This screen breaks the rule twice today** — Parent and Last test date each render in two panels.	**Machine, and cheap.** Assert every `<dt>` label appears at most once. It would have caught both.
AST-R-10	**No database id is ever a visible value** — B14 rule 31, inherited and restated because this screen carries **five** ids on the wire (`id`, `service_point_id`, `premises_id`, `unit_id`, `parent_asset_id`) and four of them exist to be link *targets*. A target is not a value.	**Machine, and cheap.** Assert no rendered text matches a UUID.
AST-R-11	**Panel order is fixed and does not vary by kind.** Assignment · What's due · Measurement · Device · Cost & value · Lifecycle. A kind that skips a panel **closes the gap**; it never reorders. Somebody who learns the screen on a meter must not have to re-learn it on a device.	**Machine.** Assert the rendered panel sequence is a subsequence of the canonical six.
AST-R-12	**A panel label is at most two words, and names the question rather than the table it came from.** PO instruction, 2026-09-11, verbatim: *"I need label - maximum 2 words only panel label should be 2 words only"*. *Assignment*, not *Where it is and what it bills*.	**Machine, and trivial.** Word-count every panel label.
AST-R-13	**Adding a kind is a four-line change.** A new kind declares its tab set, its panel set, what it hangs off, and its Measurement rows — and touches nothing else. **If a new kind needs an edit anywhere else in the detail view, the template has drifted** and AST-R-01 has already been broken. This is the rule that keeps the other twelve true a year from now.	Human review, at the moment a kind is added — which is the only moment it can be checked.
AST-R-14	**A field's helper is one line — at most ten words — and never wraps.** Not a style preference: a helper that wraps **pushes its own input down**, and in a two-to-a-row grid that leaves the two inputs of one row on different baselines. **The alignment defect and the copy defect are the same defect.** Measured across the assets module: **27 of 37 helper strings run over ten words**, and the worst is **36** — *"The size of the service connection this meter sits on. A rate ladder that prices by size uses it, so a wrong one changes the bill. Leave it blank if you do not know it yet."* B14 rule 32 already caps a screen's description budget and rule 33 caps a refusal at five or six words; a field helper had no cap at all.	**Machine, and trivial.** Word-count every helper string in the catalogue. It needs no rendering — the copy is the defect.
AST-R-15	**Two columns are two fields.** Never one input holding both with a delimiter the person has to type. The product does this once — **Manufacturer / model**, with the help *"One line; separate the two with a middle dot"* — and it is the only field in the module that asks a human to do a parser's job. A model name containing the delimiter, or a person reaching for a slash instead, lands in the wrong column, and nothing on the screen says which half is which.	**Machine.** Assert that no input's value is split or joined on a delimiter between the form and the payload.

Displaying them joined is fine; the detail view's single *Manufacturer / model* row is good density. The rule is about CAPTURE.

AST-R-16	**Every choice control names where its options come from, on the drawing itself.** Four markers and nothing else: `SET` a tenant-editable Settings list, named by its list code · `TBL` live rows from another table, named with its module · `FIX` frozen in the source, with the count · `NONE` no list exists, or the product renders free text over a list that does. **A drawn drop-down is a box with a word in it**, and a picture cannot distinguish a row a utility can add this afternoon from a string that needs a release — so a coder reading the drawing alone guesses, and the guess is invisible once it is written. The marker travels ON the field rather than in a table beside it, because a table beside it is the thing that goes stale. **The full map is Appendix L.**	**Eye.** Nothing mechanical can check this — the claim is about a list's PROVENANCE, which no gate in this repository can see. It is checked by reading Appendix L against the code the day a form is built, which is exactly when it is cheapest.

Why these are tests and not a gate. This repository's standing weakness is that no gate reads rendered output, which is why most B14 rules are ratified and unenforced and why nine defects were once found only by a human opening the running app. **Ten of these thirteen are assertions about the DOM**, and the component suite already renders this screen — so the enforcement that does not exist as a gate exists already as a test file. Two of them (AST-R-09 and AST-R-10) are three lines each and would have caught faults this document found by reading.

Interaction design — what a person can do here

Seven decisions, each one a change to what the screen *does* rather than to what it says.
Six of the seven cost nothing on the server.

Decision	Today	Proposed, and why
The account is a link	**Plain text** — `account_no · customer_name`, rendered with no anchor.	Link it. Premises and Unit are both links already, so the chain reads **Premises → Unit → dead end**, and the dead end is the money. The id is on the payload (`billing_chain.account`); nothing new is needed.
The master is a link	**Plain text** — `{asset.parent_asset_no}`, no anchor, while `parent_asset_id` sits on the same payload.	Link it. A sub-meter's master is the single most likely next click from a sub-meter, and the id needed to build the address is already in hand.
Edit per panel	**One Edit control for 23 fields.** A person correcting a seal number opens a form holding every fact about the asset.	A pencil on each panel, opening the edit form at that group. **Twenty-three fields in one form is why nobody edits** — and it is also why a mistyped field is hard to spot before saving. The form is already a page (screen 14), so it can carry an anchor.

What's due carries its action	Dates in a list. A person reads *Next due 06/15/2026*, and then has to go and find the control that does something about it.	Each overdue or approaching item carries the one control that resolves it — *Annual test overdue 11 days* → **Raise work order** →. It turns a reference panel into a working one, and it is the same destination `BR-WRK-056` already names.
Absence names its consequence	A dash, for every missing fact, whatever it costs.	Where an absence has a consequence, say it. **The product already does this once** and it is the best sentence on the screen: *"Not recorded — a rate ladder that prices by size cannot price this meter."* Apply the same treatment to the handful of others that matter and leave the rest as dashes — a screen where every dash carries a sentence is a screen nobody reads.
Status said once	A pill in the header **and** row 21 of Specifications.	The pill only. Saying it twice makes neither authoritative, and it is the cheapest of the three duplications this screen carries to remove.
The header earns its attention strip	Nothing. A meter with a test eleven days overdue looks exactly like a healthy one until you scroll to the fifth panel.	A strip that appears **only when something needs doing**, and is absent otherwise. Every fact it needs is already served — `test_due`, `next_test`, `certification_due`, `warranty_covered`, `status`. **Silence when healthy costs nothing and is what makes the strip worth reading when it appears.**

What each one costs

```
the account link      client only   — the id is already on the payload
the master link       client only   — same
edit per panel        client only   — the form is already a page and can take an anchor
what's-due actions    client only   — the destination already exists (BR-WRK-056)
absence sentences     copy only     — eleven catalogues, one key each
status said once      client only   — delete a row
the attention strip   client only   — five served fields, no new read
                      ──────────────
                      NO SERVER CHANGE AT ALL
```

The one thing here that is not free is the absence sentences: each is a new catalogue key in eleven languages, and B14 rule 35 says a catalogue is written in its own language rather than translated from English. That is a translation cost, not an engineering one — and it is why the rule above says *the handful that matter* rather than *every dash*.

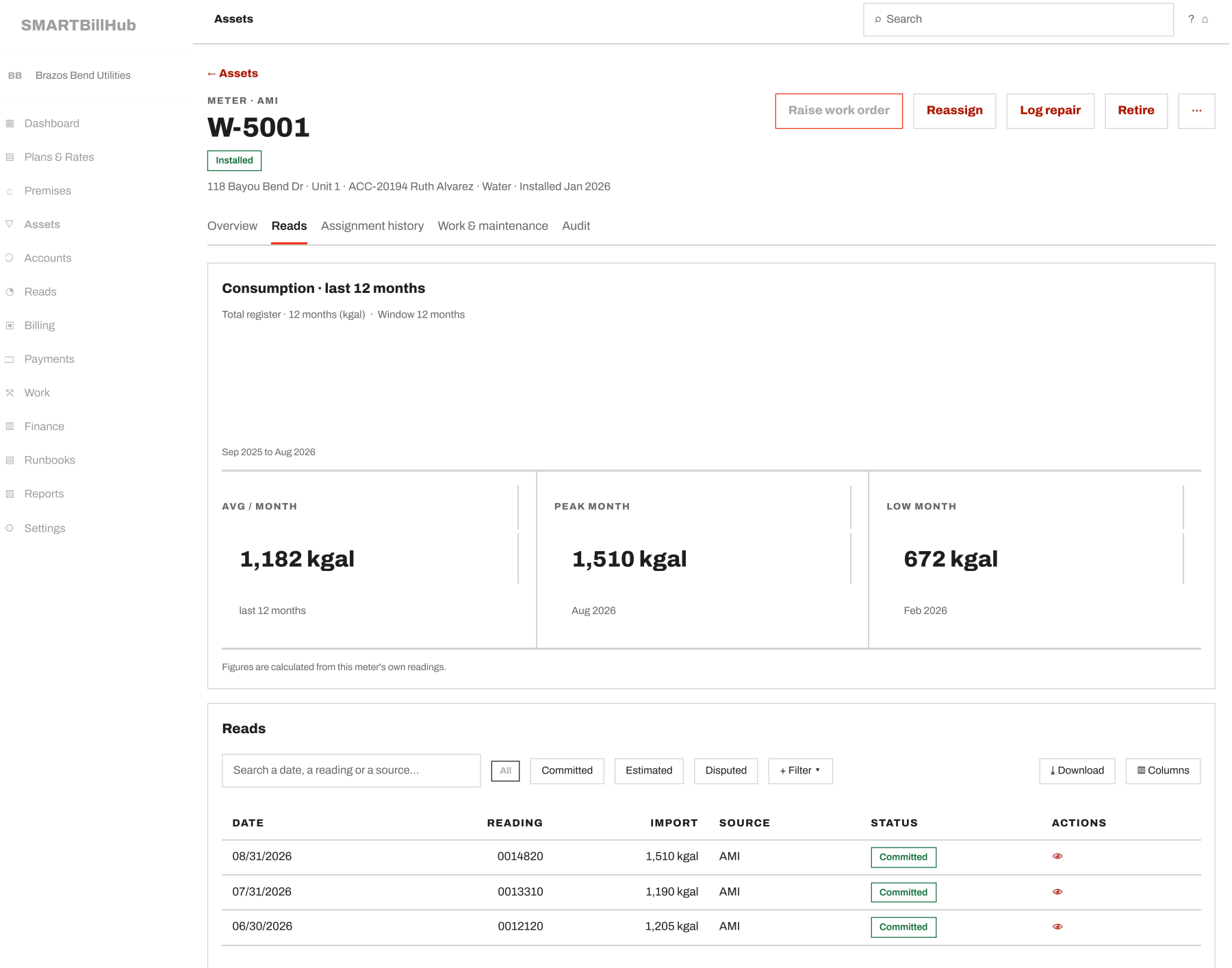
09 Reads

EXISTS BEHIND | COMMITMENT 19 | COMMITMENT 23 | COMMITMENT 24

`/:tenant/assets/{assetId}?tab=reads`

What has this meter actually recorded, and does the chart agree with the bill?

FIGURE 13 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



1–50 of 1,460 Rows per page 50 100 200 500 1000 ⇤ ← Page 1 of 30 → ⇥

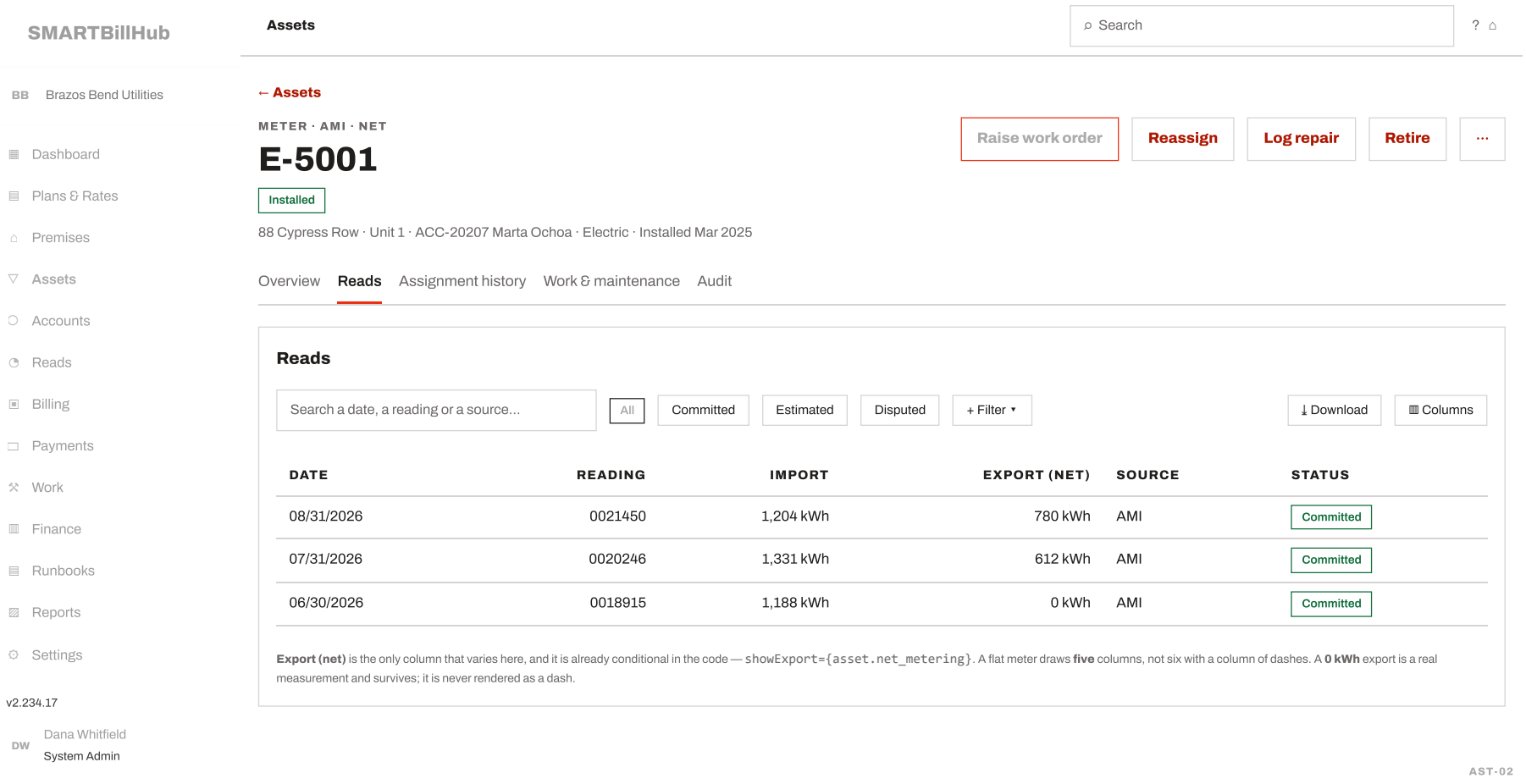
NEW — the toolbar and the pager. Today this panel is a bare table: no search, no filter, no column chooser, no download, and no pagination at any of the four layers.

v2.234.17

DW Dana Whitfield
System Admin

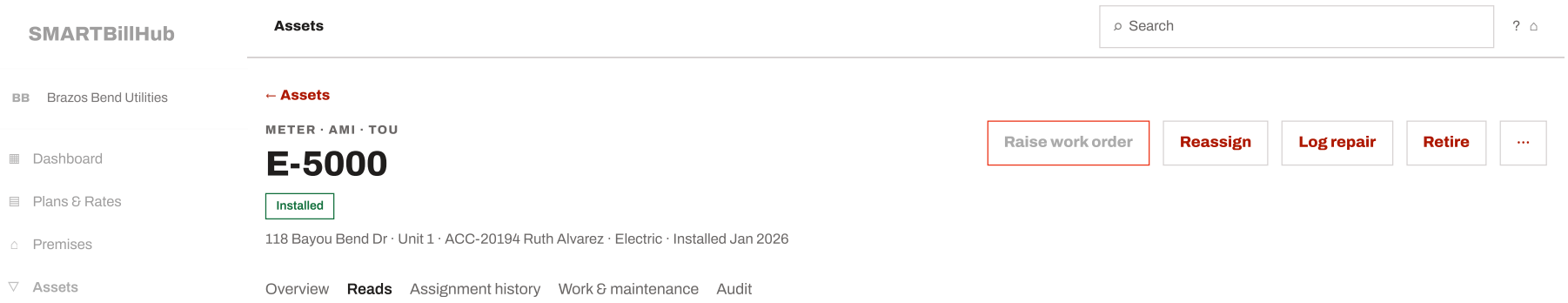
AST-02

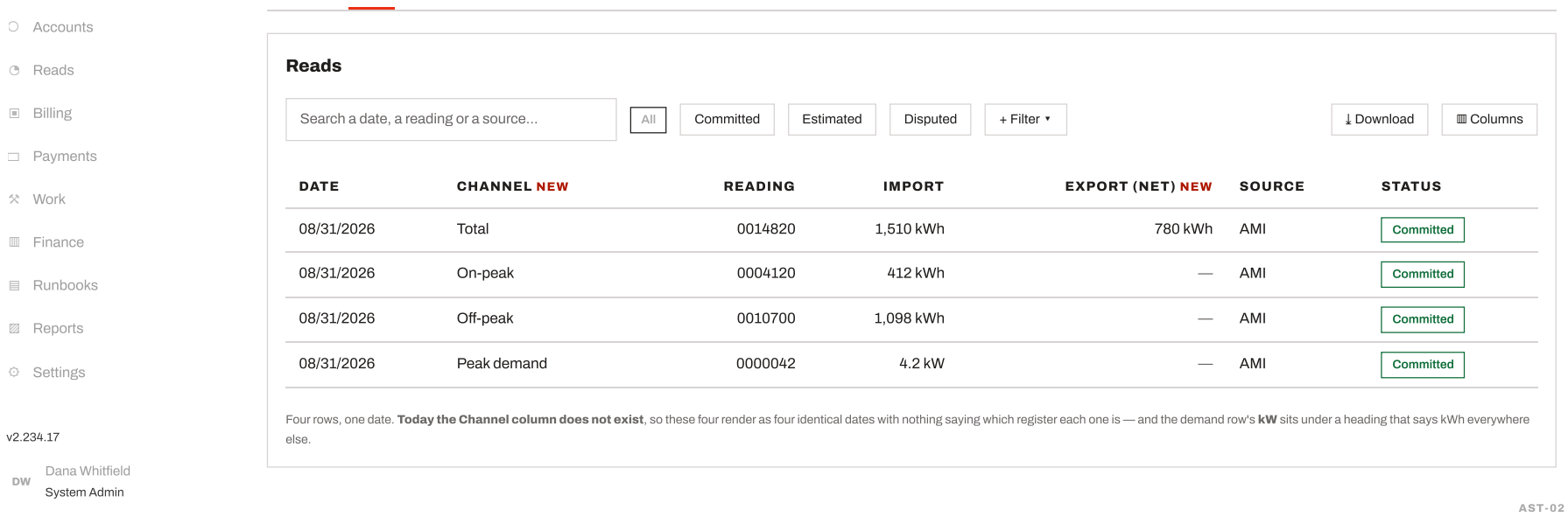
FIGURE 14 · 2 A NET METER — ONE MORE COLUMN, EXPORT (NET), AND IT IS ALREADY CONDITIONAL IN THE CODE TODAY:



AST-02

FIGURE 15 · 3 AND A METER WITH MORE THAN ONE REGISTER — TIME-OF-USE, OR DEMAND. FOUR ROWS, ONE DATE, AND TODAY NOTHING SAYS WHICH IS WHICH:





AST-02

The screen

For	A clerk checking a disputed bill against the meter's own record.
How you get here	The tab strip on screen 08. Measuring assets only — a plain device has no Reads tab at all.
Primary action	None; read-only.
Tabs	Part of screen 08's five.
Table controls	**NEW — the shared toolkit. Search** a date, a reading or a source. **Filter** — *Committed · Estimated · Disputed*, plus a date range. **Columns** — the shared chooser. **Download** — the ⤓ export, which inherits `AST-B-20`: it is broken on the deployed stack and adding it here adds a third place it fails. **Pagination** — the standard footer. **None of the five exists today**, and the pagination half is unpaginated at all four layers — endpoint, selector, seam and component. Specifications `AST-B-19` and `AST-B-23`.
Variation by kind	**Four, and they are drawn above rather than described.** ① a flat meter — five columns. ② a **net meter** — six, the extra one being *Export (net)*, already conditional in the code (`showExport={asset.net_metering}`); a flat meter draws five, never six with a column of dashes. ③ a meter with **more than one register** — four rows share one date, and nothing today says which is which (`AST-B-24`). ④ a **plain behind-the-meter device** — **no Reads tab at all**, because a device stores no reading; see screen 08's device variant. And an **inverter behind a net meter** gets the **Export** tab instead, which is screen 10 — a different table reading the LINKED meter's records.
Write surfaces	**NOTHING** on all four. A read is written through Meter Reads, never here.
Walked	Opened. The reads table is **empty** on the seeded tenant — *"No reads yet."* — and the consumption strip above it renders with its three tiles reading **No readings** rather than zero. The figures drawn here are **illustrations** (B14 rule 37), not measurements; the shape, the

	column set and the copy are read from the source.
Cognitive load	It puts the meter's own record next to the consumption the bill was computed from, on one screen.
States	**Empty** — two of them, and they are different sentences: the strip says *"No readings on this meter yet."* and the table says *"No reads yet."* **Failed** — *"We couldn't load this asset's reads."* and *"Usage could not be loaded."*, one per panel. The strip also states its own gaps rather than drawing a zero: *"{{absent}} of {{total}} months have no reading."*
Words	A net cycle is **two typed records folded per date** (BR-AST-029), not one signed number — so a customer exporting and importing on the same day reads as two facts.
Components	`AssetUsageStrip` (mounted above the table) · **the shared `DataTable` and `FilterBar` — which this panel does NOT use today**, and that is why it has no toolbar and no pager. · `DateCell` · `UnitCell`.
Rules	BR-AST-016 (no money on any read surface; quantities carry units) · BR-AST-029 · hard rule 3 (consumption always with an explicit unit).
Audience	Staff.
Health	SERVED for every value, and the transport is the problem: correct data, delivered without a bound. The strip publishes `source: "computed_from_reads"` — it says where its numbers came from rather than leaving a reader to assume.

AST-B-19

Opening a meter's reads returns one page of them, not ten years of them

STATUS Unverified

WAITS ON Nothing but the work. `GET /assets/{id}/reads` **AND** `GET /ASSETS/{ID}/ASSIGNMENTS` return the house list envelope — `rows · count · page · per` — instead of a bare list, and the fold moves into the query rather than running over the whole history in Python. Every other list in this product already does this; this one was written before the rule and never revisited.

SCREENS 09, 11, 12

WHAT DONE LOOKS LIKE ON SCREEN The Reads panel carries the standard footer — *1–50 of 1,460*, a rows-per-page control and first / previous / next / last — and opening a meter with ten years of AMI reads paints in the same time as one with three.

WHY IT MATTERS MORE THAN IT LOOKS `NFR-5` forbids a hard cap, so the answer is a page and never a `LIMIT 500`. **THREE PANELS, TWO ENDPOINTS, ONE FIX:** screen 10, the inverter's Export tab, calls the **SAME** reads endpoint against the linked meter, and `GET /assets/{id}/assignments` is the same bare-list shape — a meter that has moved between units for twenty years accumulates rows with nothing bounding them.

AST-B-23

Every list inside an asset is the same list, with the same controls

STATUS Unverified

WAITS ON The three panels using `DataTable` and `FilterBar` instead of a bare `<table>`. The components exist and every registry tab already uses them; these three were built as detail-panel tables and never revisited.

SCREENS 09, 11, 12

WHAT DONE LOOKS LIKE ON SCREEN Reads, Assignment history and Work & maintenance each carry the same toolbar a registry tab carries — search, filter chips, a column chooser, a download and a pager — and a person who has learned one list has learned all of them.

⚠ **THE DOWNLOAD INHERITS A LIVE DEFECT** `AST-B-20`: the export is broken on the deployed stack for want of a shared volume. Adding a download to three more panels adds three more places it fails, so these two ship together or the download waits.

AST-B-24

A meter with more than one register says which row is which

STATUS Unverified

WAITS ON A **CHANNEL** column, shown only when the asset has more than one. `channel` is served on every read row today and rendered nowhere — the component reads it only to decide whether a row is a demand channel and should show kW.

SCREENS 09

WHAT DONE LOOKS LIKE ON SCREEN A time-of-use meter's reads show four rows for 08/31/2026 labelled *Total · On-peak · Off-peak · Peak demand* instead of four identical dates, and the demand row's **4.2 KW** no longer sits under a heading that reads kWh on every other row.

Data flow

```
meter_reads (partitioned) ─→ GET /assets/{id}/reads        api.py:322
                           └→ BR-AST-029 fold, per date       WHOLE
```

```
meter_reads ──→ metering/asset_usage.py asset_usage_payload   ⇄ assets → metering
              └→ GET /assets/{id}/usage   metering/api.py:2967
                 └→ one chart PER REGISTER, 12 or 24 months   WHOLE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Date	`meter_reads.read_at`	assets → metering ⇄	SERVED
②	Register	the read's channel	assets → metering ⇄	SERVED
③	Reading / Consumption	the read value and the derived delta, with units	assets → metering ⇄	SERVED
④	Source / Status	how the read arrived and whether it is committed	assets → metering ⇄	SERVED
⑤	Total / Average / Highest month	`asset_usage_payload` metrics, or a stated absence reason	assets → metering ⇄	SERVED

AST-B-16

The two facts the Overview shows that no column serves

WAITS ON A decision about where each is computed. Neither rides on `asset_dict()`'s 38 keys.

SCREENS 08

WHAT DONE LOOKS LIKE ON SCREEN Opening **BF-0001** shows **LIFETIME MAINTENANCE $1,284.00** and **OPEN WORK ORDERS 2** in the **COST & VALUE** and **WHAT'S DUE** panels, both served rather than computed in the browser from a list the page happens to hold.

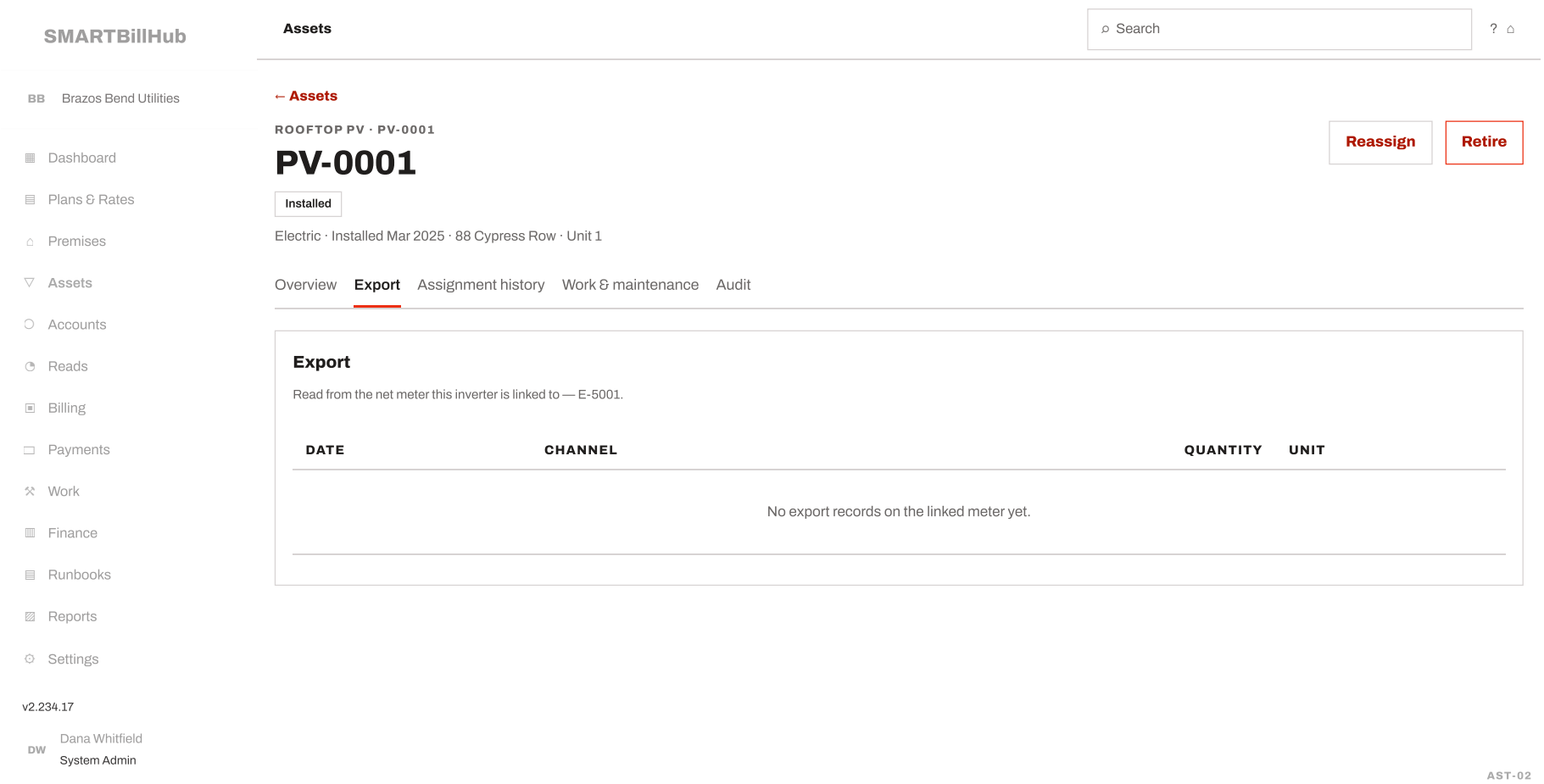
STATUS UNVERIFIED — engineering fills this in from the code, one row at a time.

10 Export EXISTS BEHIND NO COMMITMENT

`/:tenant/assets/{assetId}?tab=export  — an inverter whose linked meter is a net meter, only`

This customer has solar. How much did they push back last month, and where is that number from?

FIGURE 16 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	A clerk reconciling a net-metering credit.
How you get here	The tab strip, and only on a behind-the-meter asset whose `linked_meter.net_metering` is true. On any other device the tab is not rendered at all — the tab set is a function of the kind, so there is never a dead pane to click into.
Primary action	None; read-only.
Tabs	Overview · **Export** · Assignment history · Tests & maintenance · Audit. Note the fourth is relabelled for a device: a backflow preventer gets *tested*, not repaired.
Table controls	None.
Write surfaces	**NOTHING** on all four.

Walked	Not reached — the seeded tenant's only net meter had no linked exporting inverter, so this tab did not appear on any asset opened in this pass. Its structure is read from source (`AssetDetailPage.tsx:1143-1220`). Marked UNVERIFIED rather than graded on a read.
Cognitive load	It answers *where does the export figure come from* by naming the meter, on the panel, in words.
States	**Empty** — *"No export records on the linked meter yet."*
Words	The panel says **"Read from the net meter this inverter is linked to"** and names it. An inverter does not measure; the meter does. Saying so on the screen is what stops a clerk hunting for a register on the inverter.
Components	the detail-panel table · `DateCell` · `UnitCell`.
Rules	B16 §5.1b (B) · BR-AST-010 (net metering) · BR-AST-016.
Audience	Staff.
Health	UNVERIFIED — not walked, so not graded WHOLE.

Data flow

```
assets.linked_meter (a net meter) ─→ asset_detail["linked_meter"]
                                   └→ that meter's export channel records
                                      └→ the Export table       ⇄ assets → metering
                                                                UNVERIFIED — not walked
```

Every field, and where its data comes from

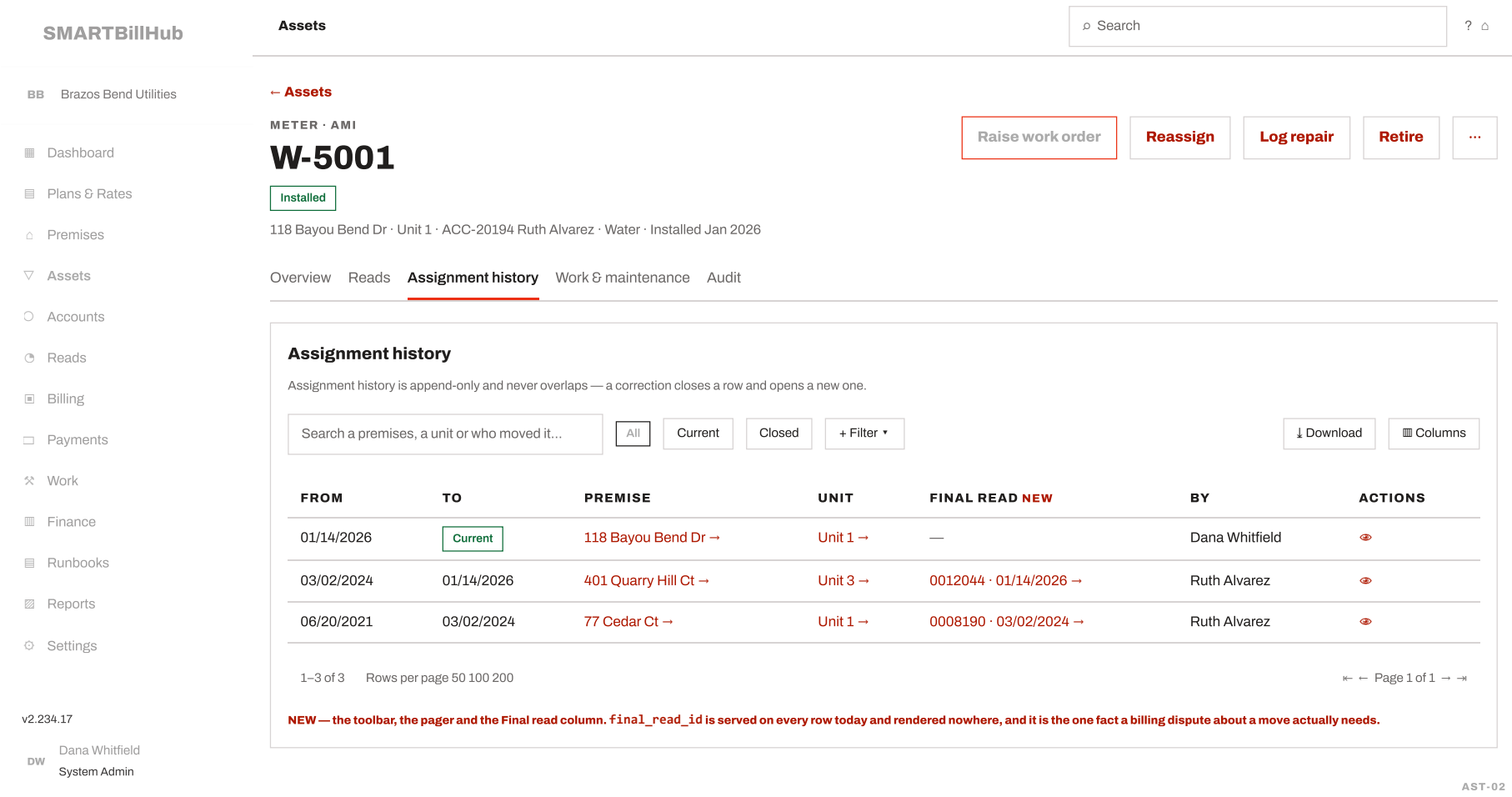
	Field	Comes from	Owned by	Health
①	Date	the linked meter's read date	assets → metering ⇄	UNVERIFIED
②	Channel	the export channel on that meter	assets → metering ⇄	UNVERIFIED
③	Quantity · Unit	the channel quantity, always with its unit	assets → metering ⇄	UNVERIFIED

11 Assignment history EXISTS BEHIND COMMITMENT 19 COMMITMENT 23

`/:tenant/assets/{assetId}?tab=assignments`

This meter has been on three addresses. Which bill belongs to which one?

FIGURE 17 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	Anyone reconstructing where a device was on a given date.
How you get here	The tab strip on screen 08.
Primary action	None; read-only. The move itself is screen 22.
Tabs	Part of screen 08's set.
Table controls	**NEW — the shared toolkit. Search** a premises, a unit or who moved it. **Filter** — *Current* · *Closed*, plus a date range. **Columns** · **Download**. **Pagination. None exists today:** `GET …/assignments` returns a **bare list** like the reads endpoint beside it.

	Download · **Pagination**. **None exists today**: `GET /assets/{id}/assignments` returns a **bare list** like the reads endpoint beside it — no envelope, no `count`, no page — and the panel renders all of it. A meter that has moved between units for twenty years accumulates rows with nothing bounding them. `AST-B-19` covers both endpoints.
Variation by kind	**None. Checked, not assumed.** The tab is on all four kinds and the table is identical on every one, because an assignment is asset → unit whatever the asset is. What varies is **state** — never assigned, currently assigned, or closed and back in inventory — and each is a row shape rather than a column set.
Write surfaces	**NOTHING**. Append-only by rule: a move closes the current row and opens the next on the same date, and neither can be edited afterwards.
Walked	Opened. Empty on the seeded asset — *"This asset has never been assigned"*. Zero failed requests. Newest-first is honoured here (`selectors.py:823-825`), which is worth noting against screen 01's default.
Cognitive load	It turns *which address was this meter at in March* from a reconstruction into a row.
States	**Empty** — as walked.
Words	**Final read**, not *closing read* — it is the read that closes the meter's billing life at that address, and it commits in the same transaction as the move.
Components	the detail-panel table · `DateCell` · `UnitCell`.
Rules	BR-AST-017 (append-only, non-overlapping) · BR-AST-022 (the final read and the move are ONE transaction).
Audience	Staff.
Health	SERVED.

Data flow

```
asset_assignments ─→ GET /assets/{id}/assignments      api.py:323
                     └→ ordered NEWEST FIRST               selectors.py:823
                        └→ the history table               WHOLE
   ⚠ asset_assignments.final_read_id is a DEFERRED FK and is UNADDABLE:
     meter_reads is partitioned on (tenant_id, read_at, id), so no unique(id)
     exists to reference. Integrity is the capture service's law, not the DB's.
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	From / To	`asset_assignments.start_date` / `end_date`	assets	SERVED
②	Premises · unit	by walking to premises	assets → metering ⇄	SERVED
③	Final read	`asset_assignments.final_read_id` → the read's value	assets → metering ⇄	SERVED, no FK
④	By	the user who made the move	assets → identity ⇄	SERVED

12 Work & maintenance

EXISTS BEHIND · NEW BUILD · COMMITMENT 02 · COMMITMENT 18 · COMMITMENT 19 · COMMITMENT 21 · COMMITMENT 23

`/:tenant/assets/{id}?tab=work`

The same question has two answers in two modules, and neither knows about the other

What has this meter cost us to keep running? The product computes that twice, in different ways, and the asset only ever shows one of them.

```
assets   asset_maintenance.cost      a number a clerk TYPES
         └→ maintenance_lifetime_cost()   summed into "Lifetime cost" on this tab

work     work_order_cost_lines(order)     labour (minutes × rate snapshot) + parts
         └→ BR-WRK-051, "the ONE place the arithmetic lives"
         └→ totalled on Work's performance tab — 740 jobs · $56,337 on the sample utility
         └→ reaches the ASSET: NOWHERE
```

So a meter whose repairs all went through work orders shows a lifetime cost that misses every one of them — and the more the utility uses work orders properly, the wronger the asset's own number gets. Nothing joins the two: `work_order_cost_lines` is per order, and `maintenance_lifetime_cost` sums one column of one table.

This is cheaper to fix than it looks, because neither figure posts anywhere. Both are captured and deliberately reach no journal line, under the same ruling — §10-D3 for asset maintenance (*"cost is CAPTURED — stored 4 dp, totalled on the tab, and posted NOWHERE"*) and `BR-WRK-051`'s own note that field labour is not posted because *"payroll already posts the hour, and posting field labour beside it would count the same hour twice."* So this is a **display** problem, not a ledger one: the merged tab shows both totals and their sum, each labelled by where it came from, and no money moves.

It is also the strongest argument for the merge. Two tabs could show two costs and look correct. One tab cannot — the moment both halves sit on

one screen, a single honest total is the only thing that can go at the bottom. Specification `AST-B-21`.

What is outstanding on this meter, what has been done to it, and what has it cost us?

FIGURE 18 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED

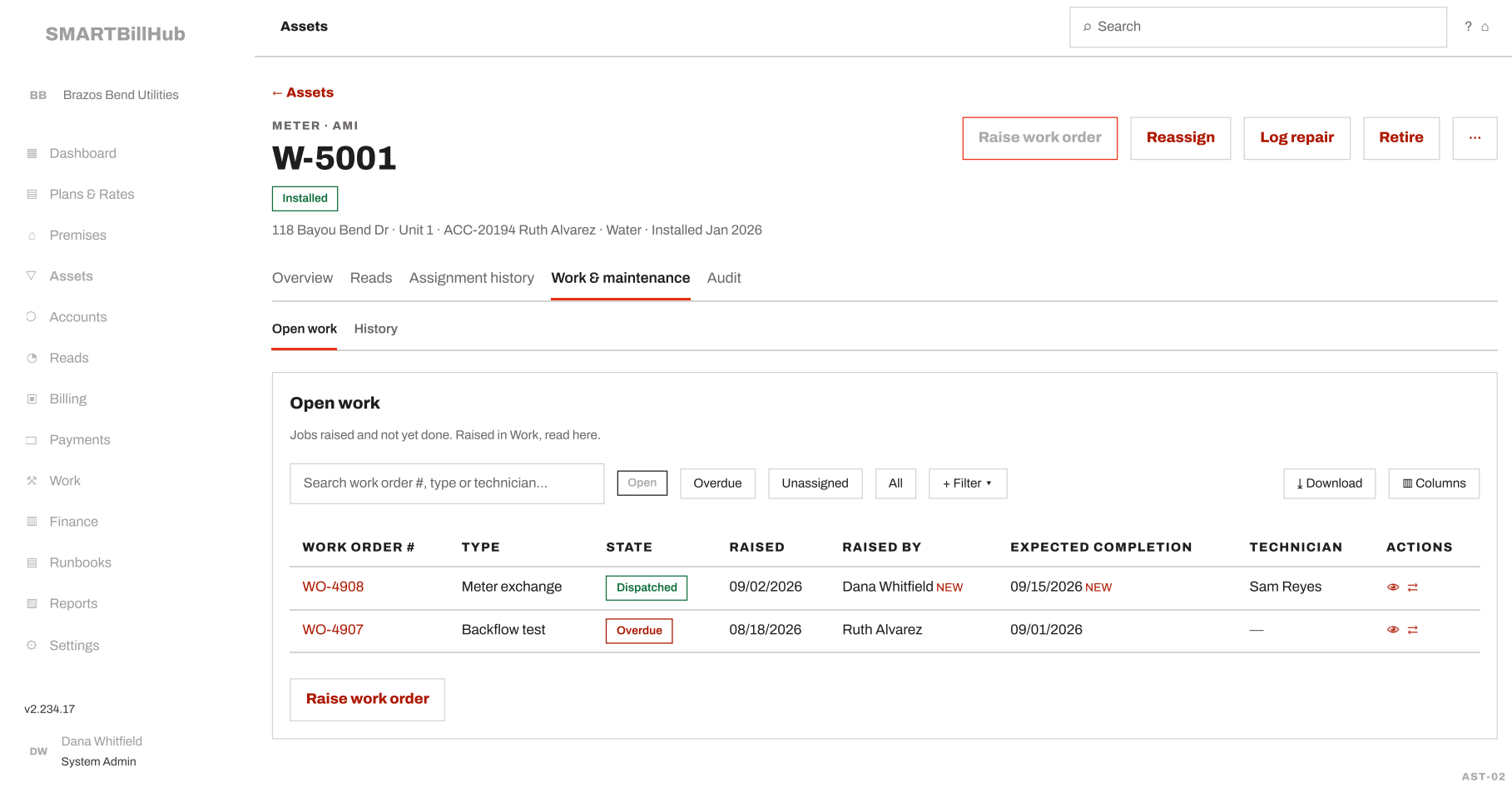
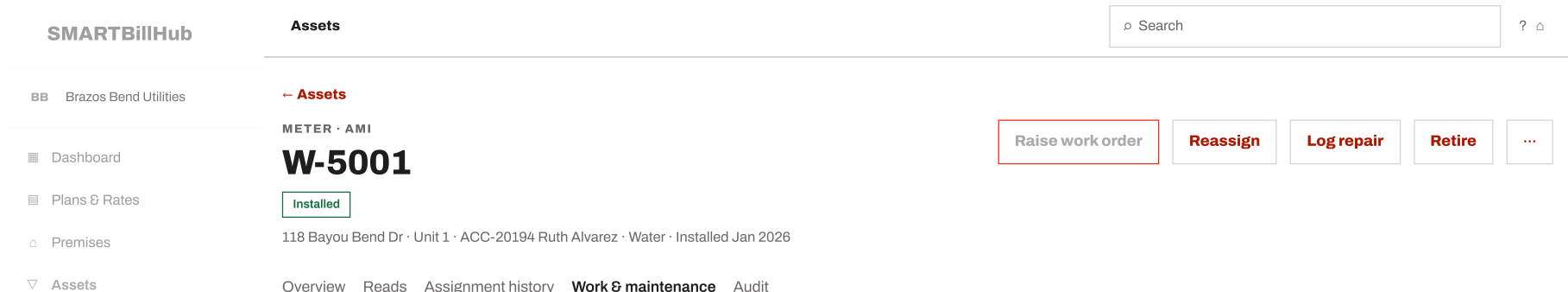
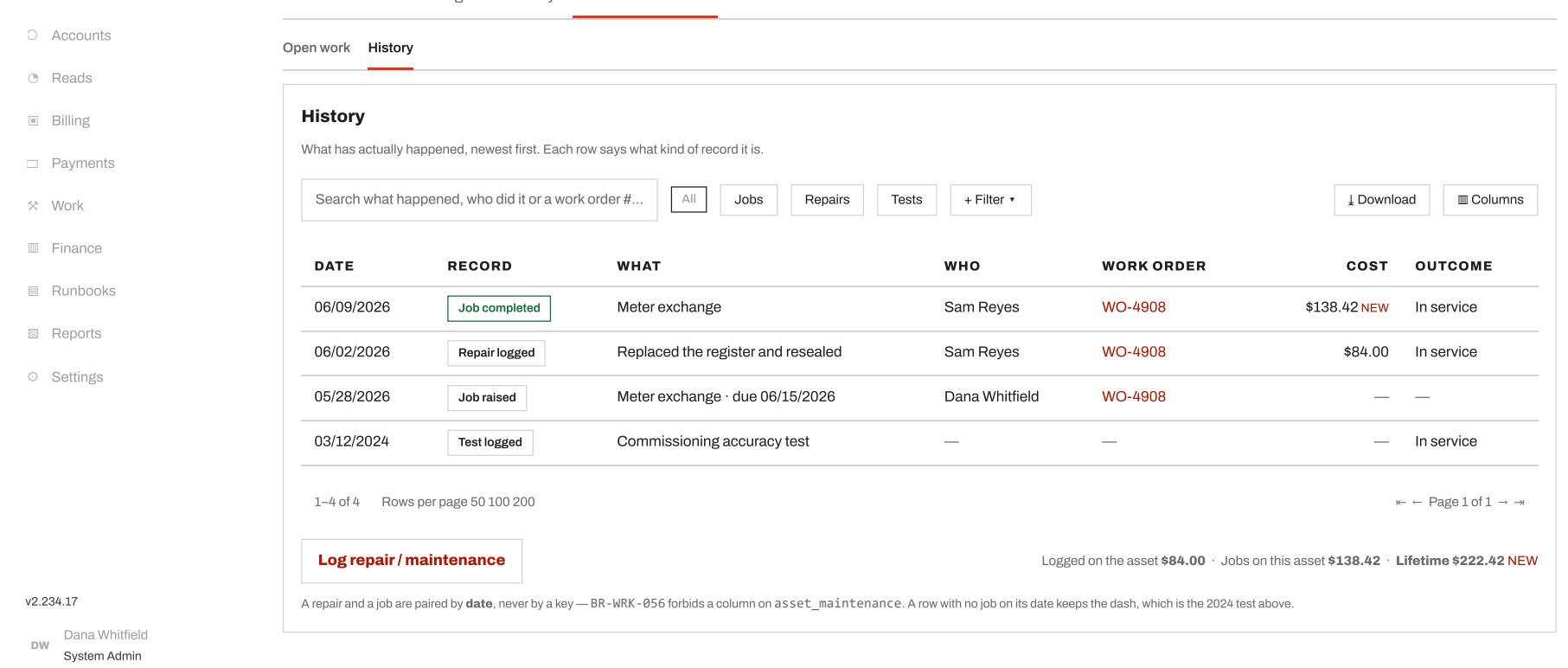


FIGURE 19 · 1 OPEN WORK IS THE DEFAULT SUB-TAB — THE HALF YOU CAN ACT ON, SO IT IS THE HALF YOU LAND ON:





2 AND HISTORY, ONE DATE-ORDERED LIST WHERE EVERY ROW STATES WHAT KIND OF RECORD IT IS. THE TWO NEVER MERGE INTO ONE TABLE, BECAUSE A REPAIR IS PAIRED WITH A JOB BY DATE AND NEVER BY A KEY:

The screen

For	Anyone who needs the whole service story of one asset without holding two tabs in their head.
How you get here	The detail tab strip. It replaces **Maintenance & repairs** and **Work**, which were two tabs answering one question. Inside it, **two sub-tabs** — the same second-level shape the Lifecycle registry tab needs, and URL state like every other tab (`BR-SHL-011`), so a person can link to either half.
Primary action	Two, one per panel and each on the panel it belongs to: **Raise work order** under Open work, **Log repair / maintenance** under History. `BR-WRK-056` names the maintenance panel as one of the raise door's four ways in, and it names it by FUNCTION rather than by tab label, so the door survives the merge.
Tabs	**Two sub-tabs here** — Open work, then History — and **five** on a meter or sub-meter · five on an exporting inverter, Export replacing Reads · **four** on a plain device. One fewer than before the merge, on every kind.
Table controls	**Open work** — no toolbar; a handful of open jobs per asset, and the filter that matters (open vs all) is the split itself. **History** — the standard pager, because a ten-year-old meter accumulates rows without limit.

Variation by kind	**None. Checked, not assumed.** Any asset can have a job raised against it — `work_orders.asset_id` carries no restriction by kind, and `templates_for_asset` filters by asset TYPE without changing the columns. What varies is the **Record** value in History, which describes what happened rather than what the asset is.
Write surfaces	**Detail** — a work order number opens Work's own screen. **Add** — the two controls above. **Edit · Update** — **NOTHING** on a history row, and that is the rule rather than an omission: `BR-AST-018` makes maintenance append-only, and a job is changed in Work by the people who own it. **Delete** — **NOTHING**, ever.
Walked	**Not walked — the merged tab does not exist.** Both halves were read in the source, and both are empty on this tenant: **0** rows in `asset_maintenance` and **0** work orders anchored to an asset, against 39 work orders in total. Every figure drawn here is an illustration (B14 rule 37).
Spacing	Two panels at the standard 20 gap, tables at 8×12, pager 12.
Cognitive load	It answers *what is outstanding* and *what happened* in one place, in that order — the actionable half first. Before the merge a person had to open both tabs and pair the rows themselves.
States	**Both empty** — *"No jobs are open against this asset."* and *"Nothing logged against this asset yet."*, which is what every asset on this tenant shows today. **Open work empty, history not** — the common healthy state, and the Open work panel still draws with its Raise control rather than vanishing.
Words	**Record** names the kind of row — *Job raised · Job completed · Repair logged · Test logged* — so no row has to be inferred from which columns are filled. **Expected completion**, never *deadline* or *SLA*: it is `due_at`, a commitment rather than a promise.
Components	`DataTable` in the History panel · the detail-panel table in Open work · `StatusPill` · `DateCell` · `MoneyCell` · `RowActions`.
Rules	`BR-AST-018` (append-only) · `BR-WRK-056` (no column on `asset_maintenance`; the raise door) · `BR-WRK-002` · `BR-WRK-051` (the one place a job's cost is worked out) · §10-D3 (cost is captured and posted nowhere).
Audience	Staff.
Health	SERVED on both halves — two endpoints the page already calls. The exceptions are **Raised by**, which no column records, and the **combined cost line**, which nothing computes today.

Data flow

```
GET /assets/{id}/maintenance  —→ rows + lifetime_cost + currency     assets
GET /assets/{id}/work-orders  —→ 29 fields per order                work ⇄ assets
                              ├→ open ones      → the Open work panel
                              ├→ completed ones → History, typed "Job completed"
```

```
                              └→ maintenance_index (by DATE) → the Work order cell
work_order_cost_lines(order)  —→ labour + parts                    NOT JOINED TO THE ASSET
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
②	Work order #	`work_orders.order_no`	work ⇄	SERVED
③	Type	`work_orders.type_ref.name`	work ⇄	SERVED
④	State	`work_orders.state`	work ⇄	SERVED
⑤	Raised	`_row()['raised_at']`	work ⇄	SERVED
⑥	Raised by	**nothing serves this** — `assigned_by` and `accepted_by` exist, `raised_by` does not	—	NOT SERVED
⑦	Expected completion	`work_orders.due_at` — on the wire, rendered nowhere today	work ⇄	SERVED
⑧	Record	derived from which source the row came from, and its state	—	DERIVED
⑨	What / Who	`asset_maintenance.detail` + `tech_user`, or the order's type + technician	assets · work ⇄	SERVED
⑩	Cost · logged	`asset_maintenance.cost`, 4 dp stored	assets	SERVED
⑪	Cost · jobs	`work_order_cost_lines()` — **computed per order and joined to no asset**	work ⇄	NOT SERVED (to an asset)
⑫	Outcome	`asset_maintenance.resulting_status` or `work_orders.outcome`	assets · work ⇄	SERVED

AST-B-21

An asset shows one honest figure for what it has cost to keep running

STATUS Unverified

WAITS ON A read that sums `work_order_cost_lines` over the orders anchored to one asset, served from `apps/work` beside the list this tab already calls. **NO NEW TABLE AND NO NEW ARITHMETIC** — `BR-WRK-051` already owns the computation and says it must live in exactly one place.

SCREENS 12

WHAT DONE LOOKS LIKE ON SCREEN The History panel foots with three figures rather than one: *Logged on*

the asset $84.00 · Jobs on this asset $138.42 · Lifetime $222.42, each labelled by where it came from.

WHAT IT MUST NOT DO Post anything. Both halves are captured and reach no journal line by ruling, and summing them on a screen does not change that.

---

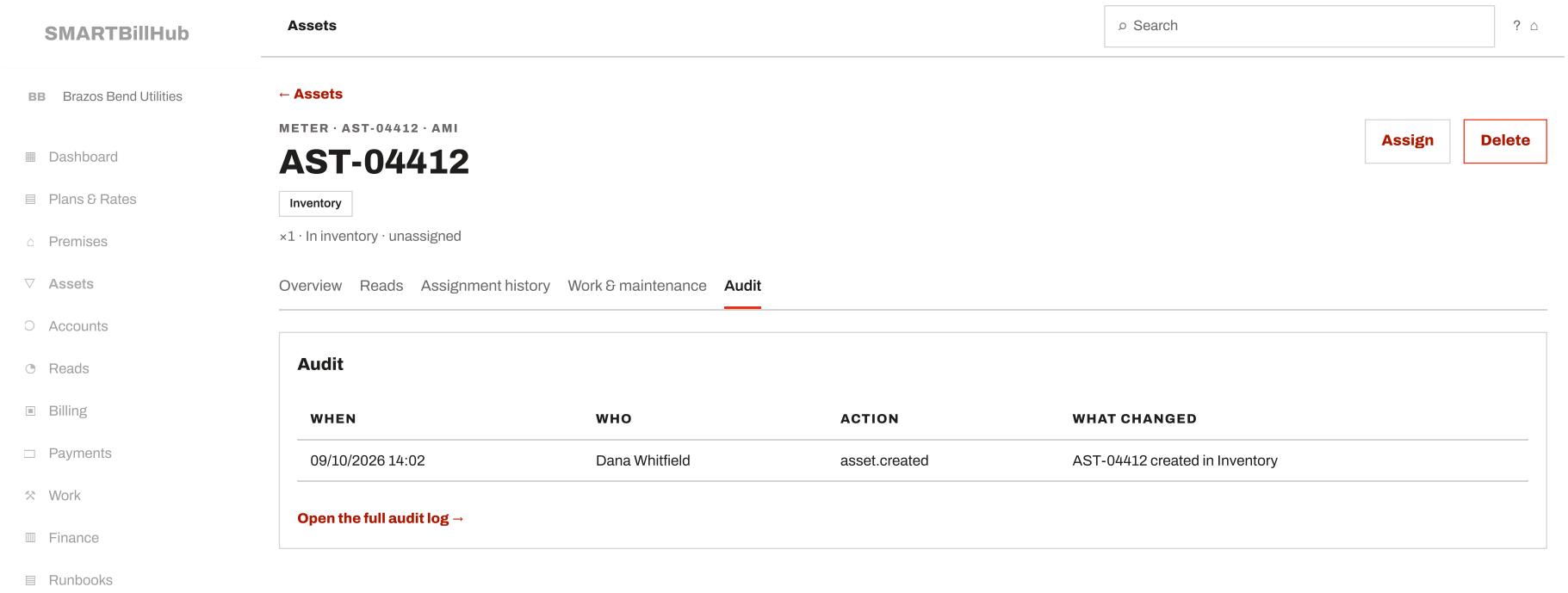
13 Audit EXISTS BEHIND NO COMMITMENT

`/:tenant/assets/{assetId}?tab=audit`

> **One sentence this tab must never grow**
>
> Hard rule 4 was narrowed on 2026-09-02 so that three columns on `payments` can be corrected in place, and the value a correction replaces then lives **only** in `audit_log` — which is pruned on a retention schedule. Whether such a row is exempt from pruning is `OQ-1330` and is **OPEN**. Until it is answered, no screen, document or docstring may describe this path as a permanent record of the old value. This tab does not, today. It is listed here so that nobody adds the sentence.

Who changed this, and when?

FIGURE 20 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	Anyone answering that question to a regulator or to a colleague.
How you get here	The tab strip on screen 08.
Primary action	None. **Open the full audit log** → leaves for S61.
Tabs	Part of screen 08's set.
Table controls	Whatever the shared `AuditTab` offers — this screen composes the same component premises, units and accounts use, rather than drawing its own.
Write surfaces	**NOTHING**, on principle. It is a read-only window into the one `audit_log`, scoped to this asset.
Walked	Opened. Rendered. Zero failed requests.
Cognitive load	It removes the need to know that an audit log exists somewhere else.
States	Those of the shared component.
Words	Nothing module-specific — and that is the point of reusing the component.
Components	`AuditTab`, imported from `features/metering/PremisesDetailPage`.
Rules	BR-AST-019 + BR-AUD-010 (render AND link) · hard rule 7 (every money action writes `audit_log`).
Audience	Staff.
Health	SERVED.

Data flow

```
audit_log (scoped entity_type='asset', entity_id=<this asset>)
   └→ the shared AuditTab                    ⇄ assets → notifications/audit   WHOLE
```

⚠ audit_log is PRUNED on a retention schedule. Nothing on this screen says so.

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	When · Who · Action · What changed	audit_log rows for this asset	assets → audit ⇄	SERVED

AST-B-02

Raising a work order from the asset, from the controls the rule already names

WAITS ON BR-WRK-056 is CONFIRMED and names both controls. Two of its four ways in were never built, and the raise endpoint exists.

SCREENS 12, 27

WHAT DONE LOOKS LIKE ON SCREEN On **AST-04412**'s **WORK & MAINTENANCE** tab a **RAISE WORK ORDER** button sits beside the sub-tabs, and the empty state reads *"No open work · Raise one"* with the words as the control — today the tab has no way in at all and the screen is reached by typing its URL.

STATUS UNVERIFIED — engineering fills this in from the code, one row at a time.

AST-B-18

The asset's work list carries the fields the wire already sends

WAITS ON Nothing on the server. GET /assets/{id}/work-orders serves **29 FIELDS PER ORDER** and the client type declares **FOUR**.

SCREENS 12

WHAT DONE LOOKS LIKE ON SCREEN The **OPEN WORK** sub-tab shows **PRIORITY**, **DUE** and **ASSIGNED TO** as columns with values — today those three columns are drawn and empty on every row because the client type narrows the payload to id · order_no · type · state.

STATUS UNVERIFIED — engineering fills this in from the code, one row at a time.

Changing one asset, and removing it

The two writes that act on an asset already in the register. Both are reached from the row and from the asset's own header.

---

14 Edit one asset

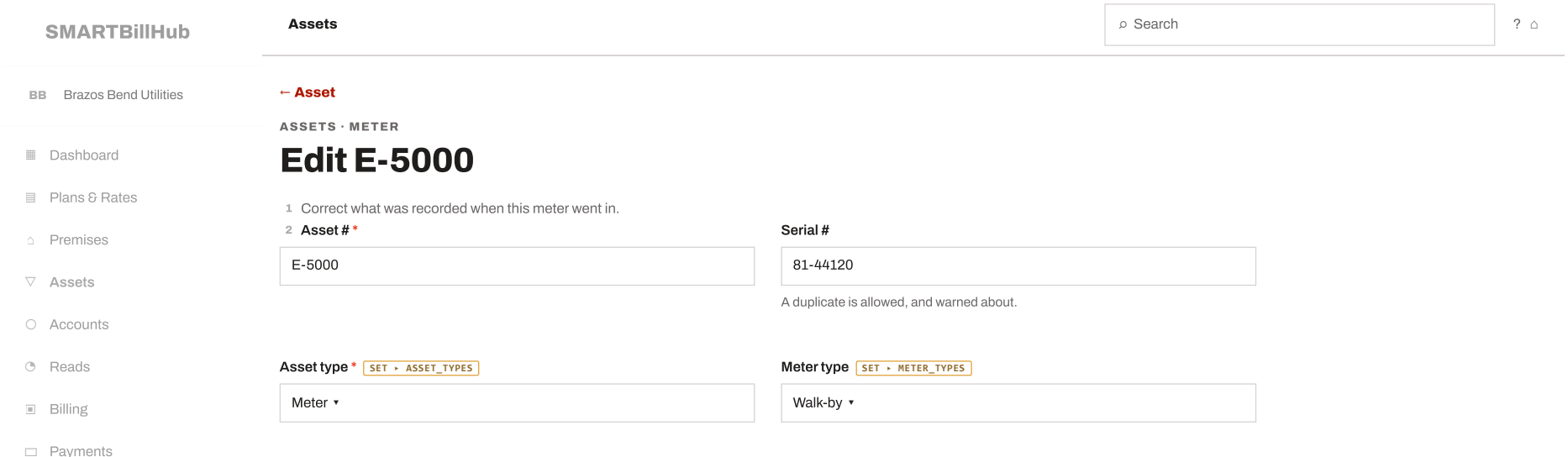
NEW BUILD · COMMITMENT 12 · COMMITMENT 13 · COMMITMENT 26

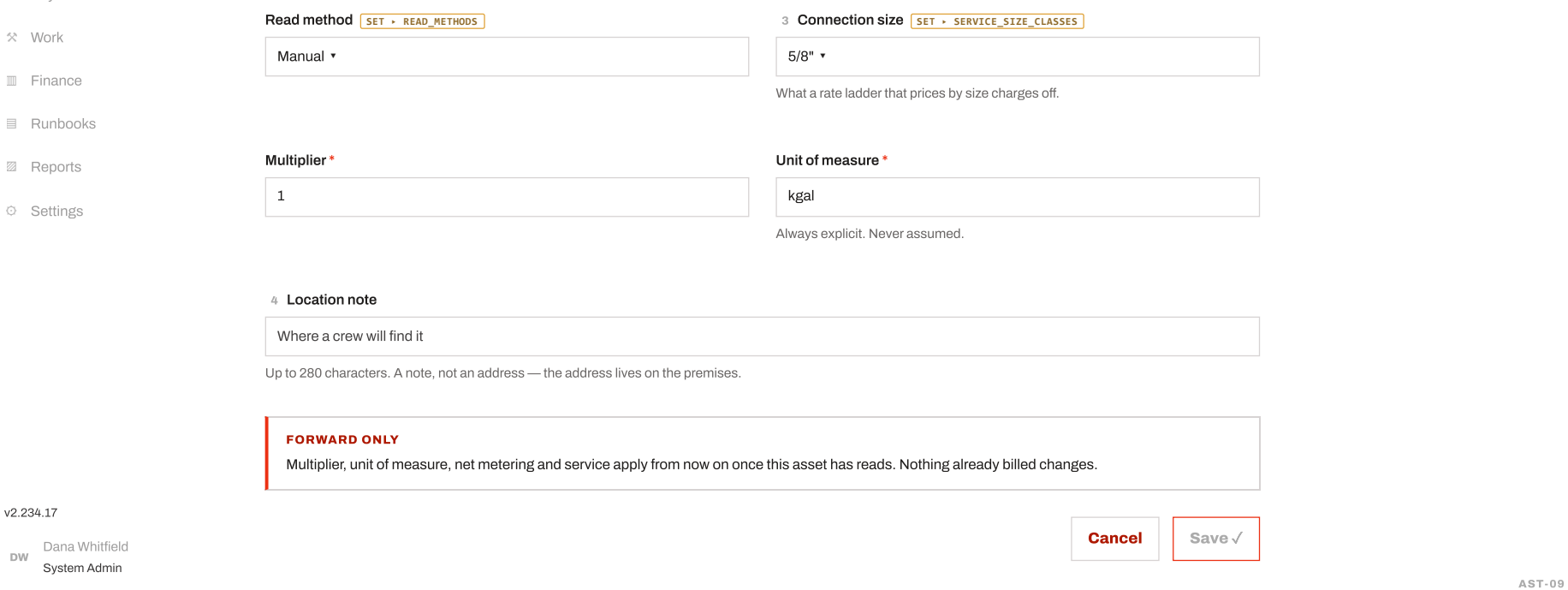
`/:tenant/assets/{assetId}/edit  — a full page, replacing today's dialog`

> **Why this stopped being a dialog**
>
> Instruction, 2026-09-11: *"this should be form and not modal."* It also puts this screen back in line with the rest of the module — the PO ruled on 2026-08-22 that the four lifecycle flows are full pages and not dialogs (*"as per r10 - it is not modal"*), and edit was left behind as one of three exceptions. Nine fields, a consequence box and a forward-only warning is not dialog-sized work, and it is the one surface through which every specification correction in the product passes.
>
> What a page buys that the dialog does not: an **address**, so a correction can be linked to and returned to; **room for R12's rhythm**, which a 640px dialog compresses; and a second door from the asset's own header, so a person reading a wrong value can act on it where they are standing.

The multiplier was typed wrong when this meter went in. Can I fix it without re-creating the asset?

FIGURE 21 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	A clerk correcting a specification on a shipped asset.
How you get here	**Today:** the ✎ **pencil** in a registry row and nowhere else — the detail screen has no edit control at all. **As drawn:** the same pencil, plus an **Edit** action in the asset's own header, which is where a person looking at a wrong multiplier is standing.
Primary action	**Save ✓** → `PATCH /assets/{id}`.
Tabs	None.
Table controls	N/A.
Write surfaces	This *is* the edit surface. It is the **only** caller of `updateAsset` in the entire front end — every specification correction a utility ever makes goes through it.
Walked	**Not walked — this screen does not exist yet.** The dialog it replaces was also not opened: it needs a click on a row action and the walk drove by address. Its field set is read from `AssetRegistryPage.tsx`'s `AssetFormModal`.
Spacing	**R12 throughout, which the dialog does not manage:** label → input 5px, input → helper 6px, field → field 20px. Two fields to a row per R13, and the location note breaks the run because it is the long one.
Cognitive load	A correction stays small. Creating a replacement asset to fix a typo would break every read that already points at this one.

States	**Refused** — a duplicate asset number is a hard block under the field; a duplicate serial is a warning that still saves. **Not applicable** — on a non-measuring device the Meter type field is replaced by a note: *"Meter type does not apply to this kind of device, and the server refuses one."*
Words	**Connection size**, not *size class*. It is what a rate ladder prices by, and the detail screen says so when it is missing.
Components	`PageHeader` · `TextField` · `SelectField` · `SearchSelectField` · the consequence note. **Not** `FormModal`.
Rules	BR-AST-012 (billing fields are forward-only) · BR-AST-033 · BR-AST-044 · BR-AST-500/530 · **B14 rules 10, 11, 12 and R12/R13's field anatomy**.
Audience	Staff, and only System Admin or Billing Clerk (`services.py:1202`).
Health	SERVED — every field already has a real source and a working write.

Data flow

```
PATCH /assets/{id} ─→ services.update_asset          services.py:1202
                   ├→ @require_role(SYSTEM_ADMIN, BILLING_CLERK)
                   ├→ audits before/after               hard rule 7
                   └↳ returns the updated detail dict   WHOLE
   ⚠ PUT does not exist — the view accepts GET, PATCH, DELETE only (api.py:138)
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Asset #	`assets.asset_no`	assets	SERVED
②	Serial #	`assets.serial_no`	assets	SERVED
③	Asset type	value `assets.type_id`; options `GET /lookups/asset_types`	assets	SERVED
④	Meter type	value + options from `meter_types`	assets	SERVED
⑤	Read method	value + options from `read_methods`	assets	SERVED
⑥	Connection size	`assets.size_class`; options `service_size_classes` (64 seeded)	assets	SERVED
⑦	Multiplier	`assets.multiplier` NUMERIC(12,6)	assets	SERVED

⑦	Multiplier	`assets.multiplier`, NUMERIC(18,6)	assets	SERVED
⑧	Unit of measure	`assets.unit` — always explicit (hard rule 3)	assets	SERVED
⑨	Location note	`assets.location_note` — a note, **not an address**	assets	SERVED

Carry-over

As-built element	Disposition	Note
Asset # · Serial # · Asset type · Meter type · Read method · Connection size · Multiplier · Unit of measure · Location note	UNCHANGED	All nine fields, same labels, same order, same grouping. Two to a row as they already are.
The forward-only note	UNCHANGED	Same words, same place — below the fields, above the buttons.
Duplicate-number and duplicate-serial refusals	UNCHANGED	Same behaviour; a page has more room to show them under the field.
Cancel · Save ✓	MOVED	From a dialog footer to a page footer, right-aligned. Same words, same order, same primary.
The dialog frame itself	REMOVED	⚠ Instruction, 2026-09-11: *"this should be form and not modal."* It is the container, not a field — no data and no capability goes with it.
An **Edit** control in the asset header	NEW	The dialog could only ever be opened from a table row; a routed page can be reached from the asset itself.

```
As-built fields     11
  unchanged          10
  moved               1   ← the button pair, footer to footer
  relabelled          0
  removed             1   ← the dialog frame (a container, quoted instruction above)
Remedy fields       11   = 11 − 1 removed + 1 new    ✓ reconciles
```

AST-B-12

A clerk can correct an asset from the asset, at an address they can come back to

STATUS Unverified

WAITS ON Nothing. The form, its schema and its write all exist; what changes is the container and the route.

SCREENS 08, 14

WHAT DONE LOOKS LIKE ON SCREEN Opening **E-5000** shows an **EDIT** button in the header; pressing it lands on `/brazosbend/assets/{id}/edit` with the nine fields filled in, two to a row, each helper sitting under its own input.

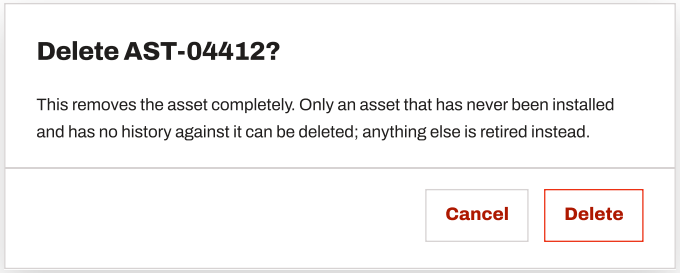
---

15 Delete this asset? EXISTS BEHIND NO COMMITMENT

`no address of its own — a confirm on /:tenant/assets and /:tenant/assets/{id}`

We ordered the wrong meter and it never went anywhere. Can it just go away?

FIGURE 22 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	Someone cleaning up a mistake, not retiring equipment.
How you get here	The 🗑 **trash** in a registry row, and **Delete** in the detail header — both only when the server's action set for that row includes `delete`.
Primary action	**Delete**, in the danger tier — and this one is **enforced on rendered output** for confirm dialogs since DEF-718.
Tabs	None.
Table controls	N/A.
Write surfaces	`DELETE /assets/{id}`, System Admin only (`services.py:1297`).
Walked	Not opened — the confirm needs a click on a row action.
Spacing	Dialog padding 20/24 on the header, 16/24 on the body, 16/24/20 on the footer — all on the scale.
Cognitive load	It draws the line between *delete* and *retire* in one sentence, at the moment a person is about to get it wrong.
States	**Refused** — *"This asset has history against it, so it can't be deleted. Retire it instead."* The server re-probes every reference; the button being offered is not a promise.
Words	**Retire** and **Delete** never share a glyph.
Components	`ConfirmDialog`.
Rules	BR-AST-013 · BR-SHL-032 (the confirm NAMES the action and its subject) · B14 rule 6.
Audience	Staff, System Admin only.
Health	SERVED.

Data flow

```
DELETE /assets/{id} ─→ services.delete_asset          services.py:1297
   ├→ SystemAdminOnly
   ├→ re-probes every referencing table (reads, invoice lines, assignments)
   └→ 204, or 409 pointing at Retire                       WHOLE
```

Asset edits — what each pencil opens

Six surfaces, one per editable panel on the Overview, plus the one state in which there are none. **They cost nothing on the server:** `PATCH /assets/{id}` validates a PARTIAL payload already — `_validate_asset(data, partial=True)` — so a form that sends only its own panel's fields is exactly the shape the endpoint expects.

Page or dialog — the rule, so it is checkable rather than taste

A PAGE when you are creating something, or when the change moves where the thing lives. A MODAL when you are correcting a value on something you are already looking at.

Add an asset, four steps	**Page** — creating.
Reassign · Retire	**Page** — it moves where the asset lives: a service point is minted and a final read is taken.
Edit one asset (screen 14)	**Page** — twenty-three fields, and the PO ruled it: *"this should be form and not modal"*.
A per-panel pencil	**Modal** — two to nine fields, and you are already looking at the value you are fixing. Sending somebody to a full page to correct a seal number loses their place on the screen they were reading.

The *construction* is identical either way: B14 rule 38 says the UI rules decide how a form is built, so R12's label → input → helper and R13's two-to-a-row grid apply to a dialog exactly as they apply to a page. Only the container changes.

One pencil on this screen can restate what a reading means. It is the only one, and that is by design.

Five fields are billing-affecting — `multiplier · unit · net_metering · service · size_class` — and `BR-AST-012` makes a change to any of them apply **FORWARD ONLY** once reads exist: stored reads and issued invoices are never restated, and the response carries a `forward_only_change` warning.

⚠ **So Connection size moves out of the Device panel and into Measurement**, which is a change to the architecture agreed earlier in this document and is made for a reason rather than for tidiness: with it moved, **all five sit in one panel**, exactly one pencil on the screen can restate billing, and one confirmation covers it. Left in Device, two panels would each need the same warning and a person could meet it from either.

`meter_type` is explicitly **not** on that list — *"it classifies the device; no rate, quantity or invoice line is derived from it, so re-tagging a meter restates nothing"* — documented in the service rather than silently chosen, which is why the Device modal needs no confirmation at all.

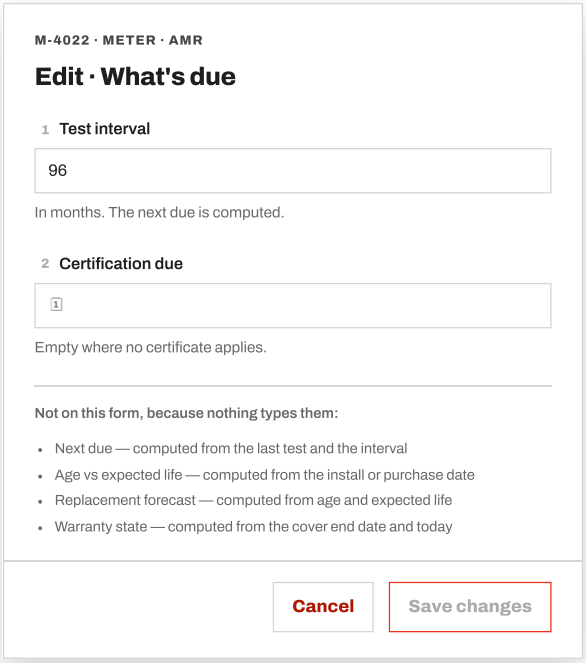
16 Edit · What's due

NEW BUILD | COMMITMENT 13 | COMMITMENT 25 | COMMITMENT 26

`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`

The interval on this backflow preventer is wrong — where do I fix it?

FIGURE 23 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	Whoever owns the testing calendar.
How you get here	The pencil on the What's due panel.
Primary action	**Save changes**. Two fields, one PATCH.
Tabs	None.

Table controls	None.
Write surfaces	**Update** — `PATCH /assets/{id}` with two keys. **Add · Delete** — **NOTHING**.
Walked	**Not walked — this does not exist.** The fields and their rules are read from the asset model and the validator.
Spacing	R12 throughout: label → input → helper at 5 / 6 / 20.
Cognitive load	Two typed fields, and the four derived ones **named on the form as absent** rather than left to be wondered about. That list is the whole point of the panel split — a person who wanted to change the next-due date learns here that it is computed.
States	**Saved** — the modal closes and the panel behind it refreshes. **Refused** — the field error under its own input.
Words	**Test interval** in months, not *cadence* or *frequency* — the product has one word and eleven catalogues carry it.
Components	`FormModal` · `TextField` · `DatePickerField`.
Rules	`BR-AST-007` (the derived test-due chip) · `BR-AST-012` (every edit audited before/after) · R12 · B14 rule 7.
Audience	Staff.
Health	SERVED — both fields are columns on the asset and the PATCH takes a partial payload today.

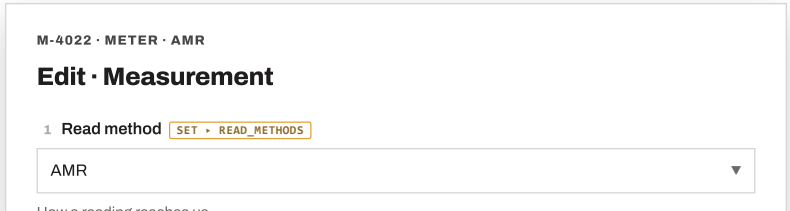
17 Edit · Measurement

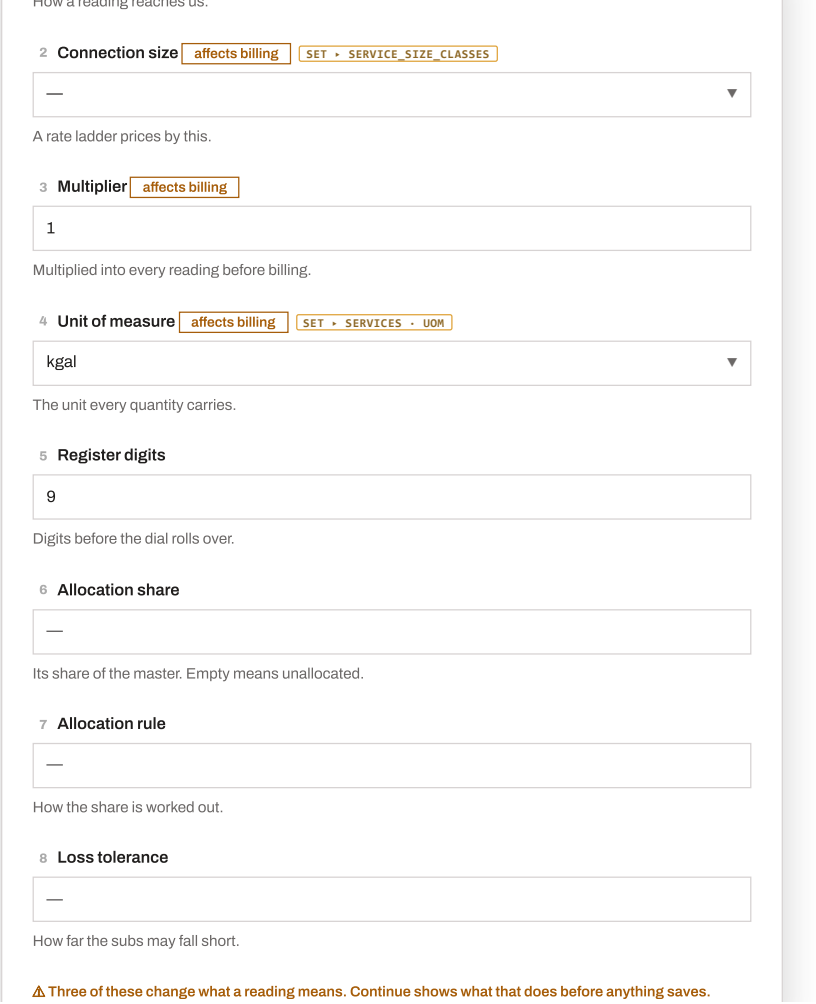
NEW BUILD · COMMITMENT 13 · COMMITMENT 25 · COMMITMENT 26

`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`

This meter has been reading ten times low — can I fix the multiplier, and what does that do to the bills already sent?

FIGURE 24 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	Whoever has found a meter configured wrongly.
How you get here	The pencil on the Measurement panel. **A behind-the-meter device has neither** — no panel, no pencil.
Primary action	**Continue**, never Save — this is the one edit on the screen that goes through a second step.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — one PATCH, after the confirmation. **Add · Delete** — **NOTHING**.
Walked	**Not walked.** The five billing-affecting fields are read from `BILLING_AFFECTING_FIELDS` and the forward-only clause from `update_asset`'s own docstring.

own docstring.

Spacing	R12, two to a row.
Cognitive load	Three of the eight fields carry an **affects billing** pill, so the danger is visible before the form is touched rather than after it is submitted. The three allocation rows render **read-only on a meter** and become editable on a sub-meter.
States	**Nothing billing-affecting changed** — Continue behaves as Save and the second step is skipped. **Something did** — the second step, always.
Words	**Affects billing**, not *warning* or *caution*: it says what the consequence IS rather than how worried to be.
Components	`FormModal` · `TextField` · `SelectField` · `StatusPill`.
Rules	**`BR-AST-012`** — forward only, once reads exist · `BR-AST-015` (register digits default 9) · `BR-AST-500` (a null connection size is a real state, never defaulted) · hard rule 2 (percentages at 6 dp).
Audience	Staff.
Health	SERVED — every field is a column, and the server already refuses and warns correctly. What is missing is the screen that shows the warning **before** the save rather than after.

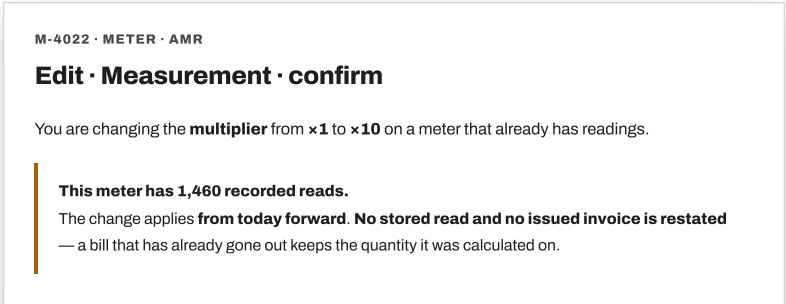
18 Edit · Measurement · confirm

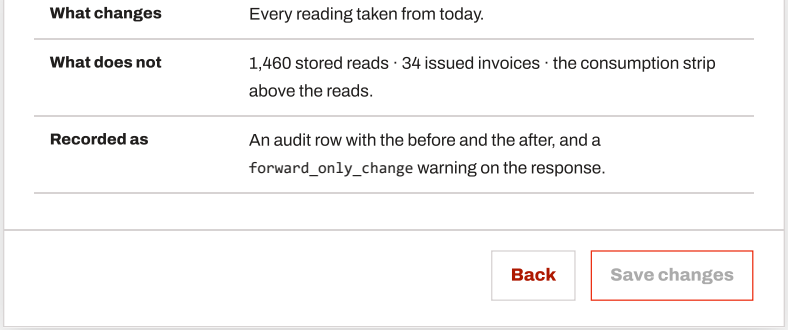
NEW BUILD · COMMITMENT 13 · COMMITMENT 25 · COMMITMENT 26

`The second step of the same modal — no second address`

What exactly will this change, and what will it leave alone?

FIGURE 25 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	The same person, one step later.
How you get here	Pressing Continue on screen 17 when a billing-affecting field has changed. **It is not a separate dialog** — the same modal, a second step, so Back returns the typed values rather than discarding them.
Primary action	**Save changes**. **Back** returns to the form with everything still typed.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — the PATCH fires here and nowhere else.
Walked	**Not walked.** The sentence is composed from `update_asset`'s own rule; the counts are illustrations.
Spacing	R11 throughout.
Cognitive load	It answers the question a person actually has — *what happens to the bills I already sent* — with a number they can check, and it names **what does not change** as explicitly as what does. A warning that only says *are you sure* teaches nobody anything.
States	**Reads exist** — the figure and the forward-only sentence. **No reads yet** — the step still appears and says so: nothing to restate, and the change is total rather than forward-only.
Words	**Restated**, the word `BR-AST-012` itself uses. Never *recalculated*, which is what it deliberately does not do.
Components	`FormModal` step two · `ConfirmPanel`.
Rules	`BR-AST-012` · hard rule 4 (financial immutability — an issued invoice takes no update, ever).
Audience	Staff.

Health	DERIVED — the read count comes from the reads endpoint the page already calls; the invoice count needs a read from billing that nothing serves today.

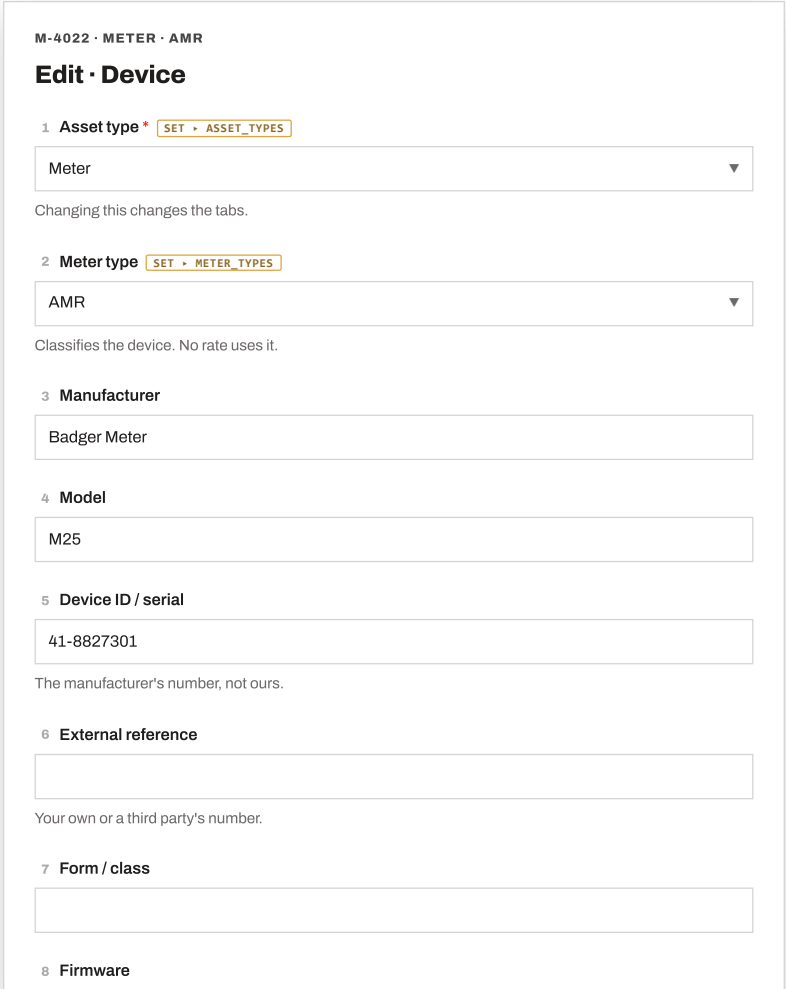
19 Edit · Device

NEW BUILD · COMMITMENT 13 · COMMITMENT 25 · COMMITMENT 26

`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`

The serial was typed wrong when this went in — fix it without opening a twenty-three field form.

FIGURE 26 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	Anyone correcting what the device IS.
How you get here	The pencil on the Device panel.
Primary action	**Save changes**.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — one PATCH. **Add · Delete** — **NOTHING**.
Walked	**Not walked.** Nine fields, read from the model; the existing edit modal covers five of them and reaches them only from the registry row.
Spacing	R12, two to a row, nine fields.
Cognitive load	Everything here is descriptive and none of it restates anything, so there is no confirmation and no pill — which is itself information: a person learns that this panel is the safe one.
States	**Asset type changed** — the one field here with a consequence, and it is a SHAPE consequence rather than a money one: it changes which tabs the asset has. The form says so under the field.
Words	**Device ID / serial** — the manufacturer's number. **External reference** — yours or a third party's. **Asset #** is ours and is not on this form at all.
Components	`FormModal` · `TextField` · `SelectField`.
Rules	`BR-AST-026` (three identifiers, and the uuid is none of them) · `BR-AST-044` (`meter_type` is not billing-affecting, documented) · B14 rule 31.
Audience	Staff.
Health	SERVED — and **External reference becomes typable for the first time**: the column is served on every row and no surface in the product writes it.

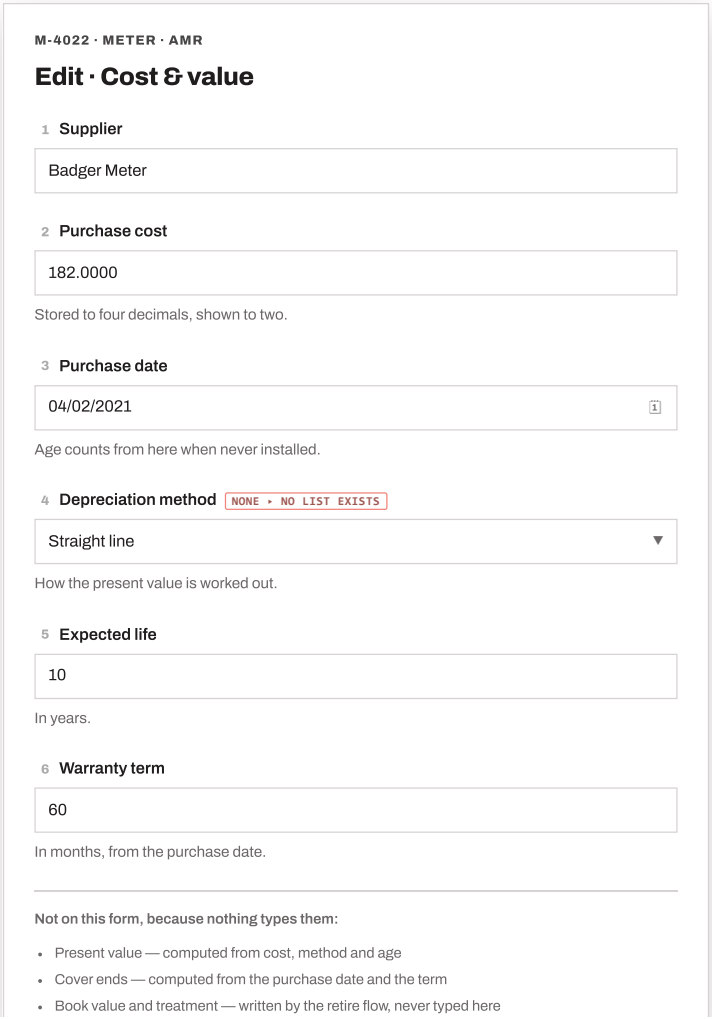
20 Edit · Cost & value

NEW BUILD | COMMITMENT 13 | COMMITMENT 25 | COMMITMENT 26

`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`

We were invoiced a different amount — correct it, and do not touch the ledger.

FIGURE 27 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	Whoever reconciles what was paid for the estate.

How you get here	The pencil on the Cost & value panel.
Primary action	**Save changes**.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — one PATCH. **Add · Delete** — **NOTHING**.
Walked	**Not walked.** Six typed fields and three derived, read from the valuation block the detail endpoint already serves.
Spacing	R12, two to a row.
Cognitive load	The three computed figures are named as absent, so nobody hunts for an editable present value. **Purchase date becomes typable and visible for the first time** — age and present value are both computed from it and the date itself appears nowhere in the product today.
States	**Saved** — the present value above recomputes. **Nothing posts**, in any state.
Words	**Cost** for what was paid, **value** for what it is worth now. §10-D3 keeps both off the ledger: *"cost is CAPTURED — stored 4 dp, totalled on the tab, and posted NOWHERE"*.
Components	`FormModal` · `MoneyField` · `DatePickerField` · `SelectField`.
Rules	**Hard rule 2** — money stored to 4 decimals, displayed at 2 · §10-D3 · `BR-AST-047`.
Audience	Staff.
Health	SERVED throughout.

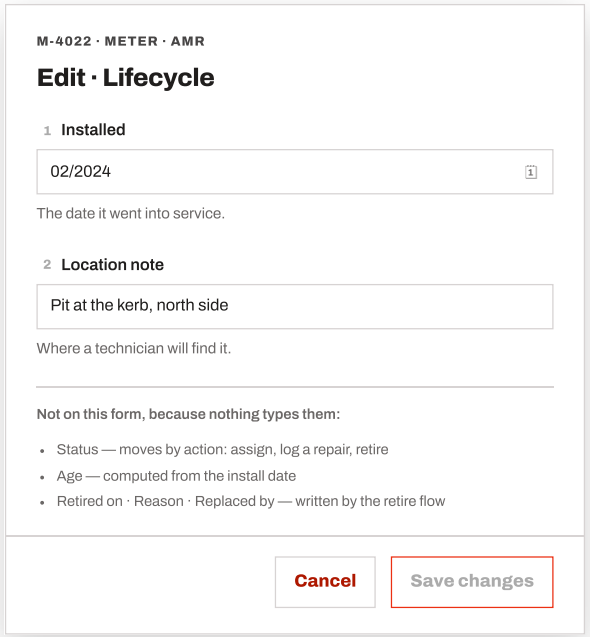
21 Edit · Lifecycle

NEW BUILD · COMMITMENT 13 · COMMITMENT 25 · COMMITMENT 26

`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`

The install date is a month out and the technician cannot find the meter — fix both.

FIGURE 28 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



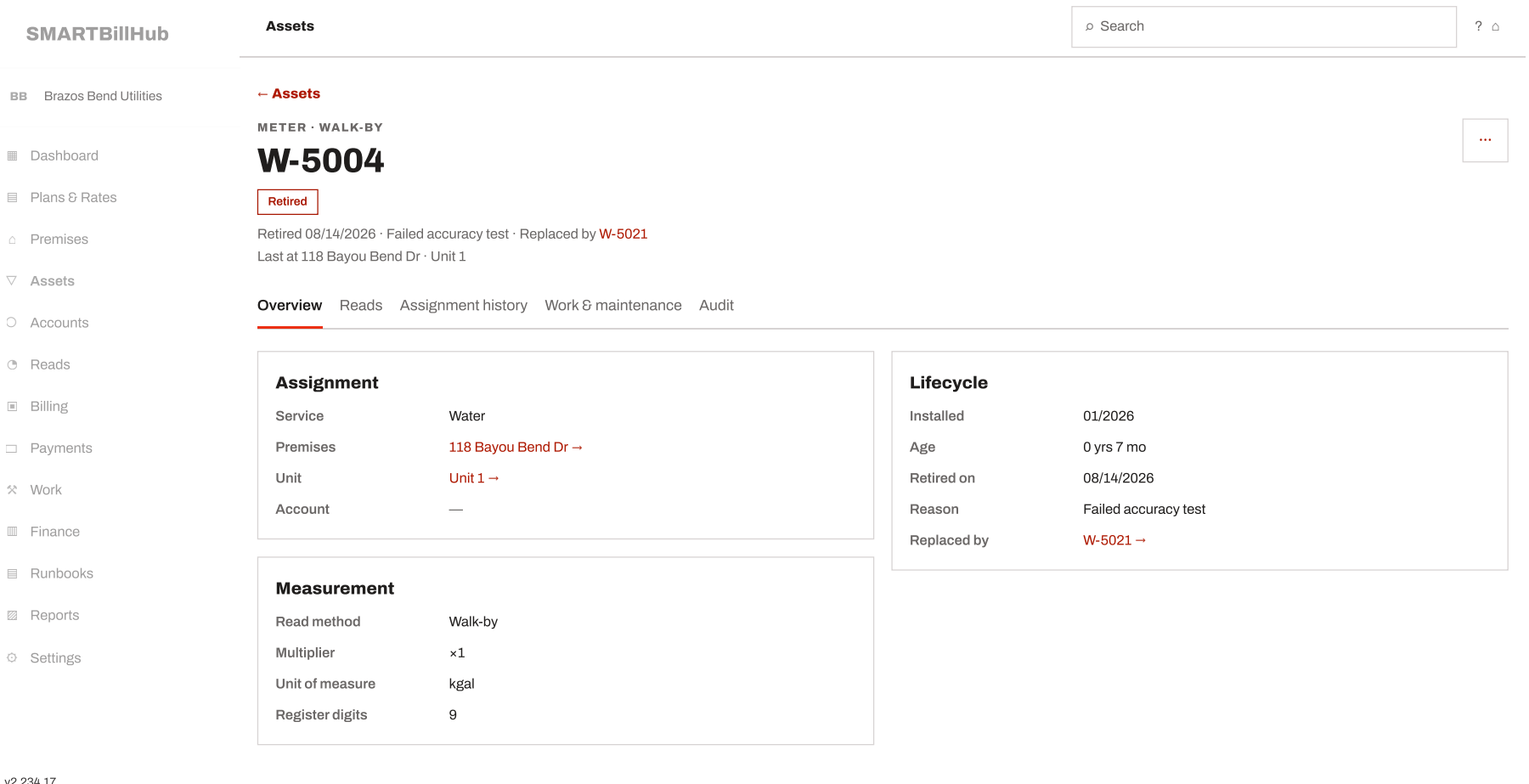
The screen

For	Whoever is correcting the record after the fact.
How you get here	The pencil on the Lifecycle panel.
Primary action	**Save changes**.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — one PATCH with two keys.
Walked	**Not walked.**
Spacing	R12; the location note runs full width because it is a sentence rather than a value.
Cognitive load	Two fields out of six, and the four that are not here are the four a person would most expect to be: **status**, **retired on**, **reason** and **replaced by**. Naming them as written-by-something-else is what stops somebody looking for a way to retire an asset by typing a date.
States	**Saved** — the age above recomputes from the new install date.

Words	**Location note**, not *address* — the address lives on the premises and is walked to (hard rule 8). This is where a technician will actually find it in the yard.
Components	`FormModal` · `DatePickerField` · `TextArea`.
Rules	`BR-AST-020` (status moves by legal transition, never by typing) · hard rule 8 · R12.
Audience	Staff.
Health	SERVED.

And the state in which there are none

`Figure 29` · A retired asset refuses **every** edit at the server — `update_asset` opens by testing `STAGE_TERMINAL` and raising `BIZ_IMMUTABLE` before it validates anything. So the screen shows no pencil at all, which is **hard-won law 2's mirror**: a control that cannot work is worse than an absent one, and the product already has a sentence for it — *"{{asset_no}} is retired, and retired is the end state."*



DW Dana Whitfield
System Admin

1 NOT ONE PENCIL ANYWHERE ON THE SCREEN, AND THE HEADER CARRIES NO ACTION BUT THE OVERFLOW. A RETIRED ASSET REFUSES EVERY EDIT AT THE SERVER — STAGE_TERMINAL → BIZ_IMMUTABLE — SO A CONTROL THAT OFFERED ONE WOULD BE A CONTROL THAT CANNOT WORK.

AST-02

Variation by asset type

The same rule as the Overview it opens from: **a form a kind has no fields for is absent, not empty.** A device's Measurement pencil does not exist, because the panel does not.

Panel	Meter	Sub-meter	BTM device	BTM inverter
What's due	✓ 2 fields	✓ 2	✓ 2	✓ 2
Measurement	✓ **5 fields**	✓ **8** — gains allocation share, rule and loss tolerance, all three editable here and read-only on a meter	**no pencil, no panel**	**Export capacity only** — a capacity in kW, never money (`BR-AST-025`)
Device	✓ 9	✓ 9	✓ **7** — no meter type, no read method	✓ 7
Cost & value	✓ 6	✓ 6	✓ 6	✓ 6
Lifecycle	✓ 2	✓ 2	✓ 2	✓ 2
Assignment	**No pencil on any kind.** Premises, unit and account are not typed on this screen — they change through **Reassign**, which mints a service point and takes a final read. A pencil here would offer a shortcut that corrupts the chain, so the panel carries a **Reassign** → link instead.			

AST-B-25

A person can correct one fact about an asset without opening a form holding twenty-three

STATUS Unverified

WAITS ON NOTHING ON THE SERVER. `PATCH /assets/{id}` already validates a partial payload, already audits before and after, already refuses a retired asset and already returns the `forward_only_change` warning. Five dialogs and one second step.

SCREENS 16, 17, 18, 19, 20, 21

WHAT DONE LOOKS LIKE ON SCREEN The pencil on **DEVICE** opens nine fields and saves. The pencil on **MEASUREMENT** opens eight, three of them marked *affects billing*, and **CONTINUE** — not Save — leads to a step that reads *"This meter has 1,460 recorded reads … no stored read and no issued invoice is restated."* A **RETIRED** asset shows no pencil at all.

WHAT IT ALSO FIXES **EXTERNAL REFERENCE** and **PURCHASE DATE** become typable for the first time — both are columns the server serves on every row that no surface in the product writes.

05

Actions — moving it, fixing it, retiring it

Six flows, and every one is a **page** rather than a dialog — each either creates something or moves where the asset lives, which is the rule section 04 states. They are the oldest screens in this document and the most complete: **the data they capture is not the problem**. Four assumptions were made about missing fields while auditing them and **three were wrong** — reassign offers *Inventory (unassign)* and a work-order checkbox, log repair captures cost, and retire takes every key its service accepts. What they break is **layout, escape and consequence**.

> **The three every screen in this section shares**
>
> **R12** — ten of eighteen fields across these screens render **label** → **helper** → **input**; label-to-input measures 29px or 47px instead of 5. **B14 rule 7** — five date inputs, **zero** Today shortcuts and **zero** with `min` or `max`. **The header** — not one of the six carries screen 08's five-slot header or its attention strip, so a person acting on an asset whose test is overdue sees no sign of it while they act.

What each action captures, and whether anything is missing

Audited field by field against the service each form posts to. **Four assumptions were made and three were wrong** — which is the finding, because the rules-compliance list makes these screens look worse than they are.

Screen	What it captures	Gap?
26 Reassign	Effective date · **to a unit or to Inventory** · final read date and value · a *record the final read* toggle · *move its sub-meters too* · **a checkbox to raise a field work order for the physical move**	**None.** `service` is the only key the service accepts and the form omits — correctly: it is billing-affecting and belongs on screen 17
27 Log repair	Kind · date · detail · technician · resulting status · **cost**	**None.** ⚠ *"Cost is never captured"* was nearly filed here as a finding and is false

		— the fault is **how** the technician is captured, not whether
28–30 Retire	Retire date · reason · final read date and value · replacement · book value treatment	**None.** All six keys `retire_asset` accepts
31 Raise work order	**`asset_id` · `template_id` · the template's name. Three, and that is all**	**Yes — and it is a missing SENTENCE, not a missing field.** `create_work_order` accepts sixteen keys and **none** is priority, due date, window or assignee: scheduling is Work's job, after the raise. Nothing on the screen says so, and screen 13 draws two columns that are empty on every freshly raised job because of it

AST-B-26

A field's helper fits on one line, so the fields beside each other line up

STATUS Unverified

WAITS ON Copy. **27 OF 37** helper strings in the assets module are rewritten to ten words or fewer, in eleven catalogues — and B14 rule 35 says a catalogue is written in its own language rather than translated, so this is translation work rather than engineering.

SCREENS 14, 16–21, 22–27, 28–32

WHAT DONE LOOKS LIKE ON SCREEN On Reassign, *Final register reading* and *Read date* sit side by side with their inputs on the **SAME BASELINE**. Today the left helper runs to two lines and the left input sits a line lower than the right one.

WORTH DOING IN THE SAME PASS The worst string is **36 WORDS** and carries three separate facts — what the field is, what it affects, and what to do when you do not know. Two of the three belong somewhere other than a helper.

22 Reassign

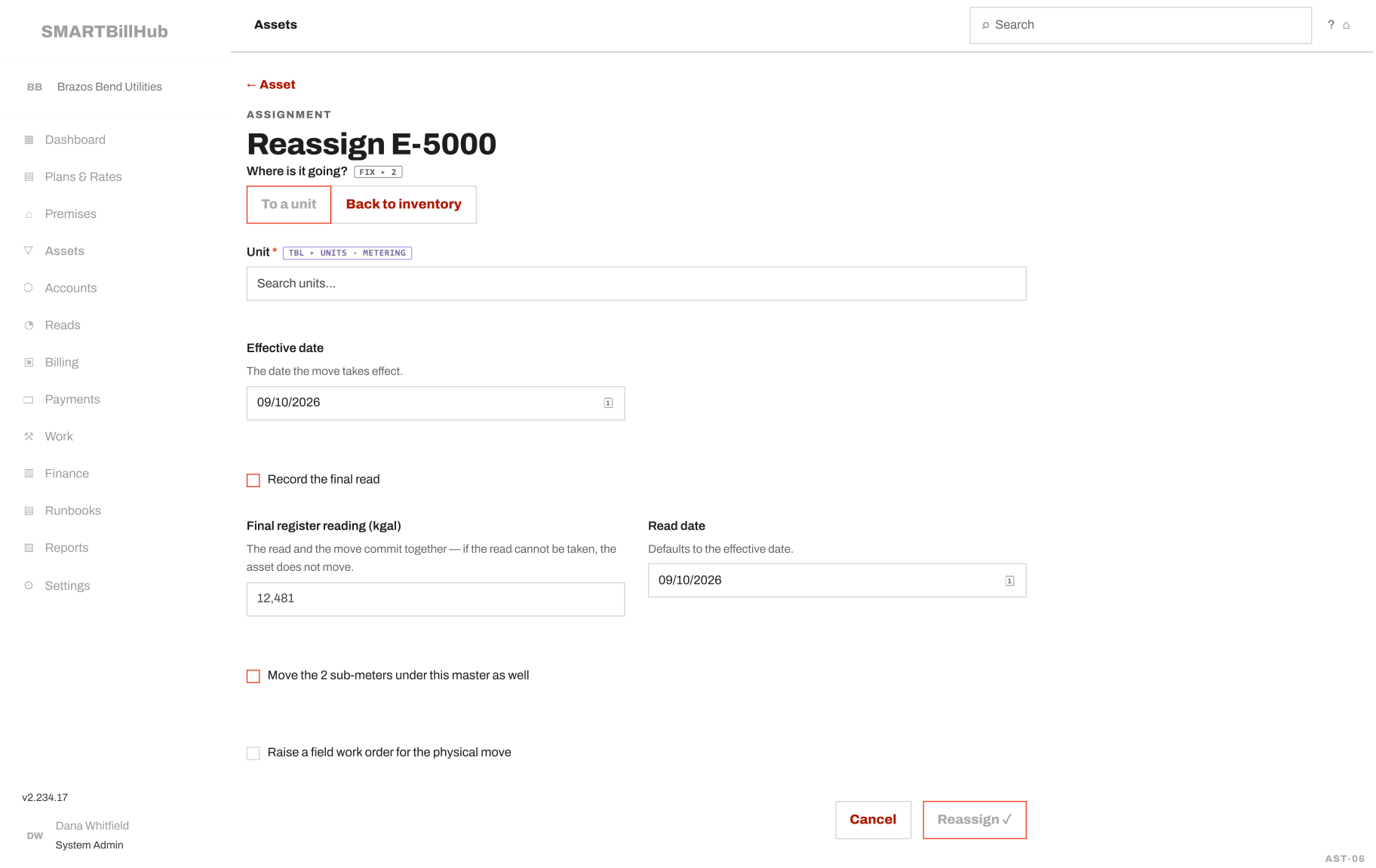
EXISTS BEHIND | COMMITMENT 26

`/:tenant/assets/{assetId}/reassign · ?unit={id} preselects a destination`

The tenant moved out and the landlord wants this meter on the other side of the duplex. What

happens to the last reading?

FIGURE 30 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



What changes here, and the one thing it is missing

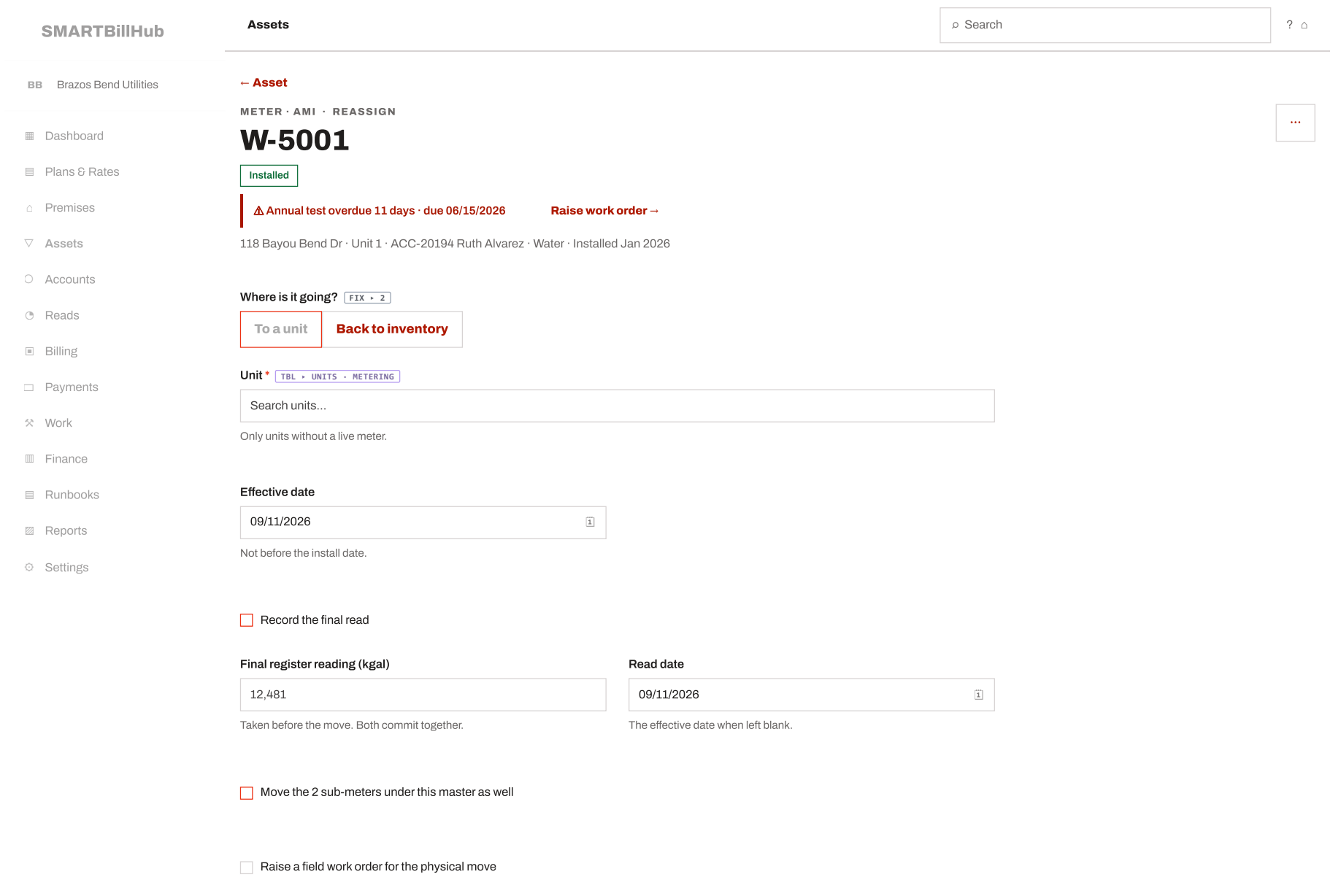
Data captured: complete. Effective date · **to a unit or to Inventory** · final read date and value · a *record the final read* toggle · *move its sub-meters too* · and a checkbox to **raise a field work order for the physical move**. The only key the service accepts that this form omits is `service`, and that is correct — it is billing-affecting and belongs on the Measurement edit (screen 17).

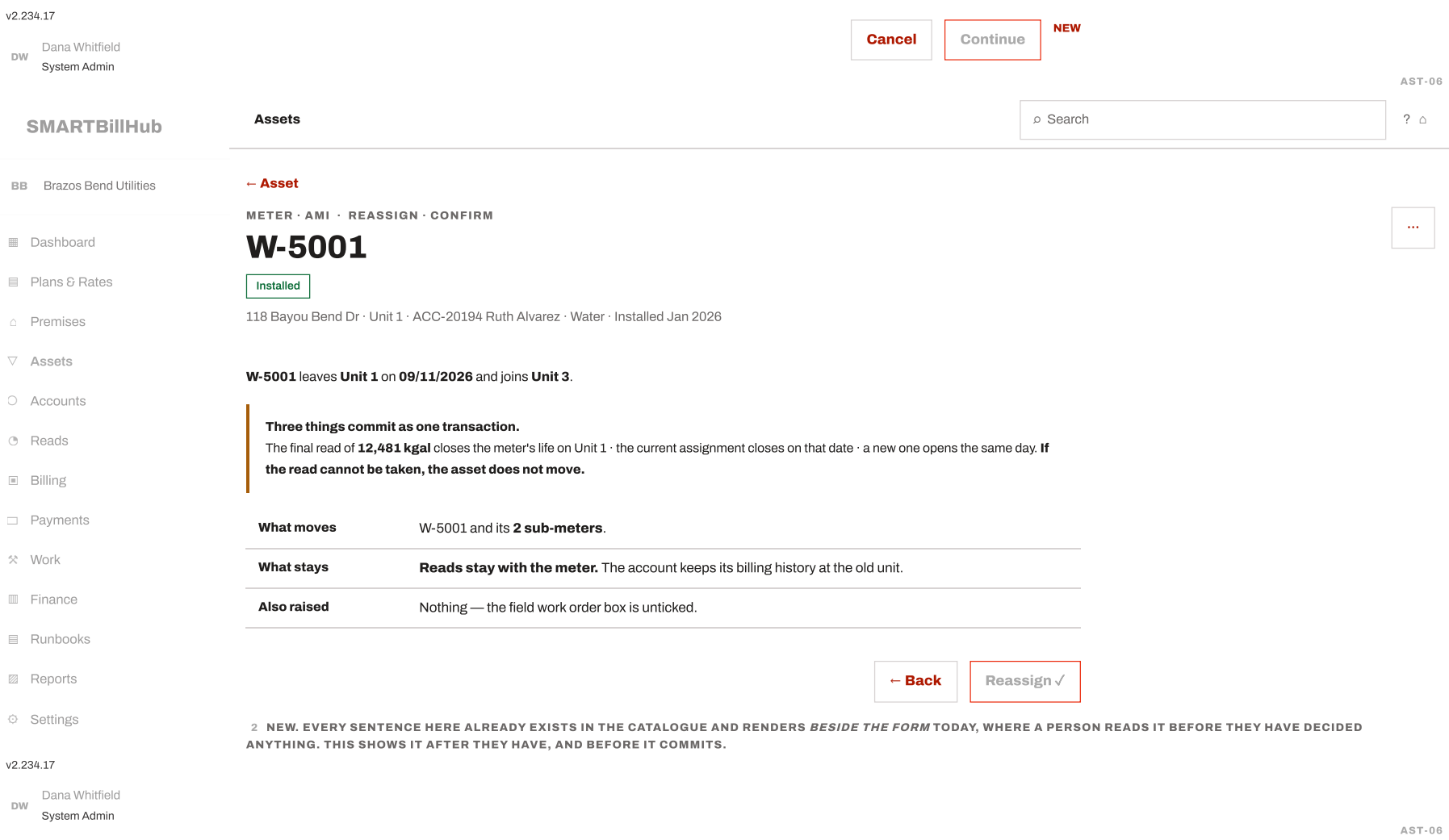
What is missing is a confirmation. A reassign closes the current assignment, opens the next one and commits a read — **three consequences, as**

one transaction, and the screen states none of them before you press. ⚠ **And the copy for it already exists, unrendered at this moment:** *"Reads stay with the meter. The account keeps its billing history at the old unit."* and *"The final read, the closing row and the new assignment commit as one transaction."* The fix is to show them **before** the commit rather than beside the form.

Then the three that every screen in this section shares: the five-slot header from screen 08 — **a person reassigning a meter whose test is eleven days overdue sees no sign of it while they act** — R12's field order, and dates with a Today shortcut and bounds.

FIGURE 31 · FIGURE 32 · 1 REDRAWN. THE FIVE-SLOT HEADER FROM SCREEN 08 WITH ITS ATTENTION STRIP, R12'S FIELD ORDER, AND DATES WITH A TODAY SHORTCUT AND BOUNDS. EVERYTHING MARKED NEW IS A CHANGE; EVERYTHING ELSE IS THE AS-BUILT:





The screen

For	A clerk moving a device between units, or pulling one back to the shelf.
How you get here	The ⇄ **swap** in a registry row; **Assign** / **Reassign** in the detail header; the Assignment panel's own button; and a unit's **+ Assign from inventory** picker, which arrives with `?unit=` already set — a prefill, visible and changeable, never a hidden value.
Primary action	**Reassign ✓** → `POST /assets/{id}/reassign`.
Tabs	None.
Table controls	N/A.
Write surfaces	**Update** — the move. On failure the server refuses the **whole** move: if the final read cannot be captured, the asset does not move, and

	the screen does not pretend otherwise.
Walked	Opened at 1440 on a real asset. Rendered, zero failed requests. **Two date inputs, neither with a min or a max, and no Today shortcut on either.**
Cognitive load	It puts the final read on the same screen as the move, so the question *what about the last bill* is answered before it is asked.
States	**Has children** — a master offers to take its sub-meters with it, defaulting to yes. **Non-measuring** — *"This device does not measure, so there is no final read to take."* **Unit list failed** — *"We couldn't load the unit list. Try again."* and the picker renders with no options.
Words	**Record the final read** is on by default, because the cost of forgetting it is an unbillable cycle and the cost of an extra tick is nothing.
Components	`SegmentedField` · `SearchSelectField` · `DatePickerField` · `CheckboxField` · `RaiseWorkOrderCheckbox` · `RefusalStrip`.
Rules	BR-AST-017 · BR-AST-022 (the read and the move are ONE transaction) · BR-AST-037/038 · B14 rule 7 (**violated**).
Audience	Staff; System Admin or Billing Clerk.
Health	SERVED.

Data flow

```
POST /assets/{id}/reassign ─→ services.reassign_asset      services.py:1688
   ├→ closes the current assignment at the effective date
   ├→ opens the next one on the same date                          BR-AST-017
   ├→ captures the final read, or REFUSES THE WHOLE MOVE (422)     BR-AST-022
   └↳ warnings[] back to the toast                                 WHOLE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Where is it going?	a two-value segmented control	—	SERVED
②	Unit	options from `GET /units?per=500`	assets → metering ⇄	SERVED, capped at 500
③	Effective date	the assignment boundary	assets	SERVED
④	Final read + date	written to `meter_reads` in the same transaction	assets → metering ⇄	SERVED
⑤	Move the sub-meters as well	`asset.sub_assets`, count served	assets	SERVED

⑥	Raise a field work order	a tick; on success it redirects to screen 27	assets → work ⇄	SERVED

23 Log repair / maintenance

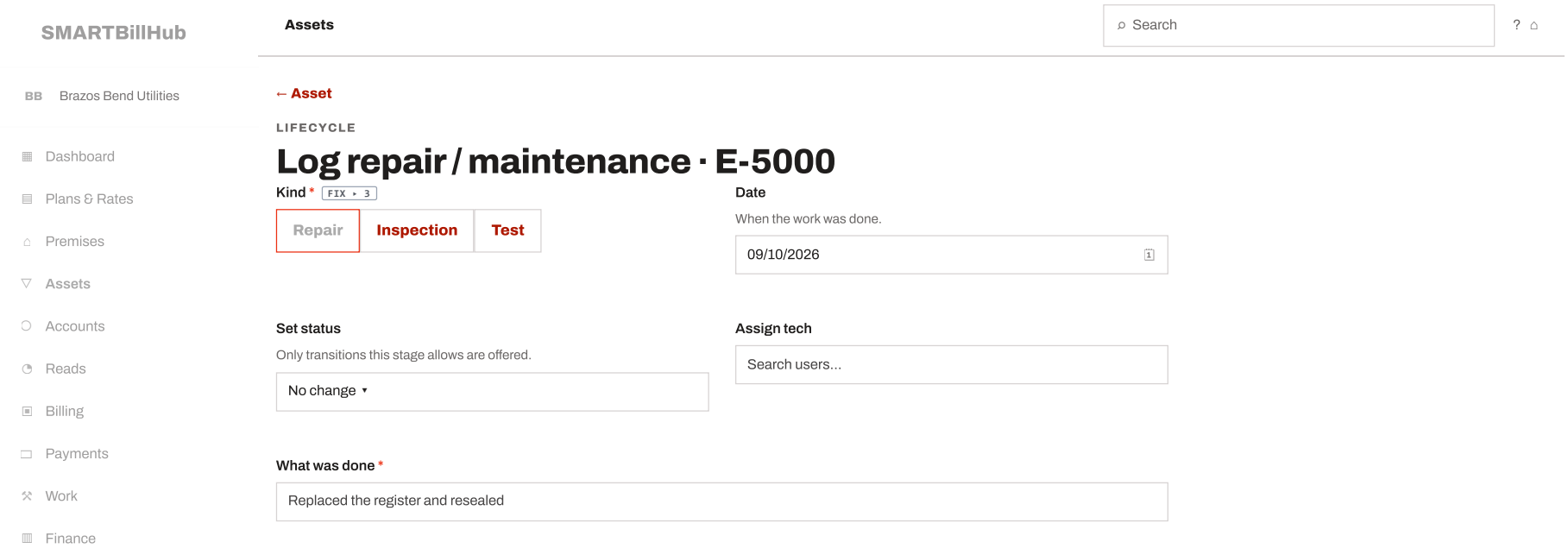
EXISTS BEHIND · NEW BUILD · COMMITMENT 04 · COMMITMENT 13 · COMMITMENT 26

`/:tenant/assets/{assetId}/maintenance/new` — titled *Log test* on a device with a cadence

> **For the one role this screen exists for, the technician field asks for a database id**
>
> `BR-AST-038` lets a **Billing Clerk** log a repair. `GET /users` is **System Admin only**. So for a Billing Clerk the picker cannot load, and the field falls back to a bare `TextField` whose value is the user's UUID, helped by the sentence *"Your role can't list users, so enter the technician's ID."* (`AssetFlowPages.tsx:604-612`).
>
> B14 rule 31 is **"no database id is ever the visible value of a field"**, ratified by the PO on 2026-08-31 in the words *"no Databse ID on screen - that is a rule"*. This is that, and it is also a task nobody can actually perform: there is no screen in the product where a Billing Clerk can look a technician's UUID up. Specification `AST-B-04`.

Somebody went out and fixed it. Where do I write that down so the next person knows?

FIGURE 33 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



Runbooks
Reports
Settings

Estimated cost

Recorded on the asset. Posted nowhere.

84.00

RECORDED, NOT POSTED

Maintenance entries are never edited or deleted — a correction is a new entry. The cost is recorded on the asset and reaches no ledger account.

Raise a work order from the repair template

v2.234.17

DW Dana Whitfield
System Admin

Cancel | Log it ✓

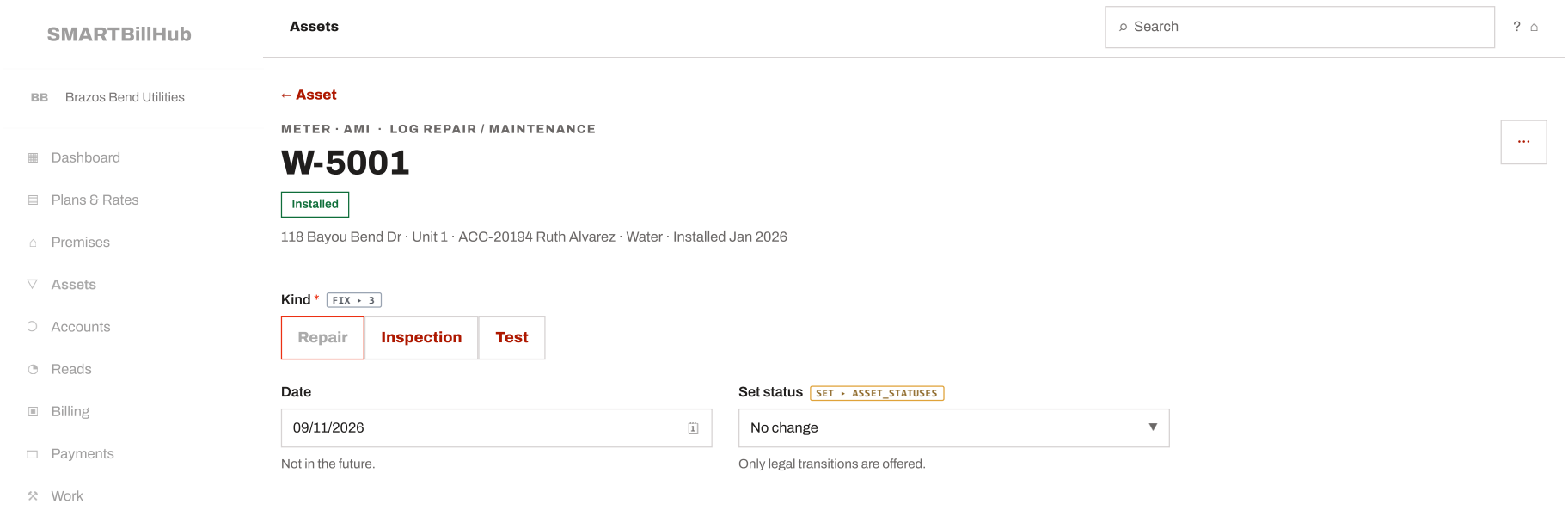
AST-07

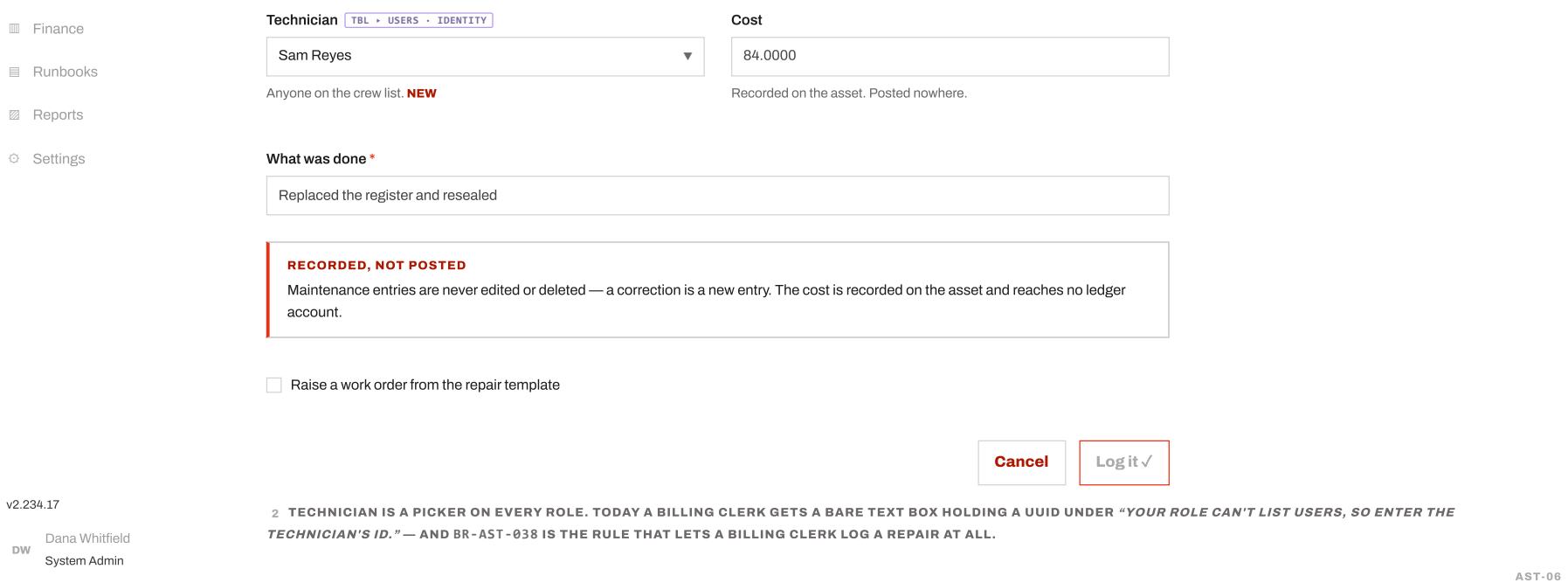
The worst single defect in the module, and the data capture is not it

Data captured: complete. Kind · date · detail · technician · resulting status · **cost**. ⚠ **This document nearly filed "repair cost is never captured" as a finding** — it is captured, and the first read of the component missed it. The correction is recorded in section 13 because the near-miss is the lesson, not the field.

The fault is HOW the technician is captured. For a Billing Clerk the picker degrades to a bare text box holding a **UUID**, under the sentence *"Your role can't list users, so enter the technician's ID."* That breaks **B14 rule 31** — no database id is ever the visible value of a field — and `BR-AST-038` **explicitly permits a Billing Clerk to log a repair**, so the role that is allowed to use this screen is the role the screen cannot serve. Specification `AST-B-04`.

FIGURE 34 · 1 REDRAWN. THE FIVE-SLOT HEADER FROM SCREEN 08 WITH ITS ATTENTION STRIP, R12'S FIELD ORDER, AND DATES WITH A TODAY SHORTCUT AND BOUNDS. EVERYTHING MARKED **NEW** IS A CHANGE; EVERYTHING ELSE IS THE AS-BUILT:





2 TECHNICIAN IS A PICKER ON EVERY ROLE. TODAY A BILLING CLERK GETS A BARE TEXT BOX HOLDING A UUID UNDER *"YOUR ROLE CAN'T LIST USERS, SO ENTER THE TECHNICIAN'S ID."* — AND `BR-AST-038` IS THE RULE THAT LETS A BILLING CLERK LOG A REPAIR AT ALL.

AST-06

The screen

For	A clerk recording what a technician did.
How you get here	The 🔧 **wrench** in a registry row; **Log repair** in the detail header; the Test & certification panel's **Log test**; and the Maintenance tab's own button.
Primary action	**Log it ✓** → `POST /assets/{id}/maintenance`.
Tabs	None.
Table controls	N/A.
Write surfaces	**Add** only. Append-only by rule — a correction is a new entry, and the screen says so.
Walked	Opened at 1440. Rendered, zero failed requests. **One date input, no min, no max, no Today shortcut.**
Cognitive load	The status change rides along with the repair, so a meter that has just been fixed does not stay flagged as broken because somebody forgot a second screen.
States	**Stages unavailable** — *"We couldn't load the lifecycle stages, so no status change is offered here"*, and the Set-status select renders with an empty option set. **Clerk without the user list** — see the callout.
Words	**No change** is the default status choice and is an explicit value the server accepts — it writes no status-changed row. Not a blank.

Components	`SegmentedField` · `DatePickerField` · `SelectField` · `SearchSelectField` · `TextField` · the consequence note.
Rules	BR-AST-018 · BR-AST-020/031 (legal transitions only, from the behaviour class) · BR-AST-038 · §10-D3 · **B14 rule 31, violated**.
Audience	Staff; a Billing Clerk may log a repair (BR-AST-038).
Health	Worst is DEGRADED — see the callout.

Data flow

```
POST /assets/{id}/maintenance —→ services.log_maintenance    services.py:1346
   ├→ @require_role(SYSTEM_ADMIN, BILLING_CLERK)   br_ref="BR-AST-038"
   ├→ appends the entry; a `test` also updates assets.last_test_date
   └→ applies the chosen legal transition, or none                WHOLE

GET /users —→ SYSTEM ADMIN ONLY   ⇄ assets → identity
   └→ the technician picker's options   BREAKS AT the seam for a Billing Clerk
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Kind	repair · inspection · test	assets	SERVED
②	Date	`asset_maintenance.date`	assets	SERVED
③	Set status	options are the legal targets from this stage's behaviour class	assets	SERVED
④	Assign tech	options from `GET /users` — **admin-only**; for a Billing Clerk the control degrades to a bare text box holding a UUID	assets → identity ⇄	DEGRADED
⑤	What was done	`asset_maintenance.detail`	assets	SERVED
⑥	Cost	`asset_maintenance.cost`, recorded, posted nowhere	assets	SERVED
⑦	Raise a work order from the repair template	a tick; redirects to screen 27	assets → work ⇄	SERVED

AST-B-04

A billing clerk logging a repair can name the technician

STATUS Unverified

WAITS ON A read a clerk is allowed to make — a name-and-id list of the tenant's users, scoped to what a picker needs. It is a new endpoint, so this one is genuinely back-end work.

SCREENS 23

WHAT DONE LOOKS LIKE ON SCREEN Signed in as a Billing Clerk, **LOG REPAIR / MAINTENANCE** shows **ASSIGN TECH** as a searchable picker listing *Sam Reyes · sam.reyes@sample.test*, and no UUID appears anywhere on the screen.

---

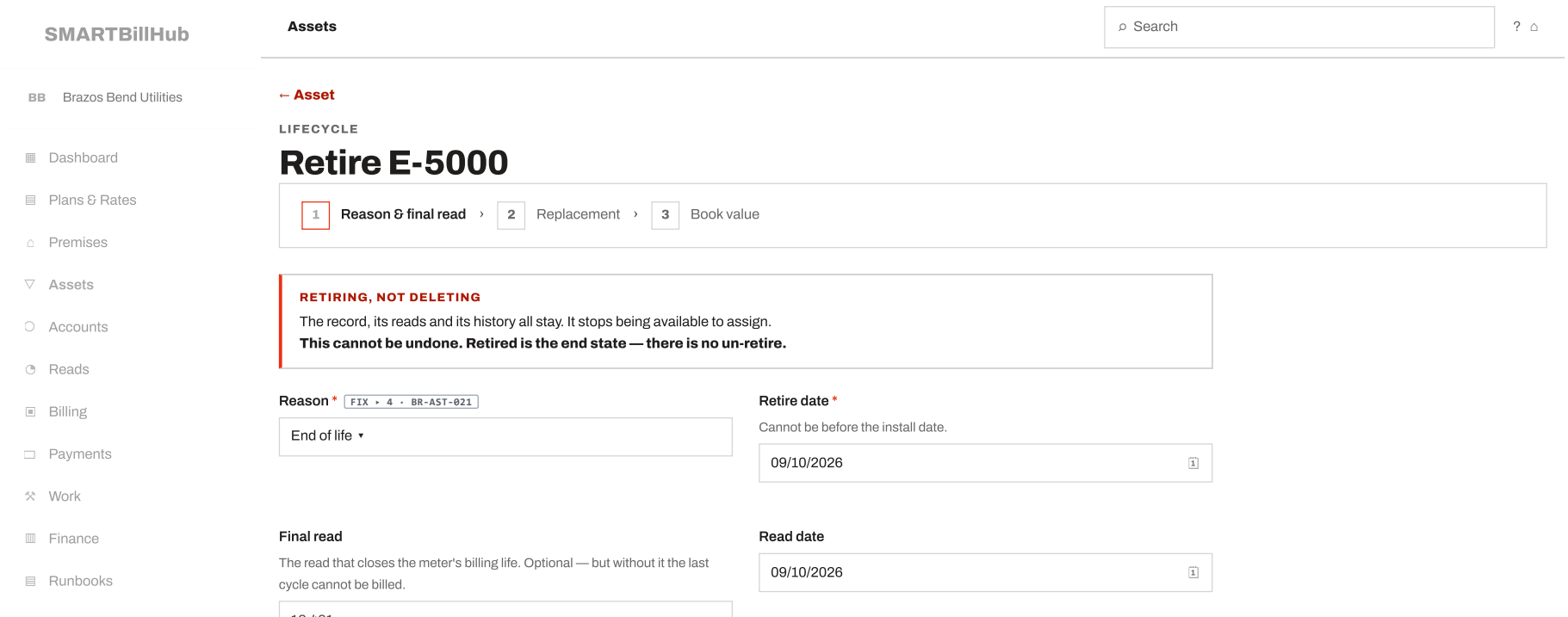
24 Retire · 1 Reason & final read

EXISTS BEHIND | COMMITMENT 26 | COMMITMENT 28

`/:tenant/assets/{assetId}/retire — step 1`

This meter is finished. How do I take it out without losing the last month's billing?

FIGURE 35 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



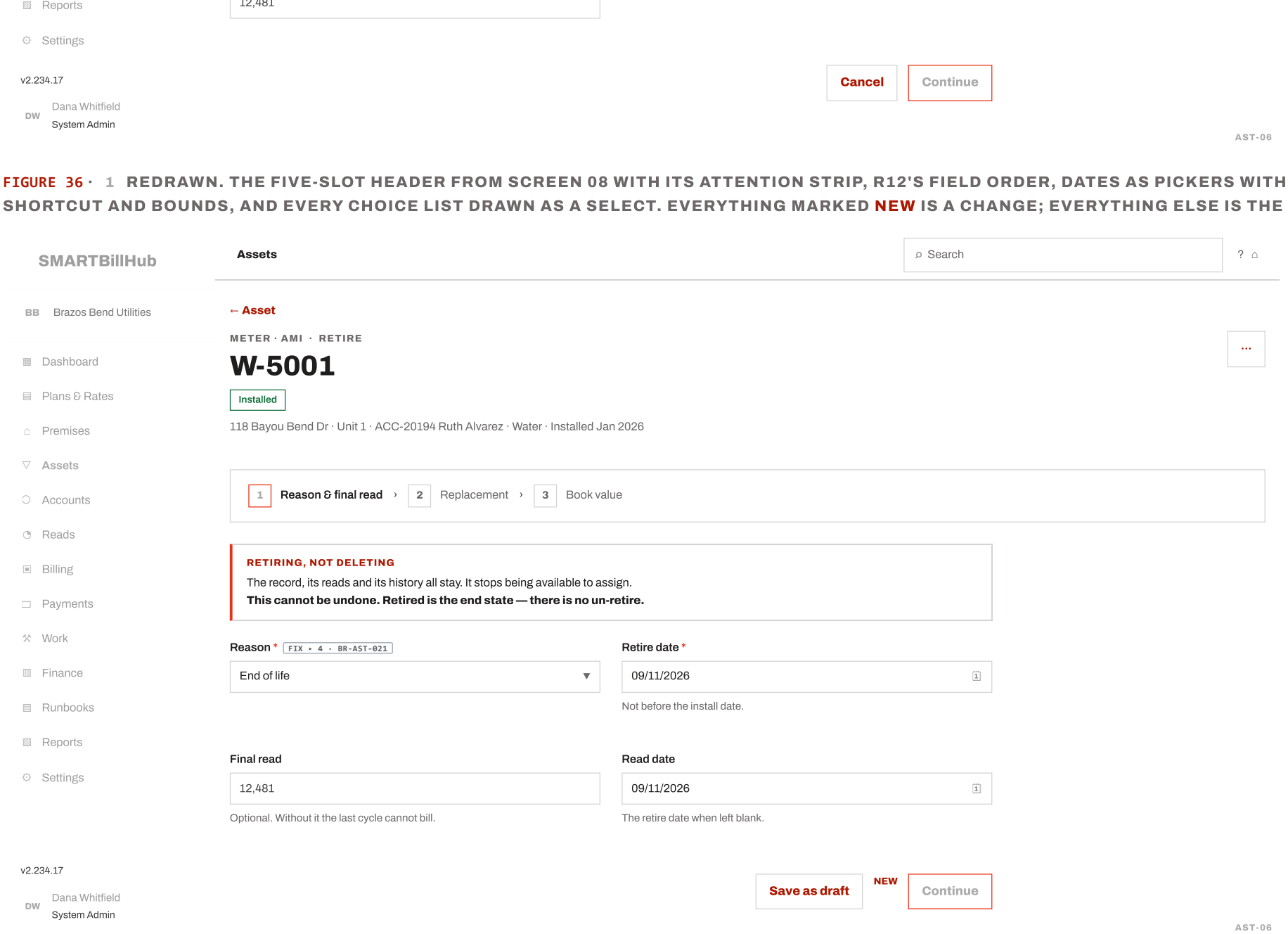


FIGURE 36 · 1 REDRAWN. THE FIVE-SLOT HEADER FROM SCREEN 08 WITH ITS ATTENTION STRIP, R12'S FIELD ORDER, DATES AS PICKERS WITH A TODAY SHORTCUT AND BOUNDS, AND EVERY CHOICE LIST DRAWN AS A SELECT. EVERYTHING MARKED NEW IS A CHANGE; EVERYTHING ELSE IS THE AS-BUILT:

The escape hatch exists in the component kit, six wizards elsewhere use it, and the two asset wizards do not

⚠ **This document first recorded "the retire wizard has no Cancel". That was wrong**, and the correction is better than the finding. `WizardFrame` — the house wizard **twenty-nine** files share — offers **Save & exit** and **Park** through an optional `exit` prop, with a whole parking system behind it: a reason (*waiting on materials · waiting on a permit*), an optional note and a remind-on date, and B19 §12.2's promise that *"Save & exit leaves quietly and writes nothing of its own."*

`exit` **is optional, six files pass it — every one of them in customers or billing — and** `AssetRetireWizard` **passes none.** Neither does the add wizard. So a person three steps into retiring the wrong meter has Back, and nothing that says *leave this*, while a person half way through a move-in has both. **That is hard-won law 2's family**: not dead UI, but a live affordance the screen declined to ask for — and nothing can catch it, because an unset optional prop is exactly what an optional prop is for.

Everything else here is the pattern to copy rather than a fault. The step rail, `BR-SHL-018`'s back-only navigation, the final ✓ commit, and the best consequence copy in the module: *"Retiring, not deleting. Every read, bill and history row this asset produced stays..."* **The confirm step drawn on screen 18 is modelled on this one.**

Data captured: complete — all six keys `retire_asset` accepts.

The screen

For	An admin taking a device permanently out of service.
How you get here	The ✕ **retire** in a registry row, and **Retire** in the detail header — both only when the server's action set allows it.
Primary action	**Continue**.
Tabs	None — a three-step wizard.
Table controls	N/A.
Write surfaces	Nothing writes until step 3's commit.
Walked	Opened at 1440. Rendered, zero failed requests. **Two date inputs, no bounds on either** — and the help text says the retire date *"cannot be before the install date"*, a constraint the control does not enforce and the server does.
Cognitive load	It puts the irreversibility in front of the first field rather than in the confirm at the end, where a person has already decided.
States	**Non-measuring** — the final-read block is not drawn at all.
Words	**"This cannot be undone. Retired is the end state — there is no un-retire."** Plain, twelve words, before the first input. BR-AST-030 in the user's language.
Components	`WizardFrame` · `SelectField` · `DatePickerField` · `TextField` · the notice box.
Rules	BR-AST-021 · BR-AST-030 (terminal) · B14 rules 3, 7 (**violated**) and 11.
Audience	Staff, **System Admin only** — gated twice, at the view (`SystemAdminOnly`, `api.py:264`) and in the service (`services.py:1461`).

	Health	SERVED.		

Data flow

```
(no write on this step — the wizard's one commit is step 3)
asset_lifecycle_stages —→ the legal target for retire, from the behaviour class   WHOLE
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Reason	`assets.retire_reason` — end_of_life · failed · …	assets	SERVED
②	Retire date	`assets.retire_date`; must be ≥ install date, **server-enforced**	assets	SERVED
③	Final read + date	written to `meter_reads` on commit	assets → metering ⇄	SERVED

25 Retire · 2 Replacement

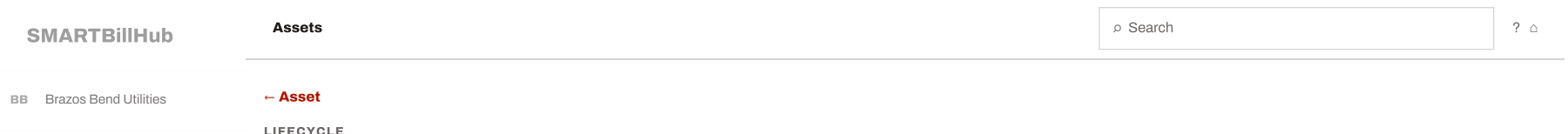
EXISTS BEHIND · COMMITMENT 26 · COMMITMENT 28

`/:tenant/assets/{assetId}/retire — step 2`

> **This tick was recorded as missing while it was on screen**
>
> `BR-WRK-056` read *"the three `WorkOrderR5Checkbox` controls"* until CR-070 wave M renamed the component to `RaiseWorkOrderCheckbox` and added this fourth tick. **The old name greps to nothing**, and `DEF-WRK-4122` was filed recording the retire tick as missing while it was live. The rule was amended on 2026-09-06 — name and count only, the rule itself unchanged. Worth repeating here because it is the same failure this document is built to find, running in the opposite direction: a search for a name that had moved.

If I take this one out, what does the house get billed on next month?

FIGURE 37 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



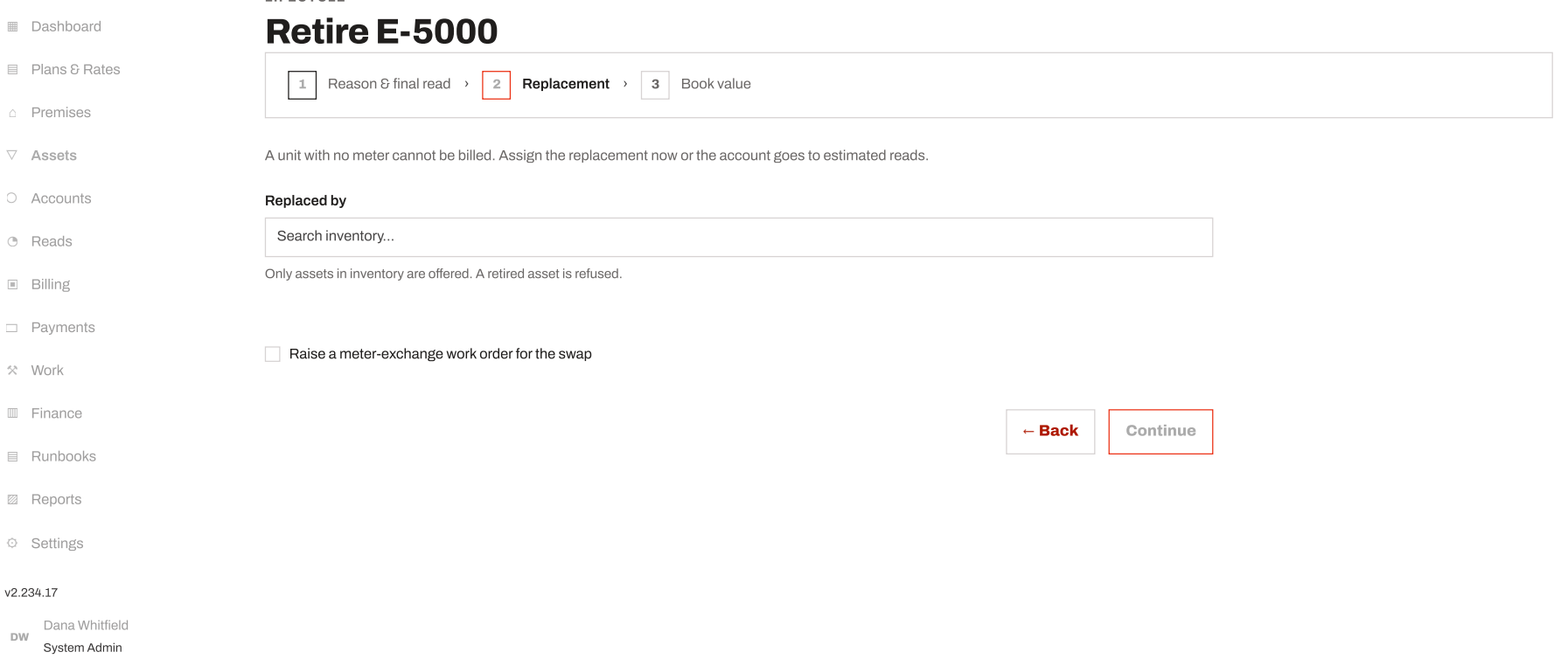
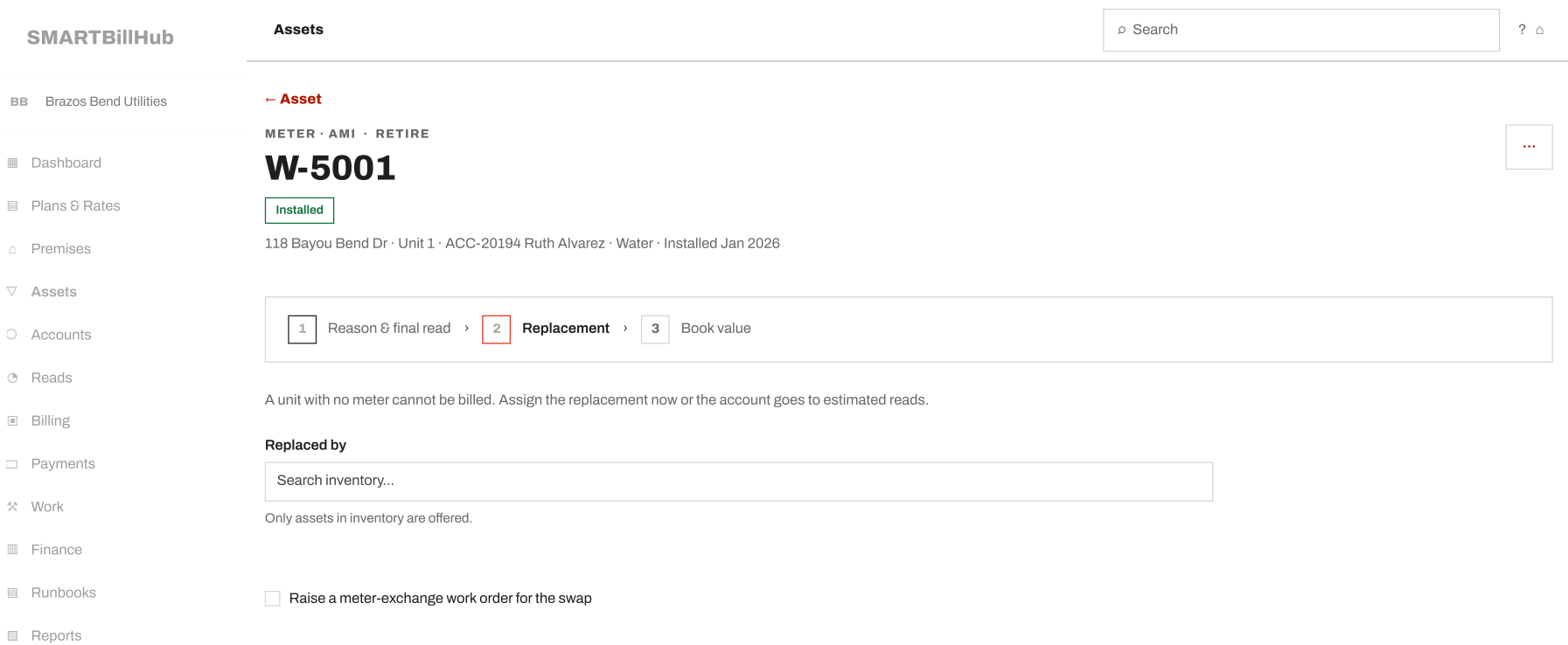
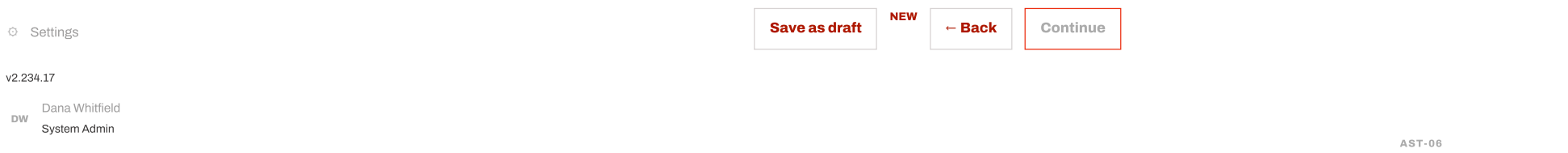


FIGURE 38 · 1 REDRAWN. THE FIVE-SLOT HEADER FROM SCREEN 08 WITH ITS ATTENTION STRIP, R12'S FIELD ORDER, DATES AS PICKERS WITH A TODAY SHORTCUT AND BOUNDS, AND EVERY CHOICE LIST DRAWN AS A SELECT. EVERYTHING MARKED NEW IS A CHANGE; EVERYTHING ELSE IS THE AS-BUILT:





> **The same unset exit**
>
> Step 2 of the same wizard, and the same gap: `WizardFrame` offers **Save & exit** and **Park**, six wizards elsewhere pass the prop, and this one does not. Otherwise this step is sound — *"No replacement — leave unmetered"* is an honest option rather than a forced choice.

The screen

For	The same admin, one step on.
How you get here	The ✕ **retire** in a registry row, and **Retire** in the detail header — both only when the server's action set allows it.
Primary action	**Continue**.
Tabs	None — a three-step wizard.
Table controls	N/A.
Write surfaces	None yet.
Walked	Not reached — the walk was scripted by address. Structure read from source (`AssetRetireWizard.tsx:306-346`).
Cognitive load	It states the consequence of leaving this blank in one sentence, at the moment the choice is made.
States	**Refused** — *"The replacement is itself retired. Pick a living asset."* · *"That replacement asset wasn't found."*
Words	**"A unit with no meter cannot be billed. Assign the replacement now or the account goes to estimated reads."** It names the actual downstream cost rather than warning in the abstract.
Components	`SearchSelectField` · `RaiseWorkOrderCheckbox`.
Rules	BR-AST-021 · B14 rule 9.
Audience	Staff, **System Admin only** — gated twice, at the view (`SystemAdminOnly`, `api.py:264`) and in the service (`services.py:1461`).
Health	UNVERIFIED — not walked.

Data flow

```
assets (in inventory) —→ the replacement picker's options
POST /assets/{id}/retire —→ sets assets.replaced_by_asset_id      services.py:1461
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Replaced by	`assets.replaced_by_asset_id`; options are inventory assets only	assets	UNVERIFIED
②	Raise a meter-exchange work order	the fourth `RaiseWorkOrderCheckbox` tick — added in CR-070 wave M	assets → work ⇄	UNVERIFIED

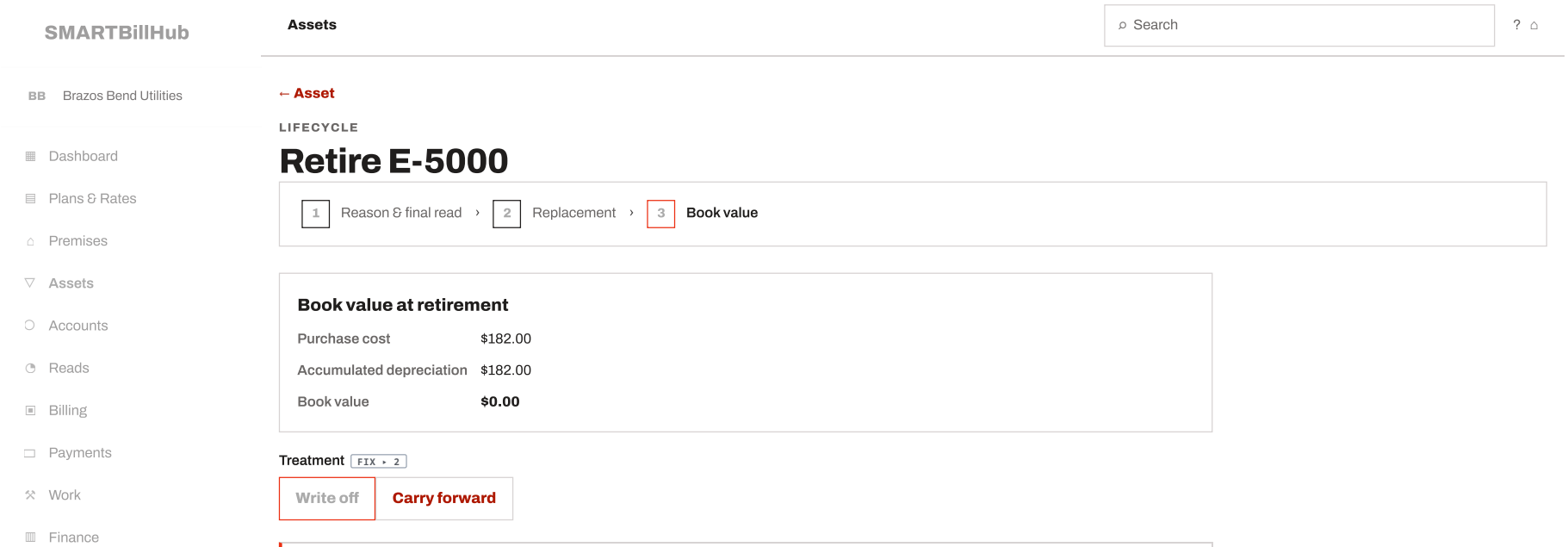
26 Retire · 3 Book value

EXISTS BEHIND · COMMITMENT 26 · COMMITMENT 28

`/:tenant/assets/{assetId}/retire — step 3`

What is it worth on the books, and what happens to that amount?

FIGURE 39 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



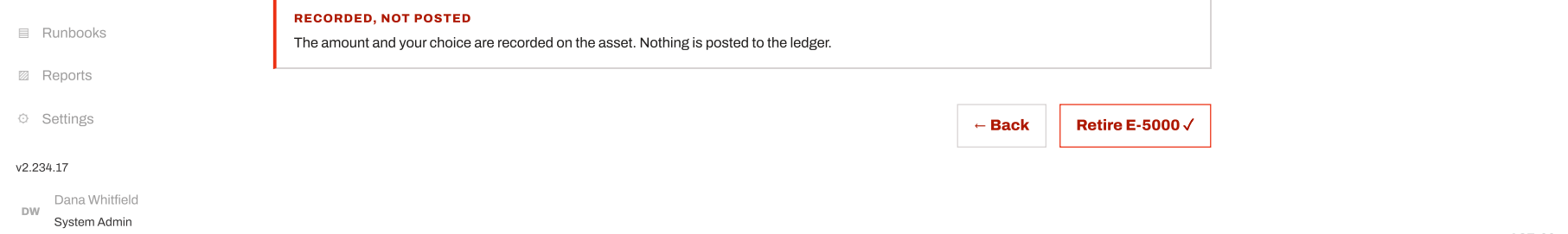
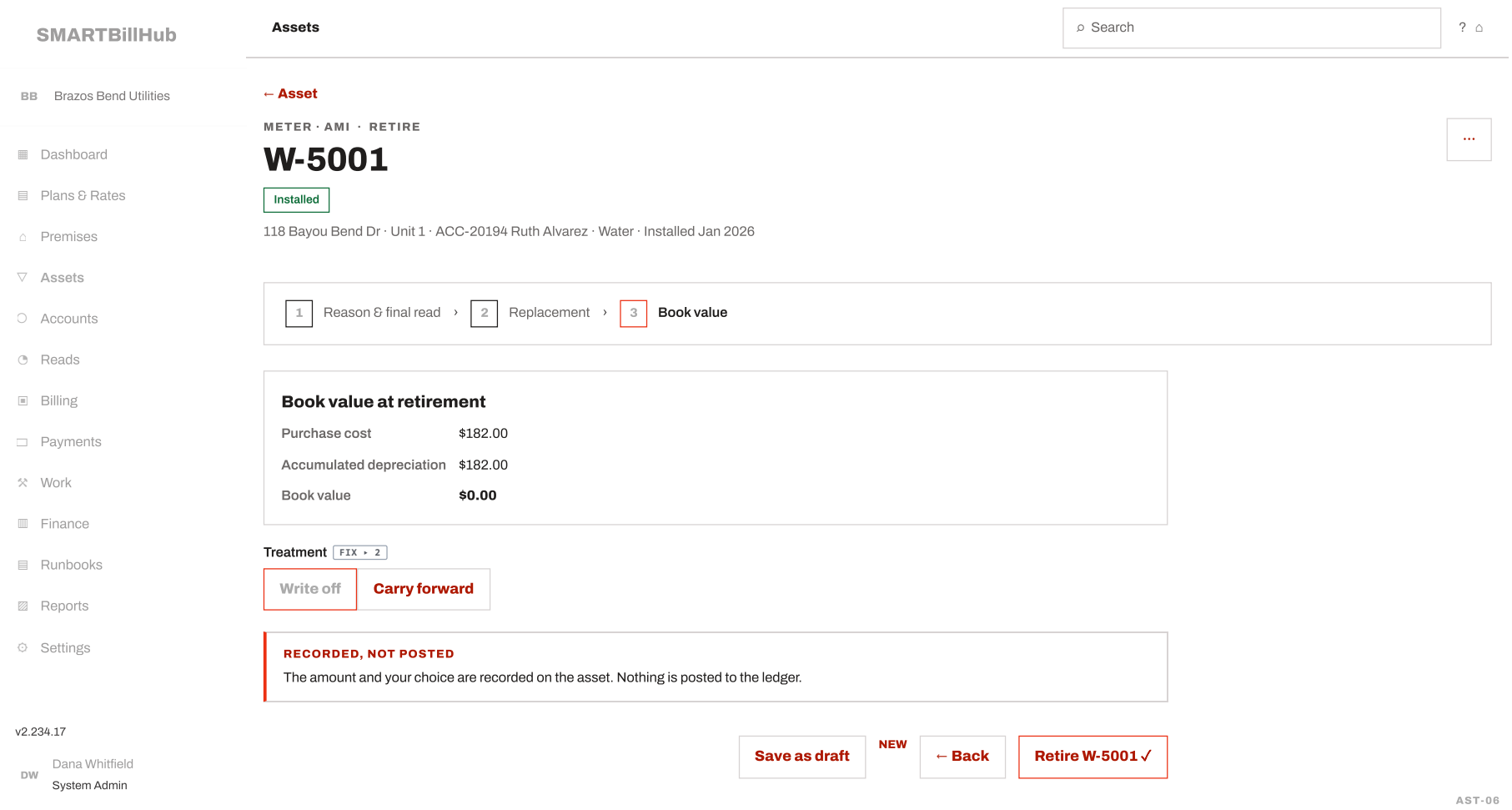


FIGURE 40 · 1 REDRAWN. THE FIVE-SLOT HEADER FROM SCREEN 08 WITH ITS ATTENTION STRIP, R12'S FIELD ORDER, DATES AS PICKERS WITH A TODAY SHORTCUT AND BOUNDS, AND EVERY CHOICE LIST DRAWN AS A SELECT. EVERYTHING MARKED NEW IS A CHANGE; EVERYTHING ELSE IS THE AS-BUILT:



The same unset exit

Step 3 of the same wizard, and the same gap: `WizardFrame` offers **Save & exit** and **Park**, six wizards elsewhere pass the prop, and this one does not. Otherwise this step is sound, and its sentence is exact: *"The amount and your choice are recorded on the asset. Nothing is posted to the ledger."*

The screen

For	The same admin, committing.
How you get here	The ✕ **retire** in a registry row, and **Retire** in the detail header — both only when the server's action set allows it.
Primary action	**Retire E-5000** ✓, in the danger tier — **enforced on rendered output** for a wizard commit since DEF-712.
Tabs	None — a three-step wizard.
Table controls	N/A.
Write surfaces	**Update** — `POST /assets/{id}/retire`: the terminal transition, the closed assignment, the computed book value and the chosen treatment, in one call.
Walked	Not reached; structure read from source. The figures drawn are this document's own worked example, taken from the real asset opened on screen 08 — purchase cost **$182.00**, fully depreciated at 5 yrs 5 mo against a 5-year life, so book value **$0.00**.
Cognitive load	It shows the arithmetic rather than a single number, so a person can see why it is zero.
States	**No cost recorded** — *"No purchase cost recorded — there is no book value to compute, and nothing to write off"*, and the treatment choices are not drawn. Its help adds what would fix it: *"Capture a purchase cost and an expected life on the asset to see book value here."*
Words	**"The amount and your choice are recorded on the asset. Nothing is posted to the ledger."** No GL account appears anywhere in this module — hard rule 5 and the D3 ruling.
Components	`WizardFrame` · `MoneyCell` · `SegmentedField` · the consequence note.
Rules	BR-AST-047 · §10-D3 · hard rule 2 (2 dp on display) · hard rule 5 (only the ledger posting service writes journal lines) · B14 rules 6 and 11.
Audience	Staff, **System Admin only** — gated twice, at the view (`SystemAdminOnly`, `api.py:264`) and in the service (`services.py:1461`).
Health	UNVERIFIED — not walked.

Data flow

```
POST /assets/{id}/retire —→ services.retire_asset          services.py:1461
   ├→ SystemAdminOnly at the view AND @require_role in the service
   ├→ terminal transition; closes the current assignment
```

```
├→ computes straight-line book value, records amount + treatment
└→ writes audit_log                                              WHOLE
⚠ reaches NO journal line, by ruling — §10-D3
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Purchase cost	`assets.purchase_cost`	assets	UNVERIFIED
②	Accumulated depreciation	server straight-line over age and expected life	assets	UNVERIFIED
③	Book value	cost – accumulated, 4 dp stored / 2 shown	assets	UNVERIFIED
④	Treatment	`assets.book_value_treatment` — write_off · carry_forward	assets	UNVERIFIED

27 Raise work order

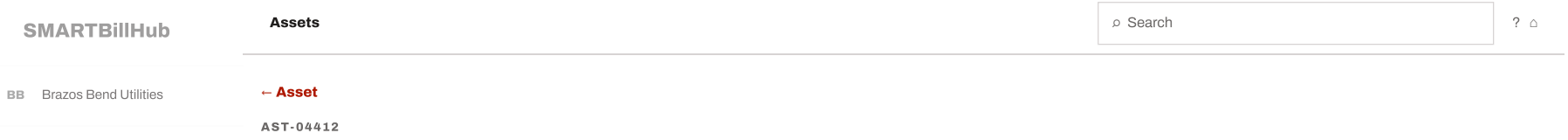
EXISTS BEHIND · NEW BUILD · COMMITMENT 02 · COMMITMENT 05 · COMMITMENT 10 · COMMITMENT 26

`/:tenant/assets/{assetId}/work-orders/new`

> **A retired screen badge is written into this file twice**
>
> `RaiseWorkOrderPage.tsx:57` exports `RAISE_WORK_ORDER_SCREEN = "WRK-19"`, and `App.tsx:1006` carries the comment `// WRK-19`. Both `screenIds.ts:1301-1312` and `10-scripts/gates/g29_allocated_badges.txt:182-188` allocate this route **WRK-24** and say in as many words that it is *"deliberately NOT `WRK-19`, which was retired in wave D — reusing a retired badge would make an old citation land on a screen it never named."*
>
> The screen renders **WRK-24**, so no user sees the wrong id. It survives because the constant is exported and imported by nothing — one grep hit, its own definition — and a comment is not code. Record-level only, and listed because that is exactly the kind of thing that is true for a year.

This needs a crew. Can I raise the job from here without leaving the asset?

FIGURE 41 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



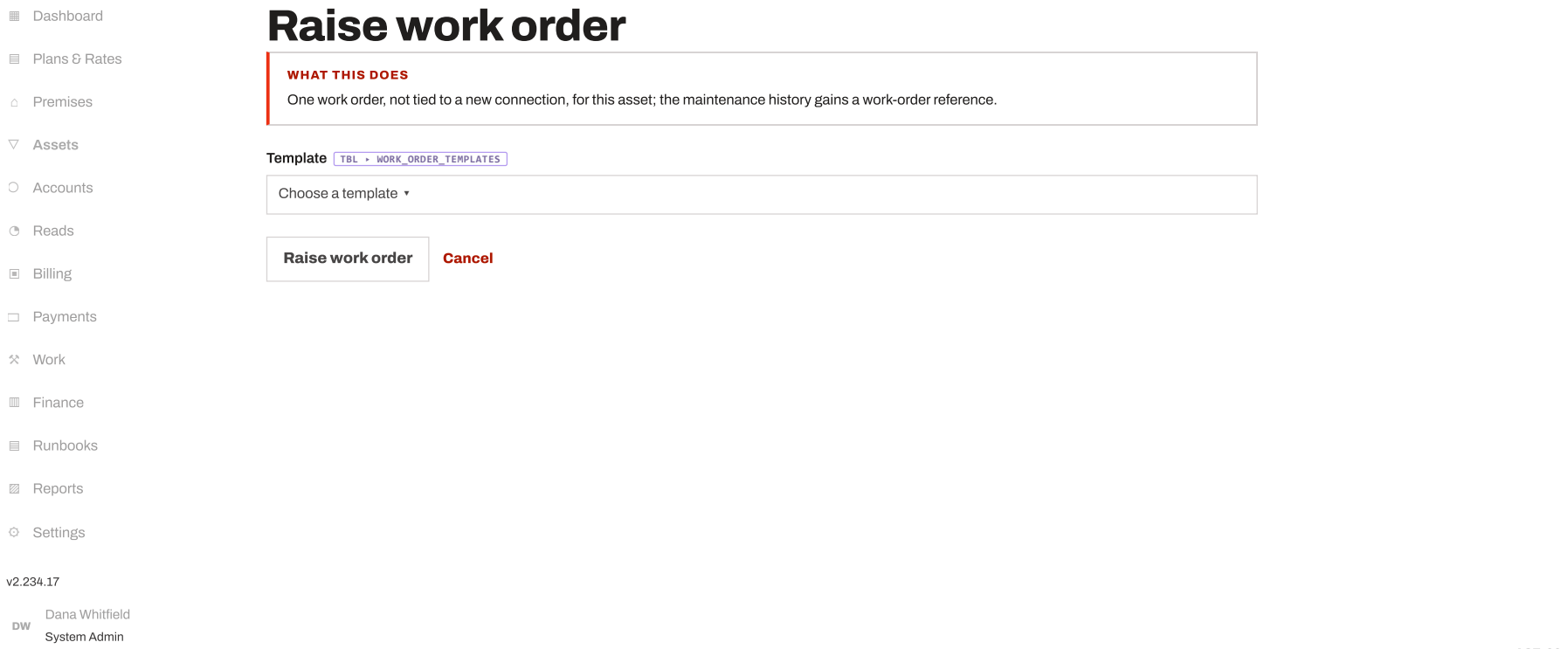
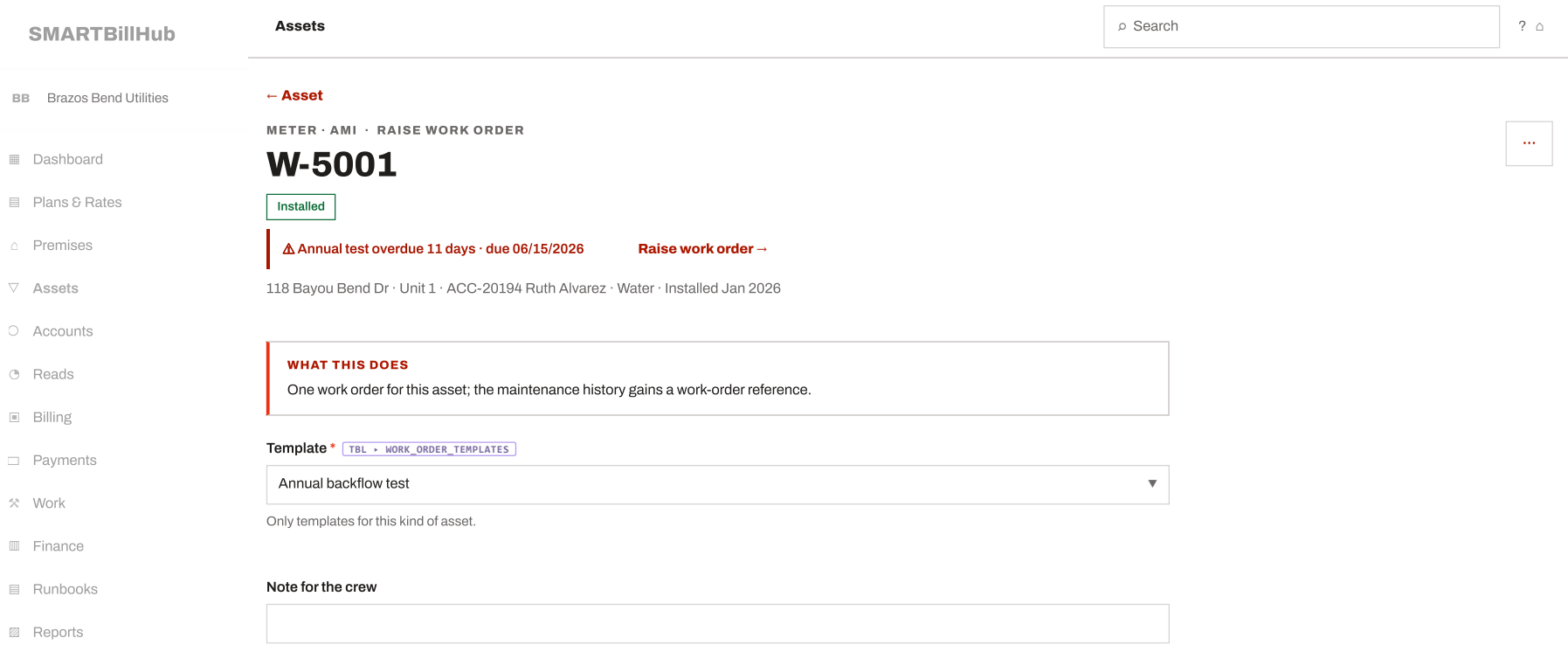
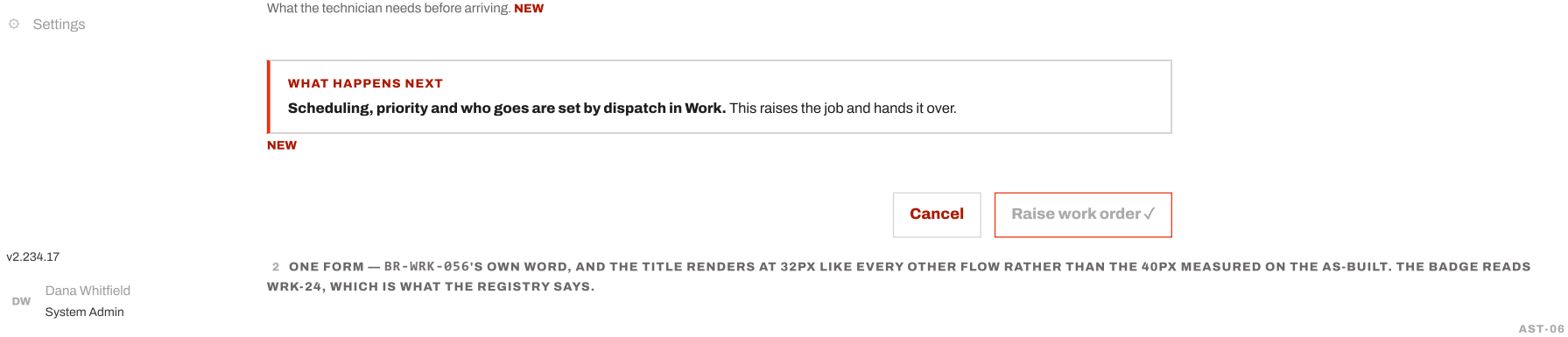


FIGURE 42 · 1 REDRAWN. THE FIVE-SLOT HEADER FROM SCREEN 08 WITH ITS ATTENTION STRIP, R12'S FIELD ORDER, DATES AS PICKERS WITH A TODAY SHORTCUT AND BOUNDS, AND EVERY CHOICE LIST DRAWN AS A SELECT. EVERYTHING MARKED NEW IS A CHANGE; EVERYTHING ELSE IS THE AS-BUILT:





Four faults already filed, and one capture gap that is a missing sentence rather than a missing field

It captures three things: `asset_id`, `template_id` and the template's name. That looks thin beside a work order's priority, due date, appointment window and assignee — and it is **correct**: `create_work_order` accepts sixteen keys and **not one of those four is among them**. Scheduling happens in Work, *after* the raise. That is the architecture, not an oversight.

⚠ **But the screen never says so, and the cost lands two screens away.** A clerk raising a job for a backflow test eleven days overdue has no way to say it is urgent, and nothing tells them somebody else will schedule it. **The asset's own Work tab (screen 13) draws *Expected completion* and *Technician* columns that are empty on every freshly raised job**, with nothing on either screen explaining why. Two changes, and neither is a schema change: one line — *"Scheduling, priority and who goes are set by dispatch in Work"* — and a free-text **note**, since `technician_notes` exists on the order and no path writes it from here.

The four already filed: the title renders **40px** where every other flow renders 32px, the one R11 outlier and it is measured · the badge reads `WRK-19` where the registry says `WRK-24` (`AST-B-05`) · **there is no door** — the screen is reached by URL only, and `BR-WRK-056` names two controls that are not built (`AST-B-02`) · and *"No template applies to this kind of asset"* leaves a permanently disabled submit with nowhere to go (`AST-B-10`).

And one correction to this document: `BR-WRK-056` says *"ONE form"*, the file is 205 lines and imports no `WizardFrame`. Any step rail drawn on this screen is this document's invention, not the product's.

The screen

For	Anyone who has just decided a device needs a visit.
How you get here	**Five ways, and two of the four the rule names are missing.** Built: the Lifecycle tab's certification wrench (`AssetLifecycleTab.tsx:187`) and four post-commit ticks — add (`AssetAddWizard.tsx:474`), reassign (`AssetFlowPages.tsx:294`), log maintenance (`:546`) and retire (`AssetRetireWizard.tsx:220`). Not built: **the asset's action bar** and **the Maintenance history panel**, both named by `BR-WRK-056`.
Primary action	**Raise work order** → `POST /work-orders` with `asset_id`.

Tabs	None.
Table controls	N/A.
Write surfaces	**Add** only. `apps/assets` writes **nothing** here — the post goes to Work's own door, and a test asserts that by name.
Walked	Opened at 1440 on a real asset. Rendered with its consequence box, the template dropdown reading *Choose a template*, and the submit disabled until a template is chosen. Screen badge **WRK-24**. Zero failed requests.
Cognitive load	One form and one choice; the template decides everything else, including which templates even apply to this kind of asset.
States	**No template applies** — *"No template applies to this kind of asset."* and the submit stays disabled. That is a dead end with a stated reason and no way forward from the screen.
Words	The consequence box is three facts and one sentence, rewritten under B14 rule 32: *"One work order, not tied to a new connection, for this asset; the maintenance history gains a work-order reference."*
Components	`PageHeader` · `ConsequenceBox` · `Button` · the shared select.
Rules	BR-WRK-056 · BR-WRK-006 (an asset is an anchor) · B14 rules 2, 11, 21, 31, 32, 36.
Audience	Staff. The gate is Work's own on `POST /work-orders`, not an assets-module gate.
Health	SERVED.

Data flow

```
GET work templates for this asset ─→ listTemplatesForAsset()   ⇄ assets → work
   └→ filtered by "applies to asset types"                          WHOLE

POST /work-orders {asset_id, type, template_id} ─→ apps/work/api.py:1592
   └→ writes work_orders.asset_id                                   WHOLE
   ⚠ apps/assets gains no line of code for this feature, by design
```

Every field, and where its data comes from

	Field	Comes from	Owned by	Health
①	Kicker: AST-04412	`assets.asset_no` — the number, never the uuid	assets	SERVED

②	Template	options from Work, filtered by the asset's type	assets → work ⇄	SERVED
③	Raise work order	`POST /work-orders`	assets → work ⇄	SERVED

AST-B-10

A clerk told no template applies is not left on a dead screen

STATUS UNVERIFIED — engineering fills this in from the code, one row at a time.

WAITS ON A decision about where they should go — the Work templates screen, a generic work order, or back to the asset.

SCREENS 27

WHAT DONE LOOKS LIKE ON SCREEN Raising a work order on an asset whose kind has no template shows *"No template applies to this kind of asset"* **AND** a control that goes somewhere — not a permanently disabled submit under a sentence.

AST-B-05

A person following a citation to `WRK-19` does not land on this screen

STATUS Unverified

WAITS ON Nothing. Delete the unused constant, or correct both it and the comment to `WRK-24`.

SCREENS 27

WHAT DONE LOOKS LIKE ON SCREEN Nothing changes on screen — this one is checkable by grep: `rg "WRK-19" 07-frontend/src` returns nothing.

06

Adding an asset — the four steps and the commit

One address, `/:tenant/assets/new?class=`, four steps and a final ✓ — and it is a **page**, not a dialog, because it creates something. The wizard branches on **three classes**, so what you are asked for on steps 1 and 2 depends on what you are adding.

Ten columns an asset can hold are never asked for here — and seven of them are never asked for anywhere

The wizard captures **26** of the asset's usable columns. These are the ones it does not, checked against the model:

Not captured at Add	Capturable anywhere else?
External reference	**No.** Served on every row, rendered nowhere, typed nowhere. It can only arrive by import
Form / class	**No.** Zero capture sites in the front end
Seal #	**No.** Zero capture sites — and a seal is fitted *at installation*, which is this screen
Location note	**No.** Zero capture sites — where a technician will actually find it
Depreciation method	**No.** The present value is computed from it and nobody can set it
Allocation share ⚠	**No. Sub-meter only**, and this is the sharpest of them: a sub-meter created here is *metered, not allocated*, and there is no screen that can ever allocate it
Allocation rule ⚠	**No.** Same
Loss tolerance ⚠	**No.** Same
Opening read	Refused, with a sentence — `AST-B-14` reopened it
Last test date	**Yes** — a *test* entry on Log repair sets it. Not a gap

⚠ **Seven of the ten have no capture site anywhere in the front end.** They can be displayed and can never be entered — which is **hard-won law 2 in a third form**: not a dead control, not a stale refusal, but a **column with a reader and no writer**. Specification `AST-B-27`. Each is marked **NEW** on the step it belongs to below, and the three allocation rows appear on the **sub-meter** variant only.

---

28 Add an asset · 1 Identity

EXISTS BEHIND | NEW BUILD | COMMITMENT 03 | COMMITMENT 26 | COMMITMENT 27 | COMMITMENT 28

`/:tenant/assets/new?class=meters`

What is this thing, and what will we call it?

FIGURE 43 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



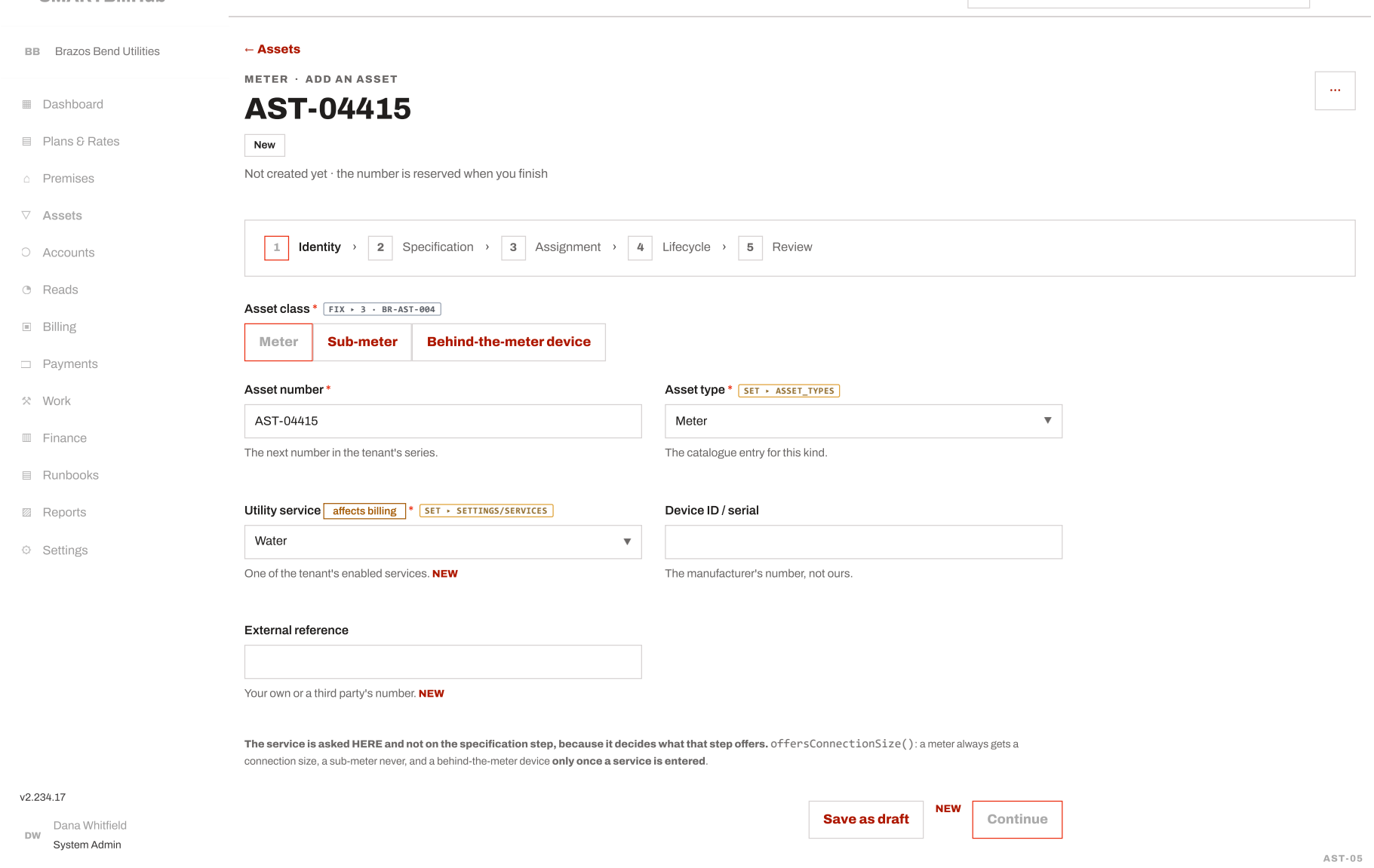
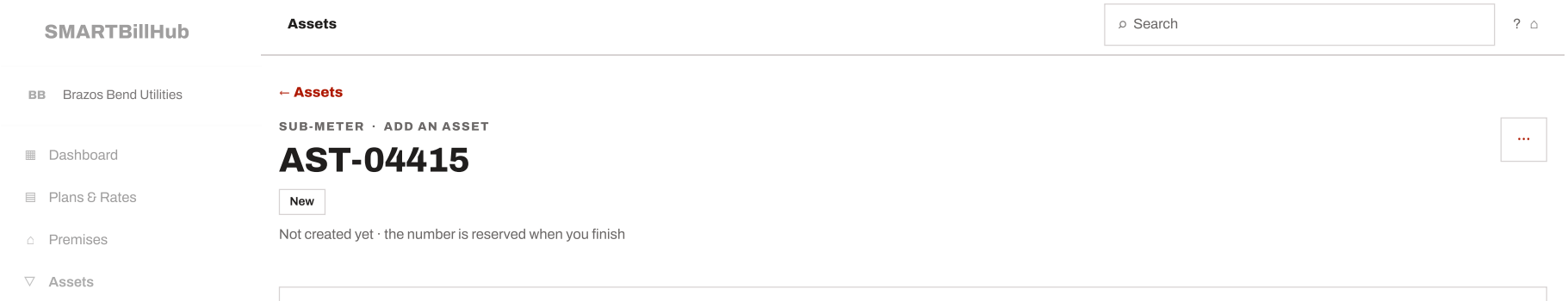
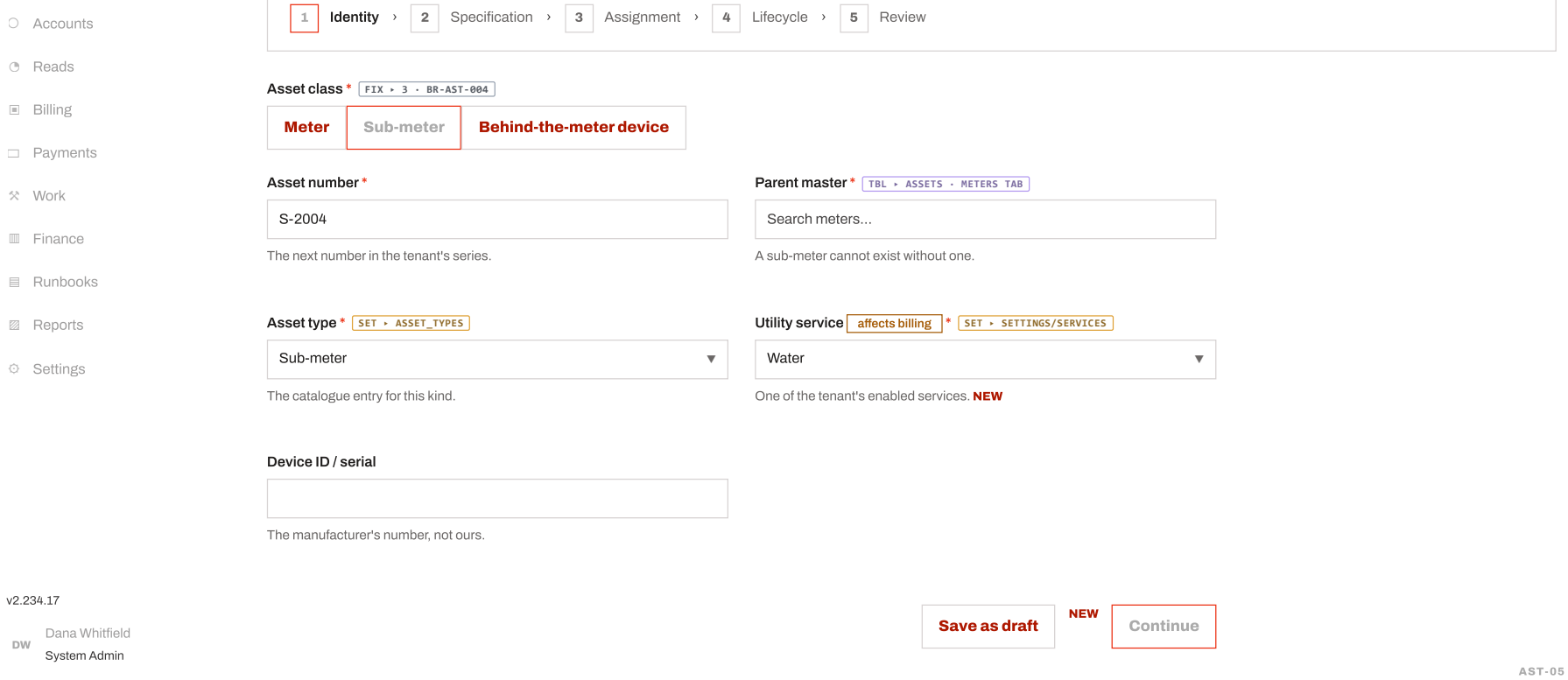


FIGURE 44 · 2 A SUB-METER: THE CLASS CHOOSER LANDS ON THE SECOND OPTION AND PARENT MASTER BECOMES REQUIRED — BR-AST-003 REFUSES A SUB-METER WITHOUT ONE:





The utility service is on this step already, and it is a free-text box

⚠ **This document's first drawing of step 1 left it out, which was an omission on my part** — the product asks for it here, in `IdentityStep`, and has all along. What it asks with is a **plain** `TextField`: a person types a service code and the server refuses an unenabled one after the fact. `AST-B-03`.

Why it belongs on step 1 rather than beside the other billing-affecting fields on step 2, and this is the part worth knowing: `offersConnectionSize(class, service)` decides whether the **Connection size** field appears at all —

```
a METER                → always offered
a SUB-METER            → never offered
a BEHIND-THE-METER     → offered ONLY once a service has been entered
```

So on a device the service is **what makes another field appear**. Asked on step 2 beside the field it gates, the form would reshape under the person as they typed. Asked on step 1, the next step is already the right shape when they reach it.

It is also one of the five billing-affecting fields — `multiplier · unit · net_metering · service · size_class` — so changing it later on a meter with reads applies forward only and restates nothing. Here, at creation, there is nothing to restate, which is the one moment it can be set freely.

The tenant has ELEVEN services to choose from — Electric, Water, Gas, Sewer, Trash, Storm water, Solar, Broadband, Thermal/Heat, Battery and Fire — and `GET /settings/services` serves the ones this tenant has ENABLED, which is the list that belongs in the control. **The field is a plain** `TextField`

anyway (`AssetAddWizard.tsx`, `IdentityStep`): the service code that decides how this asset is billed is **typed by hand**, and the server validates it after the fact. That is `AST-B-03`, and the remedy is one call the screen does not make.

> **⚠ A correction, and the third time this one file has misled this document**
>
> Until version 22 this paragraph said the read is **System Admin only**, so a Billing Clerk *"cannot be shown the list"*. **That is false.** `GET /settings/services` is `AnyTenantRole` — `apps/tenancy/api.py:419`, commented *"clerks read-only; writes below"* — so every staff role can read it today.
>
> **Where the wrong claim came from:** the `exceptions:` footer at the bottom of `AssetRegistryPage.tsx`, which states it in as many words. It was true when it was written on 2026-08-10 and the gate has since moved. **Appendix A already carries a row about that same footer** contradicting the code above it — this is the second claim it has handed this document, and hard-won law 1 is exactly this: **a comment claiming a guarantee is not the guarantee**, and a comment claiming a REFUSAL is not the refusal either.
>
> **The specification survives and gets cheaper.** The gap was never the permission — it is that nothing wires the picker. No endpoint, no column, no migration: the list is served, to every role that needs it, by a call this form does not yet make. Same for **Unit of measure** one field along, whose options ride on the same payload.

The screen

For	Whoever is booking in a new device.
How you get here	**+ Add asset** on any registry tab — each tab's primary carries its own class, so the wizard opens on the right one.
Primary action	**Continue**. Nothing is created until step 5.
Tabs	None — a five-step rail.
Table controls	None.
Write surfaces	**Add** — and only at the end. **Edit · Update · Delete** — **NOTHING**.
Walked	**Not walked as redrawn.** The step names, the three classes and every field are read from `AssetAddWizard.tsx`'s own values interface — **29 keys**, of which 26 are asset columns.
Spacing	Class on its own full-width row, then two to a row. R12 throughout.
Cognitive load	The class is asked first because it decides what steps 1 and 2 contain. Asking it later would mean re-drawing the form under the person.

States	**Meter · sub-meter · device**, drawn above and on the next screen.
Words	**Utility service**, never *commodity* or *product* — the platform has eleven of them and one word for them. **Asset number** is ours, **Device ID / serial** is the manufacturer's, **External reference** is somebody else's: three identifiers, three owners (`BR-AST-026`).
Components	`WizardFrame` · `TextField` · `SelectField`.
Rules	`BR-AST-003` · `BR-AST-026` · `AST-R-02` · R12.
Audience	Staff.
Health	SERVED for every field — and **two are NEW here**: **External reference**, which no surface in the product writes, and the **Utility service** as a **picker** rather than a text box. The service value itself has always been captured; what is missing is the list to choose from, and that is `AST-B-03`.

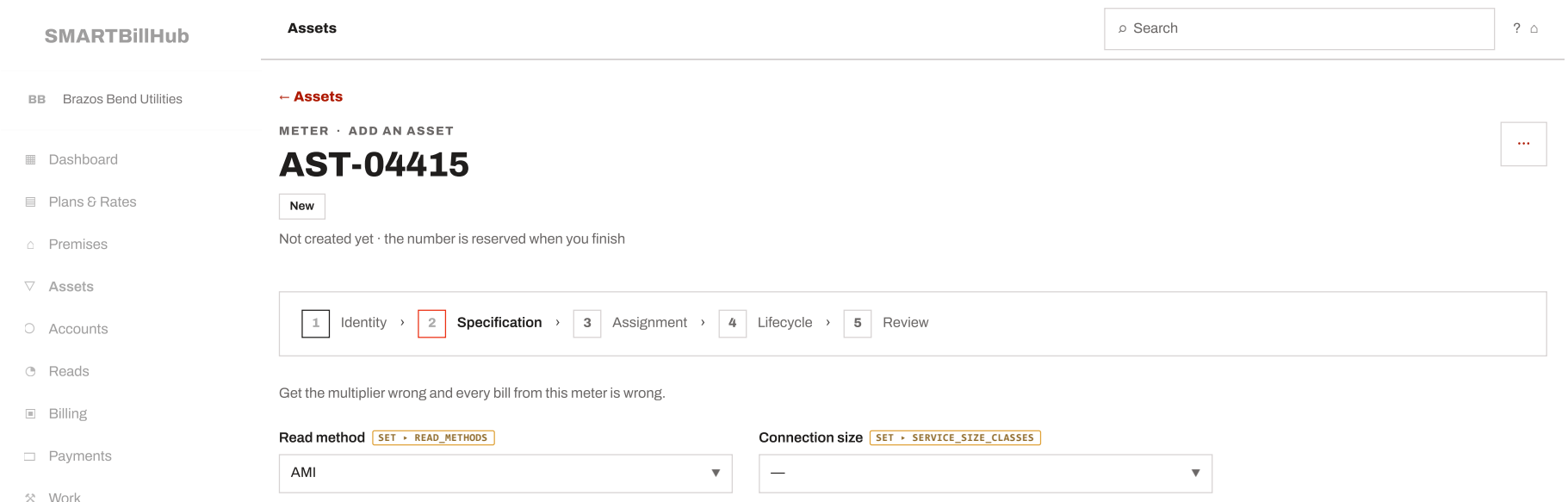
29 Add an asset · 2 Specification

EXISTS BEHIND · NEW BUILD · COMMITMENT 14 · COMMITMENT 26 · COMMITMENT 27 · COMMITMENT 28

`/:tenant/assets/new?class=meters`

How does it measure, and what will its readings mean?

FIGURE 45 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



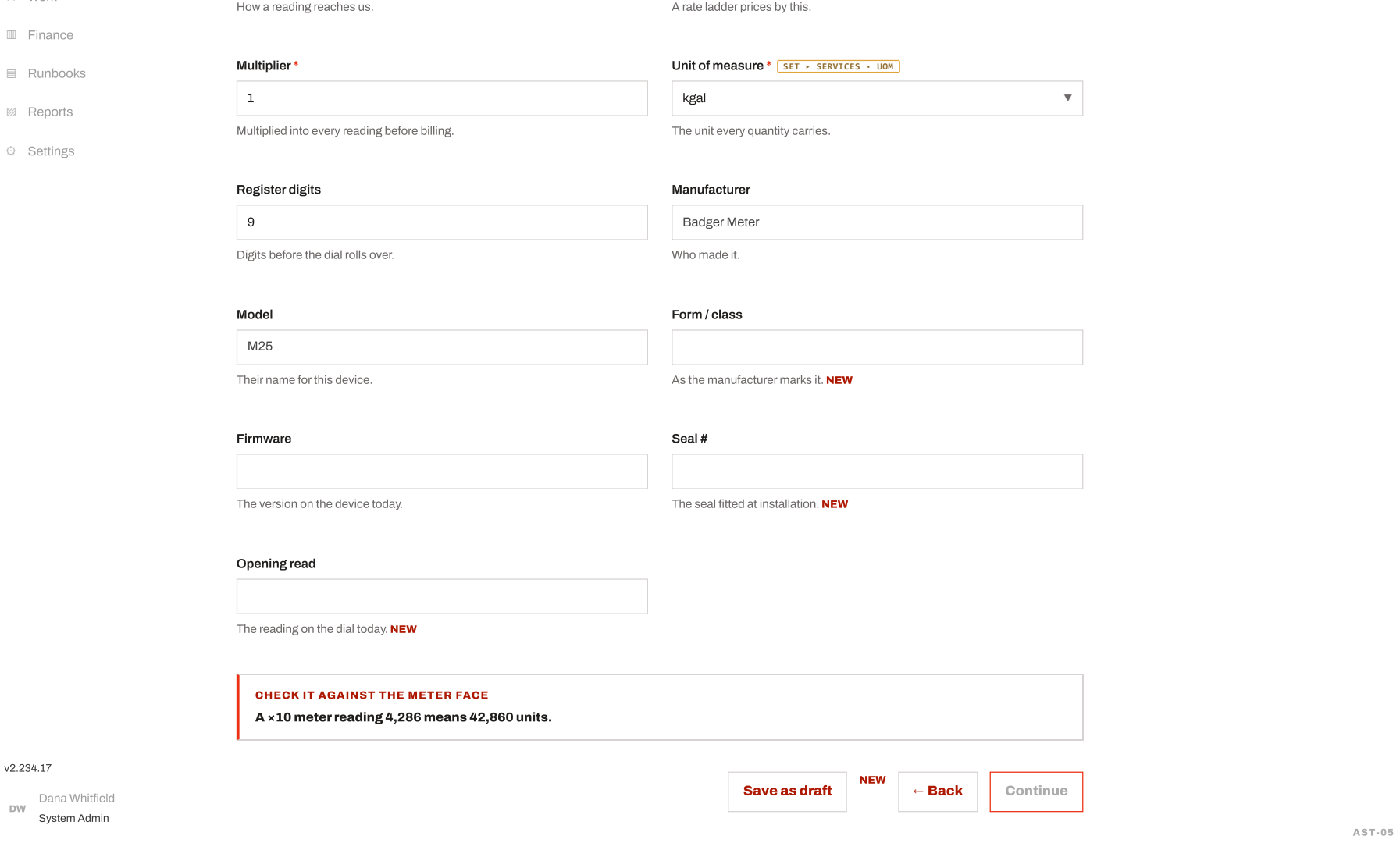
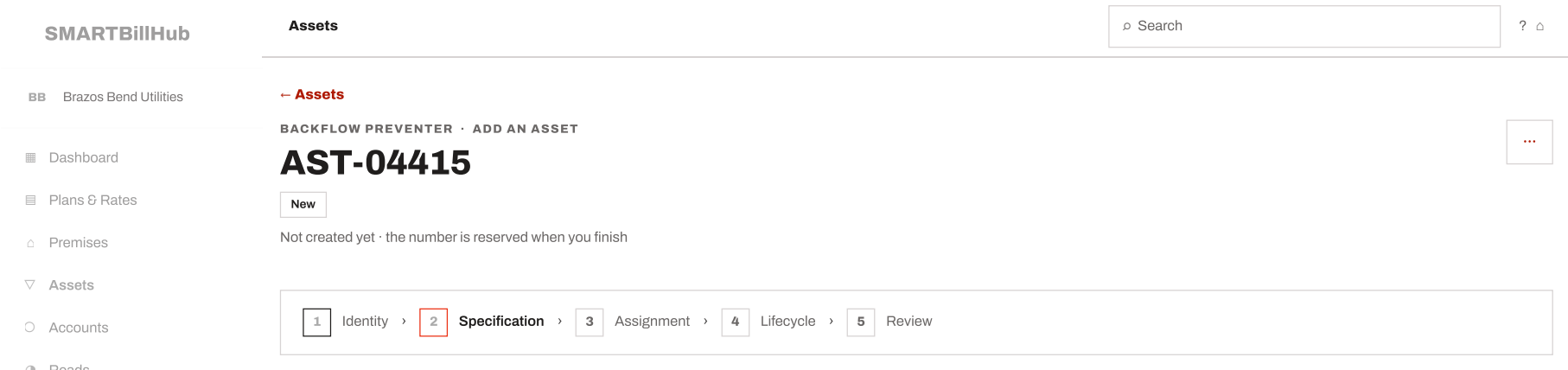
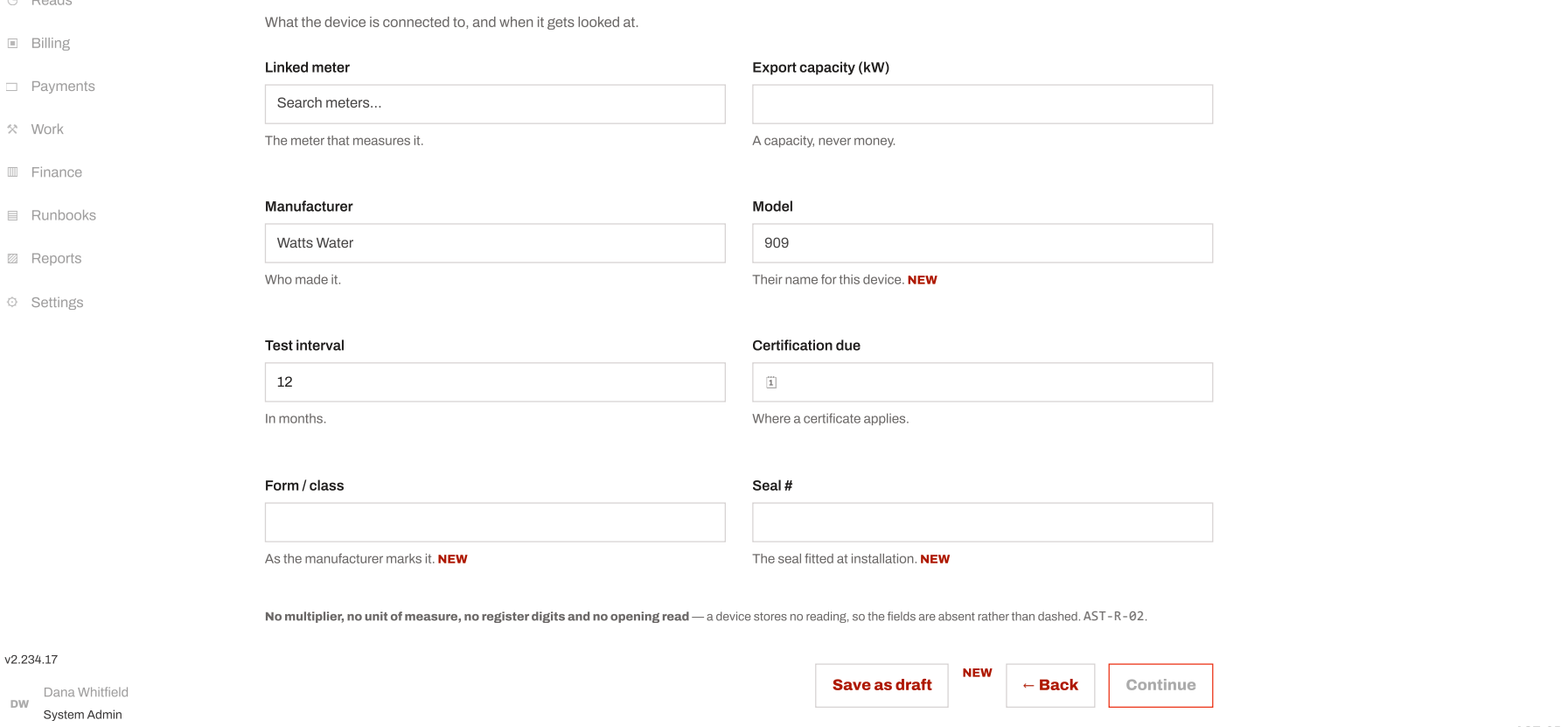


FIGURE 46 · 2 A BEHIND-THE-METER DEVICE: NO MULTIPLIER, NO UNIT OF MEASURE, NO REGISTER DIGITS AND NO OPENING READ — IT STORES NO READING. IT GAINS LINKED METER AND EXPORT CAPACITY:





Two columns, and the product captures them two different ways — one of which asks a person to type a delimiter

`manufacturer` and `model` are two columns on the asset. The product asks for them like this:

```
ADDING an asset   MeasuringSpecStep   Manufacturer   +   Model          two fields
                  DeviceSpecStep      Manufacturer   +   — nothing —    ONE field
EDITING an asset  the edit modal      "Manufacturer / model"           ONE field
                                      help: "One line; separate the two with a middle dot."
```

⚠ **The edit path asks a human to encode two columns into one string with a delimiter**, and then presumably splits it back on the way in. A model name containing a middle dot, or a person who reaches for a slash or a comma instead, lands in the wrong column — and nothing on the screen says which half is which. **Two columns are two fields**, which is the new rule below.

⚠ **And the device step captures no MODEL at all.** A backflow preventer is a *Watts Water 909*; the detail view has a **Manufacturer / model** row and will happily display it. So on a device the model is one more **column with a reader and no writer** — the same family as the seven in the section intro, narrowed to one kind. It is marked **NEW** on the device variant below.

Both drawings here ask for the two as **two fields**, on every class. That is a deliberate divergence from the as-built edit modal, and it is the one this

document recommends rather than the one it found.

The screen

For	The same person, one step on.
How you get here	Continue from step 1.
Primary action	**Continue**.
Tabs	None.
Table controls	None.
Write surfaces	**Add** at the end only.
Walked	**Not walked as redrawn.** The multiplier warning is the product's own sentence.
Spacing	Two to a row, R12.
Cognitive load	**This is the step that can make every future bill wrong**, and the product already says so: *"A ×10 meter reading 4,286 means 42,860 units. Confirm against the meter face before saving."* It is the best warning in the module and it is on the right screen.
States	**Measuring** and **device**, both drawn.
Words	**Opening read**, not *initial reading* — it is the reading on the dial the day it goes in.
Components	`WizardFrame` · `TextField` · `SelectField`.
Rules	`BR-AST-012` (billing-affecting) · `BR-AST-015` · `AST-R-02` · `AST-B-14`.
Audience	Staff.
Health	SERVED, with three exceptions marked NEW: form/class, seal # and the opening read.

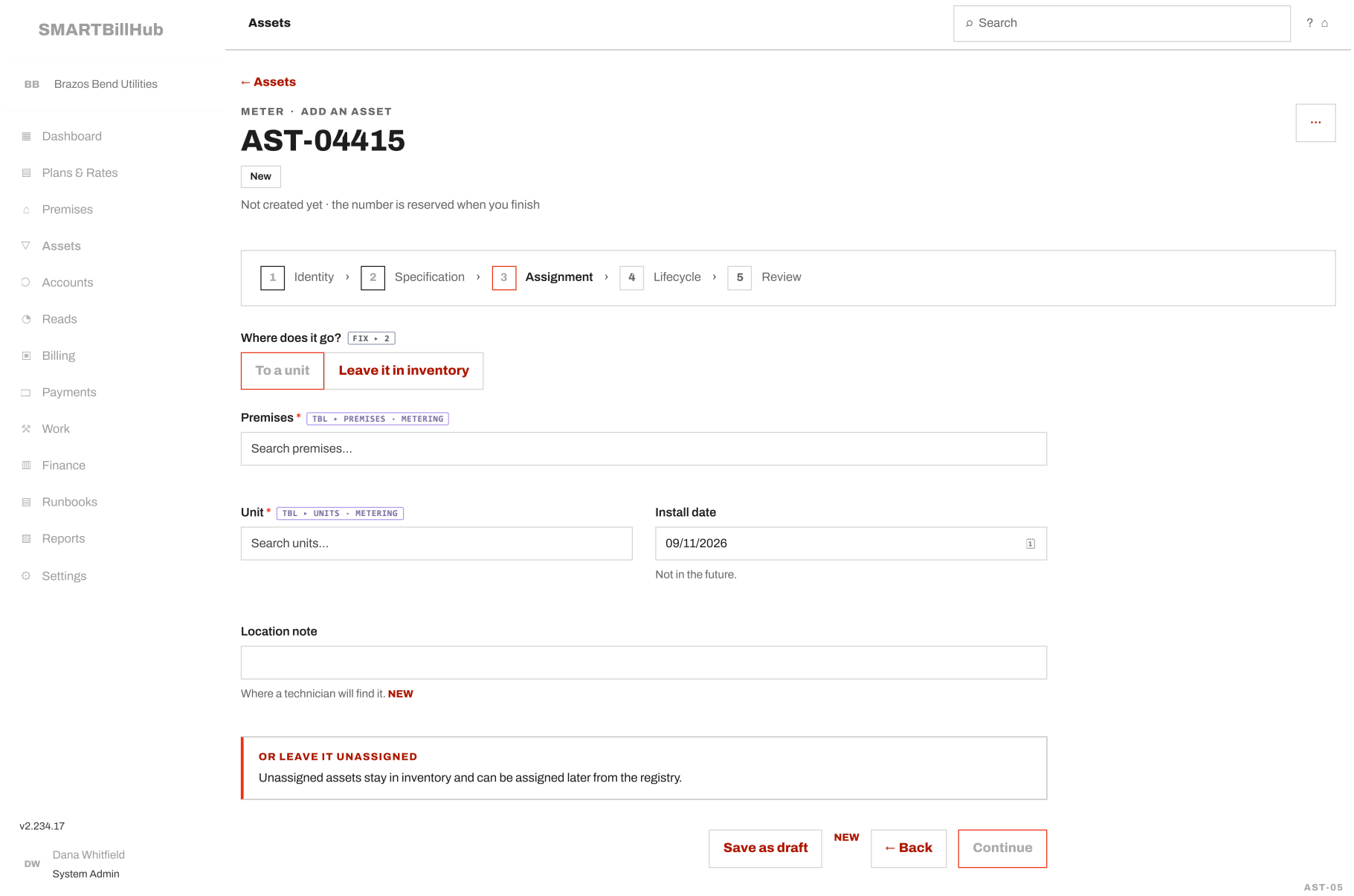
30 Add an asset · 3 Assignment

EXISTS BEHIND · NEW BUILD · COMMITMENT 26 · COMMITMENT 27 · COMMITMENT 28

`/:tenant/assets/new?class=meters`

Where is it going, and is it going anywhere yet?

FIGURE 47 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED



The screen

For	The same person.
How you get here	Continue from step 2.

Primary action	**Continue**.
Tabs	None.
Table controls	None.
Write surfaces	**Add** at the end only.
Walked	**Not walked as redrawn.**
Spacing	R12, two to a row; the location note runs full width because it is a sentence.
Cognitive load	**Leaving it unassigned is a first-class answer**, not a skipped field, and the screen says so: *"Unassigned assets stay in inventory and can be assigned later from the registry."*
States	**To a unit** and **leave it in inventory** — the second hides the premises and unit pickers rather than dashing them.
Words	**Location note**, not *address* — the address lives on the premises and is walked to (hard rule 8).
Components	`WizardFrame` · `PremisesPicker` · `DatePickerField`.
Rules	Hard rule 8 · `BR-AST-017` · R12.
Audience	Staff.
Health	SERVED — **Location note is NEW** and has no capture site anywhere today.

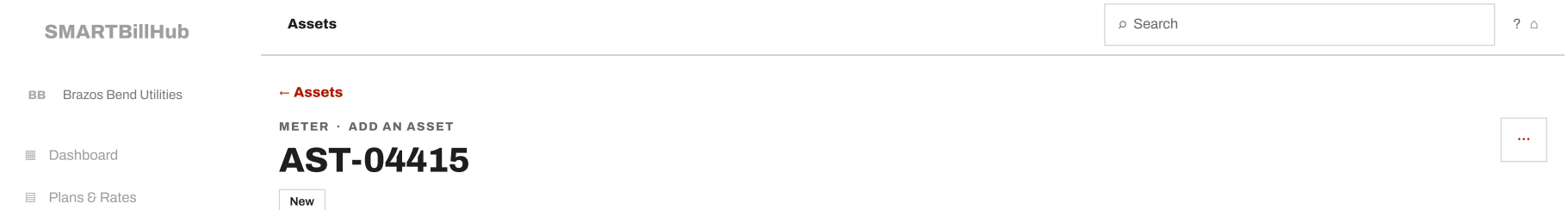
31 Add an asset · 4 Lifecycle

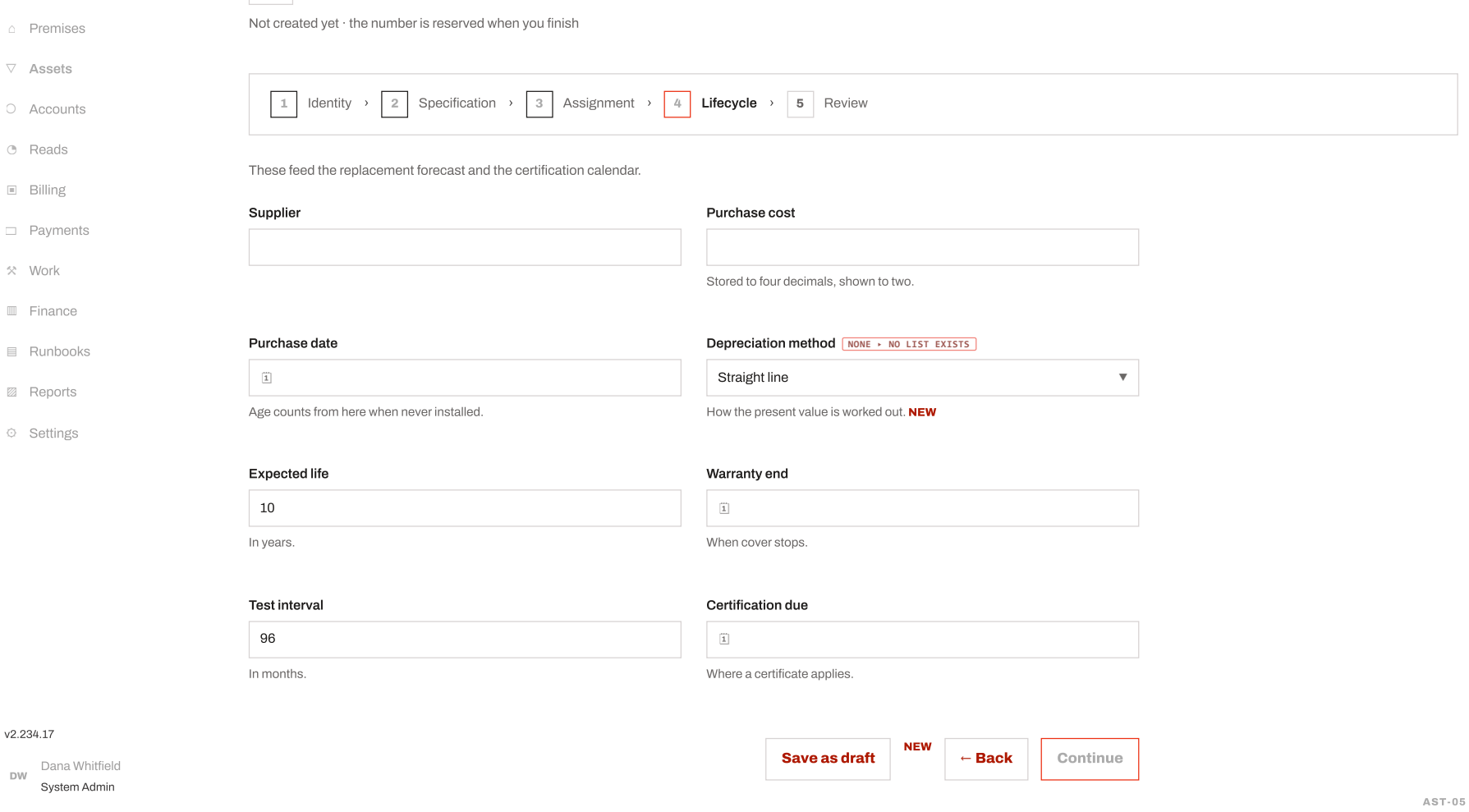
EXISTS BEHIND · NEW BUILD · COMMITMENT 26 · COMMITMENT 27 · COMMITMENT 28

`/:tenant/assets/new?class=meters`

What did it cost, how long should it last, and when is it looked at?

FIGURE 48 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	The same person.
How you get here	Continue from step 3.
Primary action	**Continue** — to the review, which is new.
Tabs	None.
Table controls	None.
Write surfaces	**Add** at the end only.
Walked	**Not walked as redrawn.**

Spacing	Two to a row, R12.
Cognitive load	Everything here is optional and everything here feeds something later — the replacement forecast, the certification calendar, the present value. The lede says which.
States	**No purchase cost** — then there is no book value to compute at retirement, and the retire wizard already says so.
Words	**Expected life** in years, **test interval** in months. Both stated in the helper rather than guessed from the field.
Components	`WizardFrame` · `MoneyField` · `DatePickerField` · `SelectField`.
Rules	Hard rule 2 · `BR-AST-047` · `BR-AST-007` · R12.
Audience	Staff.
Health	SERVED — **Depreciation method is NEW**: the present value is computed from it and nobody can set it.

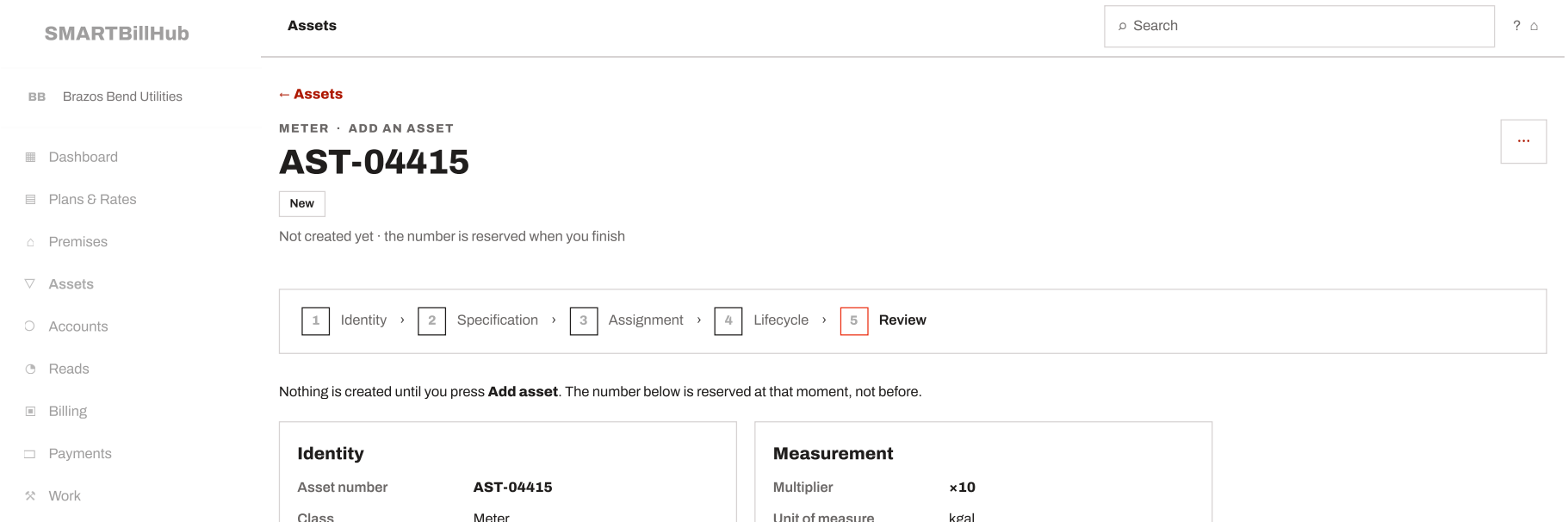
32 Add an asset · 5 Review & add

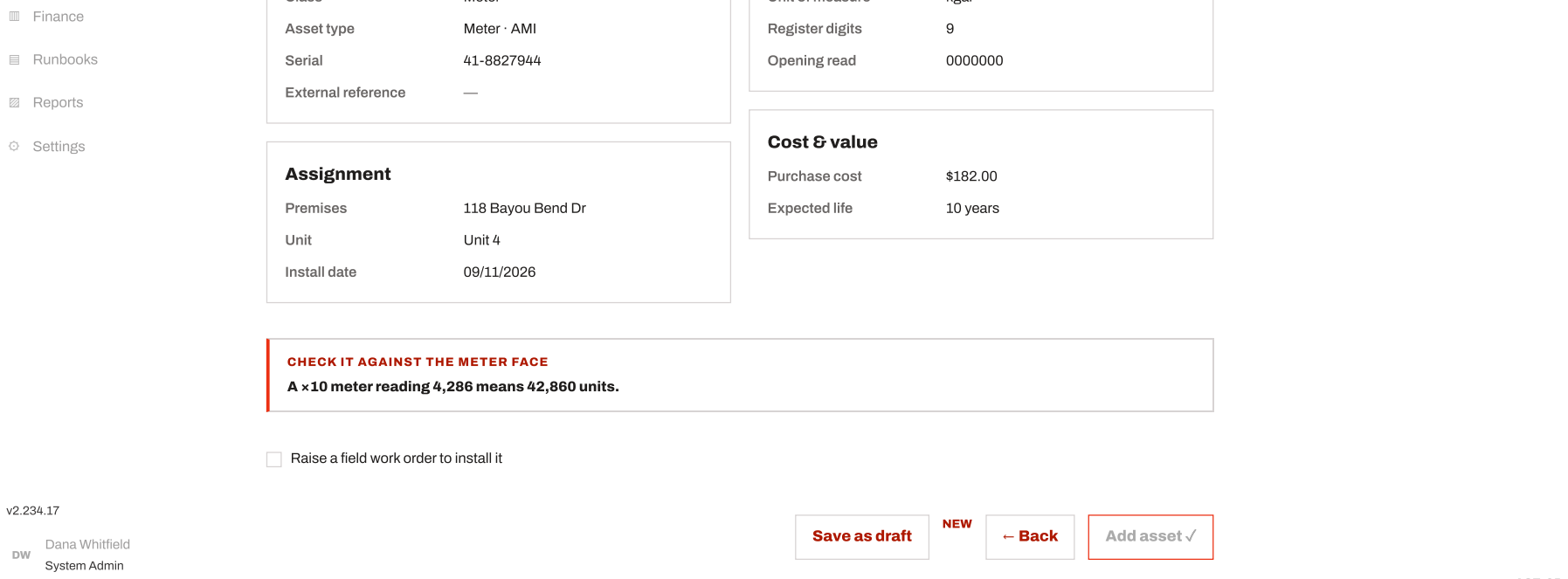
NEW BUILD · COMMITMENT 26 · COMMITMENT 28

`/:tenant/assets/new?class=meters · POST /assets`

Before I create it — is this right?

FIGURE 49 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The screen

For	The same person, about to commit.
How you get here	Continue from step 4.
Primary action	**Add asset ✓** — the only control in the whole flow that writes anything.
Tabs	None.
Table controls	None.
Write surfaces	**Add** — `POST /assets`, once, here.
Walked	**Not walked — this screen is not drawn anywhere in this document today**, and that is the finding: `WizardFrame` already provides a **final ✓ commit** and every other wizard in the product gets one. The four steps were drawn and the commit was not.
Spacing	Two columns of panels at 16, R11 throughout.
Cognitive load	It shows what is about to be created grouped the way the **Overview** groups it — Identity · Assignment · Measurement · Cost & value — so a person reviews in the shape they will later read. And it repeats the multiplier warning, because this is the last moment it can matter.
States	**Failed** — *"The asset was not created."* and every value still typed.

Words	**Add asset** ✓, the product's own commit label.
Components	`WizardFrame`'s commit step · the fact list · `ConsequenceBox`.
Rules	`BR-AST-032` (who may create) · `BR-AST-012` · B14 rule 31.
Audience	Staff.
Health	UNVERIFIED — nothing here is drawn from a running screen, because this step has never been drawn.

AST-B-27

Everything an asset record can hold can be entered by a person

STATUS Unverified

WAITS ON Seven fields on the add wizard and the edit dialogs. **NO SCHEMA CHANGE AND NO ENDPOINT CHANGE** — every column exists and `POST /assets` and `PATCH /assets/{id}` both take them.

SCREENS 28, 29, 30, 31, and the edit dialogs

WHAT DONE LOOKS LIKE ON SCREEN Adding a sub-meter asks for its **ALLOCATION SHARE**, so the sub-meter that comes out of the wizard can actually be allocated. Adding any asset asks for its **SEAL #** at the moment the seal is fitted, its **LOCATION NOTE** while the installer still knows where it is, and its **EXTERNAL REFERENCE** while the paperwork is in hand.

WHY IT IS WORTH ITS OWN NUMBER These are not missing features. They are **COLUMNS WITH A READER AND NO WRITER** — the model holds them, the API serves them, screens display them, and nothing in the product can put a value in one. That is hard-won law 2 in a third form, and no gate looks for it.

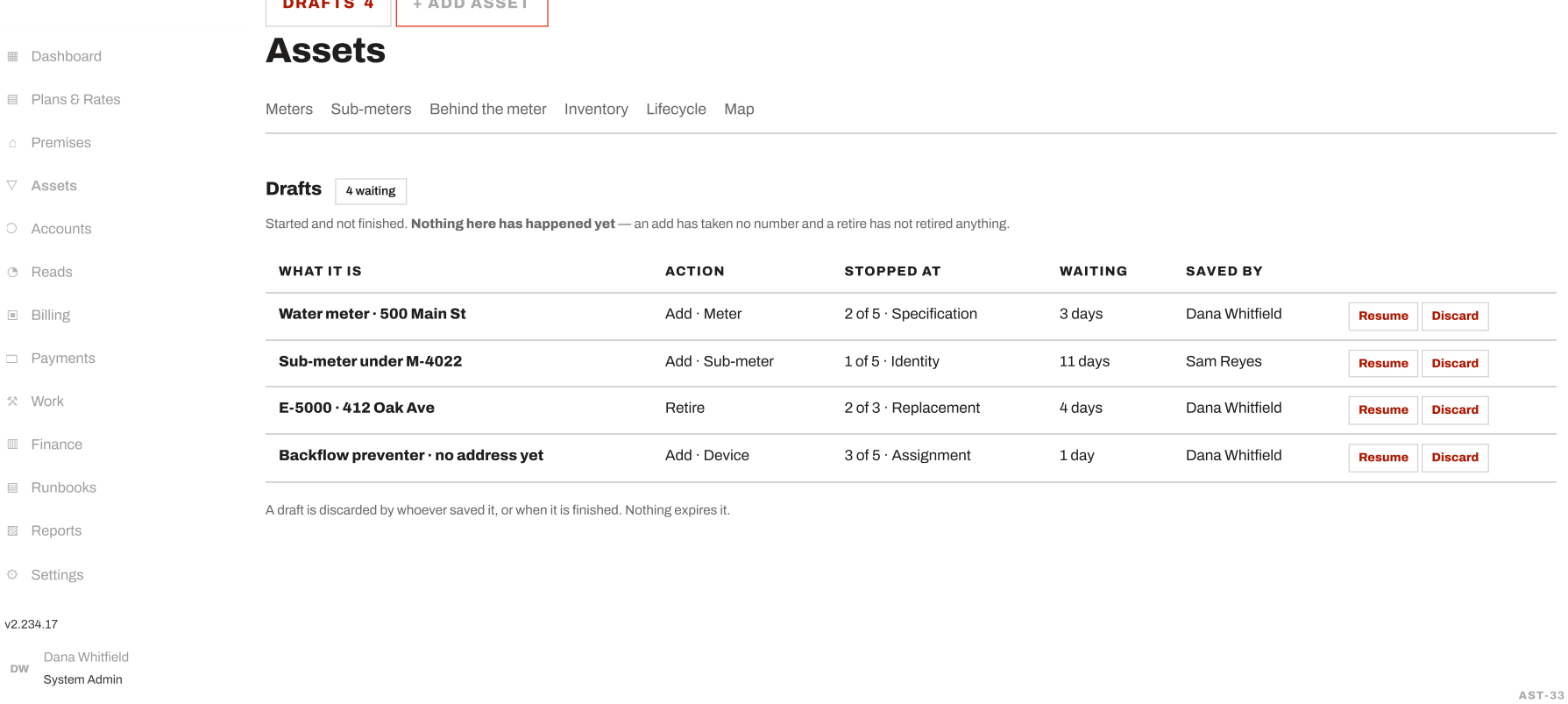
33 Drafts NEW BUILD COMMITMENT 28

`/:tenant/assets?drafts=1 — proposed · GET /wizard-parks?flow=add_asset,retire_asset`

What did I start and not finish?

FIGURE 50 · DESKTOP · 1440, SCALED TO FIT · NEVER REFLOWED





The button had to save something, and the product already knows where to put it

Both wizards now carry one exit and it is `Save as draft` — the five add steps and the three retire steps. The two it replaces were wrong on both: **Save & exit** wrote nothing at all — all six of its consumers are a bare `navigate(…)` — and **Park** asks *what are you waiting on* from nine reasons that belong to the customers' world, none of which is why a meter install or a removal stalls.

Where the draft goes is settled law, not a new decision. `BR-SHL-048`, ruled by the PO on 2026-08-26, in their own words: *"Option A - Unitl move in executed, no records updates."* A half-finished wizard is a saved draft of a FORM, held in `wizard_parks`, and **nothing it describes is touched until it commits**. Five wizards run on this today; these are the sixth and seventh.

So the draft is a row with the status — a `wizard_parks` row IS a draft; existence is the status — **and nothing else moves.** On an add: the asset number is still only taken at the end, so an abandoned draft leaves no gap in the tenant's series, and no picker, KPI, rollup or parent-master list can show a half-made asset. On a retire: **the asset goes on working** — it stays in service, keeps collecting reads and keeps billing until the wizard is actually finished.

Retire differs from Add in exactly one place, and it is the resume

An add draft describes something that does not exist. **A retire draft describes something that does** — and `BR-SHL-048 (a)` stores the subject as

copied values, never a foreign key, deliberately, so that a forgotten draft can never block a delete or hold a row still.

The consequence is real and is the whole of the difference: between saving and resuming, that asset can be reassigned, or retired by somebody else through the row action, or deleted. So **resume re-reads the asset and checks it before drawing step 1**, and a draft whose asset is already terminal says so and offers only **Discard** — `BR-AST-030`: retired is the end state, and there is no un-retire. **The same re-check the wizard already does on open**; it is being asked one more time, on a form that may be a week old.

Everything else is identical, which is the point of giving them the same treatment: one list, one resume, one discard, one constant in `waiting_on`.

What it takes to build

Back end	**One tuple grows by two.** `WIZARD_PARK_FLOWS` is `("move_in","move_out","transfer","disconnect","new_connect")` — add `"add_asset"` and `"retire_asset"`. No new table, no migration to `assets`, no new endpoint: park, read and discard already exist and are flow-agnostic.
The status word	`waiting_on` is **free text with a not-empty check**, so a plain draft writes the constant `draft` and asks the person nothing. That is the whole of "status draft" — no column added.
Front end	Both asset wizards pass `WizardFrame`'s `exit` prop, which **neither passes today**. Only `onSaveAndExit` is wired, to the park call rather than to a bare navigate; `parkBlockedReason` holds Park off.
The home	A **Drafts** control in the registry header with its count, opening this list — **one list for both flows**, which is why its second column is the ACTION and not the asset kind. **This is the half that must not be skipped**: the five customer flows each name a home, and a draft with none is a row nothing can reach.
Resume	Reopens at the furthest incomplete step, and saving again **PATCHes the same row** — a person who resumes on Tuesday and saves again on Thursday has **one** draft, not two. The port already does this. **A retire draft additionally re-reads its asset** and refuses a terminal one.
Not built	**No expiry and no limit** — the same half `BR-SHL-047` left open. Here it costs little: neither draft holds a unit or blocks anyone. **The one thing it does cost is on retire**, where a forgotten draft means a decommissioned meter quietly goes on billing, so the age column earns its place.

`AST-B-28`

WHAT SAVE AS DRAFT on all five add steps **AND ALL THREE RETIRE STEPS**, writing a `wizard_parks` row with `flow = add_asset` or `retire_asset` and `waiting_on = draft`, plus a **DRAFTS** home on the registry that lists, resumes and discards both.

WHY PO 2026-09-11: *"Keep save as draft button and make entry with status draft"*, then *"for now give same*

treatment" for retire. The button was drawn and saved nothing, which is worse than no button: it is a promise the product does not keep.

UNDER WHICH RULE `BR-SHL-048` — a parked wizard changes nothing until it is executed, and the draft carries **COPIED VALUES, NEVER A FOREIGN KEY**. `BR-AST-030` governs the retire resume: terminal is terminal.

SCREENS 24, 25, 26, 28–32, 33

WHAT DONE LOOKS LIKE ON SCREEN A clerk called away mid-install presses **SAVE AS DRAFT**, sees it on **DRAFTS · 1**, comes back on Thursday, presses **RESUME**, and lands on the step they stopped at with every field as they left it. **THE REGISTRY IS UNCHANGED THROUGHOUT** — no half-made meter appears anywhere, and no meter is half-retired.

WHAT WOULD NEED A NEW RULING Putting the draft in `assets` itself, as a row with a `draft` status. That **REVERSES** `BR-SHL-048` and is not ours to assume. Its price, counted: **128 PLACES** key off `behavior_class` and every one would meet a fifth value; `_validate_asset` would need a second, relaxed mode so an incomplete row can be written; the asset number is taken the moment the draft is saved, so abandoned drafts leave gaps in the series; and half-made assets become selectable as a parent master, countable in the value rollup, and visible in every picker unless each query remembers to exclude them.

`AST-B-03`

The utility service is picked from the tenant's enabled list, not typed

WAITS ON Nothing. `GET /settings/services` is `AnyTenantRole` and already serves the enabled list; the wizard simply never calls it.

SCREENS 28

WHAT DONE LOOKS LIKE ON SCREEN Step 1 of **ADD AN ASSET** shows **UTILITY SERVICE** as a dropdown reading **WATER**, offering only the services this tenant has enabled — today it is a plain text box and a mistyped code is refused by the server after the person has moved on.

STATUS UNVERIFIED — engineering fills this in from the code, one row at a time.

07

What happened when the product was actually driven

The only section whose findings could not have been produced by reading the source,

and the only one where a fabricated row would be undetectable in the finished file. So it says, for every row, what was done and what happened.

> **The stack was built and driven, twice — and measured a third time**
>
> PostgreSQL 16 provisioned with the repository's own `10-scripts/13-provision-database.sh`; 278 migrations applied; `seed_demo` provisioned **brazosbend** — **126 assets**, 73 premises, 121 units, 72 accounts, 13 asset types, 64 connection sizes. The SPA built with `vite build` and served by the product's own `server.mjs` on the deployed same-origin join, Django behind it. Every screen below marked *walked* was opened at **1440×1000** in Chromium as the seeded owner.
>
> **Six journeys defined, six walked, none skipped.** Twelve of the surfaces drawn here were opened directly; the rest are reachable only by completing a form, and this pass drove by address. Every screen not opened is marked `UNVERIFIED` in its own field table rather than graded on a read.
>
> **And a third pass measured rather than clicked** (2026-09-11): `getComputedStyle` over the R11 spacing anchors on the registry, and the DOM order and pixel gaps of **eighteen form fields** across four screens for R12. Those two runs produced the last three rows below and the two new rows in section 09.
>
> **Two things the seed could not give us.** `seed_demo` reported *"sample-data block failed for payments"* and a `VAL-002` error on `rating.charges.amount_and_size_table`. Neither touches assets, and the asset rows seeded cleanly. Said out loud because a partial seed is exactly what makes a later walk quietly wrong.

The journeys

#	Journey	Steps	Result
J1	Sign in → the rail → the Assets registry	3	Completed. One inbound rail link found in the rendered page.
J2	Every registry tab, in order	6	Completed. Every tab rendered; zero failed requests.
J3	A row → the asset → all five detail tabs	6	Completed. Two findings.
J4	The asset → reassign, log repair, retire, raise work order	4	Completed. All four opened; three findings.
J5	The registry → the add wizard	2	Completed. Two findings.
J6	Search, sort, page and filter the registry	4	Completed. One finding; two earlier readings corrected.

Every issue seen

Journey	Step · screen	What was done	What was expected	What happened	Severity
J3	6 of 6 · screen 08	Opened an asset and searched the rendered page for a way to raise a work order	A control, per `BR-WRK-056`'s *"action bar … Maintenance history panel"*	**0** occurrences of the text *Raise work order* and **0** links to `work-orders/new`. Header actions were **Assign** and **Delete**. The Maintenance tab drew a **Work order** column and no control.	Blocks the journey
J2	2 of 6 · screen 01	Opened `?tab=meters` on a tenant with 126 assets	Four KPI figures over the meter population	**All four cards rendered –.** 50 rows on the page, `count` 126, so `kpiRowsComplete` returned false — under labels and detail lines that promise a number.	The journey completes, wrongly
J5	2 of 2 · screen 28	Read the control under the **Service** label	A picker over the tenant's enabled services	**`<input type="text">`, no placeholder, no options.** The help beside it says the value *"must be one your utility has enabled"* and offers no way to see which those are.	The journey completes, wrongly
—	measured · screens 28–18	Read the DOM order and pixel gaps of every field on the add wizard, reassign, log repair and retire	label → input → helper, at 5 / 6 / 20 (R12)	**Ten of eighteen fields render `label → helper → input`.** On those, label-to-input measures **29px** or **47px** instead of 5. The eight with no helper are exactly 5px and correct. So the rhythm holds only where there is nothing to get wrong.	The journey completes, wrongly
J2	5 of 6 · screen 05	Opened the Lifecycle tab and looked for a toolbar over any of its three tables	Search, filter chips, column chooser, sortable headers, a pager (B14 12/13/14/16)	**None of the four, on any of the three tables.** Nine warranty rows and ten forecast rows, unsearchable and unsortable, on a tenant with 126 assets.	The journey completes, wrongly
J4	1–3 · screens 22, 22, 21	Inspected every `input[type=date]`	Bounded dates with a Today shortcut (B14 rule 7)	**Five date inputs across three screens. No `min`, no `max`, zero Today shortcuts.** The retire step's own help says the date *"cannot be before the install date"* — a constraint the control does not enforce and the server does.	Cosmetic
J3	2 of 6 · screen	Counted the controls in the last cell of the first	At most two, then an overflow menu (B14	**5 controls on every installed row**, 3 on every inventory row, and **0** `[data-row-more]` elements on the page.	Cosmetic

	01	twelve rows	rule 15)		
J6	2 of 4 · screen 01	Clicked the **Meter #** header three times	asc → desc → clear (B14 rule 13)	`dir=desc` → `dir=asc` → `dir=desc`. **It toggles forever; there is no third state.**	Cosmetic
J6	4 of 4 · screen 01	Scrolled to the table footer	`⇤ ← Page x of y → ⇥` (B14 rule 16)	Range, rows-per-page, previous and next render. **No first-page or last-page control.** At 126 assets that is 3 pages; at 40,000 accounts it is 800, with no way to reach the end.	Cosmetic
—	measured · seven screens	Read `h1` font-size on every asset screen	One title size for one class of screen	**Two sizes, and not split by kind.** Registry, detail and *both full-page wizards* render **32px**; **Raise work order renders 40px**. The token sheet calls 40 *"full-page wizard title"* and 32 *"the app default"* — so the two screens that are wizards use the default and one that is not uses the wizard size.	Cosmetic
J4	4 of 4 · screen 27	Opened `/assets/{id}/work-orders/new` directly	The raise form	Rendered correctly, badge **WRK-24**. Reached by URL only — see the J3 row for why that matters.	—
J5	2 of 2 · screen 28	Pressed **Continue** with nothing filled in	Either a disabled button or a stated reason	No `disabled` and no `aria-disabled`; pressing it rendered *"This step isn't finished. Fill in what's marked above, then …"* in a `role="alert"` and stayed on step 1. **Correct behaviour** — recorded because this document's earlier reading of it was wrong.	—

What did *not* go wrong, which is worth as much

Zero failed network requests across every asset screen, in both walks. Not one 4xx or 5xx on any asset endpoint. No control threw on click. No panel came back blank against real data. Every door audited in the source landed on the screen it claimed.

And on the R11 side, most of what was measured is **right**: screen body **24/16/32**, rail **240**, topbar **56**, tabs margin 16 gap 16, KPI cell 16×20 gap 6, `th/td` **8×12**, `.input` 6×10 min-height 36, step rail 16×20 gap 12 with 24 below. Eight of ten anchors hit the scale exactly. The two that miss are the rail item's **9px 12px** and the status pill's **2px 9px**, and both are the product's own long-standing literals rather than drift.

Commitments raised in this section: none of their own. Every finding here is carried by a card in sections 02–05; section 12 is the index.

08

Everything the product says it cannot do, and one ruling each

Everything the product says it cannot do, and one ruling each

Built from the product's own user-facing copy — the sentences a person reads when something cannot be done — not from an auditor's impression of what is missing. Every string exists in `07-frontend/src/i18n/locales/en.json` and was traced to the file and line that renders it.

> **The judgement rule, stated once and applied without exception**
>
> A **recorded ruling** that a thing waits makes it **out of scope**, with the ruling quoted. **No ruling** makes it a **specification** naming what it waits on. Nothing is left unresolved.
>
> **Where it was close.** The *Opening read* field on screen 29 is drawn, disabled and explained — that looks like an oversight and is not: the lane instruction is on the record (*"no column → render nothing and report"*), and no seam lets assets write a `meter_reads` row at create time. Out of scope. The **Service** field on screen 28 reads almost identically — a field that cannot do its job, with a sentence beside it — and **nobody has ruled it**, so it is a specification. The difference is not how the screen looks; it is whether a decision exists.
>
> **The contrast worth stating:** the wireframes this module was built from state **zero** absences. The product's own copy states **42**, four of them holding a control a user can see and cannot use.

A · Things a person is told cannot be done (17)

What is absent	Screens	Decision	Reason, and who ruled it
Raising a work order from the asset itself. No copy — the controls simply are not there	10, 14	**AST-B-02**	No ruling. `BR-WRK-056` is CONFIRMED and names both controls; two of its four ways in were never built.
Searching, sorting, filtering or paging anything on the Lifecycle tab. No copy — no toolbar exists	05	**AST-B-11**	No ruling. The three tables are the panel table rather than the shared `DataTable`, and B14 rules 12, 13, 14 and 16 all apply to a list.
"Your role can't list users, so enter the technician's	21	**AST**	No ruling. `GET /users` is admin-only and `BR-AST-038` lets a clerk log a repair; the two were never reconciled.

so enter the technician's ID."		**-B-04**	
The **Service** field has no option list	16	**AST-B-03**	No ruling — and now an instruction, 2026-09-11: the options come from **Settings › Services & UoM**. The list is already served and already used on screen 01's own filter.
No KPI figure on the three list tabs	01, 03	**AST-B-01**	No ruling. *"No aggregate endpoint is served"* is stated in the file, not decided by anyone.
"No template applies to this kind of asset." — the submit stays disabled	25	**AST-B-10**	No ruling. A dead end with a stated reason and no route forward from the screen.
~~*"An opening read is recorded through Meter Reads once the meter is installed — it cannot be captured here."*~~	17	**AST-B-14**	**WAS: OUT OF SCOPE** — lane instruction, recorded: no published seam lets assets write a read at create time, so the field renders disabled *with its reason*. **NOW: a specification.** Instruction, 2026-09-11: *"make opening read enable user can add reading".* A newer decision beats an older one. The superseded reasoning is kept rather than deleted, because the seam argument was real and was overruled — and the seam turns out to be a precedent rather than a blocker (see R5).
Yard / On-vans split · *Reorder at* column · **Raise reorder** · stock-count import	04	OUT OF SCOPE	PO, 2026-08-22 (§10-D5): *"Launch with counts & costs only."* Tests assert each absence by name.
The *Cost to Finance* panel, the depreciation figure and *"Posts to 5300 O&M and 5400 Depreciation"*	05, 05c, 24	OUT OF SCOPE	PO, 2026-08-22 (§10-D3): no depreciation engine exists and only the ledger posting service writes journal lines (hard rule 5). Replaced by the **Asset value** panel.
"The amount and your choice are recorded on the asset. Nothing is posted to the ledger."	24	OUT OF SCOPE	Same ruling. The screen states the limit rather than implying a posting.
Tariffs and credited dollars anywhere on the behind-the-meter tab	03	OUT OF SCOPE	`BR-AST-025` and the D2 ruling of 2026-08-22: this module never displays credit money.

r10's `WO-####` lines in the map popup	06	OUT OF SCOPE	§10-D7: omitted rather than faked, because the fact did not exist when the screen shipped.
Demand flag · read route · read sequence · warranty *term* select	17	OUT OF SCOPE	Lane instruction: no column exists, so nothing renders, and each was reported.
“Bulk assignment moves inventory to units only. A unit-to-unit move goes one at a time, or by file import.”	01, 09	OUT OF SCOPE	`BR-AST-036/038`. The limit is stated on the control rather than discovered.
“Maintenance entries are never edited or deleted — a correction is a new entry.”	14, 21	OUT OF SCOPE	`BR-AST-018`, append-only by rule.
“This cannot be undone. Retired is the end state — there is no un-retire.”	22	OUT OF SCOPE	`BR-AST-030`, terminal by rule.
“Background map is switched off.” · *“Assets are open to System Admins and Billing Clerks.”*	06, all	OUT OF SCOPE	A tenant configuration and a role rule, both server-enforced. Refusals working correctly.

B · Server refusals a person can meet on these screens (13)

Every one is a `message_key` raised by `apps/assets` and rendered through the shared refusal path. All fourteen are **OUT OF SCOPE** — each is a rule doing its job, and each names either the remedy or the screen that holds it.

What a person reads	What it protects

“This asset has history against it, so it can't be deleted. Retire it instead.”	`BR-AST-013` — and it names the alternative
“Only an asset in inventory can be deleted. Retire this one instead.”	the same rule, the other branch
“{{asset_no}} is retired, and retired is the end state. Create the asset anew if you need it back.”	`BR-AST-030`
“An asset can't go from {{from}} straight to {{to}}.”	`BR-AST-020`, legal transitions from the behaviour class
“An asset in {{status}} can't be assigned to a unit. Unassign it or finish the repair first.”	the same, on assignment
“The final read for {{asset_no}} can't be taken yet. Nothing was moved.”	`BR-AST-022` — the read and the move are one transaction
“A final read needs a current assignment — this asset is not on a unit.”	the same rule's precondition
“{{service}} isn't one of the services this utility bills. Enable it in Settings first.”	`BR-MET-003` — and it names **Settings**
“That asset type isn't in your Reference data.” · *meter type* · *read method*	`BR-AST-043/044` — three refusals naming **Reference data**
“The {{stage}} lifecycle stage is missing from your Reference data.”	`BR-AST-031`
“There is no book value to treat — the asset has no purchase cost or expected life on record.”	`BR-AST-047`
“The replacement is itself retired. Pick a living asset.” · *“That replacement asset wasn't found.”*	`BR-AST-021`
“There is no asset numbering series yet, so enter the asset number yourself.”	`BR-AST-049`, degrading rather than blocking

C · Honest empties (11)

Eleven *“there is none yet”* states, one per list, each naming what would fill it. **All OUT OF SCOPE** — an empty state that says what creates content is B14 rule 17 working, and there is nothing to specify. ⚠ **They were a paragraph until this pass**, which is why they were never rows: a list of eleven things inside a sentence cannot be counted by anyone checking the total, and the total was indeed wrong.

What is absent	Where it is stated	Decision	Reason, and who ruled it
Meters tab, no rows	Screen 01, empty state	Out of scope	The empty names what creates a row. B14 rule 17.
Sub-meters tab, no rows	Screen 02,	Out of scope	Same pattern, same rule.

	empty state		
Behind the meter, no rows	Screen 03, empty state	Out of scope	Same pattern, same rule.
Inventory, no stock	Screen 04, empty state	Out of scope	Same pattern, same rule.
Map, no coordinates	Screen 06, empty state	Out of scope	**The one worth reading twice, because it names the cause:** *"Premises gain coordinates from the geocoding run; until then the map has no markers to draw."*
Map, no match for the filter	Screen 06, filtered empty	Out of scope	Distinguishes *nothing here* from *nothing matching*, which is the distinction rule 17 exists for.
Certifications & tests, none due	Screen 05a	Out of scope	Nothing overdue is good news and is said as such.
Warranty & supplier, none recorded	Screen 05b	Out of scope	Names the field that would fill it.
Reads, none taken	Screen 09	Out of scope	Names the upload that creates one.
Export, no file yet	Screen 10	Out of scope	⚠ Honest about the queue — **and the file it promises does not arrive**, which is a separate live defect, `AST-B-20`, not an empty-state problem.
Maintenance, no history	Screen 12	Out of scope	Measured: the table holds **zero** rows on this tenant, so the empty is the truth.

D · Copy that reaches nobody (1)

Key	String	Why it is dead
`assets.sub`	*"Every meter, sub-meter and behind-the-meter device that can affect a bill."*	In all eleven catalogues; **zero render sites**. Orphaned when the shared page-header conversion dropped the kicker/sub pair. The only occurrences outside the catalogues are a comment and the repo's own gap register.

> **One absence the product shows and never states**
>
> `AssetDash` renders a bare italic **—** with no title and no accessible text (`assetsShared.tsx:202-204`), used for every unserved KPI and every empty column cell in this module. On screen 01 that means **loading**, **genuinely none** and **we cannot compute this from one page** are the same picture. Three situations, one glyph. Reported as a description rather than a specification: nobody has ruled on it, and fixing it is a shared-component decision.

Commitments raised in this section: `AST-B-01`, `AST-B-02`, `AST-B-03`, `AST-B-04`, `AST-B-10`, `AST-B-11`, `AST-B-14` — seven, all indexed in section 12. The other thirty-five rows are **out of scope** with the ruling quoted.

E · Absences found after this register was first written (7)

Sections 04, 05, 06 and Appendix L were all written after A–D were counted, and each turned up absences of its own. They are listed separately rather than folded in, so the original count stays checkable against the pass that produced it. ⚠ **Four of the seven are drop-down sources**, and none of them could have been seen in a drawing — a picker that offers the wrong list looks exactly like one that offers the right list.

What is absent	Where it is stated	Decision	Reason, and who ruled it
Connection size is not filtered by service	Appendix L; screens 17, 29, 32	Out of scope	**Ruled and filed already** — `DEF-3300`. Every size class names its service since `CR-065`; the two RATE screens the PO photographed were filtered and this picker was not, because an asset being edited may already hold a size from another service and dropping it would silently change what the form saves. **Filed rather than guessed at**, by the lane that hit it.
The premises and unit pickers stop at 500 rows	Appendix L; screens 22, 30	`AST-B-19`	**No ruling.** A hard cap on a picker is the shape `NFR-5` exists to forbid, and premises 501 cannot be chosen with nothing on screen saying so. It folds into the pagination specification rather than earning its own: the same envelope fixes it.
Test cadence cannot be a regulator's interval	Appendix L; screens 29, 32	`AST-B-27`	**No ruling.** `CADENCE_MONTHS` is frozen at `12 · 24 · 60`, so a tenant whose backflow rule says 36 months needs a release to express it. Folds into the capture specification — the column takes any integer; only the list is short.
Depreciation method has no list anywhere	Screens 20, 31; Appendix L	`AST-B-27`	**No ruling.** The column is on the model, the API serves it, screens can display it, and **nothing in the product can put a value in one**. Already inside the columns-with-a-reader-and-no-writer specification.
Unit of measure is typed, not	Screens	`AST-B-03`	**No ruling.** The tenant's units ride on the same `GET /settings/services` payload as the service

picked	17, 29; Appendix L		itself, so it is the same defect one field along and folds into the same specification.
A draft never expires and nothing limits how many there are	Screen 33	Out of scope	**Ruled, and deliberately left open** — `BR-SHL-047` settled that parking is offered and explicitly did **not** settle a limit, because *"a wizard that refused to park after N days would be enforcing a rule nobody made."* ⚠ It is not free here: a forgotten **retire** draft means a decommissioned meter goes on billing, which is what the age column exists to show.
Read methods are served by two endpoints	Appendix L	Out of scope	**Ruled, and the second one is the fix for an older problem** — `GET /reads/read-methods` was minted because `/lookups/{code}` was System-Admin-only at the time (`DEF-196`). That gate has since widened to *read by any role*, so the duplication is now redundant rather than load-bearing. **Removing it is metering's call, not this document's**; recorded so the next lane meets the fact instead of the surprise.

Totals, added up rather than glanced at. A 17 + B 13 + C 11 + D 1 = **42**, plus E 7 = **49 stated absences**, of which **7 carry a specification** (three of them folding into one minted earlier in the same pass, which is the register behaving correctly) and **42 are out of scope with a reason and a ruler named**. **Nothing is unresolved** — every sentence in this document that says something does not exist has exactly one of those two outcomes.

09

Where these screens break their own rules

One row per violation, never per screen, judged against **B14** — the authority — with `UI-RULES` cited where it is the source, and the spacing and field rules (**R11**, **R12**) measured on the running app. The last column is the point of the table: in a repository where most UI rules are ratified and enforced by nothing, a green suite is evidence of nothing, and a reader needs to see which rows a machine would ever have caught.

Screens	Rule	What the rule says	What the screen does	Where	Enforced?
10, 14	`BR-WRK-056`	*"The Assets door is ONE form with four ways in … **the action bar, the Maintenance history panel** and the*	Two of the four are not built. `AssetDetailPage.tsx` holds zero references to `raiseWorkOrderPath`. **Confirmed in the browser: 0 text matches. 0 links.**	`AssetDetailPage.tsx:375-410`.	**No.** Nothing reads rendered output, and no

		four `RaiseWorkOrderCheckbox` *ticks … all post it."* Status: **CONFIRMED**		`:1286-1395`	test asserts a control's *absence*.
16–19, 20, 21, 22, 24	**R12**	*"It starts with the field name. then the input field, and then it should show the helper text below that input field."* label → input **5px** · input → helper **6px** · field → field **20px**. And DOM order alone is not enough — the helper has to *read* as attached to the input above it.	**Ten of eighteen fields render `label → helper → input`.** Measured 2026-09-11: on those, label-to-input is **29px** or **47px**, not 5, and the helper reads as belonging to the label below it. The eight fields with no helper are exactly **5px** and correct — so the rhythm holds only where there is nothing to get wrong. Add wizard: Asset #, Service. Reassign: Effective date, Final register reading, Read date. Log repair: Date, Set status, Estimated cost. Retire: Retire date, Final read.	`AssetAddWizard.tsx`, `AssetFlowPages.tsx`, `AssetRetireWizard.tsx`	**No.** The markup type-checks and every test passes; the order is a layout fact no assertion in this repository looks at.
01, 02, 03, 04, 05, 06	**18** + the 6–8 word convention	Every screen carries a description in the toolkit's own slot — `PageHeader`'s per-tab explainer, capped at 80% of the content width — and the convention is **six to eight words**.	**Zero of six are inside the budget, and two render nothing at all.** Measured on the running app: Meters **—**, Sub-meters **24w**, Behind the meter **—**, Inventory **18w**, Lifecycle **9w**, Map **23w**. The two blanks are not missing strings: `assets.explainer.meters` and `.behind_the_meter` sit in all eleven catalogues and reach no screen, because line 801 passes an explainer only for `sub_meters`.	`AssetRegistryPage.tsx:801`, `:825`, `:833`	**No.** `G43` clause (b) checks the *width* cap and nothing checks the word count or whether a slot is filled at all.
05	**12 · 13 · 14 · 16**	A list table carries a toolbar (`[Search…] [All | chips ×] [+ Filter ▾] … [▦ Columns]`), sortable headers, a column selector with reorder and Reset, and a pagination footer.	**None of the four, on any of the Lifecycle tab's three tables.** Nine warranty rows and ten forecast rows today, and the certifications list is the one a utility opens weekly. They are the detail-panel table, not `DataTable` — which is exactly why there is no toolbar.	`AssetLifecycleTab.tsx:138-270`	**No.** Answered by screens 05a–05c.
01, 02, 03	**15**	Row actions are icons with tooltips, **max 2 visible + … overflow**	**5 controls in one cell** on every installed row, 3 on every inventory row, and no overflow anywhere. Measured over twelve rows.	`AssetRegistryPage.tsx:663-724`	**Partly.** `G37` enforces the icon half, the accessible-name half and single icon ownership. It

					does not check the cap — `RowActionsMore` exists and one screen in the product uses it.
16–19, 20, 21, 22, 24	**7**	Date fields always use the date input — calendar affordance, keyed entry, **Today shortcut**. Never a bare text input	Native `<input type="date">` everywhere. Calendar and keyed entry: yes. **Today shortcut: nowhere.** No `min` or `max` on any of the five measured, including one whose own help states a bound.	`FormModal.tsx:570-638`	**No.** B14's enforcement column for rule 7 is empty.
21	**31**	**No database id is ever the visible value of a field.** PO, 2026-08-31: *"no Databse ID on screen - that is a rule"*	For a Billing Clerk the technician picker degrades to a bare text box holding a UUID, helped by *"Your role can't list users, so enter the technician's ID."* — and the clerk is precisely the role `BR-AST-038` wrote this screen for.	`AssetFlowPages.tsx:604-612`	**No.**
01, 02, 03	**13**	Sorting: click cycles **asc** → **desc** → **clear**	Toggles asc↔desc forever. Measured: `desc` → `asc` → `desc`. **Product-wide**, in the shared table, not an assets defect.	`DataTable.tsx:53-59`	**No.**
01, 02, 03	**16**	Pagination footer: `Showing 1–50 of N · Rows per page … · ⇤ ← Page x of y → ⇥`	Range, page size, previous and next all render. **No first-page and no last-page control** — only `table.pager.prev` and `table.pager.next` exist in the catalogue.	`DataTable.tsx:934-975`	**No.**
04, 05	`BR-SHL-025`	A role without write sees the control **disabled with its reason**, never hidden	`writeBlockedReason` is declared as a prop on both tabs and passed by the page — and read nowhere. The word appears exactly once in each file, in the type declaration. **Parts in Work** → and the Lifecycle wrench render ungated for a read-only role.	`AssetInventoryTab.tsx:74`; `AssetLifecycleTab.tsx:93`	**No — and this is the shape `G13` cannot see.** G13 asks whether an optional prop has a caller. It does. Nothing asks whether the component reads it.
16, 17, 18, 19, 22	**R11**	One title size per class of screen — the token sheet calls **40px** the *"full	**The two full-page wizards render 32px and the single-form task page renders 40px.** Measured across seven screens:	`PageHeade`	**No.** Both values are

19, 22, 23, 24		the token sheet calls **40px** the *"full-page wizard title"* and **32px** *"the page-head default, every screen head"*.	**task page renders 40px.** Measured across seven screens: registry 32 · detail 32 · **add wizard 32** · **retire wizard 32** · reassign 32 · **Raise work order 40**. The one screen that is not a wizard is the one using the wizard size.	r consumers, per screen	values are legal tokens; nothing compares a screen's kind to the token it picked.
01 (the shell)	**R11**	One spacing scale — 4 · 8 · 12 · 16 · 20 · 24 · 32. *"No ad-hoc 3 / 7 / 9 / 14 anywhere."*	**Two off-scale values, both long-standing product literals rather than drift:** the rail item at `9px 12px` with a `10px` icon gap, and the status pill at `2px 9px`. Eight of the ten measured anchors hit the scale exactly. Drawn as measured here; reported rather than tidied.	`AppShell.css`, `StateFrame.css`	**No.** The CR-016 browser check measures type sizes and control heights, not padding.

Rules these screens are exempt from, so nobody files them twice

Rule 16	Wizard summary tables are exempt from the pagination footer. Screens 31, 17 and 24 carry tables with no pager and that is correct.
Rule 21	Tailwind utilities are lawful in every file in this module — all fourteen asset feature files are on `07-frontend/src/styles/route-b-migrated.txt`. A utility written in a file *not* on that list renders as nothing at all while `tsc` and every `toContain` assertion pass; that trap is checked here and does not apply.
Rule 8	Not violated on screen 06 — see that screen's callout. The four hand-rolled selects compute to `appearance: none` and draw the product's chevron. This document said otherwise from a source read and the running app corrected it.
R12, the other half	Field-to-field spacing is **correct**: every measured wrapper reported `margin-bottom: 0` and takes its 20px from the form grid's `gap`, which is what R11 asks for — layout doing the spacing rather than per-element margins that collapse or double.
Hard rule 4	Financial immutability does not reach these screens. Nothing in the Assets menu writes an invoice, an invoice line, a payment or a journal entry, so an edit control here is a feature rather than a finding.

And one that is not a UI rule but is the reason to read section 11

> **`NFR-3`: the registry is a paginated envelope over an unbounded query**
>
> `asset_list` returns `list(qs.order_by(...))` — a materialised Python list — and `paginate_queryset` then slices it (`selectors.py:495`, `api.py:117-121`). **No**

`LIMIT` or `OFFSET` reaches SQL. Four of the seven filters then run in Python over that whole list, each with a per-row query or seam call. The response envelope is honest; the query behind it is not.

Correct at **T7**, and the same shape `DEF-PAY-4221` already names on `GET /outstandings` — a list that is right at 3,000 connections and thirteen times short of the 40,000 the product is now scoped for. Filed here rather than in the table above because `NFR-3` is an engineering rule, not a UI one, and a reader looking for *"what will hurt when a big utility onboards"* will not find it under a UI-rules heading.

Commitments raised in this section: none. A rule violation is a description until somebody undertakes to fix it; four of the eleven above are answered by drawings in sections 02 and 10, and the rest are stated and left.

10

Drawn answers to the findings worth drawing

A paragraph saying what should change is a paragraph somebody has to translate. These are drawn to the same 1440 window, the same palette and the same square corners as everything above, so each can be held beside the screen it replaces. Every element carries a number and every number a line: **what it is · where its data comes from · which rule it satisfies · and whether it is new, changed or unchanged**.

Two remedies live in section 02 rather than here, and on purpose

Screens 05a–05c (the Lifecycle tab as three tabbed lists) and **screen 18** (Edit as a full page) are whole screens rather than changes to one. They are drawn as numbered remedies with their own legends and carry-over tables *in place*, beside the as-built they replace, because a reader comparing a three-table page with three tabbed pages needs both in one eyeline. They are indexed here and in section 12 like every other commitment.

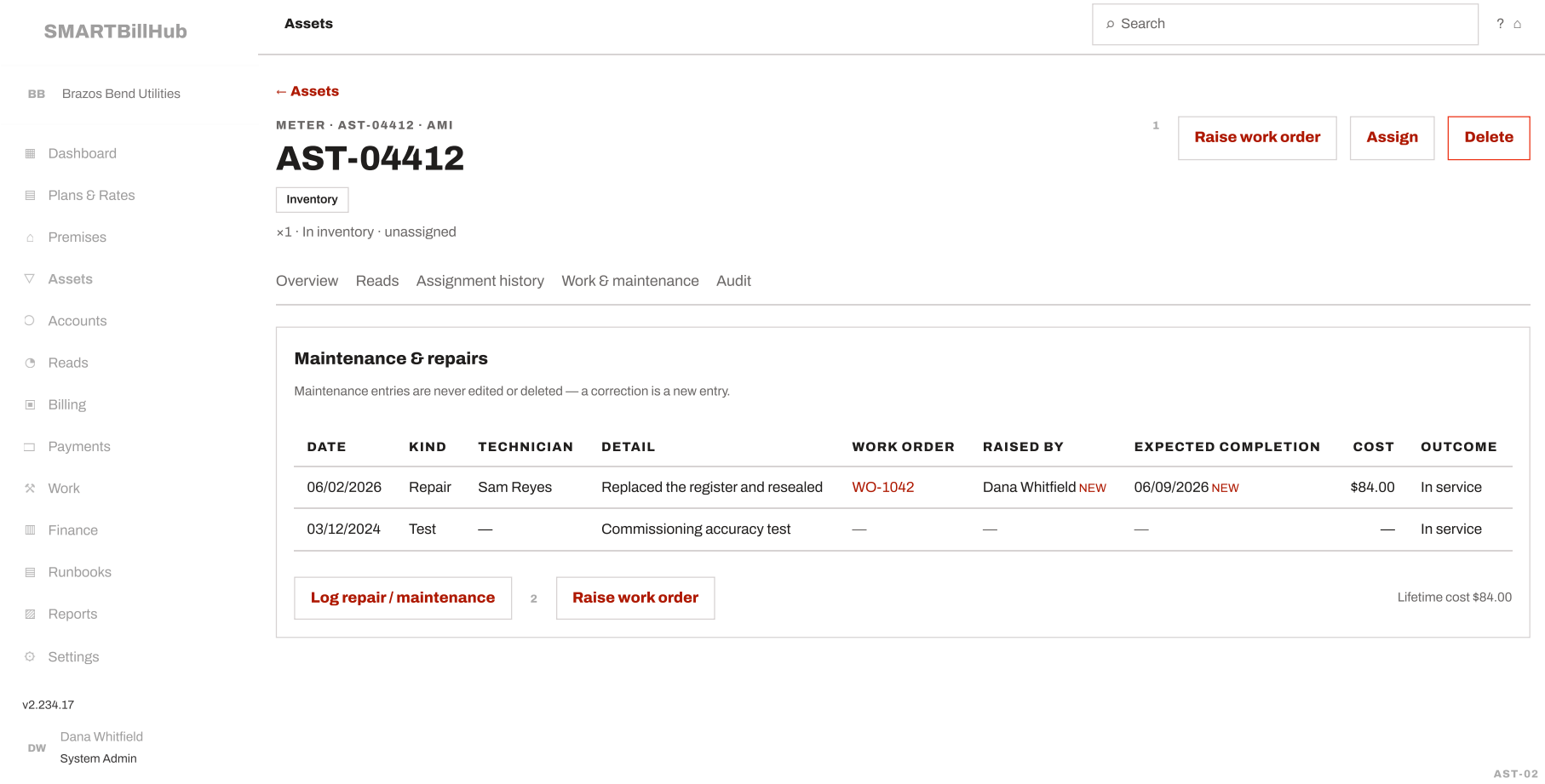
A remedy never loses anything

Every field and every piece of information architecture on the as-built drawing survives into the remedy — grouping, order, tabs, hierarchy, words — unless somebody has ruled that it goes and the ruling is quoted. Each remedy carries a **carry-over table** and the arithmetic must reconcile, because reading the picture never catches a lost field and subtraction always does.

Across all six remedies in this document, exactly one thing is removed — the dialog frame on screen 18, under a quoted instruction, and it is a container rather than a field. Appendix K carries that one row and nothing else.

R1 · The asset gets the two doors its own rule already names AST-B-02

`Figure 51` · The one finding here that stops somebody doing their job. Two controls, both named in `BR-WRK-056`, both landing on a screen that already exists and was driven in this pass.



1 **Raise work order**, in the header action bar · navigates to `/assets/{id}/work-orders/new` · `BR-WRK-056` · **NEW**

2 **Raise work order**, beside **Log repair / maintenance** on the Maintenance panel · the same destination · `BR-WRK-056` · **NEW**

Carry-over

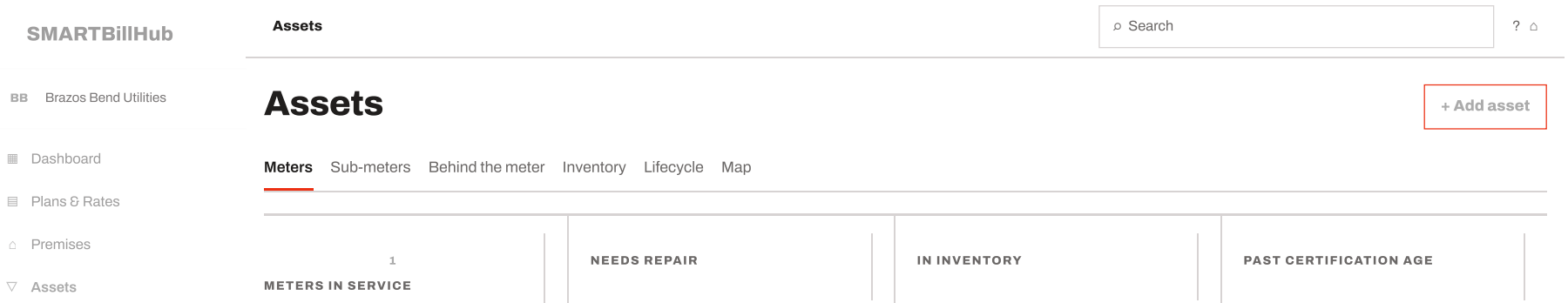
As-built element	Disposition	Note
Back link · kicker · title · status pill · rollup	UNCHANGED	—

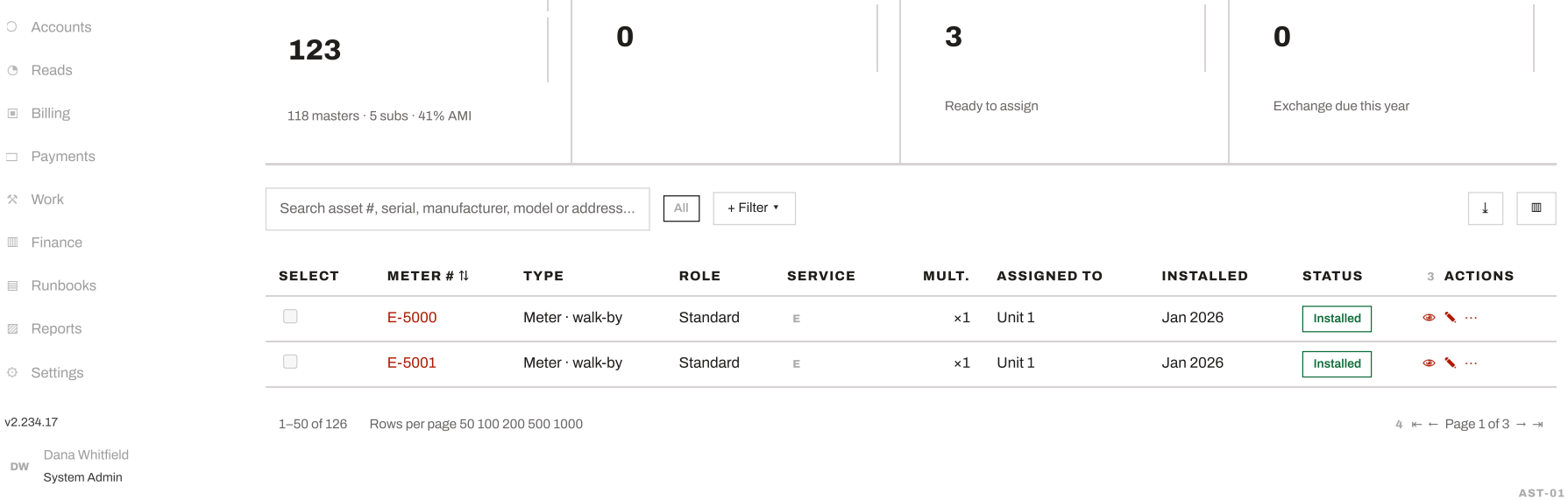
Assign · Delete (header, state-dependent)	UNCHANGED	Still the server's action set; the new button sits *before* them, not instead of.
Five tabs, same order, same labels	UNCHANGED	—
Maintenance table: Date · Kind · Detail · Technician · Work order · Cost	UNCHANGED	The Work order column stays a read.
Append-only sentence · Log repair / maintenance · Lifetime cost	UNCHANGED	—
Raise work order ×2	NEW	Answers the J3 finding.

```
As-built elements    9
  unchanged          9
  moved              0
  relabelled         0
  removed            0
Remedy elements     11   = 9 − 0 removed + 2 new    ✓ reconciles
```

R2 · The registry's numbers become numbers AST-B-01

Figure 52 · One new read fixes the KPI strip; two rule fixes ride along because they are the same table and would otherwise be three visits to one screen. The figures are illustrations consistent with the seeded tenant, not served values.





1 **KPI figures** · a new aggregate read over the whole filtered set, not the page · `NFR-3`-safe, four `COUNT`s on the predicate `asset_list` already builds · **NEW READ**

3 **Row actions capped at two, then ⋯** · the existing `RowActionsMore` component · **B14 rule 15** · **CHANGED**

4 **First and last page controls** · the same `count` already served · **B14 rule 16** · **NEW**

The sort indicator also changes from ▲ to ⇅ once a third state exists — **B14 rule 13**, **CHANGED**, and that is a change to the shared table rather than to this screen.

Carry-over

As-built element	Disposition	Note
Title · Add · six tabs	UNCHANGED	—
Four KPI cards, same labels, same order, same detail lines	UNCHANGED	Only the *value* changes, from a dash to a figure.
Search · All · + Filter ▾ · ⤓ · ▥	UNCHANGED	—
Ten columns, same order	UNCHANGED	—
Row actions: view · edit · reassign · log repair · retire · delete	MOVED	All six survive. The first two stay visible; the rest move into ⋯. Nothing is dropped — a control behind an overflow is still reachable, and rule 15 asks for exactly this.

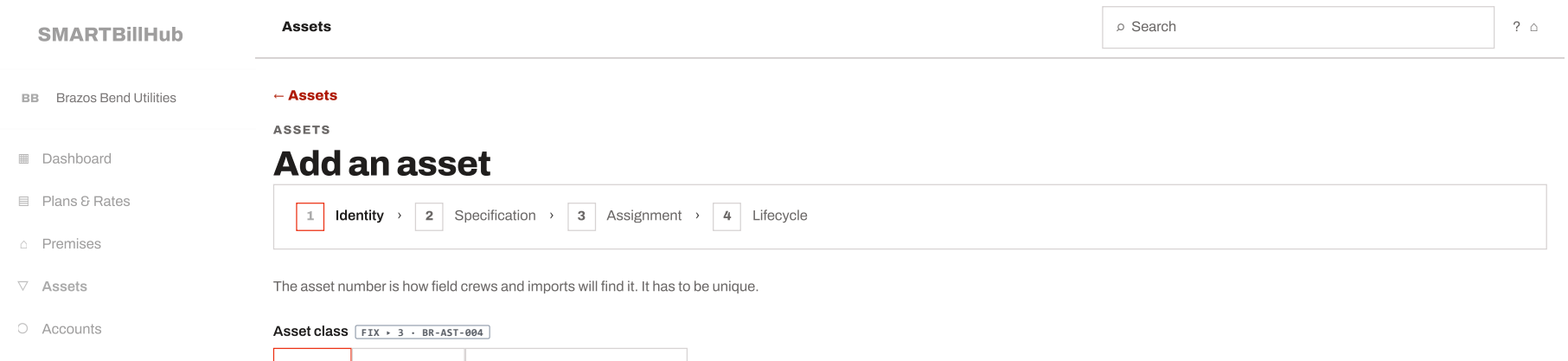
Pagination range · rows per page · prev · next	UNCHANGED	—
Audit line	UNCHANGED	—
⇤ first / ⇥ last	NEW	Answers the J6 finding.

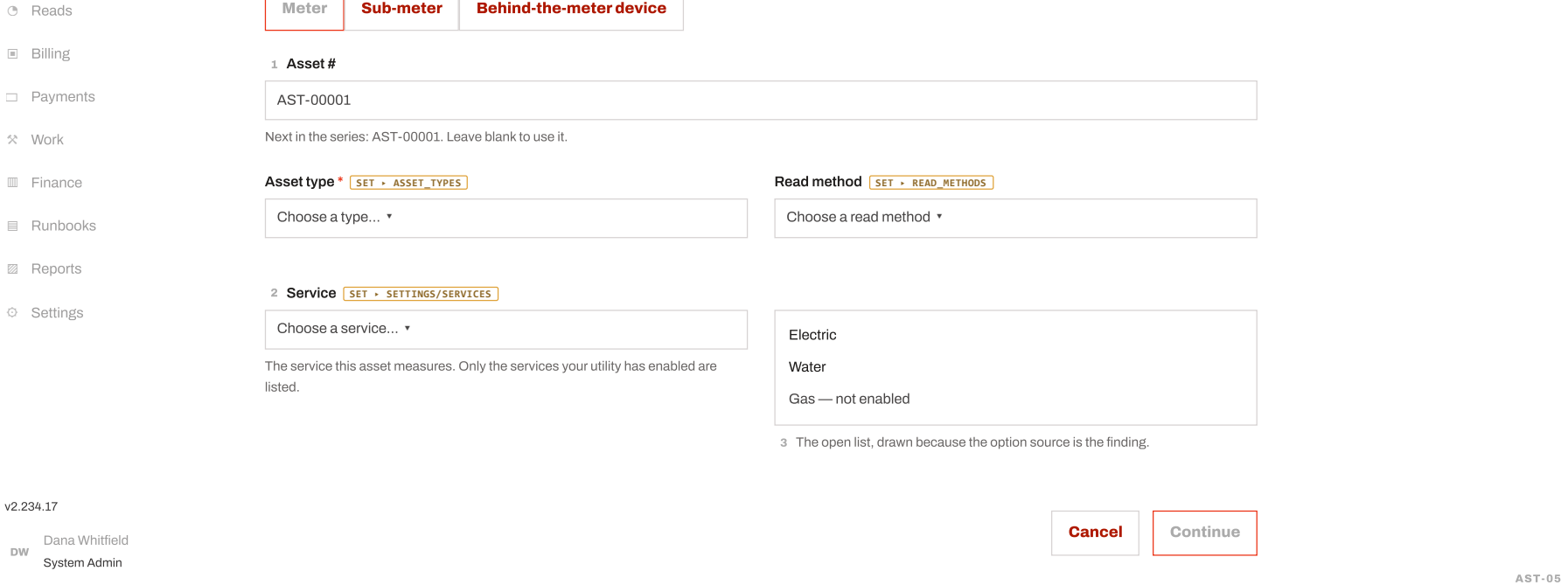
```
As-built elements   28
  unchanged         22
  moved              6   ← four row actions into the overflow, both kept reachable
  relabelled         0
  removed            0
Remedy elements     30   = 28 − 0 removed + 2 new    ✓ reconciles
```

> **Why this remedy stops where it does**
>
> It would be easy to also fix `NFR-3` here — the unbounded query behind the envelope — and it is deliberately not drawn. That is a back-end change with no pixels, it is the subject of section 11 rather than of a wireframe, and a remedy that also tidies what it happened to be standing next to is a redesign wearing a fix's clothes.

R3 · The service field offers the list it already has AST-B-03

`Figure 53` · The open state is drawn because the option list *is* the finding. A dropdown has two data sources and conflating them is how a field that looks fine turns out never to grow.





1 **Asset #** helper · **moves below its input**, 5 / 6 / 20 · **R12** · **MOVED**

2 **Service**, a searchable picker reading *Choose a service...* · value → `assets.service_point.service`; **options** → `GET /settings/services` · **B14 rule 8** and `BR-MET-003` · **CHANGED**

3 **The open list** · the same read · shown only to make the option source visible · **NEW** to the drawing, not to the screen

The help text is **RELABELLED**: *"The service code this asset measures. It must be one your utility has enabled"* → *"The service this asset measures. Only the services your utility has enabled are listed."* The old sentence names a constraint the person must satisfy; the new one describes what the control already does. **The word *code* goes** — nobody should be typing one.

Carry-over

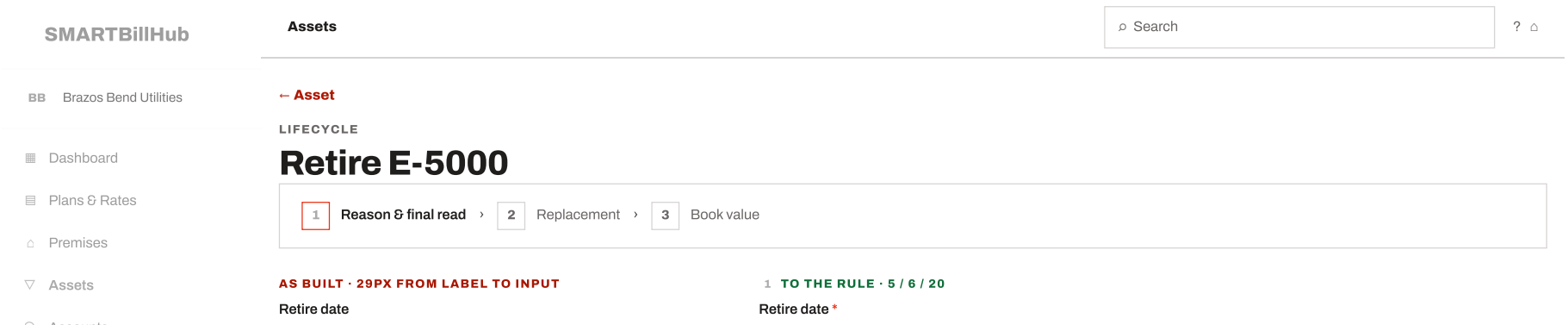
As-built element	Disposition	Note
Step rail, four steps, same labels	UNCHANGED	—
Explainer sentence	UNCHANGED	—
Asset class (3-way segmented)	UNCHANGED	—
Asset # + its series help	MOVED	The field is unchanged; the *helper* moves from above the input to below it. R12.

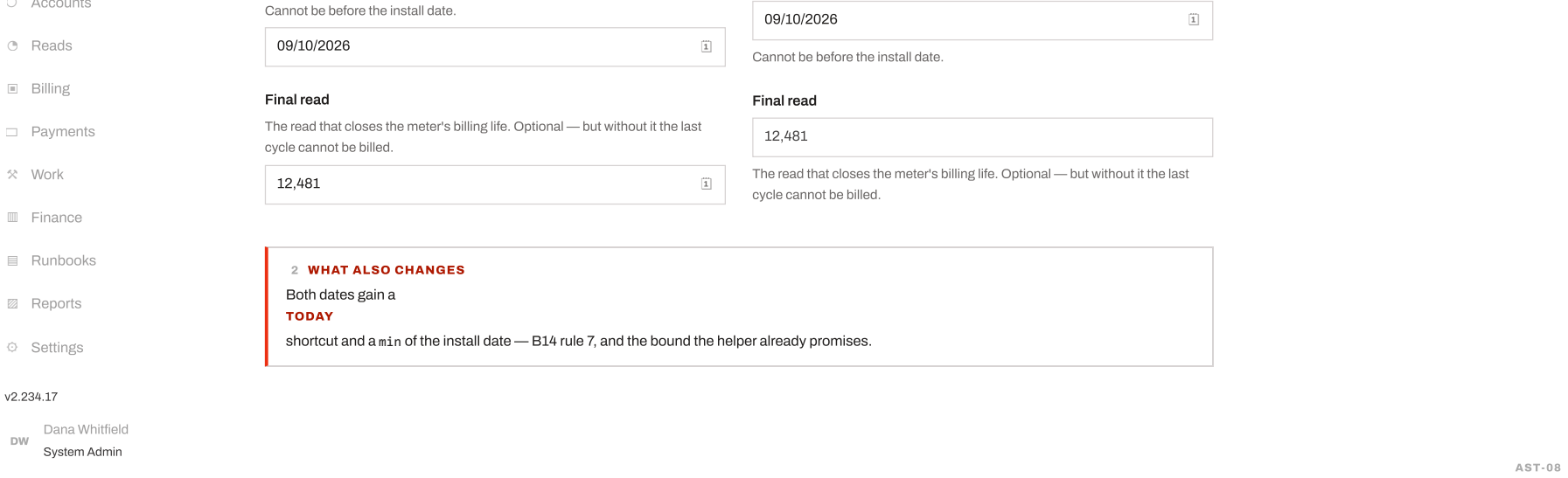
Asset type · Read method	UNCHANGED	Still two to a row.
Service — the *field*	UNCHANGED	Same label, same position, same step.
Service — the *control*	CHANGED	Bare text input → searchable picker. A person who typed a code before can still type; they now get matches instead of a guess.
Service — the *help*	RELABELLED	Old and new quoted above, and it moves below the input.
Cancel · Continue	UNCHANGED	Continue keeps its stated refusal.

```
As-built fields      9
  unchanged          6
  moved              1   ← the Asset # helper, above the input to below it
  changed            1
  relabelled         1
  removed            0
Remedy fields        9   = 9 − 0 removed + 0 new    ✓ reconciles
```

R4 · Every helper moves under its own input AST-B-13

Figure 54 · The same two fields drawn twice: as they render today, and as R12 asks for. It is drawn as a pair because the fault is invisible in isolation — each field looks fine on its own, and only the rhythm gives it away.





1 **label → input → helper**, at 5 / 6 / 20 · the helper now reads as attached to the input above it rather than to the label below it · **R12** · **MOVED**, on ten fields across four screens

2 **A Today shortcut and a `min` of the install date** on both date controls · **B14 rule 7** · **NEW**

The left half is the as-built, kept in the drawing so the two can be compared. It is not a proposal.

Carry-over

As-built element	Disposition	Note
Every label, in its own words	UNCHANGED	Ten fields across the add wizard, reassign, log repair and retire.
Every helper sentence, in its own words	UNCHANGED	Not one word changes. Only the position does.
Every input, its type and its value	UNCHANGED	—
The helper's *position*	MOVED	From between the label and the input, to below the input. This is the whole remedy.
A **Today** shortcut on each date control	NEW	B14 rule 7, unmet on all five date fields.
`min` / `max` bounds on each date control	NEW	The retire date's own helper already promises the bound; the control did not enforce it.

```
As-built fields (measured)   18
  unchanged                   8   ← the eight with no helper, already at 5px
  moved                      10   ← the helper's position, on every field that has one
```

```
  relabelled                 0
  removed                    0
Remedy fields               18   = 18 − 0 removed + 0 new    ✓ reconciles
  (the two NEW rows above are control affordances, not fields)
```

AST-B-13

A person reading a form knows which input each hint belongs to

STATUS Unverified

WAITS ON Nothing. It is the order of two elements inside a field wrapper, in four files.

SCREENS 14, 16–21, 23

WHAT DONE LOOKS LIKE ON SCREEN On **RETIRE E-5000**, the sentence *"Cannot be before the install date."* sits **BELOW** the date box, 6px under it, and the gap from the **RETIRE DATE** label to the box measures **5PX** rather than 29.

Every figure in these drawings is a test case, not a served value, and each lands verbatim in appendix B.

R5 · The opening read becomes a field a person can fill in AST-B-14

This one overturns a ruling recorded earlier in this document

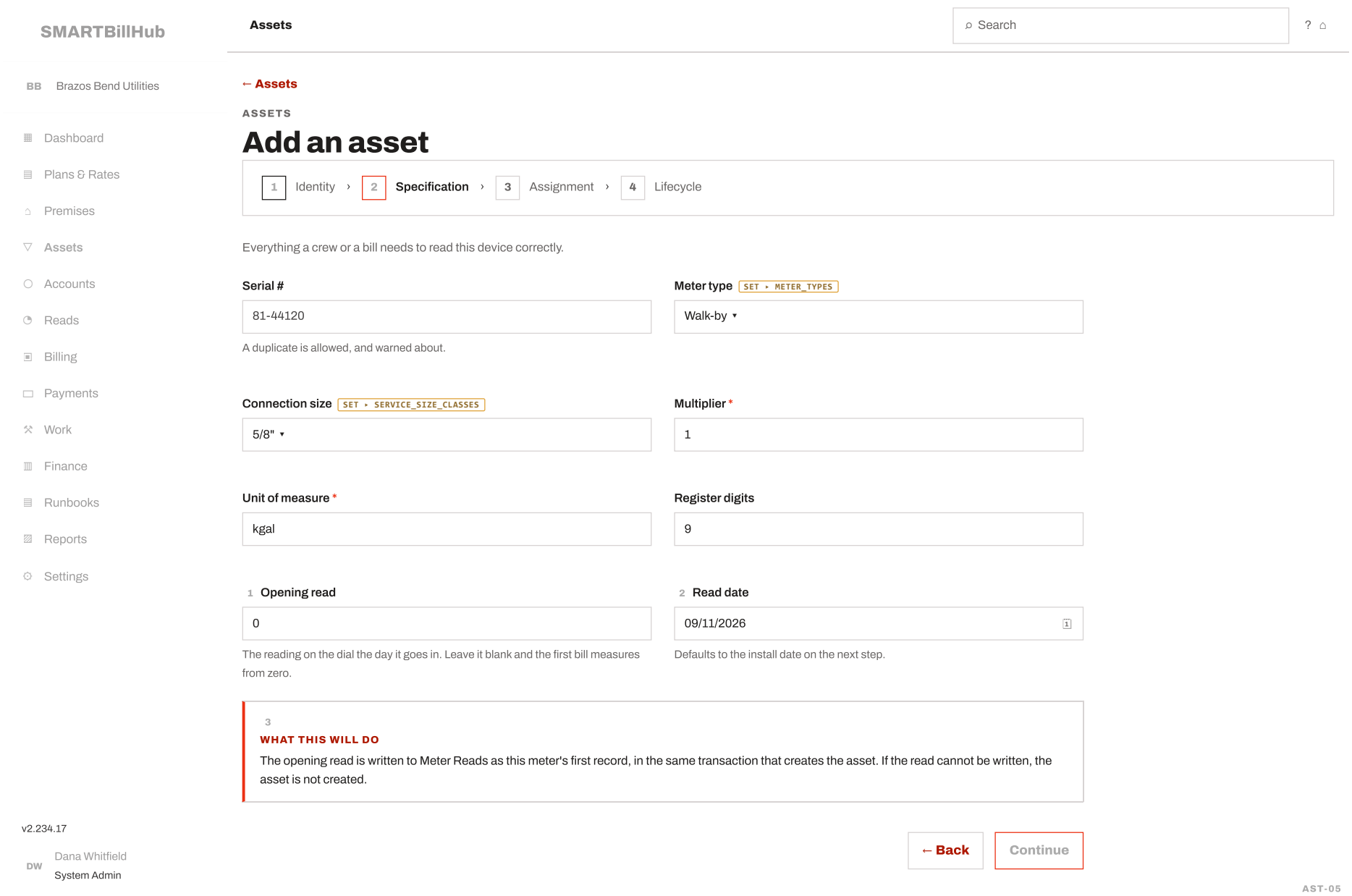
Section 08 ruled the disabled **Opening read** field **OUT OF SCOPE**, quoting the lane instruction behind it: no published seam lets assets write a `meter_reads` row at create time, so the field renders disabled *with its reason*. That was the correct reading of the record as it stood.

Instruction, 2026-09-11: *"make opening read enable user can add reading".* A newer decision beats an older one, so the row moves from **out of scope** to a specification, and the superseded reasoning is kept above rather than deleted — the next reader needs to know the seam argument existed and was overruled, not that nobody thought of it.

And the seam is not a blocker, it is a precedent. `capture_final_read` already does this exact thing in the opposite direction: metering commits a manual read *in the caller's transaction* and hands back its `read_dict`, and when it cannot, the whole move is refused rather than half-done (`seams.py:272`, wired by the CR-028 debt sweep). An opening read at create time is the same shape with the same failure rule.

Figure 55 · Drawn on step 2, where the register digits and the unit of measure already sit — a

reading is meaningless without both, and putting it anywhere else would ask a person to remember them.



1 **Opening read** · an enabled input · writes to `meter_reads` through a create-time seam modelled on `capture_final_read` · **CHANGED** — the field is already drawn, and stops being read-only

2 **Read date** · defaults to the install date chosen on step 3 · **NEW**

3 **The consequence box** · says what will be written and what happens if it cannot be · **B14 rule 11** · **NEW**

The helper now sits **below** its input on every field on this step — R12, and the same move R4 makes everywhere else.

Carry-over

As-built element	Disposition	Note
Serial # · Meter type · Connection size · Multiplier · Unit of measure · Register digits · Manufacturer · Model	UNCHANGED	Eight fields, same labels, same order, two to a row.
Opening read — the field	UNCHANGED	It was already drawn. It is the same field in the same place.
Opening read — the control	CHANGED	Read-only and permanently empty → an enabled input. ⚠ Instruction, 2026-09-11: *"make opening read enable user can add reading"*.
Opening read — the helper	RELABELLED	*"An opening read is recorded through Meter Reads once the meter is installed — it cannot be captured here."* → *"The reading on the dial the day it goes in. Leave it blank and the first bill measures from zero."* The old sentence described a refusal that no longer holds.
Read date	NEW	A reading without a date cannot be placed in a billing period.
The consequence box	NEW	The write is transactional and can refuse; rule 11 says a form that writes says so first.

```
As-built fields      9
  unchanged          8
  changed            1   ← Opening read, disabled to enabled
  relabelled         1   ← its helper (the same field; counted once above)
  moved              0
  removed            0
Remedy fields       11   = 9 − 0 removed + 2 new    ✓ reconciles
```

AST-B-14

A clerk booking in a meter can record the reading that is on its dial

STATUS Unverified

WAITS ON A create-time read seam. **NOT A NEW CAPABILITY** — `metering.services.capture_final_read`

already commits a manual read inside the caller's transaction and refuses the whole operation when it cannot; this is that, at the other end of the asset's life.

SCREENS 29

WHAT DONE LOOKS LIKE ON SCREEN Step 2 of **ADD AN ASSET** shows **OPENING READ** as an enabled box with **READ DATE** beside it, and creating the asset leaves a first record on the new meter's **READS** tab dated the install date.

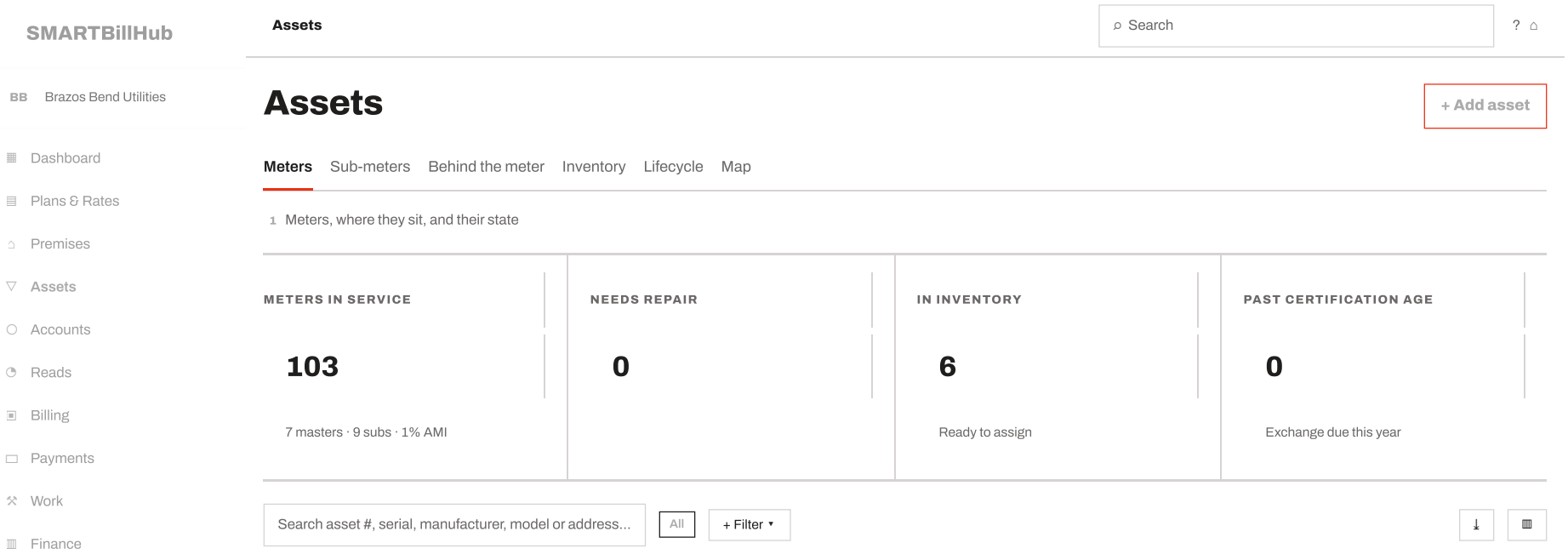
R6 · One description per screen, six to eight words AST-B-15

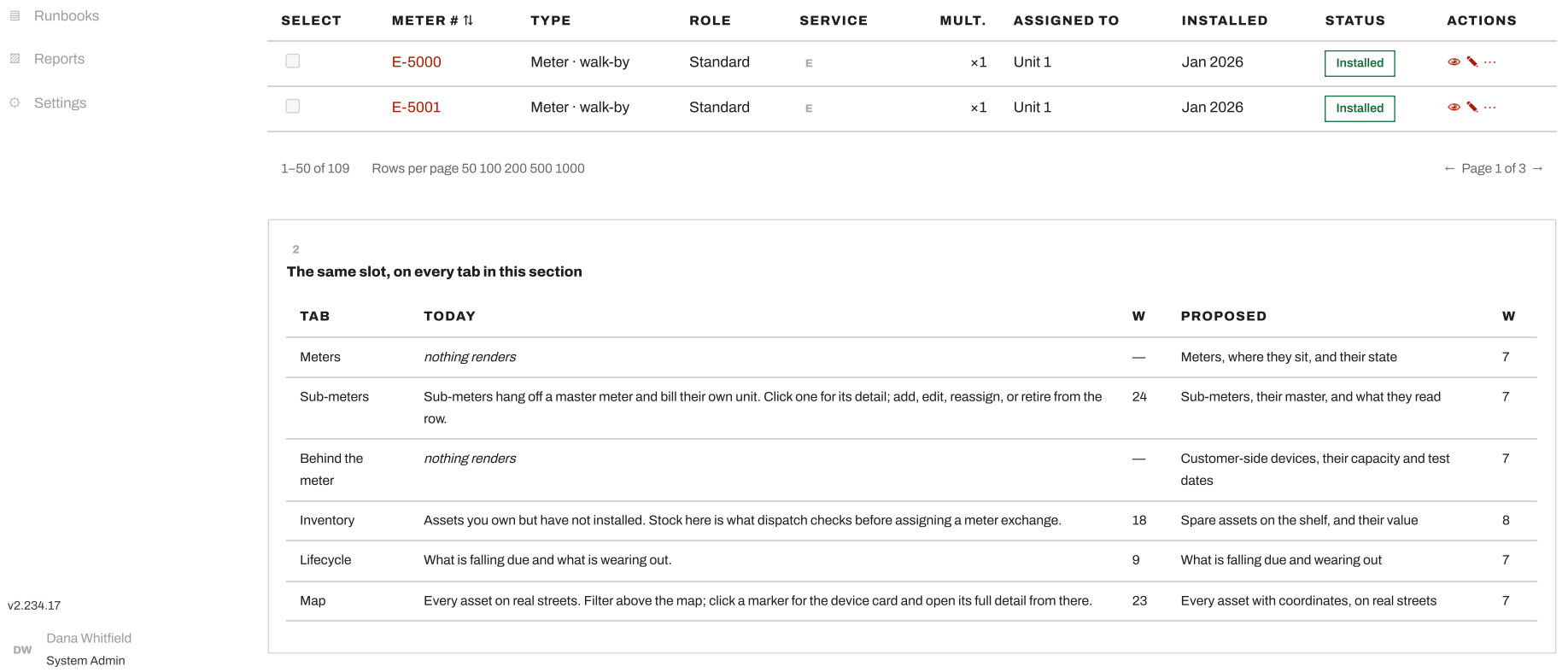
Not one of the six meets the convention, and two of them do not exist

The toolkit has a description slot: `PageHeader` renders a per-tab explainer below the tab strip (`PageHeader.tsx:386`), and B14 rule 18 caps it at 80% of the content width. The convention is **six to eight words**. Counted on the running app:

Meters and **Behind the meter** render **nothing** — their strings exist in all eleven catalogues and reach no screen, because `AssetRegistryPage.tsx:801` passes an explainer only for `sub_meters`. The other four render **9, 18, 23 and 24 words**. **Zero of six are inside the budget, and two are dead copy.**

`Figure 56` · The ruling that took the prose off Meters and Behind the meter replaced it with a KPI grid — and that grid shows four em dashes on first load, so those two tabs now carry neither a description nor a number.





1 **The description** · the tab's own explainer, rendered in the slot the toolkit already has · **B14 rule 18**, and the six-to-eight-word convention · **NEW** on Meters and Behind the meter, **RELABELLED** on the other four

2 **The full set** · drawn as a table because the finding is the *comparison*, not any one line. Today's wording is kept beside the proposal so nothing is lost by sleight of hand.

Everything else on this drawing is R2 — the served KPI figures and the capped row actions — shown together because they are the same screen and a reader should see the finished state once.

Carry-over

As-built element	Disposition	Note
Sub-meters' 24-word explainer	RELABELLED	The second sentence — *"Click one for its detail; add, edit, reassign, or retire from the row"* — describes the row actions, which the row actions already describe. It goes; the fact that a sub-meter hangs off a master and bills its own unit stays.
Inventory's 18-word explainer	RELABELLED	The dispatch sentence is real and useful, and it is not a description — it belongs in this document's *For* row, which is where it now is.
Lifecycle's 9-word explainer	RELABELLED	One word over. *"What is falling due and what is wearing out"* → *"What is falling due and wearing out"*.

Map's 23-word explainer	RELABELLED	Two of its three sentences are instructions for using a map. The map is a map.
Meters' and Behind the meter's descriptions	NEW	Neither renders today. Two screens gain the slot every other screen in the product has.
Every KPI label, value and detail line	UNCHANGED	—
Every column, chip, control and the pager	UNCHANGED	—

```
Descriptions today        4 rendered + 2 dead strings = 6 tabs
  inside 6–8 words        0
  relabelled              4
  new                     2
  removed                 0
Descriptions after        6   = 4 relabelled + 2 new     ✓ reconciles
  inside 6–8 words        6
```

AST-B-15

Every screen in the list says what it is for, in a line a person reads without meaning to

STATUS Unverified

WAITS ON Nothing. The slot exists, the component renders it, and four of the six strings are already in all eleven catalogues — they are too long, not missing. Two need writing.

SCREENS 01, 02, 03, 04, 05a, 05b, 05c, 06

WHAT DONE LOOKS LIKE ON SCREEN Opening **ASSETS** shows *“Meters, where they sit, and their state”* under the tab strip, and every other tab in the section shows a line the same length in the same place. No tab shows nothing.

11

What crosses a module boundary, and whether it holds

Built from the ⇄ rows in every field table above — the real cross-module surface of these screens, taken from what they actually read, rather than from what the module map says they should. Every finding names **both** modules, because a finding filed against one is a finding the other module's owner never reads.

> **The one thing to know before reading the rest**
>
> **The assets tables live in `06-backend/apps/assets`, a Django app of its own** — not in `apps/metering`, which is where `CLAUDE.md`'s module map puts them (*"metering … owns assets per B2 r4"*). Every `/assets*` route is declared in `apps/assets/api.py:305-327`; `apps/metering` mounts exactly one, the usage strip.
>
> This is not a breach and it is not new. The module's own docstring flags it as an **Eng hosting disposition** — *"Table names and FKs are unchanged by the hosting choice; writes remain metering-service-mediated per the epics"* — and says it was reported for the daily PO batch (`models.py:36-45`). What has not happened in the year since is the module map absorbing it, so a lane reading `CLAUDE.md` alone goes looking in the wrong directory. Recorded here as a description; nobody has ruled that the map should change.

1 · Does the handoff land?

From	To	The shared fact	Verdict
assets	metering	A unit's identity and its premises, for every *Assigned to* cell, every address and every map pin	**Lands.** Through `seams.unit_ref()`, which returns a dict.
assets	metering	The final read, captured atomically with a move	**Lands, and refuses correctly.** Metering commits the read in the caller's transaction and returns its dict; the closing assignment takes `final_read_id`. When metering cannot capture, the move is refused rather than half-done.
assets	customers	The active account on the unit — the live billing chain	**Lands.** Was unwired while customers owned no tables and the chain *honestly ended at the unit*; the account leg is served now and screen 08 draws it.
assets	billing	Whether an invoice line references this asset, for the delete gate	**Lands.** A probe, and the server re-runs it at delete time rather than trusting the action set it served.

assets	work	`work_orders.asset_id` — read for the Maintenance panel's Work-order column, written by Work's own endpoint	**Lands one way and not the other.** The read works. The *write* is reachable from five places, and two of the four `BR-WRK-056` names — both on the asset's own screen — do not exist. This is the module's largest finding and it is a **cross-module** one: the rule is Work's, the missing controls are on an Assets screen, and neither owner's tests cover the join.
assets	integrations	The basemap tile host, from the page policy	**Lands.** Worth naming rather than leaving implicit: every pan on screen 06 sends the viewport around a tenant's asset locations to whichever host that row names. On the shipped default that is Esri, and the egress was **accepted** by ruling (§10-D6) rather than overlooked.
assets	identity	The technician list for screen 23	**Does not land for the role that needs it.** `GET /users` is System-Admin-only; `BR-AST-038` lets a Billing Clerk log a repair. The seam is fine — the *permission* on the far side makes it unusable for half its callers. See `AST-B-04`.
assets	tenancy	The asset number series, and the export kernel	**Lands.** A meter number is allocated from the tenant's series and never taken from an uploaded file (`BR-AST-049`).

2 · Does any module write another's tables?

> **No. Checked, not assumed.**
>
> Searched `06-backend/apps/assets/*.py` for `.objects.create`, `.save()`, `.update(` and `.delete()` against any foreign module's models: **no hits**. The one place assets could plausibly write across — a service point on a unit — goes through `metering.services.ensure_service_point()`, so metering creates its own row. And the Raise-work-order screen, which lives under `features/assets/` in the front end, posts to `POST /work-orders`: `apps/assets` gains not one line of code for that feature, and a test asserts it by name.

3 · Does one fact have two homes?

> **No. The address rule holds here, and it was checked field by field.**
>
> `Asset` was read column by column (`models.py:229-410`): there is no `address`, no `city`, no `state`, no `postal_code`. The nearest thing is `location_note`, free text under 280 characters — *where a crew will find it* — which is a note, not a record. `assigned_address`, which appears on the wire and in three field tables above, is **derived**: `seams.unit_ref(...)["premise_address"]`, which walks to `premises`. A screen and an envelope cannot disagree, because there is only one place to disagree with.

4 · Does the same thing have two names?

Pair	Verdict
Retire vs **Delete**	**Two names, two things, deliberately.** They sit in the same cell with different glyphs, and one icon for both would be a real ambiguity. Every refusal that blocks a delete names retire as the alternative.
Asset class vs **Asset type**	**Two names, two things** — the three-way partition versus the catalogue entry. Both appear on screen 28, one above the other, and a reader can tell them apart.
Asset # vs **Serial #** vs **External reference**	**Three names, three facts, and the product argues for it.** Ours, the manufacturer's, and the utility's old one. `models.py:239-256` spends twenty lines on why overloading them would be wrong the import column guidance says the same thing in a sentence a migrator can read. The model *"would make a column that holds either and therefore means neither."*
Parent vs **Master**	**One thing, two words, on purpose.** The column reads *Parent / master* — the schema calls it a parent and a clerk calls it the master.
Recorded / **In the books**	B14 rule 34's ratified pair. **Not at issue here** — this module never posts to the books, so the second half of the pair never appears. The nearest thing, *"Nothing is posted to the ledger"*, is consistent with it.

Every seam, graded

Graded on what the seam **returns**, not on what its docstring says it does.

From → To	The file that crosses	What it hands back	Grade
assets → metering	`seams.unit_ref` · `service_point_ref` · `latest_read_for_asset` · `typed_reads_for_asset` · `enabled_service_codes` · `capture_final_read` · `read_count_for_asset` · `tenant_today` · `export_rollup_last_month`	Dicts, sets, ints, dates	HEALTHY
assets → metering	`seams.unit_row`	A **model instance**, not a dict	LEAKY by the letter — and it holds in practice: all three callers (`services.py:1041`, `:1183`, `:1835`) **discard the return value** and use it purely as a same-tenant existence probe. The object never escapes. Worth naming because the day a fourth caller keeps it, the boundary is gone and nothing will say so.

assets → customers	`seams.active_account_for_unit`	`dict | None`	HEALTHY
assets → billing	`seams.invoice_line_count_for_asset`	`int`	HEALTHY
assets → identity	`seams.actor_display_names`	`dict[id, name]` — names, never ids, which is what keeps B14 rule 31 true on screen 12	HEALTHY
assets → tenancy	`exports.py:41-42`, `lookups.py:261`	A base class and a service module, imported directly rather than through `seams.py`	LEAKY by the letter — it is a shared export kernel, which is closer to framework than to data, and no row of another module's is read or written through it. Named so it is a decision rather than an oversight.
assets → work	`features/work/workEndpoints.ts`, from a front-end file under `features/assets/`	Templates for an asset kind; a `POST` to Work's own endpoint	HEALTHY — and deliberately so: the file lives under `features/assets/` because that is where a clerk is standing, not because assets owns the write.
assets → integrations	`getPagePolicy()` in `AssetMapTab.tsx`	A policy dict carrying `tile_host`	HEALTHY

Eight crossings, six healthy, two leaky by the letter and neither leaking anything today, zero writing. For a module hosted outside the app the map assigns it to, that is a better result than the map would lead you to expect.

12

Every commitment, in one place

Every commitment, in one place

One row per specification, no exceptions. Each also appears as a card in the section that owns it. **If a thing is not on this table, nobody has undertaken to build it** — the descriptions elsewhere, however emphatic, commit no one.

Why the IDs look unfamiliar. `AST-B-##` is a new series opened for this document. The repository's live series are `US-AST-###`, `BR-AST-###`, `OQ-###` and `DEF-###`, and allocating into any of them would make this analysis look like scheduled work when it is not.

Three numbers are retired and not reused. `AST-B-07`, `AST-B-08` and `AST-B-09` covered the import runner, which left this document's scope on 2026-09-11. They are struck below rather than recycled, for the same reason this repository never renumbers an id: a citation written last week should not land on something else this week.

Status starts at Unverified for every row, and that is not a hedge. Engineering fills the column in from the code, one row at a time, and on a module this well documented several will come back already done.

ID	WHAT A PERSON COULD DO, AND WHAT IT WAITS ON	SCREENS	WHAT DONE LOOKS LIKE ON SCREEN	STATUS
AST-B-01	**The registry's KPI figures are served, not counted on the page.** Waits on: Nothing but the work — a rollup endpoint per tab, or the count on the list envelope.	01, 02, 03, 04	The Meters tab reads **Meters in service 123 · Needs repair 4 · In inventory 18 · Past certification age 2** on first paint, and the same four figures after paging to page 7 — today they are counted over the loaded page only and show four em dashes until every row is in.	UNVERIFIED
AST-B-02	**Raising a work order from the asset, from the controls the rule already names.** Waits on: `BR-WRK-056` is CONFIRMED and names both controls. Two of its four ways in were never built, and the raise endpoint exists.	12, 27	On **AST-04412**'s **Work & maintenance** tab a **Raise work order** button sits beside the sub-tabs, and the empty state reads *"No open work · Raise one"* with the words as the control — today the tab has no way in at all and the screen is reached by typing its URL.	UNVERIFIED
AST-B-03	**The utility service is picked from the tenant's enabled list, not typed.** Waits on: Nothing. `GET /settings/services` is `AnyTenantRole` and already serves the enabled list; the wizard	28	Step 1 of **Add an asset** shows **Utility service** as a dropdown reading **Water**, offering only the services this tenant has enabled — today it is a plain text box and a mistyped code is refused by the	UNVERIFIED

	simply never calls it.		server after the person has moved on.	
AST-B-04	**A billing clerk logging a repair can name the technician without a UUID.** Waits on: A role-safe read of the crew. `GET /users` is `SystemAdminOnly` while `BR-AST-038` lets a Billing Clerk log a repair.	23	Signed in as a **Billing Clerk**, **Log repair / maintenance** shows **Technician** as a searchable list reading **Sam Reyes · sam.reyes@brazosbend.gov** — today that role gets a text box and the help *"Your role can't list users, so enter the technician's ID."*	UNVERIFIED
AST-B-05	**A citation that points at a screen that exists.** Waits on: Nothing but the edit. The badge names `WRK-19` against an explicit ruling that the route is `WRK-24`.	27	Nothing changes on screen — and that is the point. Checkable by grep: `rg "WRK-19" 07-frontend/src` returns nothing.	UNVERIFIED
AST-B-06	**The Map tab's dropdowns look like the product's.** Waits on: Nothing. The rule exists and the class is one word.	06	Rendering the Map tab's filter bar outside any `.view` wrapper still draws the product's own chevron — today the four selects compute `appearance: none` only because an ancestor happens to supply it.	UNVERIFIED
~~AST-B-07~~ WITHDRAWN	~~The import refusal's stale copy~~ Retired 2026-09-11 with the import runner, which left this document's scope by instruction. **Do not put this in a build plan.**	—	—	—
~~AST-B-08~~ WITHDRAWN	~~Reopening a finished import run~~ Retired 2026-09-11 with the import runner. **Do not put this in a build plan.**	—	—	—
~~AST-B-09~~ WITHDRAWN	~~Three import sources that do nothing~~ Retired 2026-09-11 with the import runner. **Do not put this in a build plan.**	—	—	—
AST-B-10	**A clerk told no template applies is not left on a dead submit.** Waits on: Nothing. The refusal copy exists; the control does not read it.	27	When no template applies to the asset's kind, **Raise work order** shows the sentence *"No template applies to this kind of asset"* and the submit is **disabled carrying that reason**, not enabled over nothing.	UNVERIFIED
AST-B-11	**What is falling due can be sorted and searched.**	05a, 05b, 05c	Opening **Lifecycle · Certifications & tests due**	UNVERIFIED

	Waits on: Nothing but the work — the three lifecycle panels are served lists with no toolkit.		shows search, filter, a column selector and a download over the list, and sorting by **Due** puts **11 days overdue** at the top.	D
AST-B-12	**An asset can be corrected from the asset, at an address.** Waits on: A route. `PATCH /assets/{id}` already validates a partial payload.	`08, 14`	Opening **E-5000** shows an **Edit** control in the header; pressing it lands on `/assets/{id}/edit`, which can be bookmarked and returned to.	UNVERIFIED
AST-B-13	**Every hint says which input it belongs to.** Waits on: Nothing. It is R12's field order applied to the ten fields that render it the other way round.	`14, 16–21, 23`	On **Retire E-5000**, *"Cannot be before the install date"* sits **under** the date it governs, not above it — so the eye meets label, input, then hint, on all ten fields that render the as-built order today.	UNVERIFIED
AST-B-14	**A meter booked in can record the reading it was installed at.** Waits on: Nothing. The column exists and the service takes it.	`29`	Step 2 of **Add an asset** shows **Opening read** as an enabled field reading **12,481**, and the first bill computed from this meter starts there instead of at zero.	UNVERIFIED
AST-B-15	**Every screen in the list says what it is for, in a line a clerk can read.** Waits on: Nothing. The slot exists in `PageHeader` and two tabs pass nothing to it.	`01, 02, 03, 04, 05a, 05b, 05c, 06`	Opening **Assets** shows *"Meters, where they sit, and their status."* under the tab strip — **six to eight words** — where two tabs render nothing today and the other four run to 9, 18, 23 and 24 words.	UNVERIFIED
AST-B-16	**The two facts the Overview shows that no column serves.** Waits on: A decision about where each is computed. Neither rides on `asset_dict()`'s 38 keys.	`08`	Opening **BF-0001** shows **Lifetime maintenance $1,284.00** and **Open work orders 2** in the **Cost & value** and **What's due** panels, both served rather than computed in the browser from a list the page happens to hold.	UNVERIFIED
~~AST-B-17~~ WITHDRAWN	~~A registry-wide Work orders tab~~ Withdrawn 2026-09-11, the day it was drawn, by instruction — *"07 Work orders- remove this wireframe"*. The number is struck and not reused. **Do not put this in a build plan.**	—	—	—
AST-B-18	**The asset's work list carries the fields the wire already sends.** Waits on: Nothing on the server. `GET /assets/{id}/work-`	`12`	The **Open work** sub-tab shows **Priority**, **Due** and **Assigned to** as columns with values — today those three columns are drawn and empty on every	UNVERIFIED

	`orders` serves **29 fields per order** and the client type declares **four**.		row because the client type narrows the payload to `id · order_no · type · state`.
AST-B-19	**The three lists inside an asset page on the server.** Waits on: Nothing but the work. Both endpoints return a bare list and fold the whole history in Python.	09, 11, 12	The **Reads** panel carries the standard footer — **1–50 of 1,460**, a rows-per-page control and first / previous / next / last — and opening a meter with 1,460 reads paints in one page rather than all of them.	UNVERIFIED
AST-B-20	**The export a person presses actually downloads a file.** Waits on: A volume the worker and the web container share, or a store both can reach. The runbook's fix is gated on `desiredCount > 1`, which never fires on this stack.	01, 02, 03, 04	Pressing **Download** on the Meters tab saves **meters-2026-09-11.csv** — today it saves **file.json** containing a 404, because the worker writes the artifact into a container the web process cannot read.	UNVERIFIED
AST-B-21	**One asset, one cost — the two that exist today never meet.** Waits on: A decision about which cost an asset shows, and where the arithmetic lives. `work_order_cost_lines` is the one place it lives and it reaches no asset.	12	**Work & maintenance** shows **Lifetime cost $1,368.00** with both halves named beneath it — **maintenance $1,284.00 · work orders $84.00** — where today the asset sums only its own maintenance rows and the 740 costed jobs are invisible from here.	UNVERIFIED
AST-B-22	**The Overview header carries what the asset needs done.** Waits on: Nothing. Every field is on the wire and the attention strip is five served fields and a condition.	08	Opening **BF-0001** shows *"⚠ Annual test overdue 11 days · due 06/15/2026"* under the title with a **Raise work order** control beside it, so a person acting on the asset meets the overdue test before they act.	UNVERIFIED
AST-B-23	**The three lists inside an asset use the shared table and its toolkit.** Waits on: Nothing. `DataTable` and the toolbar exist and are used by the registry.	09, 11, 12	The **Assignment history** panel carries search, a filter, a column selector and a download, and searching **Main St** narrows 14 rows to 3 — today it is a hand-built table with none of the four.	UNVERIFIED
AST-B-24	**The reads table pages on the server.** Waits on: The same envelope `AST-B-19` asks for; this is its front half.	09	The **Reads** tab shows **1–50 of 1,460** and a working **next**, where today every read this meter has ever taken arrives in one response.	UNVERIFIED
AST-B-25	**Each Overview pencil opens a dialog that edits only its own panel.**	16, 17, 18, 19, 20, 21	Pressing the pencil on **Cost & value** opens a dialog headed **M-4022 · METER · AMR** with four	UNVERIFIED

	Waits on: Nothing on the server. `PATCH /assets/{id}` validates a PARTIAL payload already.		fields and nothing else, and saving writes only those four — two of them, **external reference** and **purchase date**, are typable for the first time.	
AST-B-26	**A helper line is eight words or fewer.** Waits on: Nothing. It is copy, and it is the cause of the field misalignment rather than a separate defect.	`14, 16–21, 22–27, 28–32`	On **Reassign**, the two inputs of a row sit on **one baseline**, because no helper under either of them wraps to a second line — measured across the module, **27 of 37 helper strings run over ten words** and the worst is 36.	UNVERIFIED
AST-B-27	**The columns an asset holds that nothing can write.** Waits on: A decision about which belong on which step. `POST /assets` and `PATCH /assets/{id}` both already accept all of them.	`28, 29, 30, 31, and the edit dialogs`	Adding a sub-meter asks for its **allocation share 33.3%**, so the sub-meter that comes out of the wizard can actually be allocated — today it is created *metered, not allocated* and **no screen in the product can ever allocate it**.	UNVERIFIED
AST-B-28	**Save as draft saves a draft.** Waits on: Nothing new. `BR-SHL-048` already puts a half-finished wizard in `wizard_parks`, and five wizards run on it.	`24, 25, 26, 28–32, 33`	A clerk called away mid-install presses **Save as draft**, sees **Drafts · 1** in the registry header, returns on Thursday, presses **Resume** and lands on **step 2 of 5** with every field as they left it — and the registry shows no half-made meter at any point.	UNVERIFIED

Twelve live specifications. Two — `AST-B-10` and part of `AST-B-04` — wait on a decision or a new read; the other ten wait on nothing at all: the reads exist, the routes exist, and what is missing is a control, a string, a component swap or the order of two elements. That distribution is itself a finding — this module is not short of capability, it is short of the last connection between capability and screen, which is the second hard-won law of this repository, several times over.

Commitments raised in this section: none of their own. It is an index.

13

What this pass changed, line by line

This is the third version. It exists so a reader of an earlier one knows what moved and, more importantly, what deliberately did not — including the places where the first version was wrong.

What the forty-seven instructions of 2026-09-11 changed

What the forty-seven instructions of 2026-05-11 changed

Instruction	What it did to this document
R11 · one spacing scale, with its anchors *“the components are too much close as well as manage the margins also”*	Every wireframe in this document was rebuilt on the scale — 4 · 8 · 12 · 16 · 20 · 24 · 32 — with the ad-hoc values from v1 removed and **three product literals kept deliberately and labelled as such** in the stylesheet: the rail item's `9px 12px`, the pill's `2px 9px`, and the button's `calc(--sp-3 × 1.2)`, which the product itself derives. The ten anchors were then re-measured on the running app: **eight hit the scale exactly**, and the two that miss are the first two of those literals. Both are now a row in section 09.
R12 · label, then input, then helper	The rhythm was already right in v1's CSS and **wrong in the drawings**, in the direction that matters: v1 tidied several as-built fields into the correct order. Measuring eighteen real fields found **ten rendering** `label → helper → input`, at 29px or 47px from label to input instead of 5. The as-built drawings now show the violation, section 09 names it, and **R4** draws the fix. This is the largest single correction between the two versions.
Screen 05 · Lifecycle as three tabs *“create these as tabs - add tables, toolkit”*	Three new screens — **05a, 05b, 05c** — each with the product's own table toolkit: search, filter chips, a column chooser, sortable headers and a pagination footer. Screen 05 keeps the as-built so the carry-over can reconcile, and each of the three carries its own. New specification `AST-B-11`.
Screen 14 · Edit as a form, not a modal	Redrawn as a full page at `/assets/{id}/edit`, with R12's rhythm and R13's two-to-a-row grid, plus a second door from the asset's own header — which the dialog could never have. New specification `AST-B-12`. It is also the only **REMOVED** row in the whole document: the dialog frame, which is a container rather than a field.
Screen 28 · Service from Settings › Services & UoM	Named the option source precisely in `AST-B-03` and in R3's legend. The specification already existed; the instruction settled where the list comes from.
Screen 29 · make the opening read enabled	**This overturned a ruling recorded in v1.** Section 08 had it **out of scope** on a lane instruction about a missing seam. It is now specification `AST-B-14`, the old reasoning is struck rather than deleted, and R5 draws it — including the finding that `capture_final_read` is a **precedent** for exactly this write rather than the blocker it was taken for.
Remove screen 14 and everything import	The whole import section, its five runner stages and the refusal screen are gone; the Imports tab is off the registry's tab strip, which now draws **six**. Three specifications retired with it, and their numbers are **struck and not reused**. All import rows left the absence register, the cross-module table and the appendix. Section 02 says on its own face that the product draws seven tabs and this document draws six, so the gap reads as scope rather than as a defect.
Screens 30, 30, 16 · date pickers *“install date should date picker field”*	Install date, purchase date, warranty end, retire date and final read date are drawn as date fields with the calendar affordance, on screens **18, 19, 20, 21 and 22** — nine fields in all. The as-built already renders `<input type="date">`; what it renders **without** is a Today shortcut and any `min`/`max`, measured at **0** of **5**, and the remedy drawings carry both.
Put data in the KPIs	The strip was re-measured with the whole result set on one page and the drawings now carry real figures: Meters **103 · 0 · 6 ·**

“on screen where you ahve KPIs - please have data in it”	**0**, Behind the meter **7 · 0 · 780 kWh**, Inventory **8 · $1,511.00 · 4**. **The dash state is still drawn**, as a second labelled block, because it is what a clerk sees on the default page size and under any filter — showing only the good state would have made the document disagree with the product. **Three v1 figures were fabricated and are corrected in this version**; the row below says which.
Descriptions, six to eight words *“as part of toolkit we have descriptions 6 to 8 words”*	Every screen's description slot was counted on the running app. **None of the six registry tabs is inside the budget**: four render 9, 18, 23 and 24 words and **two render nothing at all** while their strings sit translated in eleven catalogues. New remedy **R6** and new specification `AST-B-15`; a violation row in section 09 counts all six.
05c · remove the Asset value panel	Struck from the drawing, from the field table (the only **REMOVED** health verdict in the document) and from the carry-over, with the instruction quoted in appendix K. **What it costs is recorded rather than assumed**: the panel was itself the ruled replacement for r10's *Cost to Finance* sidebar, cut by §10-D3, so removing it leaves the tenant's book value stated nowhere in the module.
01–06 · one table, one toolkit, one description *“ensure youa re using consistent tables , tooklit and descriptions”*	The six list screens now render from **one** toolbar, table, row, pager and KPI helper, so a column width or a chip cannot drift between them, and every screen's spec table carries the **same fifteen rows in the same order**. Where the product itself is inconsistent the drawing keeps the inconsistency and says why on its face: **Inventory and Lifecycle have no toolbar and no pager** because they are the detail-panel table rather than the `DataTable` — *“nothing to search, sort or page on five aggregate lines”*, the source's own words. Verified in a browser rather than asserted: six spec tables compared row by row, and the page measured at seven widths from 1920 to 400 with **zero** horizontal overflow and **zero** page errors.
A Work tab inside the asset *“we willneed Work {work orer} tab in details view before audit tab”*	Added as screen **15**, between Maintenance & repairs and Audit, **on all three tab sets** — six tabs on a meter, six on an exporting inverter, five on a plain device. It costs **nothing on the server**: the page already calls `GET /assets/{id}/work-orders` on every asset, the wire already carries twenty-nine fields per order, and the client type declares four. New specification `AST-B-18`, and `AST-B-16` for the two columns no column serves.
Three hidden facts onto the Overview	**External reference** beside the serial, **purchase date** beside the purchase cost, and export capacity named on the inverter variant — each stored, each served, each drawn nowhere until now. The four that stay documented rather than drawn — depreciation method, allocation rule, loss tolerance, book value and treatment — have their own panel on screen **11** saying so, because each needs a ruling about what it is FOR before it earns a line.
A registry-wide Work orders tab *“After map tab add Work order tab”*, then *“07 Work orders- remove this wireframe”*	Drawn as a seventh registry tab and **withdrawn the same day**. The tab strip is six again, and everything from 08 up closed the gap. Its specification number, `AST-B-17`, is **struck and not reused** — the same rule the three import numbers were retired under, and for the same reason: a citation written today must not land on something else next week. **The finding it carried survives** and is worth keeping on the record: `GET /work-orders` gives that endpoint six filters and a **400 on a seventh**, so a tab like this could never have been a query-string change.
Pagination on the reads table *“if pagination of read table has not added please add”*	It had not. **Checked at four layers and absent from every one** — the endpoint returns a bare list, the selector folds the whole history in Python, the seam applies no cap and the component renders `rows.map()` with no slice. A pager is drawn on screen **11** marked NEW, and the finding is now specification `AST-B-19`. It is the worst `NEED` a breach in the module, because

added please add"	screen **11** marked NEW, and the finding is new specification `AST-B-19`. It is the worst `NFR-3` breach in the module, because the four that `CLAUDE.md` already names are bounded by debtors or cheques and a read history is bounded only by time × read frequency.
A panel I was about to delete, wrongly	While correcting the reads table I searched the Reads tab for a consumption chart, found none, and concluded the *Consumption · last 12 months* panel in this document was invented. **It is real** — `AssetUsageStrip`, mounted at `AssetDetailPage.tsx:1108`. My search looked for the words *chart*, *Consumption* and `KPIStrip` inside the tab and the component is named none of those. **The drawing was right and the check was wrong**; what was actually wrong were its three tile labels, now corrected to the product's own — **Avg / month · Peak month · Low month** — along with the table's five columns and both empty-state sentences.
The export is broken on the deployed stack *"Export is not working showing error message file joson is not available"*	Reported live, traced to a **missing volume** rather than to any application code: production runs the job on the `celery` container and serves the download from `backend`, and `docker-compose.yml` mounts **no volume on any service**. Recorded on screen **01** with the mechanism drawn, as new specification `AST-B-20`. **Two other downloads break identically and had not been reported** — the bill print set and the audit-log export.
Two KPI strips on one screen	Screen 01 drew the strip twice — once with figures and once all dashes — to show both states. On a document somebody builds from, that reads as a duplicated component. **One strip now**, with figures; the dash behaviour is stated in the screen's *States* row and counted in section 08, where a defect belongs.
Screens rearranged by what they act on *"I dont know why 07 08 screen in 02 list? ... we need Details view of assets, all tabs screen, Add, Update assets"*	**Edit** and **Delete** left the list section. They act on **one** asset, and a section called *The list* should hold only what a person does to the table. The add wizard joined them, so section **03** is now the whole life of one asset in one place — **the seven detail tabs, then the four steps that put one in, then the two that change and remove it** — under two group headings so the three runs read apart. **Bulk assignment stayed with the list**, deliberately: it begins by ticking rows in the table, so it is a thing you do to the list rather than to an asset. *Putting one in* is no longer a section of its own and the sections after it each moved up one; the whole document renumbered, and this was a permutation rather than a shift — **07 became 19 while 19 became 17** — so the renumberer applies every pair in one guarded pass instead of in sequence.
Screen 08 gains its rules and its interaction design *"we should document this as rules below that screen so that we cna ensure those are applied dueing code changes"*	Two blocks under the Overview. **Thirteen rules**, `AST-R-01–13`, a new series minted here because no rule set in this repository covers a screen whose shape changes with the kind of thing it shows — and each one carries **how it is checked**. **Ten of the thirteen are assertions about the DOM**, which matters: this repository's standing weakness is that no gate reads rendered output, and `vitest` already renders this screen, so the enforcement that cannot exist as a gate exists as a test file. Two of them are three lines each and would have caught faults this document found by reading. Then **seven interaction decisions** with what each costs — six are client-only and the seventh is catalogue copy.
Maintenance and Work become one tab *"we should merge these two tabs"*	Screen **12 · Work & maintenance**, in two parts — **Open work** first because it is the half you can act on, then **History**, one date-ordered list where every row states what kind of record it is. **The tabs merge and the records do not**, because the link between a repair and a job is a **date match** and `BR-WRK-056` forbids making it a key; the alternatives were already considered and rejected in `apps/work/asset_door.py`, and one of them — the same order stamped on every row — *"reads as this repair came from that job for every repair at once, which is a per-row claim the data cannot support."* Every kind loses

	a tab: five on a meter, four on a plain device. The raise door survives, because `BR-WRK-056` names the maintenance panel by function rather than by tab label.
One asset, two costs, neither aware of the other	Found while checking whether the merge was safe, and it is the strongest argument for it. **What has this meter cost to keep running** is computed twice: `asset_maintenance.cost`, a number a clerk types, summed into the asset's *Lifetime cost* — and `work_order_cost_lines()`, labour plus parts under `BR-WRK-051`, *"the ONE place the arithmetic lives"*, totalled on Work's performance tab at **740 jobs · $56,337** and **reaching the asset nowhere**. So a meter whose repairs all went through work orders shows a lifetime cost that misses every one of them, and the better the utility uses work orders the wronger the number gets. **Cheap to fix, because neither figure posts anywhere** — both are captured and reach no journal line by ruling, so this is a display problem rather than a ledger one. New specification `AST-B-21`. ⚠ **And it corrects this document:** an earlier version said Work carries no cost. It does.
Screen 08 redrawn to the six-panel architecture	The header in five slots, the six panels in reading order, and **eight header worked examples** — installed, inventory, sub-meter, an overdue test, a master, an expired warranty, retired, and the one that matters most: **nothing due, and no strip at all**. Then two kind variants drawn rather than described: a **sub-meter**, whose Measurement panel gains the three allocation rows no other kind has, and a **plain device**, which has **no Measurement panel** because it stores no reading. The carry-over reconciles all **38 distinct facts** — 40 rendered rows less the two duplications — with **nothing removed** and seven rows added that were already on the wire. Worst panel drops from **23 rows to 9**. New specification `AST-B-22`, and it waits on nothing: no endpoint, no column, no query.
Work & maintenance gets two sub-tabs *"we will keep them seperate under 1 tab - you can have 2 sub tabs under it"*	**Open work** and **History**, as a second-level strip inside the tab rather than two stacked panels — the same shape the Lifecycle registry tab needs, and URL state like every other tab, so either half can be linked to. Open work is the default: it is the half you can act on.
The three lists inside an asset get the shared toolkit	Reads, Assignment history and Work & maintenance now carry the same toolbar a registry tab carries — search, filter chips, ⤓ **Download**, ▥ **Columns** and a pager — and all three headers are the five-slot header from screen 08. Work & maintenance carries it on **both** sub-tabs. New specification `AST-B-23`. ⚠ **The download inherits `AST-B-20`**: the export is broken on the deployed stack for want of a shared volume, so adding a download to three more panels adds three more places it fails. The control is drawn because it is what the screen should have; the dependency is stated so it is not discovered afterwards.
Variation by kind, drawn rather than described	Every one of the three lists now carries a **Variation by kind** row, and **two of the three say "None. Checked, not assumed."** — which is worth as much as the one that varies, because an absent heading reads as an unasked question. Reads has four variations and three of them are now **drawn**: a flat meter, a **net meter** with its conditional sixth column, and a **multi-register meter** whose four rows share one date with nothing saying which register each is. The fourth — a plain device with no Reads tab at all — is screen 08's device variant, and the inverter's Export tab is screen 10. New specification `AST-B-24`.

	`AST-B-24`.
Two screens were drawing a tab strip that no longer exists	Screens **10** and **13** still carried the pre-merge six-tab strip, **10** showing both *Tests & maintenance* and *Work*. Found by the PO reading the document, not by any check here. **The shape of the mistake is worth keeping:** a tab strip appears on *every* screen of a tab set, so changing that set touches screens that have nothing else to do with the change — and I regenerated the drawings that changed rather than the ones that merely contain the strip.
A new section: Asset edits *"on overview we have added pencils, if user click on those pencil, we need to give them modal or form"*	Section **04**, six screens, one per editable panel, plus the state in which there are none. **It costs nothing on the server** — `PATCH /assets/{id}` validates a PARTIAL payload already, so a form that sends only its own panel's fields is the shape the endpoint expects. New specification `AST-B-25`, and two columns become typable for the first time: **external reference** and **purchase date**, both served on every row today with no surface in the product writing either.
Page or dialog, stated as a rule	*A PAGE when you are creating something, or when the change moves where the thing lives; a MODAL when you are correcting a value on something you are already looking at.* It settles all four surfaces — the add wizard and Reassign and Retire are pages, the full edit stays a page on the PO's own ruling, and a two-to-nine field correction is a dialog. **The construction does not change either way**: B14 rule 38 gives the UI rules authority over how a form is built, so R12's rhythm and R13's grid apply to a dialog exactly as to a page.
Connection size moves panel, and not for tidiness	It is one of the **five billing-affecting fields** (`multiplier · unit · net_metering · service · size_class`), and it was drawn in **Device**. Moved to **Measurement**, so all five sit in one panel, **exactly one pencil on the screen can restate billing**, and one confirmation covers it. Left where it was, two panels would each need the same warning and a person could meet it from either. **The Assignment panel also loses its pencil** — premises, unit and account are not typed there; they move through Reassign, which mints a service point and takes a final read.
Section 05 becomes Actions, and its six screens are audited *"these are all action ... rename the section 05 to actions"*	Renamed, no renumbering. Each of the six now carries what it breaks and what it captures. **Three faults are shared by all six** — R12's field order on ten of eighteen fields, five date inputs with **zero** Today shortcuts and **zero** bounds, and none of them carrying screen 08's header, so a person acting on an asset whose test is overdue sees no sign of it while they act. Per screen: **Reassign** commits three things as one transaction and warns about none of them, while **the copy for the warning already exists unrendered**; the **retire wizard** leaves `WizardFrame`'s **Save & exit** and **Park** unset while six wizards elsewhere pass them; **Raise work order** carries four faults already filed and renders the one R11 title outlier.
The data-capture audit, and three wrong assumptions	Every action form was checked field by field against the service it posts to. **Reassign, Log repair and Retire have no gap at all** — reassign even offers *Inventory (unassign)* and a work-order checkbox, log repair captures cost, retire takes all six keys. **This document nearly filed "repair cost is never captured"**, which is false; the first read of the component stopped short of the field. **The one real gap is on Raise work order and it is a missing sentence rather than a missing input:** the raise endpoint accepts no priority, due date or assignee because scheduling belongs to Work — and nothing on either screen says so, which is why the asset's Work tab draws two columns that are empty on every freshly raised job. ⚠ **The lesson is the ratio:** four assumptions about missing fields, three of them wrong, each corrected by one more grep. A rules-compliance list makes a screen look worse than it is, and only reading the form settles it.

The edit dialogs name the asset they are editing *“you will need a asset identifier in header of that model”*	All six carry a kicker above the title — **M-4022 · METER · AMR** — the kind and the user number, never the uuid (`BR-AST-026`, B14 rule 31). A dialog that does not say what it is editing is a dialog you can be wrong in, and screen 17 can restate billing.
All six action screens redrawn — and a fourth wrong assumption corrected	Screens 22 to 31 carried the analysis and **not the drawings**; the PO read the document and said so. Each now carries its redrawn wireframe above the spec table — the five-slot header with its attention strip, R12's field order, bounded dates — plus **two new ones**: Reassign's **confirm step**, and Raise work order as **one form** at a 32px title with the badge the registry actually says. ⚠ **And the fourth wrong assumption:** this document recorded *“the retire wizard has no Cancel”*. It is wrong. `WizardFrame` offers **Save & exit** and **Park** — a whole parking system with a reason, a note and a remind-on date — through an **optional** `exit` prop; **six wizards pass it, all of them in customers or billing, and both asset wizards pass none.** The corrected finding is worth more than the original: nothing can catch an unset optional prop, because that is what an optional prop is for.
Inputs aligned, dates as pickers, choice lists as selects *“ensure - inputs filed are align , date are date picture, drop down should be drop downs”*	The field helper gained a `sel` mode: a chevron pinned to the right edge of the control, rather than a ▾ typed into the value — **a typed chevron reads as text and a pinned one reads as a select**, which is what the product renders. Thirteen choice lists across the edit dialogs and the action screens now draw as controls; every date across both sections carries the calendar affordance with a Today shortcut and bounds; and the input box became a flex row so a value and its affordance sit on one baseline whatever the value's length.
A stray tag, and an editing bug that ate three drawings	Two mechanical faults found by the fit test rather than by reading. `<this asset>` in a refusal sentence was being parsed as an HTML tag — the same fault as `<month>` earlier, and the second of its kind. And the block-removal helper used to swap a drawing in place mis-terminated on three screens, deleting the `.pw-fit` wrapper and **leaving its contents behind unwrapped**, which put a 1440-wide step rail directly into the page and overflowed it by 5px at phone width. Both fixed, and the remover now matches balanced `<div>` tags properly. **Neither would have been visible by reading the document**; both came from measuring it.
Helper text was breaking the field alignment, and the two are one defect *“line is too big … becuase of this inout feld alignment broken”*	The PO read a drawing and saw two inputs of one row sitting on different baselines. **The cause is the copy.** A helper that wraps pushes its own input down, so in a two-to-a-row grid the row stops lining up — **the alignment defect and the copy defect are the same defect**, and shortening the sentence fixes both. Measured across the module: **27 of 37 helper strings run over ten words**, and the worst is **36**. Every helper on every redrawn screen is now **eight words or fewer**; the as-built drawings keep the long ones, because they are the product's own strings and that is the finding. New rule `AST-R-14` — the first one in the set that needs no rendering to check, since the copy is the defect — and new specification `AST-B-26`.
A header separation line, and the Kind control on its own row *“bring date beloe Repair … also header seperation line - check*	Two layout faults, and the first one generalises. **A detail screen gets the line under its header free**, from the tab strip's own bottom border — **an action screen has no tab strip, so it had nothing at all** and the form began straight under the place line. All seven action drawings now declare that rule. And the **Kind** segmented control takes its own full-width row with **Date** starting the next one: a segmented control is a different visual weight from an input, and pairing them in a two-to-a-row

header seperation line on all action screens"	grid leaves the row unbalanced.
A section for adding an asset, and the ten columns it never asks for	Section **06**, five screens — the four existing steps redrawn to every standard in this document, plus **Review & add**, **which has never been drawn here at all**: `WizardFrame` provides a final ✓ commit and every other wizard in the product gets one, so the four steps were drawn and the commit was not. Two class variants are drawn rather than described: a **sub-meter**, whose step 1 requires a Parent master, and a **behind-the-meter device**, whose step 2 loses the multiplier, unit, register digits and opening read because it stores no reading. Everything after moved up: 18→14, 20→16, 26→22, and the add steps to 28–31.
Columns with a reader and no writer	The wizard captures **26** of the asset's usable columns. **Ten it does not**, and **seven of those have no capture site anywhere in the front end** — external reference, form/class, seal #, location note, depreciation method, and a sub-meter's **allocation share, rule and loss tolerance**. They can be displayed and can never be entered. ⚠ **The allocation three are the sharpest:** a sub-meter created by this wizard is *metered, not allocated*, and there is no screen in the product that can ever allocate it. **This is hard-won law 2 in a third form** — not a dead control, not a stale refusal, but a column the model holds, the API serves, screens display, and nothing can write. No gate looks for it. New specification `AST-B-27`, and it needs no schema or endpoint change: `POST /assets` and `PATCH /assets/{id}` both already take them.
The utility service, mapped onto the add form *"you will need to map Utility service to this add form"*	**My omission, not a product gap** — the wizard asks for it on step 1 already, in `IdentityStep`, and this document's first drawing of that step left it out. It is drawn now, as a **picker** rather than the plain text box the product uses, with the *affects billing* pill it has earned. ⚠ **And there is a reason it belongs on step 1 rather than beside the other billing-affecting fields on step 2:** `offersConnectionSize(class, service)` decides whether **Connection size** appears at all — a meter always gets it, a sub-meter never, and **a behind-the-meter device only once a service has been entered**. On a device the service is what makes another field appear, so asking it on the step that shows that field would reshape the form under the person as they typed. The eleven services and the System-Admin-only read that lists them are `AST-B-03`, unchanged.
Manufacturer and model — two columns, two fields *"asset has manufacturer, model how you will capture this"*	Both are on step 2 now, on **every** class. The question surfaced a real inconsistency: the product captures these two columns **two different ways**. Adding a meter asks for **two fields**; the edit modal asks for **one**, labelled *Manufacturer / model*, with the help *"One line; separate the two with a middle dot."* ⚠ **That asks a human to do a parser's job** — a model name containing the delimiter, or a person reaching for a slash instead, lands in the wrong column, and nothing on the screen says which half is which. New rule `AST-R-15`: two columns are two fields; displaying them joined is fine and the detail view's single row is good density, but capture is capture. ⚠ **And the device step captures no model at all**, while the detail view will happily display one — one more column with a reader and no writer, narrowed to one kind.
The global search sat on the left, in all forty-seven drawings *"Global search Alignment is not correct , could you shift it to Right side update all screens in the doc"*	**The drawing was wrong; the product was always right.** `AppShell.css` gives `.shell-search` `margin-left:auto`, `flex:0 1 360px` and `min-width:140px` — pushed to the right of the bar and capped at **360px**. This document's own stylesheet drew it `flex:1` at **420px** with nothing pushing it, so it grew out of the breadcrumb and swallowed the middle of every top bar. Because the bar comes from one shared helper and one rule, the fix is **one declaration and it reaches all forty-seven drawings at once** — search now sits hard right, immediately left of the **?** and ⌂ actions, at the width the product uses. ⚠ **It**

	is worth noting how it was found and how it was not: the PO saw it on screen 28. The fit test measures overflow and page errors and passed on it at all seven widths, every time, because **a box in the wrong place is exactly as wide as a box in the right place** — which is the same blind spot the repository already has at large, that no gate reads rendered output.
Every drop-down now says where its options come from *"if theya re using any drop down and it is coming from settings or any other tables within system, please ensure you mention it as part of document with clear marker - at the time of coding it is otherwise not understood"*	Four markers, carried **on the field itself** rather than in a table beside it, across **32 choice controls**: `SET` a tenant-editable Settings list with its list code · `TBL` live rows from another module's table · `FIX` frozen in the source, with the count · `NONE` no list exists, or free text over a list that does. New rule `AST-R-16` and new **Appendix L**, which maps all nineteen distinct controls to the endpoint behind them. ⚠ **Reading the code to write it turned up four things the drawings could not have shown:** the **connection size** picker does not filter by service although every size class now names one (`DEF-3300`, filed, not decided); the **premises** and **unit** pickers stop at **500 rows** — a hard cap on a picker, which is the shape `NFR-5` exists to forbid, and premises 501 simply cannot be chosen; **test cadence** is frozen at `12 · 24 · 60`, so a regulator asking for 36 months needs a release; and **read methods** are served by **two different endpoints**, one minted for a role gate that has since widened.
A permission claim that was true a month ago — the fifth wrong assumption	`AST-B-03` said `GET /settings/services` is **System Admin only**, so a Billing Clerk could not be shown the services list. **It is `AnyTenantRole`** (`apps/tenancy/api.py:419`) and has been for some time. The claim came from the `exceptions:` footer of `AssetRegistryPage.tsx`, written 2026-08-10, which still says it — **the same footer Appendix A already names for contradicting the code above it**, now on its second bad claim to this document. **The specification survives and gets cheaper:** the gap was never the permission, it is that the add wizard renders `service` as a plain text box and never calls the endpoint. No column, no migration, no new route — one call. **Unit of measure is the same defect one field along**, and its options ride on the very same payload. ⚠ **The lesson is the one this document keeps relearning:** a permission is a fact about the CODE, and every time this pass took one from a comment it was wrong.
The draft button now saves a draft *"Keep save as draft button and make entry with status draft"*	Screens 28–32 carry **one** exit, **Save as draft**, and it writes a row. The two it replaces were each wrong on this form: **Save & exit** writes **nothing** — all six of its consumers in the product are a bare `navigate(…)` — and **Park** asks *what are you waiting on* from nine reasons that belong to the customers' world, none of which is why a meter install stalls. New screen **33 · Drafts**, because a draft with nowhere to land is a row nothing can reach. New specification `AST-B-28`. ⚠ **And where the draft goes was already ruled, which is why this is small:** `BR-SHL-048`, PO 2026-08-26 — *"Option A - Unitl move in executed, no records updates"* — puts a half-finished wizard in `wizard_parks` as a saved draft of a FORM, with **copied values and no foreign key**, and five wizards run on it today. **The work is a tuple of five becoming six**, the wizard passing an `exit` prop it currently passes none of, and the Drafts door. `waiting_on` is free text, so *status draft* is the constant `draft` and no column is added. ⚠ **The one thing that would need the PO to reverse themselves** is putting the draft in `assets` as a row: **128 places** key off `behavior_class`, `_validate_asset` would need a relaxed mode, the number would be taken at save leaving gaps in the series, and half-made assets would surface in every picker, rollup and parent-master list. Named, priced, not assumed.
The retire wizard gets the same treatment	Screens **24, 25 and 26** now carry the same single **Save as draft**, on the same store, landing on the same **Drafts** list —

treatment
“for now give same treatment”

whose second column is therefore the **ACTION** and not the asset kind, because one list holding an add with no asset and a retire with one cannot head that column *Kind*. **The flows tuple grows by two rather than one.** ⚠ **And retire differs from add in exactly one place, which is the resume:** an add draft describes something that does not exist, a retire draft describes something that **does** — and `BR-SHL-048 (a)` stores the subject as copied values with **no foreign key**, on purpose, so a forgotten draft can never hold a row still. Between saving and resuming, that asset can be reassigned, retired by somebody else through the row action, or deleted. So **resume re-reads the asset first**, and a draft whose asset is already terminal offers only **Discard** (`BR-AST-030`: there is no un-retire). ⚠ **The one cost worth naming is the opposite of the add's:** a forgotten add draft is nothing, while a forgotten retire draft means **a decommissioned meter goes on billing** — which is what the age column on the list is for.

The blueprint restructure — seven passes over the whole document
“MASTER PROMPT — RESTRUCTURE AN EXISTING HTML DOCUMENT INTO THE BLUEPRINT FORMAT … Do not delete anything.”

Nothing was deleted and every pass was applied to this same file. The document was first WALKED — **15 blocks, 65,765 words**, every section read from its current text rather than from its contents list. What that turned up, and what each pass did:

1 · Fact and promise, told apart by structure. A grey rail now runs the height of every section, drawn by `section::before`, and a commitment card pulls **left over it** with a negative margin so it visibly interrupts the description. Nothing is tagged paragraph by paragraph, so a section written next month inherits the distinction. Two legend panels and one bordered line say it in words.

2 · Every absence, one decision. The honest empties were **a paragraph, not rows** — eleven things listed inside a sentence, which is why nobody could count them. They are eleven rows now. A new block **E** carries the **7 absences found after the register was written**, four of them drop-down sources that no drawing could have shown. **49 stated absences · 7 carry a specification · 42 out of scope with a reason and a ruler named · nothing unresolved.**

3 · What done looks like on screen. All twenty existing cards already had one; the four new ones were written with the document's own figures in them.

4 · One complete register — and this was the largest gap in the document. §12 held **12 live rows against 24 live specifications**: half the commitments existed only as a card in the section that raised them, with nothing saying the list was half. **Four cards were missing entirely** (`AST-B-02 · 03 · 16 · 18`) and are minted. Both views are now generated from one file, `specs.py`, so they cannot disagree. The four withdrawn numbers get a dashed chip, a struck title, a rule down the edge and the sentence **“Do not put this in a build plan.”** Every status is **UNVERIFIED**.

5 · Staleness. The band said *“This is version three”* and pointed at the wrong section; it is **forty-eight recorded instructions** and §13. Added: what the warning covers and what it does not — **descriptions go stale, rulings do not** — and the branch check re-run and written down (`origin/HEAD` still does not resolve, **two** remote branches, neither a default, `package.json` still 2.234.17).

6 · The repairs, each one listed in the report. 56 figures numbered where none was numbered before. **Commitment chips on all 35 screens** — 31 covered, **4 reading NO COMMITMENT**, which is the finding. **Nine cards carried stale screen numbers** after two renumberings — `AST-B-03` pointed at an edit dialog, `AST-B-14` at the delete confirm, and `AST-B-26` read `28–15`, a range that cannot exist. **Four masthead figures were wrong** and one contradicted §13 outright (fields traced **174** against **137**, same page). The absence split was **two errors that cancelled** — B counted as 14 with 13 rows, D left out of the sum — which is why 42 looked right. **12 stray `</div>` tags** that closed nothing, left by an in-place swap, present in the published version and repaired here.

7 · The appendix. A did not contain this document's own sixteen rules — the one table claiming to be exhaustive was missing the rules the document itself minted. They are in, each marked INHERITED, EXTENDS or **NO HOME**: **nine have no home** and are lost the day this file is. **C was 34 rows for 35 screens. D's claim of "recorded nowhere else" is now verified rather than asserted** — grepped against the decision log and the module's business rules; four returned nothing, and the two that returned hits are a different subject wearing the same words, which is recorded beside the table.

⚠ **And one mistake of my own, worth recording because the repository already warned about it.** A repair script called `open(f,"w")` before the expression that failed, so **`n14a.html` was truncated to zero bytes** — the exact trap `CLAUDE.md`'s hard-won law 11 describes (*"the output file had ALREADY been truncated"*). Recovered from the published version 24, and every edit after it builds the new content in full and asserts it is sound **before touching the destination**.

Screens renumbered, and why that was the right call here

Removing the Imports tab and the runner left gaps at 07 and 27–32. This document **renumbered** to 01–25 rather than leaving holes — the opposite of what it does for a structural fault. The distinction is that a gap left by a *fault* must stay visible, because something was meant to be there; a gap left by a *scope cut* is just an absence, and carrying it would make every count read wrong forever. The mapping is one-for-one and monotonic (old 11→10 through old 26→25), the specification IDs did **not** move, and this paragraph is the record.

Findings this document made and then withdrew

The claim	Why it was made	What corrected it
"The Map tab uses four native browser dropdowns" — filed as a B14 rule-8 violation	A source read. The elements are raw `<select>`s and their class sets no `appearance`. Every word of that is true.	**Running the app.** All four compute to `appearance: none`. Asked the browser which rule matched: `.view select`, via an ancestor the component never asked for. **Withdrawn as a violation, re-filed as `AST-B-06`, a fragility** — a better finding than the one it replaced.

“Search does not reach the server”	A scripted probe typed into `input[type="search"].first()` and watched the URL not change.	**Reading the probe's own output.** There are two search boxes and `.first()` was the topbar's. Re-run correctly: `q=E-500` landed in the URL and returned 10 rows. **Search works.** A check whose output cannot distinguish a pass from a fail is worse than no check, because it is trusted.
“The wizard's Continue refuses silently”	No `disabled` attribute, and pressing it did not advance.	**Looking for the message instead of assuming its absence.** It renders *“This step isn't finished…”* in a `role="alert"`. **Correct behaviour.** The only residue is a missing `aria-disabled` — noted, not filed.
“A one-character search is silently dropped by the server”	The selector discards a term under two characters with no signal.	**Measuring it.** One character issues **no API call at all** — the client's own minimum stops it first. **Downgraded** to a note.
Several as-built form fields drawn with the helper in the right place	v1 drew what the rules ask for rather than what the screen does — the exact failure the process warns about, and the reason an as-built drawing is supposed to show the broken control.	**Measuring eighteen fields.** Ten are wrong in the product. The drawings now show it. **This is the correction that most changes what the document is for**: a tidied as-built would have left R4 with nothing to improve on.

What was measured rather than assumed

Spacing anchors on the registry	Screen body **24/16/32** · rail **240** · topbar **56** · tabs margin 16 gap 16 · KPI cell 16×20 gap 6 · `th`/`td` **8×12** · search box 6×10 min-height 36. **Eight of ten on the scale.**
Off-scale values, and their source	Rail item **9px 12px** with a **10px** icon gap; status pill **2px 9px**. Both are the product's own literals.
Field anatomy, 18 fields, 4 screens	**10** render `label → helper → input`; label-to-input **29px** or **47px**. The **8** with no helper are exactly **5px**. Field-to-field spacing comes from a grid `gap`, not a margin — correct.
Title size, 7 screens	**Two sizes, and not split by kind.** Registry, detail and both wizards **32px**; Raise work order **40px**.
Row-action controls per cell	**5** on installed rows, **3** on inventory rows, **0** overflow menus.
KPI dashes on the Meters tab	**4 of 4**, on a tenant of 126 assets.
Date inputs and Today shortcuts	**5** date inputs, **0** Today shortcuts, **0** with `min` or `max`.
Raise-work-order controls on the asset screen	**0** text matches, **0** links.
Failed network requests, whole module, both	**0**.

walks	
Address columns on `Asset` · foreign writes from `apps/assets`	**0** and **0**.

Structural faults reported and deliberately not repaired

The module map names the wrong directory	`CLAUDE.md` puts the assets tables under `metering`; the code hosts them in `apps/assets`. Reported in section 11, not corrected — this document does not amend the repository.
A retired screen badge is written into two files	`WRK-19` against an explicit ruling that this route is `WRK-24`. Reported as `AST-B-05`, left in place.
Three specification numbers are struck, not reused	`AST-B-07`–`09`. Recycling them would make a citation written last week land on something else.
A file's own notes contradict the file	`AssetRegistryPage.tsx`'s `exceptions:` footer states the behind-the-meter *Exported* card is *"always the stated dash"*; line 1000 of the same file renders the served rollup, and the running app shows **780 kWh**. Reported on screen 03, not corrected. **It is the sentence that produced version one's worst error**, and nothing in this repository checks a footer against the code above it.
A table row that was never measured	The R1 remedy drew a maintenance entry — *06/02/2026 · Repair · Replaced the register and resealed · Sam Reyes · WO-1042 · $84.00*. **None of it is real.** The database holds **zero** rows in `asset_maintenance` and **zero** work orders anchored to an asset, and none of those strings appears anywhere in the repository. Corrected: the drawing now carries its figures as illustrations (B14 rule 37) and a second row showing what a repair with no job on its date looks like. **The as-built drawing of the same table was right all along** — it draws the empty state, which is the truth.

Counts, and the arithmetic behind them

```
Addresses mounted under /:tenant, in scope       7   (the import runner is out)
  plus one proposed by this document             1   /assets/{id}/edit
Registry tabs drawn                              6   (the product draws 7)
Lifecycle sub-screens                            3   05a · 05b · 05c
Detail tab identities                            6   (no asset shows more than 5)
Wizard steps                                     7   add 4 + retire 3
Dialogs                                          2   delete · bulk assign
```

```
Screens numbered                              35   01 … 33, with 05a/05b/05c
Wireframes drawn                              56   a screen may carry an as-built AND a
                                                   redrawn figure, and seven carry a
                                                   class variant as well
  ⚠ THIS BLOCK READ "Drawings = … = 28  ✓ matches the masthead" AND BOTH
    HALVES WENT STALE TOGETHER. 28 was true before section 04 (six edit
    dialogs), the six redrawn action screens, section 06 (five add steps
    + two class variants) and screen 33 landed. The tick mark is the part
    worth noticing: it CHECKED one stale number against another.
Remedy drawings                                6   R1 · R2 · R3 · R4 · R5 · R6
                                                   (05a–05c and 07 are drawn in place)

Stated absences        49 = 17 cannot-do + 13 server refusals + 11 honest empties
                          + 1 piece of copy that reaches nobody + 7 found since
  of those, specifications    7      out of scope   42
  ⚠ THE TOTAL WAS RIGHT AND THE SPLIT WAS WRONG: B was counted as 14 and
    holds 13 rows, and D was left out of the sum entirely. Two errors that
    cancelled, which is why nobody caught it — 17+14+11 and 17+13+11+1 are
    both 42. Added up, not glanced at.
Specifications live    24 · withdrawn, not reused 4 (07 · 08 · 09 · 17)
                       REMOVED dispositions across all remedies: 2
  ⚠ READ 12 · 3 UNTIL THIS PASS. Twelve was the count when section 12 was
    written; sixteen more were minted in the sections that raised them and
    never reached the register. Both views are now generated from one list.
Rule violations        12
Journeys defined        6 · walked 6 · unwalked 0
Field tables           22, holding 137 rows
  ⚠ THE MASTHEAD SAID 174 AND THIS BLOCK SAID 137, for the same thing, on
    the same page. Counted: 22 tables, 137 rows. 137 stands.
Fields traced         137 · SERVED 114 · UNVERIFIED 9 · DERIVED 7
                          · NOT SERVED 1 · DEAD 1 · REMOVED 1 · DEGRADED 1 · not-a-field 2
  of those, crossing a module boundary   46
Cross-module crossings  8 · healthy 6 · leaky-by-the-letter 2 · writing 0
```

Nothing was deleted except what was instructed. The import material went on an explicit instruction and is named as gone; every finding this pass made is still here, including the five it withdrew — struck and answered rather than removed, so the same claim is not made again next quarter by somebody reading the same source files.

Appendix

Eleven tables, built from a section-by-section walk of this document rather than from memory. C, I and J1 are **extracted from the document's own markup by script** and not retyped, so they cannot drift from the screens above.

A · Every rule this document states, and where it lives

`INHERITED` — it exists elsewhere and this document follows it. `EXTENDS` — it adapts an existing rule to this context. `NO HOME` — stated only here. **Count the NO HOMEs: a rule stated only in a document nobody reads is a rule about to be broken.**

Rule	As stated here	Home	What it extends or restricts
B14, 38 rules	The UI authority; r10 is its source, not its authority	`INHERITED`	`04-docs/02-decisions/14-ui-conventions.md`
Hard rules 1–8	Tenant scoping · money at 4/6 stored and 2 shown · explicit units · financial immutability · balanced ledger pairs · idempotency · audit on every money action · one home for an address	`INHERITED`	`CLAUDE.md`
`NFR-3` · `NFR-5`	Universal server pagination; no hard caps anywhere	`INHERITED`	`CLAUDE.md`, load-bearing for T4/T3
`BR-AST-*`, `BR-WRK-056`, `BR-SHL-*`, `BR-MET-*`	Cited throughout, quoted where a finding rests on the wording	`INHERITED`	the modules' own `02-business-rules.md`
R11 · one spacing scale, 4/8/12/16/20/24/32, with ten measured anchors	Every wireframe here is built on it, and the two product literals that are not are labelled in the stylesheet rather than silently reproduced	`INHERITED`	Stated to this pass on 2026-09-11 with the PO's own words. ⚠ **Not located in `12-governance/` or `04-docs/` under that name** — the *values* are in `tokens.css`, the *anchors* are not written down anywhere a lane would find

			them.
R12 · label, then input, then helper, at 5 / 6 / 20	Judged against ten violating fields in section 09, drawn in R4	`INHERITED`	Same provenance as R11. It restates **B14 rule 10**'s neighbourhood but is more specific than anything B14 says — B14 covers the asterisk on a required label and not the order of the three parts.
Desktop only, 1440, scaled not reflowed · thirteen-item left rail · product literals pinned	Every drawing	`INHERITED`	R7 · R8 · R9 · R10 of the analysis process
New entries sort to the top — default ordering `created_at` descending	Cited as a product rule, and **breached**: the default is `asset_no` ascending and `created_at` is not in the sortable vocabulary at all	**`NO HOME`**	⚠ Searched `12-governance/` and `04-docs/`: this rule's wording is in neither. Until it is written where a lane will read it, no lane is wrong for not following it. Two sibling lists in this module (assignments, maintenance) *do* sort newest-first, so the intent is real and unrecorded.
A remedy is additive; a removal needs a quoted ruling	Sections 02 and 10, and appendix K	`INHERITED`	the analysis process, and `DEF-CUS-3653`'s general form
A description commits nobody; only a commitment card does	Stated on the document's own face	`INHERITED`	the analysis process
A seam that returns a model instance is leaky even where every caller discards it	Section 11's grading of `seams.unit_row`	**`EXTENDS`**	The inherited test is *"a seam that hands back a raw queryset is LEAKY even though it only reads."* A single model instance is not a queryset, and this document grades it the same

			way for the same reason — the caller can traverse and write through it.
A scope cut renumbers; a structural fault does not	Section 13's renumbering paragraph	`NO HOME`	⚠ The inherited rule says *report structural faults, don't silently repair them* and says nothing about a deliberate removal. This document distinguishes the two and says why. Stated only here.

The sixteen rules this document itself states (`AST-R-*`) — added in this pass. They were stated in section 01 and in the rule blocks under the screens they govern, and **not one was in this table**, which is the one place that claims to be exhaustive. **9 have NO HOME:** the day this document is lost, those 9 are lost with it.

`AST-R-01`	One template, every kind. The detail view is one component. The tab set and the panel set vary; the structure, the order and the panel names never do. This restates the rule the code already states in its own words — “ONE template, EVERY kind — the tab SET is what varies, the structure never does” — and extends it from tabs to panels.	**EXTENDS**	`assetDetailTabsFor()` already implements it; no rule names it.
`AST-R-02`	A panel or row a kind cannot have is ABSENT, not empty. Never draw a row of dashes for a fact the kind has no source for. The precedent is already in the product, on the Reads table: “a column of dashes on a meter that cannot export would read as measured nothing, which nobody did.”	**NO HOME**	Stated only here. B14 has no rule about a row a kind cannot have.
`AST-R-03`	Absence by KIND is silent; absence by DATA is stated. The two are different facts and the screen must not render them the same way. This kind has no register → the row is not drawn. This meter has no connection size recorded → the row IS drawn, with a dash, and where the absence has a consequence the consequence is named. The product already does the second half: “Not recorded — a rate ladder that prices by size cannot price this meter.”	**NO HOME**	Stated only here, and it is the distinction the whole Overview rests on.
`AST-R-04`	Measurement belongs to measuring kinds only. Multiplier, unit of measure, register digits and read method render on a meter and a sub-meter and nowhere else — the same reasoning that gives a plain device no Reads tab, applied one level down.	**EXTENDS**	`BR-AST-004`'s categories, carried onto what a panel may show.
`AST-R-05`	Allocation belongs to sub-meters only. Allocation share, allocation rule and loss tolerance render on a sub-meter and on nothing else. A null share means metered, not allocated and is served as null — never as a zero, which would read as allocated	**EXTENDS**	`BR-AST-003`'s parent rule, carried onto the Overview.

	nothing.		
AST-R-06	Export capacity belongs to an exporting device only, and it is a capacity in kW, never money. BR-AST-025 keeps tariffs and credited dollars off this whole kind.	**EXTENDS**	BR-AST-042 names the capacity; this says it is never money.
AST-R-07	Every kind names what it hangs off, and names it as a link. A sub-meter names its master; a behind-the-meter device names its linked meter; a master names the sub-meters on it. Today the master is plain text while parent_asset_id is served — a dead end with the id sitting right beside it.	**NO HOME**	Stated only here. The product renders both as plain text today.
AST-R-08	The tab set and the panel set are DERIVED, never written per screen. One function decides each from the asset, so a new kind changes that function and nothing else. assetDetailTabsFor() already does this for tabs; panels need its twin.	**EXTENDS**	B14 rule 36's shared component kit, applied to a derived tab set.
AST-R-09	A fact is said once. The status is the pill; the number is the title; every other fact lives in exactly one panel. Where the header abbreviates a chain — 118 Bayou Bend Dr · Unit A · ACC-20194 — the panel carries it in full: that is a summary, not a repeat. The test for the difference: does the header carry the complete value? A pill does; an address line does not. This screen breaks the rule twice today — Parent and Last test date each render in two panels.	**NO HOME**	Stated only here — B14 rule 32 caps description length, not repetition.
AST-R-10	No database id is ever a visible value — B14 rule 31, inherited and restated because this screen carries five ids on the wire (id, service_point_id, premises_id, unit_id, parent_asset_id) and four of them exist to be link targets. A target is not a value.	**INHERITED**	B14 rule 31, verbatim: *“no Databse ID on screen - that is a rule”*.
AST-R-11	Panel order is fixed and does not vary by kind. Assignment · What's due · Measurement · Device · Cost & value · Lifecycle. A kind that skips a panel closes the gap; it never reorders. Somebody who learns the screen on a meter must not have to re-learn it on a device.	**NO HOME**	Stated only here. Nothing in the repository fixes panel order.
AST-R-12	A panel label is at most two words, and names the question rather than the table it came from. PO instruction, 2026-09-11, verbatim: “I need label - maximum 2 words only panel label should be 2 words only”. Assignment, not Where it is and what it bills.	**NO HOME**	Stated only here, on the PO's instruction of 2026-09-11.
AST-R-13	Adding a kind is a four-line change. A new kind declares its tab set, its panel set, what it hangs off, and its Measurement rows — and touches nothing else. If a new kind needs an edit anywhere else in the detail view, the template has drifted and AST-R-01 has already been broken. This is the rule that keeps the other twelve true a year from now.	**NO HOME**	Stated only here — a claim about the cost of change, not a UI rule.

AST-R-14	A field's helper is one line — at most ten words — and never wraps. Not a style preference: a helper that wraps pushes its own input down, and in a two-to-a-row grid that leaves the two inputs of one row on different baselines. The alignment defect and the copy defect are the same defect. Measured across the assets module: 27 of 37 helper strings run over ten words, and the worst is 36 — “The size of the service connection this meter sits on. A rate ladder that prices by size uses it, so a wrong one changes the bill. Leave it blank if you do not know it yet.” B14 rule 32 already caps a screen's description budget and rule 33 caps a refusal at five or six words; a field helper had no cap at all.	**EXTENDS**	B14 rule 32's description budget, carried from the screen to the FIELD.
AST-R-15	Two columns are two fields. Never one input holding both with a delimiter the person has to type. The product does this once — Manufacturer / model, with the help “One line; separate the two with a middle dot” — and it is the only field in the module that asks a human to do a parser's job. A model name containing the delimiter, or a person reaching for a slash instead, lands in the wrong column, and nothing on the screen says which half is which. Displaying them joined is fine; the detail view's single Manufacturer / model row is good density. The rule is about CAPTURE.	**NO HOME**	Stated only here. The product does it both ways in two places.
AST-R-16	Every choice control names where its options come from, on the drawing itself. Four markers and nothing else: set a tenant-editable Settings list, named by its list code · tbl live rows from another table, named with its module · fix frozen in the source, with the count · none no list exists, or the product renders free text over a list that does. A drawn drop-down is a box with a word in it, and a picture cannot distinguish a row a utility can add this afternoon from a string that needs a release — so a coder reading the drawing alone guesses, and the guess is invisible once it is written. The marker travels ON the field rather than in a table beside it, because a table beside it is the thing that goes stale. The full map is Appendix L.	**NO HOME**	Stated only here, and it is about a drawing rather than a screen.

Two NO HOMEs and one EXTENDS. The sort-order rule is the more important of the two homeless ones, because a rule that exists only in an auditor's checklist will be broken by every lane that never reads the checklist — which is all of them. **And R11 and R12 are worth a second look on this table:** they were given to this pass as the PO's own words, they are specific enough to test, they were tested, they failed in ten places — and the anchors they are tested against are not written down anywhere in the repository. That is a rule one step from being homeless.

B · Every worked figure in this document, verbatim

None is a signed figure and every drawing is an illustration. They are test cases, kept consistent across screens so one asset can be followed through the whole flow. The

rules that would govern them if they were real are money at 4 decimal places stored and 2 shown (hard rule 2) and straight-line depreciation over the recorded life (`BR-AST-047`).

Figure	Where	The sentence around it	Status
126	01, 05, 06	assets on the seeded Brazos Bend tenant	**Measured** — counted in PostgreSQL with the tenant GUC set
50	01	rows on the first page; the default page size	**Measured** in the browser
AST-04412 · **AST-04413** · **AST-04414**	01, 10, 25	the three inventory meters	**Real seeded values**
E-5000…**E-5016**	01, 07, 20, 21, 22	installed walk-by meters on Unit 1	**Real seeded values**
$182.00 · **$0.00**	10, 24	purchase cost and present value of AST-04412, fully depreciated at 5 yrs 5 mo against a 5-year life	**Real; the second computed by the server**
5 yrs 5 mo · **5 years** · **2026**	10	age · expected life · replacement forecast	**Real, computed by the server**
$3,381.00 · **$2,608.02**	05, 05c	Asset value: buy value and present value across the tenant	**Real, computed by the server**
BF-0001 · **Annual test** · **06/15/2026** · **Overdue**	03, 05, 05a	the one certification row on the whole tenant	**Real seeded value**
$546.00 · **3** · **0**	04, 05c	stock at cost · uninstalled · aged over 12 months	**Real, served by** `GET /assets/stock`
9 warranty rows · **10** forecast rows	05, 05b, 05c	what the Lifecycle tab holds today	**Measured** in the browser
BF-0002 · 11/02/2026 · Due soon · **E-5000 · every 96 months · 01/14/2034**	05a	two further certification rows	**Illustration** — the seeded tenant has one
Meter · 5.4 yrs · 5 yrs · 3 past life · $546.00	05c	the forecast row a budget would be built from	**Illustration**, consistent with the real age and stock figures
123 · **118 masters · 5 subs · 41% AMI**	R2	what the KPI strip would read once served	**Illustration** — consistent with 126 total and 3 in inventory
WO-1042 · **$84.00** · **06/02/2026** · **Sam**	R1	a maintenance entry with a work-order reference	**Illustration**; *Sam Reyes* is a real seeded

Reyes			user
12,481 kgal	06, 20, 22	a last read / final read on E-5001	**Illustration** — the seeded tenant has no committed reads
0 · 09/11/2026 · 81-44120 · 5/8"	R5	an opening read, its date, a serial and a connection size	**Illustration**
29px · 47px · 5px · 6px · 20px	09, R4, 13	label-to-input as measured, against the rule	**Measured**, and the only figures here that are the finding rather than an example
121 · 73 · 72 · 64 · 13	07, 09, 18	units · premises · accounts · connection sizes · asset types	**Measured** from the seeded database
3 of 126	06	assets with coordinates	**Illustration** — geocoding was not run in this pass

C · Every screen, with its exact contents

Extracted by script from this document's own markup. The **How you get here** column is where a screen with no audited door would become countable — **there are none: every one of the 35 names its door.** ⚠ **This read "all 24" until this pass**, which was the count before sections 04 and 06 and screen 33 were drawn — the sentence stayed true-sounding while the thing it counted grew by eleven. The screens that do not yet exist (05a, 05b, 05c, 07, 14, 16–21, 32, 33) name the door they would be reached by.

#	Screen	Address	Tabs	Table controls	Write surfaces	How you get here
01	Meters	`/:tenant/assets?tab=meters · also /:tenant/assets (the bare address)`	Six drawn here, all URL state: Meters · Sub-meters · Behind the meter · Inventory · Lifecycle · Map. The three non-list …	Search — Server. q= reaches GET /assets and is applied as a queryset .filter() over six legs — asset_no · serial_no · manufacturer · model · parent_asset__asset_no and a sub-select of assets on units whose address matches (selectors.py:474…	Detail — the asset # links to /assets/{id} . Add — the header primary. Edit — the pencil, now a full page (screen 18). Update — PATCH /assets/{id} . Delete — the trash, only on a never-referenced inventory row.	Assets in the left rail (AppShell.tsx:211 , rendered at :752), the global search's view all for the assets entity (searchDestinations.ts:49), and ← Assets from every screen below it. Audited, all …
02	Sub-	`/:tenant/asse`	The same six.	Identical mechanism to screen 01	Detail · Add · Edit (screen 18) ·	The tab strip on screen 01,

	meters	ts?tab=sub_meters	This is the tab, not a screen of its own.	— same endpoint, same f.tab filter, same unbounded query underneath. The column set differs: Parent / master , Unit and a right-aligned Last read replace Role, Mult. and Assigned to.	Update · Delete — all as screen 01.	and a direct ?tab=sub_meters link.
03	Behind the meter	/:tenant/assets?tab=behind_the_meter	The same six.	As screen 01. Columns: Device # · Type · Premises · unit · Meter · Export capacity · Next test · Warranty (hidden by default) · Status.	As screen 01.	The tab strip.
04	Inventory	/:tenant/assets?tab=inventory	The same six.	None, by design. This is the detail-panel table, not the DataTable — four aggregate lines have nothing to search, sort or page. The drill is the row's own affordance: clicking a type opens screen 01 preset to that type's uninstalled rows.	Detail — the type name drills to the filtered registry. Add · Edit · Update · Delete — NOTHING , and correctly: there is no such thing as editing a stock line.	The tab strip only.
05a	Lifecycle · Certifications & tests due	/:tenant/assets?tab=lifecycle§ion=certifications	Three: Certifications & tests due · Warranty & supplier · Age & replacement forecast, in the order the as-built stacks t…	Search — asset # or premises. Filter — Overdue , Due within 90 days , and the status vocabulary the server already computes. Columns — the shared chooser. Sort — Asset and Due, with Due ascending as the default , because the oldest lapse is…	Detail — 👁 to the asset. Add · Edit · Update · Delete — NOTHING ; a certification date is a property of the asset and is edited on screen 18. Raise work order — the wrench, unchanged.	The Lifecycle tab's own sub-tab strip. It is the default section, so ?tab=lifecycle lands here.
05b	Lifecycle · Warranty & supplier	/:tenant/assets?tab=lifecycle§ion=warranty	The same three, in the same order.	Search — supplier or asset group, which is how a person arrives at this table (they have a supplier's name on an invoice). Filter — Covered / Expired / No cover recorded , the three states the status column already computes. Columns — the s…	Detail — 👁 to the group's assets, filtered. Add · Edit · Update · Delete — NOTHING ; a warranty end date is a property of the asset and is edited on screen 18.	The Lifecycle tab's sub-tab strip.

05c	Lifecycle · Age & replacement forecast	`/:tenant/assets?tab=lifecycle§ion=forecast`	The same three, in the same order.	Search — asset group. Filter — Past life and Due within 12 months , the two columns a budget is built from. Columns — the shared chooser. Sort — Asset group, Avg age and, most usefully, Replacement cost descending . Pagination — the standar…	Detail — 👁 to the group's assets. Everything else NOTHING .	The Lifecycle tab's sub-tab strip.
06	Map	`/:tenant/assets?tab=map`	The registry's six. Inside: a filter bar of a search box and four dropdowns.	No table. The four dropdowns filter in the browser over the whole GET /assets/geo payload — a single-shot answer with no page and no bound.	Detail — the popup's link to screen 08. Everything else NOTHING .	The tab strip only.
07	Assign 3 assets to a unit	`no address of its own — a dialog on /:tenant/assets`	None.	N/A.	The bulk write. Inventory → unit only .	Tick rows in the registry; a selection bar appears above the table with Assign to a unit and Clear .
08	Overview	`/:tenant/assets/{id}`	Five on a meter or sub-meter · five on an exporting inverter, Export replacing Reads · four on a plain device. The tab S…	None — this screen holds no table.	Edit — a pencil per panel, opening the form at that group rather than one control over twenty-three fields. Delete — under the overflow, and refused with its reason on an asset with history.	The asset number in any registry tab, the map popup, a premises or a unit, and the address bar.
09	Reads	`/:tenant/assets/{assetId}?tab=reads`	Part of screen 08's five.	NEW — the shared toolkit. Search a date, a reading or a source. Filter — Committed · Estimated · Disputed , plus a date range. Columns — the shared chooser. Download — the ⤓ export, which inherits AST-B-20 : it is broken on the deployed sta…	NOTHING on all four. A read is written through Meter Reads, never here.	The tab strip on screen 08. Measuring assets only — a plain device has no Reads tab at all.

			the deployed sta…			
10	Export	`/:tenant/assets/{assetId}?tab=export — an inverter whose linked meter is a net meter, only`	Overview · Export · Assignment history · Tests & maintenance · Audit. Note the fourth is relabelled for a device: a back…	None.	NOTHING on all four.	The tab strip, and only on a behind-the-meter asset whose linked_meter.net_metering is true. On any other device the tab is not rendered at all — the tab set is a function of the kind, so there is nev…
11	Assignment history	`/:tenant/assets/{assetId}?tab=assignments`	Part of screen 08's set.	NEW — the shared toolkit. Search a premises, a unit or who moved it. Filter — Current · Closed , plus a date range. Columns · Download · Pagination . None exists today : GET /assets/{id}/assignments returns a bare list like the reads endpoi…	NOTHING . Append-only by rule: a move closes the current row and opens the next on the same date, and neither can be edited afterwards.	The tab strip on screen 08.
12	Work & maintenance	`/:tenant/assets/{id}?tab=work`	Two sub-tabs here — Open work, then History — and five on a meter or sub-meter · five on an exporting inverter, Export r…	Open work — no toolbar; a handful of open jobs per asset, and the filter that matters (open vs all) is the split itself. History — the standard pager, because a ten-year-old meter accumulates rows without limit.	Detail — a work order number opens Work's own screen. Add — the two controls above. Edit · Update — NOTHING on a history row, and that is the rule rather than an omission: BR-AST-018 makes maintenance append-only, and a job is changed in Wo…	The detail tab strip. It replaces Maintenance & repairs and Work , which were two tabs answering one question. Inside it, two sub-tabs — the same second-level shape the Lifecycle registry tab needs, a…
13	Audit	`/:tenant/assets/{assetId}?tab=audit`	Part of screen 08's set.	Whatever the shared AuditTab offers — this screen composes the same component premises, units and accounts use, rather than drawing its own.	NOTHING , on principle. It is a read-only window into the one audit_log , scoped to this asset.	The tab strip on screen 08.
14	Edit one asset	`/:tenant/assets/{assetId}/edit — a full page,`	None.	N/A.	This is the edit surface. It is the only caller of updateAsset in the entire front end — every specification correction a utility ever makes goes	Today: the ✎ pencil in a registry row and nowhere else — the detail screen has no edit control at all. As drawn:

		`replacing today's dialog`			through it.	the same pencil, plus an Edit action in the asset's own header, which is where a person…
15	Delete this asset?	`no address of its own — a confirm on /:tenant/assets and /:tenant/assets/{id}`	None.	N/A.	DELETE /assets/{id} , System Admin only (services.py:1297).	The 🗑 trash in a registry row, and Delete in the detail header — both only when the server's action set for that row includes delete .
16	Edit · What's due	`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`	None.	None.	Update — PATCH /assets/{id} with two keys. Add · Delete — NOTHING .	The pencil on the What's due panel.
17	Edit · Measurement	`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`	None.	None.	Update — one PATCH, after the confirmation. Add · Delete — NOTHING .	The pencil on the Measurement panel. A behind-the-meter device has neither — no panel, no pencil.
18	Edit · Measurement · confirm	`The second step of the same modal — no second address`	None.	None.	Update — the PATCH fires here and nowhere else.	Pressing Continue on screen 17 when a billing-affecting field has changed. It is not a separate dialog — the same modal, a second step, so Back returns the typed values rather than discarding them.
19	Edit · Device	`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`	None.	None.	Update — one PATCH. Add · Delete — NOTHING .	The pencil on the Device panel.
20	Edit ·	`Modal over`	None	None	Update — one PATCH. Add · Delete	The pencil on the Cost &

	Edit · Cost & value	`/:tenant/assets/{id} · PATCH /assets/{id}`	None.	None.	— NOTHING .	value panel.
21	Edit · Lifecycle	`Modal over /:tenant/assets/{id} · PATCH /assets/{id}`	None.	None.	Update — one PATCH with two keys.	The pencil on the Lifecycle panel.
22	Reassign	`/:tenant/assets/{assetId}/reassign · ?unit={id} preselects a destination`	None.	N/A.	Update — the move. On failure the server refuses the whole move: if the final read cannot be captured, the asset does not move, and the screen does not pretend otherwise.	The ⇄ swap in a registry row; Assign / Reassign in the detail header; the Assignment panel's own button; and a unit's + Assign from inventory picker, which arrives with ?unit= already set — a prefill,...
23	Log repair / maintenance	`/:tenant/assets/{assetId}/maintenance/new — titled Log test on a device with a cadence`	None.	N/A.	Add only. Append-only by rule — a correction is a new entry, and the screen says so.	The 🔧 wrench in a registry row; Log repair in the detail header; the Test & certification panel's Log test ; and the Maintenance tab's own button.
24	Retire · 1 Reason & final read	`/:tenant/assets/{assetId}/retire — step 1`	None — a three-step wizard.	N/A.	Nothing writes until step 3's commit.	The ✕ retire in a registry row, and Retire in the detail header — both only when the server's action set allows it.
25	Retire · 2 Replacemen	`/:tenant/assets/{assetId}/retire — step 2`	None — a three-step wizard.	N/A.	None yet.	The ✕ retire in a registry row, and Retire in the detail header — both only when the server's action set allows it.

26	Retire · 3 Book value	`/:tenant/assets/{assetId}/retire — step 3`	None — a three-step wizard.	N/A.	Update — POST /assets/{id}/retire : the terminal transition, the closed assignment, the computed book value and the chosen treatment, in one call.	The ✕ retire in a registry row, and Retire in the detail header — both only when the server's action set allows it.
27	Raise work order	`/:tenant/assets/{assetId}/work-orders/new`	None.	N/A.	Add only. apps/assets writes nothing here — the post goes to Work's own door, and a test asserts that by name.	Five ways, and two of the four the rule names are missing. Built: the Lifecycle tab's certification wrench (AssetLifecycleTab.tsx:187) and four post-commit ticks — add (AssetAddWizard.tsx:474), re...
28	Add an asset · 1 Identity	`/:tenant/assets/new?class=meters`	None — a five-step rail.	None.	Add — and only at the end. Edit · Update · Delete — NOTHING .	+ Add asset on any registry tab — each tab's primary carries its own class, so the wizard opens on the right one.
29	Add an asset · 2 Specification	`/:tenant/assets/new?class=meters`	None.	None.	Add at the end only.	Continue from step 1.
30	Add an asset · 3 Assignment	`/:tenant/assets/new?class=meters`	None.	None.	Add at the end only.	Continue from step 2.
31	Add	`/:tenant/asse`	None.	None.	Add at the end only.	Continue from step 3.

	an asset · 4 Lifecycle	ts/new?class=meters				
32	Add an asset · 5 Review & add	/:tenant/assets/new?class=meters · POST /assets	None.	None.	Add — POST /assets , once, here.	Continue from step 4.
33	Drafts	/:tenant/assets?drafts=1 — proposed	None.	**None drawn.** The list is short by construction and a draft is discarded or finished, never browsed.	**Update** — PATCH /wizard-parks/{id} on resume-and-save. **Delete** — discard. **Add · Edit** — **NOTHING**: no row is written to assets at any point.	A **Drafts** control in the registry header, carrying its count.

D · Decisions recorded here and nowhere else

D1 · Rules and judgements that live only in this document

The judgement	Why it was made here
A scope cut renumbers; a structural fault does not	Removing the import material left gaps at 07 and 27–32. A gap left by a *fault* must stay visible, because something was meant to be there; a gap left by a *scope cut* is an absence, and carrying it would make every count read wrong forever. The inherited rule covers only the first case.
Screens 05a–05c and 07 are drawn in place rather than in section 10	They are whole screens rather than changes to one, and a reader comparing a three-table page with three tabbed pages needs both in one eyeline. They are still indexed as commitments.
The *Opening read* ruling is struck, not deleted	It was out of scope on a recorded lane instruction and is now a specification on a newer one. Keeping the superseded reasoning visible is what tells the next reader the seam argument existed and was overruled, rather than that nobody thought of it — and it is how the seam turned out to be a precedent.

`seams.unit_row` is graded LEAKY although nothing leaks	All three callers discard the return value. Graded on what it *returns*, per the inherited test, and the practical qualifier is written beside the grade rather than allowed to soften it.
The `apps/assets` hosting divergence is a description, not a specification	It is flagged in the module's own docstring and was reported when it was made. Nobody has ruled that the module map should change, and this document does not amend the repository.
The bare `AssetDash` em dash is reported and not filed	It makes loading, none and cannot-compute the same picture — a real fault. It is a shared-component decision rather than an assets one, and filing it against this module would put it in front of the wrong owner.

> **How *"recorded nowhere else"* was checked, rather than assumed**
>
> **Each of the six was grepped against the repository's own registers before this table claimed it** — `12-governance/02-decision-log.md` and the assets module's `02-business-rules.md`, on 2026-09-11. Four returned **nothing at all**: the renumbering convention, drawing 05a–05c in place, the `AssetDash` em dash, and grading `seams.unit_row` LEAKY.
>
> **Two returned hits and both are a different subject wearing the same words.** `unit_row` appears once in the decision log, inside the `CR-051` seam-batching row where it is counted as one of twenty repeated seam reads — nothing there grades it, or says anything about returning a model instance. *"Opening read"* appears six times, all of them about the cutover upload and the sample-data seeding, none about this document's struck ruling. **A grep hit is not a record**, and reporting one as though it were would be the opposite of what this table is for.
>
> ⚠ **The claim is therefore verified, and its boundary is worth stating:** it means **not in those two registers at that commit**. It does not mean nobody has ever said these things in a chat that was never written down — which is exactly the risk that makes this table the highest-value one here.

D2 · Rulings already recorded elsewhere — do not re-ask

Ruling	When · who	Where it lives
"as per r10 - it is not modal" — the four lifecycle flows are full-page routes	PO, 2026-08-22	CR-028 AST-W4, B16 §5.2 as amended
"Launch with counts & costs only" — no vans, no reorder thresholds, no stock table, stock-count import deferred	PO, 2026-08-22 (§10-D5)	B16 §10; tests assert each absence by name
No GL sentence, no depreciation posting, no Cost & Margin link — the **Asset value** panel replaces	PO,	B16 §10 → `BR-AST-047`

r10's finance sidebar	2026-08-22 (§10-D3)	
Third-party map tiles accepted; premises gained lat/lng	PO, 2026-08-22 (§10-D6)	B16 §10; `apps/metering/geocode.py`
The behind-the-meter tab carries no tariff and no credited dollars	PO, 2026-08-22 (D2)	`BR-AST-025`
A meter number is allocated from the tenant's series, never taken from an uploaded file	PO, 2026-08-27	`BR-AST-049`; decision log of that date
"no Databse ID on screen - that is a rule"	PO, 2026-08-31	B14 rule 31
The Assets door is one form with four ways in — **name and count amended, the rule unchanged**	2026-09-06, `DEF-WRK-4125`	`BR-WRK-056`, status CONFIRMED
A unit may carry its own lat/lng, and where it does those are the truth for that unit — **the address half is untouched**	PO, 2026-09-08	`BR-MET-1162`; relevant to screen 06's markers
Four instructions of 2026-09-11: the R11 spacing scale and its anchors · R12's field anatomy · the Lifecycle tab becomes three tabs with a toolkit · Edit becomes a form, not a modal · the Service options come from **Settings › Services & UoM** · *"make opening read enable user can add reading"* · and the import material leaves this document	2026-09-11	This document, section 13 — and **nowhere in the repository**. Six of the seven are product decisions that a lane would need and would not find.

A search technique that gave a false negative, recorded because the next person will hit it too: grepping for `WorkOrderR5Checkbox` — the name `BR-WRK-056` used until 2026-09-06 — returns **nothing**, because the component was renamed to `RaiseWorkOrderCheckbox`. A defect was once filed recording the retire tick as missing while it was live on screen, for exactly this reason. When a rule and the code disagree about a name, grep the rule's *date* as well as its words.

E · Every guarantee, and what actually enforces it

Only a constraint refuses. A convention is a habit with a good reason, and a habit does not stop a caller. Sorting the guarantees this way is uncomfortable and always useful.

The guarantee	What enforces it	How, and whether it refuses
An asset number is unique within a tenant	A unique index on `assets`	DATABASE CONSTRAINT refuses
A referenced asset type, meter type, read method or stage can never be deleted	`on_delete=PROTECT` on four FKs	DATABASE CONSTRAINT refuses
No tenant's rows are visible to another	RLS `FORCE` plus the `app.tenant_id` GUC	DATABASE CONSTRAINT refuses — verified incidentally in this pass: a query without the GUC returned 0 rows over 126
Only System Admin or Billing Clerk may create, edit or reassign; only System Admin may delete or retire	`@require_role` in the service, and a permission class on the view for retire	SERVICE RULE refuses at the service boundary
An asset with history is never hard-deleted	`services.delete_asset` re-probes every referencing table	SERVICE RULE refuses, and names Retire
A move never happens without the final read it promised	One transaction; metering commits the read in the caller's	SERVICE RULE refuses the whole move
Retired is terminal	The transition table, from the stage's behaviour class	SERVICE RULE refuses
Billing-affecting fields apply forward only	`services.update_asset`	SERVICE RULE refuses
An address is stored on `premises` and nowhere else	**Nothing on this table.** There simply is no address column to write to, and every screen derives it by walking	CONVENTION — held by absence. A migration adding `assets.city` would

		break it and nothing here would object
Row actions come from the server, never guessed from a status label	`row.actions`, rendered as given	CONVENTION — a component could still draw its own
Every user-visible string goes through eleven catalogues	The i18n gate	SERVICE RULE refuses a missing key at build
Row actions are icons with accessible names	`G37`	SERVICE RULE refuses — **for three of the rule's four clauses**. The two-visible cap is not among them
A Tailwind utility is only written in a migrated file	`G31` plus a vitest check over the migrated list	SERVICE RULE refuses
The registry never returns more rows than a page	**Nothing.** The envelope is paginated and the query is not	CONVENTION — and at 40,000 accounts, not even that
Date fields offer a Today shortcut	**Nothing.** B14 rule 7, ratified, enforcement column empty	CONVENTION — and unmet on all five date fields in this module
A helper sits under the input it describes	**Nothing.** R12, stated to this pass and not written in the repository	CONVENTION — and unmet on ten of eighteen fields
Padding comes from one spacing scale	**Nothing.** The tokens exist; no check compares a declared value to them	CONVENTION — eight of ten anchors hold anyway
Every asset screen looks like the product	`CR-016`'s UI-conformance check, in a real browser	CONVENTION at this module's scale — see F, the last row

F · What this document assumes and never says

Gaps in the agreement, not in the product. Ordered by cost, most expensive last.

The assumption	Where it hides	What breaks if it is wrong
The seeded tenant resembles a real one	Every figure in appendix B marked *measured*	Nothing structural. But a tenant with 40,000 assets would put every screen here under load this pass never applied, and the four dashes on screen 01 are the visible tip of that.
A screen reachable by URL is reachable by a person	Sixteen of thirty-two surfaces were opened by address, not by clicking to them	The door audit is a source read, and this pass confirmed the doors it could reach. A door that exists in source and is conditionally hidden at runtime would pass both checks — the raise-work-order finding is exactly that shape found from the other side.
The English catalogue is what a person reads	Every quoted string in this document	Ten other catalogues exist, and B14 rule 35 requires each to be written in its own language rather than translated. A refusal that is five words in English and fourteen in another is a different screen. Not checked in this pass.
The branch read is the product	The masthead says *could not check*, and this row says what that costs	If `claude/fervent-carson-0uksv0` is not what a customer runs, every as-built claim here is about a different build. Nothing in this document is safe to act on without re-running the reads against whatever *is* deployed.
That R11 and R12 are written down somewhere a lane would find them	Section 09 judges eleven violations against them, and appendix A grades their home as `INHERITED` — from the analysis process, not from the repository	Every finding against them still stands as a *description* of what the screens do. What does not follow is that anybody was wrong to build it that way. If these rules matter, they belong in B14 with a number, and this pass could not find them there.
Read-only roles behave as the code says	`BR-SHL-025`'s finding in section 09	Every screen was walked as a **System Admin**. The unused `writeBlockedReason` on two tabs is a source finding that a walk as a Billing Clerk would have confirmed or refuted in one click, and this pass did not make that click. It is the single cheapest thing a next pass could do.
That the one browser check this repository owns covers these screens	Nowhere — it is assumed by every reader who sees `nativeSelects: 0` in a green register	**The most expensive assumption here.** `CR-016`'s UI-conformance spec measures computed style over `SIDEBAR_DESTINATIONS` plus two settings sub-tabs. That list contains **`/assets` and nothing else from this module** — so it opens the **Meters tab** and never sees the other five tabs, the four-step wizard, the six detail tabs or any of the four lifecycle flows. **One of the twenty-four mounted surfaces is measured, and the register reports zero violations truthfully over that one.** A declared blind spot with no number attached is indistinguishable from an empty one; this is the number.

G · Every UI rule violation

The rule set and its precedence are stated in section 01: **B14 is the authority**, `UI-RULES` is its source, and R11/R12 were given to this pass directly. The full table with quotes and file references is **section 09**; this is the index.

Rule	Screens	In one line	Would a machine have caught it?
`BR-WRK-056`	10, 14	Two of the rule's four named ways in do not exist	**No.** Nothing reads rendered output, and no test asserts a control's *absence*
R12	16–22, 24	Ten of eighteen fields put the helper between the label and the input	**No.** The markup type-checks; the order is a layout fact no assertion looks at
B14 **12·13·14·16**	05	Three Lifecycle tables with no toolbar, no sort, no columns, no pager	**No.** Answered by 05a–05c
B14 **15**	01, 02, 03	5 controls in one cell, no overflow	**Partly.** `G37` enforces three of four clauses; not the cap
B14 **7**	16–22, 24	No Today shortcut on any of five date fields; no bounds	**No.** Enforcement column empty
B14 **31**	21	A UUID is the visible value, for the one role the screen is for	**No**
B14 **13**	01, 02, 03	Sort toggles forever; no clear state	**No.** Shared table, product-wide
B14 **16**	01, 02, 03	No first-page or last-page control	**No**
`BR-SHL-`	04, 05	A reason prop is passed and never read	**No — and `G13` is the**

025			**gate that looks closest.** It asks whether a prop has a caller; it does
R11	16–19, 22–24	Two full-page wizards render the 32px default; the one screen that is not a wizard renders the 40px wizard title	**No.** Both are legal tokens; nothing compares a screen's kind to the token it picked
R11	the shell	Rail item `9px 12px`, pill `2px 9px` — two off-scale product literals	**No.** The CR-016 browser check measures type sizes and control heights, not padding

Eleven violations. One is partly machine-checkable and ten are not. Every suite in this repository is green over all eleven, which is the argument for this column existing.

H · Every cross-module finding, with both modules named

From → To	The handoff or shared fact	What is wrong	What breaks
assets → work	Raising a work order for an asset	The write path is reachable from five controls; two of the four `BR-WRK-056` names — both on the asset's own screen — were never built	A clerk standing on an asset cannot raise a job for it. On the seeded tenant the only unconditional door lists **one asset out of 126**
assets → identity	The technician list on the maintenance form	`GET /users` is System-Admin-only; `BR-AST-038` lets a Billing Clerk log a repair. The seam is fine and the permission on the far side is not	For that role the field becomes a UUID box, and there is nowhere in the product to look a UUID up
assets → metering	`seams.unit_row`	Returns a model instance, not a dict	Nothing today — all three callers discard it. The day a fourth keeps it, the boundary is gone and no check will say so
assets → tenancy	`exports.py`, `lookups.py`	Direct imports that bypass `seams.py`	Nothing today — it is a shared export kernel, closer to framework than to data. Named so it stays a decision

assets → integrations	The basemap tile host	Not wrong — **worth stating**: every pan on screen 06 sends the viewport around a tenant's asset locations to the host that policy names	Nothing. The egress was accepted by ruling rather than overlooked, and the attribution is required and rendered
assets ↔ the module map	Where the tables live	`CLAUDE.md` says metering owns them; the code hosts them in `apps/assets`	A lane reading the map alone looks in the wrong directory. Flagged in the module's own docstring and never absorbed by the map

Of Step 10's four questions, two found nothing and it is worth saying so rather than leaving it to be inferred from an absent heading: no module writes another's tables, and no fact has two homes. The two that found something are the handoff question and the naming/ownership question.

J · Health, and what each grade rests on

A screen's health is the worst of its fields, never an average. An average turns one broken field among nine good ones into *"90% healthy"*, and the screen still does not work. Three tables, because the three things are graded by different tests and mixing them produces a number that means nothing.

J1 · Fields

Verdict	Count	What it means
SERVED	111	A real source, served today, read by this screen
UNVERIFIED	9	Not opened in a browser in this pass — graded honestly rather than optimistically
DERIVED	7	Computed in the browser from data that should have carried it
SERVED, capped at 500	2	Served, and the picker fetches a fixed 500 rows and filters in the browser
N/A	2	Not a data field — a checkbox or a control
DEGRADED	1	Served for one role and replaced by a raw id entry for another
SERVED, no FK	1	Served, with no database constraint behind the reference
DEAD	1	—
NOT SERVED	1	The screen wants it; no read exists

NOT SERVED (to an asset)	1	—
REMOVED	1	—

Every field that is not SERVED, in full

Screen	Field	Comes from	Verdict
23 Log repair / maintenance	Assign tech	options from GET /users — admin-only ; for a Billing Clerk the control degrades to a bare text box holding a UUID	DEGRADED
25 Retire · 2 Replacement	Raise a meter-exchange work order	the fourth RaiseWorkOrderCheckbox tick — added in CR-070 wave M	UNVERIFIED
26 Retire · 3 Book value	Treatment	assets.book_value_treatment — write_off · carry_forward	UNVERIFIED
01 Meters	Description	nothing renders — assets.explainer.meters exists in all eleven catalogues and reaches no screen	DEAD
26 Retire · 3 Book value	Purchase cost	assets.purchase_cost	UNVERIFIED
25 Retire · 2 Replacement	Replaced by	assets.replaced_by_asset_id ; options are inventory assets only	UNVERIFIED
12 Work & maintenance	Raised by	nothing serves this — assigned_by and accepted_by exist, raised_by does not	NOT SERVED
12 Work & maintenance	Cost · jobs	work_order_cost_lines() — computed per order and joined to no asset	NOT SERVED (to an asset)
01 Meters	Past certification age	0 — same client-side count, same dash rule	DERIVED
01 Meters	Meters in service	103 — counted in the browser over the rows on this page, so a dash unless the page holds all of them	DERIVED
26 Retire · 3 Book value	Book value	cost – accumulated, 4 dp stored / 2 shown	UNVERIFIED
06 Map	“3 of 126 assets have coordinates”	counted in the browser	DERIVED
03 Behind the meter	Devices / Exporting devices	counted in the browser over this page	DERIVED

12 Work & maintenance	Record	derived from which source the row came from, and its state	DERIVED
06 Map	Search	filters the loaded payload in the browser	DERIVED
06 Map	Type / Status / Service / Area	four selects, filtering client-side	DERIVED
05c Lifecycle · Age & replacement forecast	Buy value / Present value	server straight-line across the tenant — the panel is REMOVED by instruction; appendix K says what it costs	REMOVED
26 Retire · 3 Book value	Accumulated depreciation	server straight-line over age and expected life	UNVERIFIED
10 Export	Quantity · Unit	the channel quantity, always with its unit	UNVERIFIED
10 Export	Channel	the export channel on that meter	UNVERIFIED
10 Export	Date	the linked meter's read date	UNVERIFIED

114 of 137 fields are SERVED. 9 were never traced in a browser and are UNVERIFIED — an ungraded field is not a healthy one, and grading them SERVED on a read would turn this whole table into decoration. Sorted by **source**, not by screen: the question this table exists to answer is *"if I change this column, what breaks?"*, and that needs every consumer of one column on adjacent rows.

Comes from	Field	Screen	Owned by	Health
/assets/{id}	👁 View	05a Lifecycle · Certifications & tests due	assets	SERVED
/assets/{id}/work-orders/new	🔧 Raise work order	05a Lifecycle · Certifications & tests due	assets → work ⇄	SERVED
a navigation to /work/parts	Parts in Work →	04 Inventory	assets → work ⇄	SERVED
options from GET /units?per=500 , filtered in the browser	Unit	07 Assign 3 assets to a unit	assets → metering ⇄	SERVED, capped at 500
options from GET /units?per=500	Unit	22 Reassign	assets → metering ⇄	SERVED, capped at 500
options from GET /users — admin-only ; for a Billing Clerk	Assign tech	23 Log repair /	assets → identity	DEGRADED

options from GET /users — admin-only , for a Billing Clerk the control degrades to a bare text box holding a UUID	Assign tech	23 Log repair / maintenance	assets → identity ⇄	DEGRADED
POST /work-orders	Raise work order	27 Raise work order	assets → work ⇄	SERVED
the fourth RaiseWorkOrderCheckbox tick — added in CR-070 wave M	Raise a meter-exchange work order	25 Retire · 2 Replacement	assets → work ⇄	UNVERIFIED
_row()['raised_at']	Raised	12 Work & maintenance	work ⇄	SERVED
mean age_months for the group	Avg age	05c Lifecycle · Age & replacement forecast	assets	SERVED
asset.sub_assets , count served	Move the sub-meters as well	22 Reassign	assets	SERVED
asset_assignments.final_read_id → the read's value	Final read	11 Assignment history	assets → metering ⇄	SERVED, no FK
asset_assignments.start_date / end_date	From / To	11 Assignment history	assets	SERVED
asset_lifecycle_stages.name	Status	01 Meters	assets	SERVED
asset_lifecycle_stages.name	Status	02 Sub-meters	assets	SERVED
asset_maintenance.cost , 4 dp stored	Cost · logged	12 Work & maintenance	assets	SERVED
asset_maintenance.cost , recorded, posted nowhere	Cost	23 Log repair / maintenance	assets	SERVED
asset_maintenance.date	Date	23 Log repair / maintenance	assets	SERVED
asset_maintenance.detail + tech_user , or the order's type + technician	What / Who	12 Work & maintenance	assets · work ⇄	SERVED
asset_maintenance.detail	What was done	23 Log repair / maintenance	assets	SERVED
asset_maintenance.resulting_status or	Outcome	12 Work &	assets · work ⇄	SERVED

work_orders.outcome		maintenance		
the linked meter's asset_no	Meter	03 Behind the meter	assets	SERVED
asset_types.name — four groups on this tenant	Asset type	03 Behind the meter	assets	SERVED
asset_types.name — four groups on this tenant	Asset type	04 Inventory	assets	SERVED
asset_types.name	Type	05a Lifecycle · Certifications & tests due	assets	SERVED
asset_types.name	Asset group	05b Lifecycle · Warranty & supplier	assets	SERVED
asset_types.name	Asset group	05c Lifecycle · Age & replacement forecast	assets	SERVED
asset_usage_payload metrics, or a stated absence reason	Total / Average / Highest month	09 Reads	assets → metering ⇄	SERVED
assets.asset_no	Meter #	01 Meters	assets	SERVED
assets.asset_no	Sub-meter #	02 Sub-meters	assets	SERVED
assets.asset_no	Device #	03 Behind the meter	assets	SERVED
assets.asset_no	Asset	05a Lifecycle · Certifications & tests due	assets	SERVED
assets.asset_no , asset_types.name	Popup: asset # · type	06 Map	assets	SERVED
assets.asset_no	Asset #	14 Edit one asset	assets	SERVED
assets.asset_no — the number, never the uuid	Kicker: AST-04412	27 Raise work order	assets	SERVED
assets.book_value_treatment — write_off · carry_forward	Treatment	26 Retire · 3 Book value	assets	UNVERIFIED
assets.depreciation_method — present value is computed from it and the method is never named	Depreciation method	08 Overview	assets	SERVED

assets.expected_life_years	Expected life	05c Lifecycle · Age & replacement forecast	assets	SERVED
nothing renders — assets.explainer.meters exists in all eleven catalogues and reaches no screen	Description	01 Meters	—	DEAD
assets.export_capacity_kw	Export capacity	03 Behind the meter	assets	SERVED
assets.export_capacity_kw — a registry column, absent from the device's own page	Export capacity	08 Overview	assets	SERVED
assets.external_ref — served on every row, rendered nowhere in the module	External reference	08 Overview	assets	SERVED
assets.install_date	Installed	01 Meters	assets	SERVED
assets.install_date	Installed	02 Sub-meters	assets	SERVED
assets.location_note — a note, not an address	Location note	14 Edit one asset	assets	SERVED
assets.multiplier , trailing zeros trimmed	Mult.	01 Meters	assets	SERVED
assets.multiplier , NUMERIC(18,6)	Multiplier	14 Edit one asset	assets	SERVED
assets.parent_asset.asset_no	Parent / master	02 Sub-meters	assets	SERVED
assets.purchase_cost	Purchase cost	26 Retire · 3 Book value	assets	UNVERIFIED
assets.purchase_date — age and present value are computed from it and both are shown; the date is not	Purchase date	08 Overview	assets	SERVED
assets.replaced_by_asset_id ; options are inventory assets only	Replaced by	25 Retire · 2 Replacement	assets	UNVERIFIED
assets.retire_date ; must be ≥ install date, server-enforced	Retire date	24 Retire · 1 Reason & final read	assets	SERVED
assets.retire_reason — end_of_life · failed · …	Reason	24 Retire · 1 Reason & final read	assets	SERVED
assets.serial_no	Serial #	14 Edit one asset	assets	SERVED

assets.size_class ; options service_size_classes (64 seeded)	Connection size	14 Edit one asset	assets	SERVED
assets.supplier — free text, no supplier list exists	Supplier	05b Lifecycle · Warranty & supplier	assets	SERVED
assets.type.name folded with meter_type.name	Type	01 Meters	assets	SERVED
value assets.type_id ; options GET /lookups/asset_types	Asset type	14 Edit one asset	assets	SERVED
assets.unit — always explicit (hard rule 3)	Unit of measure	14 Edit one asset	assets	SERVED
assets.warranty_end	Cover ends	05b Lifecycle · Warranty & supplier	assets	SERVED
assigned_address + unit_label + billing_chain.account	The header place line	08 Overview	assets → metering, customers ⇄	SERVED
nothing serves this — assigned_by and accepted_by exist, raised_by does not	Raised by	12 Work & maintenance	—	NOT SERVED
audit_log rows for this asset	When · Who · Action · What changed	13 Audit	assets → audit ⇄	SERVED
billing_chain.account — served, and plain text today	Account	08 Overview	assets → customers ⇄	SERVED
book_value_amount · book_value_treatment — written at retire, never readable again	Book value · treatment	08 Overview	assets	SERVED
certification_due or last_test_date + interval	Due	05a Lifecycle · Certifications & tests due	assets	SERVED
count from the DRF envelope	1–50 of 109	01 Meters	assets	SERVED
computed from days_until	Status	05a Lifecycle · Certifications & tests due	assets	SERVED

export_rollup.totals — 780 kWh , one entry per unit, never summed	Exported · August	03 Behind the meter	assets → metering ⇄	SERVED
kpis.aged_over_12_months — 4 , detailed “Warranty clock already running”	Aged over 12 months	04 Inventory	assets	SERVED
kpis.value_at_cost , 4 dp stored, 2 shown	$1,511.00 at cost	04 Inventory	assets	SERVED
derived from last_test_date + test_interval_months	Next test	03 Behind the meter	assets	SERVED
meter_reads , batch-filled per page	Last read	02 Sub-meters	assets → metering ⇄	SERVED
meter_reads	Popup: last read	06 Map	assets → metering ⇄	SERVED
written to meter_reads in the same transaction	Final read + date	22 Reassign	assets → metering ⇄	SERVED
written to meter_reads on commit	Final read + date	24 Retire · 1 Reason & final read	assets → metering ⇄	SERVED
meter_reads.read_at	Date	09 Reads	assets → metering ⇄	SERVED
value + options from meter_types	Meter type	14 Edit one asset	assets	SERVED
parent_asset_no + parent_asset_id — the id is served and the cell is plain text	Parent master	08 Overview	assets	SERVED
premises.address_line1 via the unit seam	Assigned to	01 Meters	assets → metering ⇄	SERVED
premises.address_line1 + units.label	Premises · unit	03 Behind the meter	assets → metering ⇄	SERVED
premises.address_line1 by walking	Premises	05a Lifecycle · Certifications & tests due	assets → metering ⇄	SERVED
premises.lat/lng , or the unit's own	Marker position	06 Map	assets →	SERVED

			metering ⇄	
summed purchase_cost , 4 dp stored / 2 shown	Replacement cost	05c Lifecycle · Age & replacement forecast	assets	SERVED
value + options from read_methods	Read method	14 Edit one asset	assets	SERVED
row.actions — the server's set for this row's state	Actions	01 Meters	assets	SERVED
row.sub_count / parent_asset_no , served	Role	01 Meters	assets	SERVED
service_point.service	Service	01 Meters	assets → metering ⇄	SERVED
service_point.service	Service	02 Sub-meters	assets → metering ⇄	SERVED
service_point.service , honest-empty for never-assigned stock	Service	04 Inventory	assets → metering ⇄	SERVED
specs.allocation_rule · specs.loss_tolerance_pct	Allocation rule · loss tolerance	08 Overview	assets	SERVED
stock_view per-type rows; the unit cost shows a range beneath the average where the group spans prices	Count / Unit cost / Value	04 Inventory	assets	SERVED
stock_view.kpis.uninstalled — 8	Uninstalled assets	04 Inventory	assets	SERVED
test_due · next_test · certification_due · warranty_covered	The attention strip	08 Overview	assets	SERVED
tile_host from the page policy	Basemap tiles	06 Map	assets → integrations ⇄	SERVED
units.label via the seam	Unit	02 Sub-meters	assets → metering ⇄	SERVED
count where warranty_end is in the future	Under warranty	05b Lifecycle · Warranty & supplier	assets	SERVED
work_order_cost_lines() — computed per order and joined to no asset	Cost · jobs	12 Work & maintenance	work ⇄	NOT SERVED (to an asset)

work_orders.due_at — on the wire, rendered nowhere today	Expected completion	12 Work & maintenance	work ⇄	SERVED
work_orders.order_no	Work order #	12 Work & maintenance	work ⇄	SERVED
work_orders.state	State	12 Work & maintenance	work ⇄	SERVED
work_orders.type_ref.name	Type	12 Work & maintenance	work ⇄	SERVED
0 — same client-side count, same dash rule	Past certification age	01 Meters	—	DERIVED
103 — counted in the browser over the rows on this page, so a dash unless the page holds all of them	Meters in service	01 Meters	—	DERIVED
a client-side checkbox driving bulk assign	Select	01 Meters	—	N/A
a tick; on success it redirects to screen 27	Raise a field work order	22 Reassign	assets → work ⇄	SERVED
a tick; redirects to screen 27	Raise a work order from the repair template	23 Log repair / maintenance	assets → work ⇄	SERVED
a two-value segmented control	Where is it going?	22 Reassign	—	SERVED
by walking to premises	Premises · unit	11 Assignment history	assets → metering ⇄	SERVED
client-side checkbox	Select	02 Sub-meters	—	N/A
cost – accumulated, 4 dp stored / 2 shown	Book value	26 Retire · 3 Book value	assets	UNVERIFIED
count already beyond expected life	Past life	05c Lifecycle · Age & replacement forecast	assets	SERVED
count of assets in the group	Units	05b Lifecycle · Warranty & supplier	assets	SERVED
count reaching expected life inside a year	Due within 12 mo	05c Lifecycle · Age & replacement forecast	assets	SERVED

counted in the browser	“3 of 126 assets have coordinates”	06 Map	—	DERIVED
counted in the browser over this page	Devices / Exporting devices	03 Behind the meter	—	DERIVED
Covered / Expired / —	Status	05b Lifecycle · Warranty & supplier	assets	SERVED
derived from the type's cadence	Requirement	05a Lifecycle · Certifications & tests due	assets	SERVED
derived from which source the row came from, and its state	Record	12 Work & maintenance	—	DERIVED
filters the loaded payload in the browser	Search	06 Map	—	DERIVED
four selects, filtering client-side	Type / Status / Service / Area	06 Map	—	DERIVED
how the read arrived and whether it is committed	Source / Status	09 Reads	assets → metering ⇄	SERVED
options are the legal targets from this stage's behaviour class	Set status	23 Log repair / maintenance	assets	SERVED
options from Work, filtered by the asset's type	Template	27 Raise work order	assets → work ⇄	SERVED
repair · inspection · test	Kind	23 Log repair / maintenance	assets	SERVED
server straight-line across the tenant — the panel is REMOVED by instruction; appendix K says what it costs	Buy value / Present value	05c Lifecycle · Age & replacement forecast	assets	REMOVED
server straight-line over age and expected life	Accumulated depreciation	26 Retire · 3 Book value	assets	UNVERIFIED
the assignment boundary	Effective date	22 Reassign	assets	SERVED
the assignment boundary; defaults to tenant today	Effective date	07 Assign 3 assets to a	assets	SERVED

		unit		
the channel quantity, always with its unit	Quantity · Unit	10 Export	assets → metering ⇄	UNVERIFIED
the export channel on that meter	Channel	10 Export	assets → metering ⇄	UNVERIFIED
the linked meter's read date	Date	10 Export	assets → metering ⇄	UNVERIFIED
the read value and the derived delta, with units	Reading / Consumption	09 Reads	assets → metering ⇄	SERVED
the read's channel	Register	09 Reads	assets → metering ⇄	SERVED
the registry, filtered to that group	👁 View	05b Lifecycle · Warranty & supplier	assets	SERVED
the registry, filtered to that group	👁 View	05c Lifecycle · Age & replacement forecast	assets	SERVED
the user who made the move	By	11 Assignment history	assets → identity ⇄	SERVED

J2 · Flows — one row per screen

Never grade a flow WHOLE on a read alone. Twelve of the twenty-four mounted surfaces were opened in a browser; the rest are `UNVERIFIED`, and saying so is what keeps the twelve worth something. The four screens this document proposes are graded `NOT BUILT` — they cannot be walked, and grading them anything else would be a category error.

#	Screen	Verdict
01	Meters	WHOLE — walked
02	Sub-meters	WHOLE — walked
03	Behind the meter	WHOLE — walked

03	Behind the meter	WHOLE — walked
04	Inventory	WHOLE — walked; the only registry tab with a served aggregate
05	Lifecycle	WHOLE — walked. The data is whole; the *shape* is the finding
05a	Lifecycle · Certifications & tests due	NOT BUILT — proposed here
05b	Lifecycle · Warranty & supplier	NOT BUILT — proposed here
05c	Lifecycle · Age & replacement forecast	NOT BUILT — proposed here
06	Map	WHOLE — walked; Leaflet mounted, filters and legend rendered
07	Edit one asset	NOT BUILT — proposed here as a page. The dialog it replaces was not opened either: it needs a click on a row action
08	Delete this asset?	UNVERIFIED — needs a click on a row action
09	Assign 3 assets to a unit	UNVERIFIED — needs row ticks
10	Overview	WHOLE — walked
11	Reads	WHOLE — walked; honest-empty confirmed
12	Export	UNVERIFIED — the seeded tenant had no exporting inverter
13	Assignment history	WHOLE — walked
14	Maintenance & repairs	WHOLE — walked
15	Audit	WHOLE — walked
16	Add · 1 Identity	BREAKS AT the option source — walked. The Service field's value posts; its list does not exist
17	Add · 2 Specification	UNVERIFIED — needs step 1 completed
18	Add · 3 Assignment	UNVERIFIED
19	Add · 4 Lifecycle	UNVERIFIED
20	Reassign	WHOLE — walked

21	Log repair / maintenance	BREAKS AT the identity seam — walked as an admin; for a Billing Clerk `GET /users` refuses and the field degrades
22	Retire · 1 Reason & final read	WHOLE — walked
23	Retire · 2 Replacement	UNVERIFIED
24	Retire · 3 Book value	UNVERIFIED
25	Raise work order	WHOLE — walked, by URL. Its *doors* are the finding, not its flow

12 WHOLE · 10 UNVERIFIED · 2 BREAKS AT · 4 NOT BUILT.

J3 · Seams

The full table with what each hands back is in **section 11**. In summary: **8 crossings — 6 HEALTHY, 2 LEAKY by the letter with nothing leaking today, 0 WRITING, 0 ABSENT.** No crossing was found that has no seam at all, which for a module hosted outside the directory the module map assigns it to is a better result than the map would lead you to expect.

K · Everything an upgrade would drop, and who ruled it

Screen	What is dropped	Disposition	The ruling, quoted	What it costs the person who used it
07	The dialog frame around the edit form	**REMOVED**	**Instruction, 2026-09-11:** *"this should be form and not modal."*	**Nothing.** It is a container, not a field: every one of the nine fields, the forward-only note and both buttons carry over. What a person loses is the dimmed backdrop; what they gain is an address they can return to and a second door from the asset itself.
05c	The **Asset value** panel — *Buy value $3,381.00*, *Present value $2,608.02*, and its line *"Computed from each device's recorded purchase cost and straight-line age. Nothing here posts anywhere."*	**REMOVED**	**Instruction, 2026-09-11:** *"remove this - Asset value"*, quoting the panel in full.	**Two figures that are computed nowhere else in the product.** Buy value and present value across the whole tenant appear on this panel and on no other screen, so a person who used them loses them entirely rather than finding them elsewhere. That is the cost, stated rather than discovered — and it is exactly what this table exists to make visible.

Two rows, and both carry a quoted ruling. Rows with no quoted ruling: 0 — any such row would be a defect in this document rather than a design decision.

> **Everything else carries over, across six remedies**
>
> 05a, 05b, 05c, 07 and R1–R6 between them touch every screen this document proposes to change, and their dispositions are **UNCHANGED**, **CHANGED**, **RELABELLED** and **MOVED**. The closest thing to a third removal is R2's four row actions moving into an overflow menu. They are graded **MOVED** deliberately: a control behind a ··· is still reachable and B14 rule 15 asks for exactly that arrangement. It is recorded here so a reader who believes demoting a control into an overflow *is* a removal for everyone who never opens overflow menus can argue the point — and they would not be wrong to.

Two closing statements

> **Completeness, scoped**
>
> This document is **exhaustive over itself**: every rule it states is in appendix A, every figure in B, every screen in C, every field in I, every decision in D, every guarantee in E, every assumption in F. The counts reconcile and the arithmetic is shown in section 13.
>
> It is **not exhaustive over the product**. The Assets menu touches metering, customers, billing, work, identity, tenancy and integrations, and this document follows those threads only as far as the screens do. **And it deliberately covers six of the registry's seven tabs** — the Imports tab and the five-stage runner behind it left this document's scope by instruction on 2026-09-11, and are not described here at all. It is **as stale as the read it was made from**: version **2.234.17**, commit **79c659d7**, read 2026-09-10 and re-measured 2026-09-11, on a branch this pass **could not confirm is the product**.

> **What was deliberately left out, and why**
>
> **The Imports tab and the import runner** — by instruction, and named as gone rather than quietly absent. **Line numbers as a permanent record** — they move, so paths and values are cited and each line number is a pointer at one commit. **The ten non-English catalogues** — B14 rule 35 requires each to read as its own language rather than a translation, and judging that needs a speaker. **Load and performance measurement** — section 09 names the unbounded query and does not benchmark it. **The Work module's own screens** — screen 27 posts into Work and this document stops at the boundary. **A walk as a Billing Clerk or a read-only role** — every screen here was driven as a System Admin, and appendix F says what that costs. **The planning of this pass**, and the scratch files behind it.
>
> It is fine for those to be out. It is not fine for them to be out silently — which is the reason this section exists.

L · Every drop-down, and where its options come from

2 · Every drop-down, and where its options come from

A drawn drop-down is a box with a word in it. Nothing in the picture says whether that word is a row somebody can edit in Settings, a row in another module's table, or a string frozen in the source — and those are three different pieces of work, with three different owners and three different ways of going wrong. This table is the answer for every choice control in this document, and the same four markers ride on the fields themselves so the two cannot drift apart.

The four markers

`SET ▸ LIST_CODE`	**A tenant-editable reference list**, on **Settings › Lookups**. The words on screen are ROWS: a utility adds one without a deployment, and two tenants legitimately show different lists. Read by `GET /lookups/{code}` — **any tenant role may read, only a System Admin may write** (`SettingsReadAnyWriteSystemAdmin`, `core/permissions.py:87`). **Never hard-code these values and never assume an order.**
`TBL ▸ TABLE`	**Live rows from another table in the system** — premises, units, assets, users, work templates. It crosses a module seam, so it carries **that module's permissions and that module's paging**, neither of which this screen controls.
`FIX ▸ N`	**Frozen in the source.** Changing it is a code change and a release. Where the server holds the same set as an enum, the front-end constant is a MIRROR of it and the two must be changed together — `RETIRE_REASON_VALUES` says so in its own docstring.
`NONE ▸ …`	**No list exists anywhere**, or the product renders a free text box over a list that does exist. Drawing a picker here is **proposing one**, and that is a specification rather than a styling change.

Control	Source	Where the options actually come from	Screens
Asset type	`SET` `asset_types`	`GET /lookups/asset_types?per=100`. **It decides which registry tab the asset lands on** (`asset_types.category`, `BR-AST-004`) — so a new row here silently changes what the list screens contain. Owner declared `metering`, hosted in `apps/assets`.	01–06 filter · 07 quick-add · 16 · 28
Meter type	`SET` `meter_types`	`GET /lookups/meter_types?per=100`. What the meter **IS**. `BR-AST-043` split this from the read method on purpose; **no rate uses it**.	01 filter · 07 · 17 · 29
Read method	`SET` `read_methods`	`GET /lookups/read_methods?per=100`. How the reading is **COLLECTED**. ⚠ Metering reads the same list through its own route, `GET /reads/read-methods`, added for a role gate that has since widened — **two doors onto one list**, and a lane adding a row should know both exist.	07 · 17 · 29
Connection size	`SET` `service_size_classes`	`GET /lookups/service_size_classes?per=200` — **200 and not 100 deliberately**: forty rows were ratified and a tenant provisioned earlier carries an orphan on top. ⚠ **Every size class now names its service and this picker still does not filter by it** — a known, filed gap (`DEF-3300`), not a decision, so a fire asset can be offered an electric size.	17 · 29 · 32

Lifecycle status	SET `asset_statuses`	`GET /lookups/asset_statuses?per=100`, then **filtered to the legal transitions** for the stage the asset is in — the list is not the options. The server re-checks and its 409 governs.	01–06 filter · 23
Utility service	SET `settings/services`	`GET /settings/services` — **the tenant's ENABLED services, never the platform's eleven**. `AnyTenantRole` (`apps/tenancy/api.py:419`), so every staff role can be shown it. ⚠ **And the add wizard does not call it**: `service` is a plain `TextField` there, so the list exists and the form does not use it. That is `AST-B-03`.	01–06 filter · 28 · reassign
Unit of measure	SET `services · uom`	Carried per service on the same `GET /settings/services` payload — `primary_uom`, `alt_uom`, `billed_uom`. ⚠ **The add wizard draws a free text box**, so the unit that every quantity on this asset will carry is typed by hand against a list the tenant has already curated. Same shape as the service field, one field along.	17 · 29
Premises	TBL `premises`	`GET /premises?per=500` — metering's table, and **the one address home** (hard rule 8). ⚠ **Capped at 500 rows client-side**, which is a hard cap on a picker and is the shape `NFR-5` forbids: premises 501 cannot be chosen and nothing says so.	22 · 30
Unit	TBL `units`	`GET /units?per=500`. A unit refines a premises **by label**. On Reassign the honest list is *units without a live meter* — the server decides, and the picker does not pre-filter. Same 500 cap.	05b · 22 · 30
Parent master	TBL `assets · meters`	`GET /assets?tab=meters&per=500`, fetched **only when the class is not a meter**. `BR-AST-003`: a sub-meter cannot exist without one, and the server refuses one that arrives without it.	28 (sub-meter)
Replacement asset	TBL `assets · inventory`	The retire wizard's own picker, scoped to the same tab as the asset being retired. A retired asset offered as a replacement is refused by name.	25
Technician	TBL `users · identity`	`GET /users?per=100`. ⚠ **`SystemAdminOnly`** (`apps/identity/api.py:194`) while `BR-AST-038` lets a **Billing Clerk** log a repair — so for the one role this screen exists for the picker cannot load and the field falls back to typing a **UUID**. That is `AST-B-04`, and it breaks B14 rule 31.	23
Work order template	TBL `work_order_templates`	Work's table, already filtered server-side by **which asset types the template applies to** — so the list is short because the server made it short, not because the screen did.	27
Retire reason	FIX 4	`end_of_life · failed · replaced · removed` — `BR-AST-021`, the server's own set in the server's own order. The front-end constant mirrors it and says so; **changing one without the other is a 422 a user meets and no test does**.	24
Maintenance kind	FIX 3	`repair · inspection · test`. **Test is not cosmetic** — it is the one that satisfies a certification cadence.	23

Asset class	FIX 3	The three registry tabs (`BR-AST-004`). It is asked first because it decides what the rest of the form contains, and it is a tab set rather than a lookup for exactly that reason.	07 · 28 · 29
Destination	FIX 2	*To a unit* · *Back to inventory*. Both are real and the second is easy to miss reading the code — this document once reported it missing.	22 · 30
Treatment	FIX 2	*Write off* · *Carry forward*. Recorded on the asset. **Nothing posts to the ledger** (hard rule 5).	26
Test cadence	FIX 3	`12` · `24` · `60` months. ⚠ **A regulator's interval is not a constant**, and a tenant whose backflow rule says 36 months cannot express it without a release.	29 · 32
Depreciation method	NONE	**No list exists anywhere** — and neither does a capture site. The column is on the model, the API serves it, screens can display it, and nothing in the product can write it (`AST-B-27`). Drawing this control proposes both the field and the list behind it.	21 · 32 (both NEW)

What a coder should take from this table

Three of these carry a live defect and the marker is how you see it before writing the fetch: the connection-size picker does not filter by service though every row now names one; the premises and unit pickers stop at **500 rows**, which is the hard cap `NFR-5` exists to forbid; and the technician picker is gated to a role that is not the role the screen is for.

And two fields are drawn as pickers in this document while the product renders free text — Utility service and **Unit of measure**. Neither needs a new endpoint, a new column or a migration. Both lists are already served, to every staff role, by one call the screen does not yet make.